# Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

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**Abstract**

Refractive Gravity (RefG) develops a relational account of measurement in which clocks, rulers, light, and observers are physical constituents of the system being measured. A single pre-spatial foundation gives rise, through resonant self-distinction, to operational space, time, matter, and gravitational geometry. Stable massive matter is represented by locally self-confined nonlinear wave structures, or oscillons, whose dynamics establish pressure deficits in the manifested medium. In the low-energy covariant formulation, the isolated-system sector, the cosmological phase source, the strong-field common-scaling medium, and galactic $k$-essence dynamics are unified within a single regular Grand Unified Covariant Action combining relational-coframe teleparallel (TEGR) and Einstein–Hilbert geometry, the canonical complex oscillon field, the Maxwell field, the collective complex condensate field $\Psi_C = \sqrt{x}\, e^{i\theta_C}$, and the spatiotemporal medium deformation invariants ($\Phi \equiv -\frac{c_0}{\mu_{C0}}\theta_C$, $M_*^4 = \frac{a_0^2}{16\pi G}$). When the collective background field is unexcited ($\Psi_C \equiv 0$, $\Lambda_F = 0$), this unified action reduces identically and without singularity on any curved metric to the isolated-system action $S_{\rm iso}$; on the active condensate branch ($|\Psi_C|^2 > 0$), it recovers the barotropic cosmological source $T^C_{\mu\nu}$. In static isotropic geometry, the clock factor $p_T = N$ and the external coordinate footprint of a ruler $p_L = A_{\rm iso}^{-1}$ are governed by independent field equations, while coinciding at leading weak-field order ($p_T \simeq p_L \simeq p$) to recover the common-scaling $p/p^2$ law and the complete standard 1PN response, as well as exact gravitational-wave propagation on the common light cone ($v_{\rm gw} \equiv c_0$, GW170817), two transverse-traceless polarizations ($h_+, h_\times$), Peters–Mathews eccentric quadrupole factors $f(e)$ and $g(e)$ with zero dipole radiation ($\dot{P}_b^{\rm dipole} \equiv 0$), and precision orbital decay for PSR B1913+16 (obs/RefG $= 0.9983 \pm 0.0017$), PSR J0737-3039A/B (obs/RefG $= 0.999953 \pm 0.000063$, $0.74\sigma$), and PSR J1738+0333 ($0.53\sigma$). At the action level, the framework establishes a gravitating trapped-pocket regime and an exact two-charge first law and inertia theorem ($M_I = M_{\rm ADM}(Q, Q_C)$); a horizonless exponential strong-field exterior matched $C^2$-smoothly to an action-derived regular interior core supported by central matter (via the common-scale Ricci total-divergence theorem $\sqrt{-g}\, R \equiv 2\nabla^2 H$), together with its rotating Kerr-like frame-dragging law $\omega_+(r) = \frac{3a_* c_0}{2m}[1 - (1 + \frac{2m}{r} + \frac{2m^2}{r^2})e^{-2m/r}]$ ($C^2$-smoothly matched to a regular core free of ring singularities and closed timelike curves), exact golden-ratio innermost stable circular orbit $r^{(0)}_{\rm ISCO} = (3 + \sqrt{5})m = 2\phi^2 m \approx 5.2361\, m$ ($\eta^{(0)}_{\rm acc} = 5.479\%$, rising to $\eta^{\rm pro}_{\rm acc} = 11.99\%$ at $a_* = 0.9$), quantitative static and spinning EHT shadow diameter ($b_c^{\rm RefG} = 2em$, $\delta_{\rm sh} = +4.63\%$, with spinning shadows $d_{\rm sh} = 53.45\,\mu$as $[0.68\sigma]$ for Sgr A* and $40.76\,\mu$as $[0.41\sigma]$ for M87*), eikonal quasinormal-mode quality-factor invariance ($\Omega^{\rm RefG}_{\rm ph} = \lambda^{\rm RefG}_{\rm L} = \frac{c_0}{2em} \Rightarrow Q^{\rm RefG}_{\ell n} \equiv Q^{\rm GR}_{\ell n}$), finite tortoise-cavity gravitational-wave echo delay ($\mathcal{I}_{\rm echo}(3/2) = 15.5553$, $\Delta t_{\rm echo} = 4.750$ ms for the GW150914 remnant), dynamical collapse arrest at $R = r_c$ with strictly positive restoring stiffness $U''_{\rm eff}(1) = q(1+q) > 0$ ($\omega_{\rm bounce} = \sqrt{q(1+q)}\, c_0/r_c$), universal boundary equation-of-state and sound-speed fixation of the canonical compactness $q^* = 2m/r_c = 3/2$ ($r_c = \frac{4}{3}m = \frac{2}{3}r^+_{\rm ph}$, $n_{\rm min} = 3$, $f^{(\rm m)}_{\rm core}(3/2) = 5.863\%$) from

causal Zel'dovich stiff-fluid saturation $\bar{w}_{\rm m}(1^-) = \frac{c^2_{s,\rm eff}(1^-)}{c_0^2} = \frac{q}{3(2-q)} = 1$, and a zero-free-parameter $SU(3)_c$ Y-junction nucleon equation of state ($a_2 = \frac{m_\Delta - m_N}{9} = 32.564$ MeV, $a_3 = \frac{a_2}{\sqrt{3}} = 18.801$ MeV, $c_3 = \frac{\alpha_s}{3} = 0.09840$) that yields the Tolman–Oppenheimer–Volkoff (TOV) maximum neutron-star mass $M_{\rm TOV}^{\rm max} = 2.1856\, M_\odot$ ($R_{1.4} = 12.57$ km [$0.18\sigma$], $R_{2.08} = 11.84$ km [$0.56\sigma$], tidal deformability $\Lambda_{1.4} = 586.9 \le 720$) and geometric Y-junction percolation ($r_{\rm touch} = 0.5004$ fm $\approx L_Y^{(N)} = 0.5150$ fm) into the horizonless core; a parameter-free cosmological derivation of the galactic acceleration scale $a_0^{(\rm RefG)} = \frac{3}{5\pi} c_0 H_0 \approx 1.2506 \times 10^{-10}$ m/s$^2$ together with a zero-parameter rational law $\mu_{\rm RefG}(x) = \frac{x+x^2}{1+x+x^2}$ that reproduces the Baryonic Tully–Fisher Relation and weak gravitational lensing while screening anomalous accelerations in the Solar System; and a potential-depth threshold mechanism ($Q_C$) that yields an analytical $n = 1$ Lane–Emden condensate core in galaxy clusters and accounts for the weak-lensing separation in the Bullet Cluster; an exact analytical and numerical derivation on the operational $A = a/p = a^{5/2}$ cosmological geometry of primordial nucleosynthesis (BBN) coframe clock invariance ($dX_n/dA = [\Gamma^{(0)}_{n\leftrightarrow p}(X_n^{\rm eq} - X_n) - X_n/\tau_n]/(AH_{\rm op})$, $\partial(dX_n/dA)/\partial p \equiv 0$, $\Delta N_{\rm eff}^{\rm RefG} \equiv 0$, $Y_p = 0.2456$, $({\rm D/H})_p = 2.512 \times 10^{-5}$), the recombination sound horizon ($r_s(z_*) = 144.29$ Mpc), the drag-epoch BAO standard ruler ($r_d = 146.94$ Mpc $= 98.98\, h^{-1}$ Mpc $\approx 100\, h^{-1}$ Mpc), the CMB acoustic scale and first three peaks ($100\theta_* = 1.0404$, $\ell_A = 301.97$, $\ell_1 = 220.8, \ell_2 = 538.0, \ell_3 = 811.6$), Thomson scattering and Silk diffusion damping coframe invariance ($\partial\dot{\kappa}/\partial p = \partial(dk_D^{-2}/dA)/\partial p \equiv 0$, $k_{D,\rm eff} = 0.1298$ Mpc$^{-1}$, $\ell_{D,\rm eff} = 1800.3$), baryon-loaded $TT/TE/EE$ angular power spectra ($R_{12} = \mathcal{D}^{TT}_{\ell_1}/\mathcal{D}^{TT}_{\ell_2} = 2.2616$, $R_{32} = 0.9899$, first $TE$ adiabatic anticorrelation trough at $\ell_{TE,\rm tr,1} = 164.5$), the Mészáros matter transfer function $T(k)$, the linear matter power spectrum $P(k,0)$ ($k_{\rm eq} = 0.01039$ Mpc$^{-1} = 0.0154\, h\,$Mpc$^{-1}$, turnover peak $k_{\rm peak} = 0.0167\, h\,$Mpc$^{-1}$), and the zero-free-late-parameter clustering amplitudes $\sigma_8^{\rm RefG} = 0.8098$ ($0.22\sigma$) and $S_8^{\rm RefG} = 0.8300$ ($0.15\sigma$), the Gauss hypergeometric solution for sub-horizon linear structure growth ($D_+(A) = A\, {}_2F_1(\frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}} A^3)$, growth index $\gamma(\Omega_m) = \frac{6}{11} + \frac{15}{2057}(1 - \Omega_m)$, $f\sigma_8(0) = 0.4270$), and the quantitative process-time galaxy assembly model at $z \in [8, 20]$ ($\tau_{\rm mat}(z)/t_{\rm mat}(z) = \frac{5}{3}(1+z)^{3/5}$, $\Delta\tau(20 \to 10) = 1.461$ Gyr vs. $\Delta t = 292.6$ Myr, assembling $M_* = 10^{10}\, M_\odot$ at $z = 10$ with normal baryon efficiency $\epsilon_*^{\rm RefG} = 0.218 < 0.25$); as well as the emergence of continuum 3+1 Einstein–Hilbert gravity from the discrete 3D nodal graph $\Gamma = \mathbb{Z}^3$ via Sakharov–Seeley–DeWitt spectral heat-kernel induction ($\ell_* = 2\sqrt{\pi}\, \ell_{\rm Pl} \approx 3.5449\, \ell_{\rm Pl}$, with exact odd-order LIV cancellation $\delta_1^{\rm LIV} \equiv 0$ and $|\Delta c/c|_{100\ \rm TeV} = 1.05 \times 10^{-28} \ll 10^{-17}$), exact Bogoliubov–de Gennes (BdG) spectral factorization of oscillon normal modes on any static isotropic coframe ($\nu_n^{(t)}(p_T, p_L; Q) = s_n(Q)\, \Omega_t(p_T; Q)$ with $\partial s_n/\partial p_T = \partial s_n/\partial p_L \equiv 0$), local $U(1)$ gauge coupling ($J^\mu_{\rm em} = q_{\rm em} J^\mu_O$, $\left.\frac{dE_{\rm tot}}{dQ}\right|_{q_{\rm em}} = \Omega_\infty$) with screened $1/r^2$ Coulomb flux, exact coframe invariance of the vacuum wave impedance and fine-structure constant ($Z_{\rm eff} = Z_0$, $\alpha(p_T, p_L) = \alpha_0 = \frac{Z_0}{2R_K}$) together with its topological-geometric impedance derivation ($\alpha^{-1}_{\rm geom} = 4\pi^3 + \pi^2 + \pi = 137.036304$, $\alpha^{-1}_{\rm RefG} = 137.035999092$ [0.061 ppb, $0.40\sigma$], electroweak running $\hat{\alpha}^{-1}_{\overline{\rm MS}}(m_Z) = 127.951$ [$0.024\sigma$]), the 2-loop topological instanton action and parameter-free derivation of Newton's gravitational constant ($S_{\rm inst}^{\rm RefG} = \frac{3}{8}\alpha^{-1}_{\rm RefG} + \frac{7}{16\pi} + \frac{1}{9\pi^2 \alpha^{-1}_{\rm RefG}} = 51.527842$, $G_{\rm RefG} = \frac{\hbar c_0}{m_e^2} e^{-2S_{\rm inst}^{\rm RefG}} = 6.674268 \times 10^{-11}$ m$^3$ kg$^{-1}$ s$^{-2}$ [$-0.21\sigma$, 4.78 ppm]), the Spin$(3,1)$ double cover and Möbius $Z_2$ holonomy ($U(2\pi) = -I_4$, $U(4\pi) = +I_4$, $\mathbf{S}^2 = \frac{3}{4}\hbar^2 I_4$, $: H_D :_{\rm Fermi} \ge 0$), the tree-level Dirac gyromagnetic factor $g_0 = 2$ and exact coframe clock transposition of Dirac and Larmor frequencies ($\omega_D^{(t)} = p_T \omega_{D,0}$, $\omega_L^{(t)} = p_T \omega_{L,0}$), the Compton-self-similar core 1-loop QED cancellation theorem and 2-loop Koide $C_3$ radiative pole-mass bridge ($\Delta\theta_{\rm exp}/\theta_0 = -c_2(\alpha_0/\pi)^2$, $c_2 = 0.14606 \approx \frac{4}{27} = \theta_0 K_0$), and the tri-axial $\mathfrak{su}(3)_c$ Gell-Mann color algebra ($C_2(\mathbf{3}) = \frac{4}{3}$, $C_2(\mathbf{8}) = 3$, $C_2(\varepsilon_{abc} q^a q^b q^c) = 0$) with the Cornell flux-tube confinement potential $V_{\rm Cornell}(r) = -\frac{4}{3}\frac{\alpha_s \hbar c_0}{r} + \sigma_{\rm str} r$, strong coupling and QCD scales ($\alpha_s(m_Z) = \theta_d(1 + \frac{1}{16}) = \frac{17}{144} = 0.118056$ [$+0.062\sigma$], $\Lambda^{(5)}_{\overline{\rm MS}} = 209.34$ MeV $\approx K_0 \mu_\ell^2 = 209.24$ MeV [0.048%], $\sqrt{\sigma_{\rm str}^{\rm RefG}} = \frac{73}{54}\mu_\ell^2 = 424.29$ MeV $\Rightarrow \sigma_{\rm str} = 0.18002$ GeV$^2$ [0.011%]), unique anomaly-cancelled fractional quark charges ($Q_u = +2/3, Q_d = -1/3$), the 120° Fermat–Torricelli–Steiner Y-junction baryon mass origin (99.04% field energy vs. 0.96% valence quark rest mass, $M_0 = 1085.46$ MeV, $m_\Delta - m_N = +293.08$ MeV), $SU(6)$ nucleon magnetic moments and axial coupling ($\mu_n/\mu_p = -2/3$, $g_A^{\rm rel} = 1.2716$), QCD+QED neutron–proton mass splitting ($\Delta m_{np} = +1.29333$ MeV $> m_e$), electroweak $SU(2)_L \times U(1)_Y$ / neutron beta-decay closure ($\det M^2_{\rm neutral} \equiv 0 \Rightarrow m_\gamma^2 \equiv 0$, $f_0(\epsilon_0) = 1.6361$, $\tau_n^{\rm pred} = 879.9$ s), the harmonic $C_3$ holonomy sequence ($\theta_\ell = \frac{2}{9}, \theta_d = \frac{1}{9}, \theta_u = \frac{2}{27}$) that yields the six quark masses ($m_u = 2.47$ MeV$, m_d = 4.54$ MeV$, m_s = 94.0$ MeV$, m_c = 1.266$ GeV$, m_b = 4.180$ GeV$, m_t = 172.67$ GeV, $K_d = \frac{79}{108}, K_u = \frac{275}{324}$), the unitary $3 \times 3$ CKM mixing matrix

($\theta_{12}^{\mathrm{CKM}} = \frac{55}{243}, |V_{cb}| = \frac{2}{27\sqrt{3}}, |V_{ub}| = \frac{8}{2187}, \delta_{\mathrm{CP}}^{\mathrm{CKM}} = \frac{\pi}{3} + \frac{1}{9}, J_{\mathrm{CP}}^{\mathrm{CKM}} = 3.13 \times 10^{-5}$ [$0.09\sigma$]), the PMNS leptonic mixing matrix ($\sin\theta_{13}^{\mathrm{PMNS}} = \frac{4}{27}, \sin^2\theta_{12}^{\mathrm{PMNS}} = \frac{25}{81}, \sin^2\theta_{23}^{\mathrm{PMNS}} = \frac{89}{162}, \delta_{\mathrm{CP}}^{\mathrm{PMNS}} = \pi + \frac{2}{9}$), the 11th-order $C_3$ holonomy neutrino mass hierarchy ($\mu_\nu \simeq 3^{-11}\mu_\ell$, $\Delta m_{21}^2 = 7.46 \times 10^{-5}$ eV$^2$, $\Delta m_{31}^2 = 2.53 \times 10^{-3}$ eV$^2$, $\sum m_\nu = 0.0593$ eV $< 0.072$ eV), and the 137-e-fold ($\frac{3}{8} + \frac{5}{8} = 1$) Planck-to-BAO spectral bridge ($\ln(r_d/\ell_{\mathrm{Pl}}) = 136.88 \approx \alpha_{\mathrm{RefG}}^{-1}$). On a fixed Minkowski background, canonical quantization of the quadratic oscillon sector about its strict vacuum yields positive-energy particle and antiparticle states in bosonic Fock space with mass $m$ and spin zero. In the cosmological description, increasing inter-node separation lowers the background pressure and material scale, and internal measurements determine the operational scale factor $A = a/p$. This monograph presents the conceptual architecture and physical content of the theory; the final section summarizes the established results, their domains of validity, and the open problems of the research program.

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# Preface

Refractive Gravity (RefG) takes a relational principle of measurement as its starting point: clocks, rulers, light, and observers belong to the same physical world, and measurements are made using material standards within that world. On this basis, the monograph develops a unified ontological account of a single foundation and the geometry, matter, and time that emerge from it. In the specified low-energy limit, the same relational coframe supports Einstein–Hilbert geometry, while the quadratic sector of the oscillon action about its strict vacuum is a free complex scalar quantum field on that coframe. The text presents the conceptual analysis and physical interpretation; the published formalism and calculations are available in the public scientific archive on Zenodo [Lotishvili, 2026a,b].

The subjects addressed in this monograph are at different stages of development. Some results have been derived and tested formally; others are specified physical constructions or directions for further mathematical development within the same framework. The final section, *Current Status of the Research Program*, summarizes the results, their conditions of validity, and the unresolved problems.

# Introduction

General relativity and quantum field theory are the two principal frameworks of modern fundamental physics. Einstein's general relativity describes gravity as the dynamical geometry of spacetime. Quantum field theory usually describes matter and gauge interactions through operator-valued fields on a prescribed background.

Both theories have extensive experimental support, but they use different underlying structures. Spacetime geometry is dynamical in general relativity, whereas the usual formulation of quantum field theory assumes a specified background geometry and time parameter.

Several unresolved problems bear on the construction of a unified description:

1. **The Singularity Problem:** Classical descriptions of black holes and the cosmological origin lead to geodesic incompleteness and, in important cases, divergent curvature, indicating a limit of validity of the classical description.

2. **Astrophysical and Cosmological Anomalies:** Within the standard cosmological model, galactic rotation curves, the Radial Acceleration Relation (RAR), and the Baryonic Tully–Fisher Relation (BTFR) are described through dark-matter halos together with baryonic galaxy-formation physics, while accelerated cosmic expansion is attributed to a vacuum-energy cosmological-constant term. A naive continuation of field theory to a Planck-scale cutoff may exceed the observed vacuum density by roughly 120 orders of magnitude.

3. **Structural and Hierarchical Problems:** Quantum-measurement randomness and the arrow of time lack a unified structural explanation. The pronounced hierarchy of charged-lepton and quark masses, the values of dimensionless constants—including $\alpha \approx 1/137$—and the weakness of gravity relative to the electroweak scale ($M_{\mathrm{Pl}}/v_{\mathrm{EW}} \sim 10^{17}$) enter the standard framework as empirical features without a common explanation.

The search for a common physical foundation has taken several forms. Einstein pursued a geometric unification of electromagnetism and matter throughout his later life. John Archibald Wheeler investigated the relations among geometry, matter, and information through the concepts of *Mass without Mass* and *It from Bit* [Wheeler, 1955, 1990].

Andrei Sakharov proposed an elastic interpretation of gravity induced by quantum vacuum fluctuations [Sakharov, 1967]. Ted Jacobson derived Einstein's equations from a thermodynamic equation of state for local horizons [Jacobson, 1995]. Erik Verlinde described gravitational attraction as a holographic entropic force [Verlinde, 2011], while Roger Penrose related the causal structure of spacetime to gravitational collapse through his singularity theorem [Penrose, 1965].

These approaches use different mathematical formalisms. The ontological question examined here is whether space, time, gravity, and the particle spectrum can be described as dynamical manifestations of a single underlying foundation.

The monograph develops the conceptual and ontological framework of **Refractive Gravity** (RefG) to address this question.

## Operational Foundations and Three Fundamental Principles

Established physical formalisms generally assume specified measurement procedures and structures, including smooth coordinates, a time axis, and field operators. RefG extends Percy Bridgman's operationalism [Bridgman, 1927] by examining the physical prerequisites of measurement itself: the processes that establish local distinguishability and thereby make material rulers and clocks possible.

In 1905, Einstein rejected absolute time and defined simultaneity operationally using clocks and light signals [Einstein, 1905]. RefG seeks to describe the physical mechanism that produces the stable clock, material particle, and propagating signal on which such measurements depend.

The framework is based on three fundamental postulates:

1. **Monistic Ontological Foundation:** Nature has one universal foundation from which spatial relations and physical timekeeping emerge. Its resonant self-distinction produces a manifested medium whose phase, pressure, shear, rotational, and topological modes correspond to elementary fields, forces, and particles.

2. **The Oscillon Nature of Matter:** A stable massive material unit is an *oscillon*: a locally self-confined, self-sustaining nonlinear wave configuration of the manifested medium. Massless carrier modes are freely propagating waves in the same medium. Intense internal oscillation establishes a persistent static pressure-deficit profile around the oscillon.

3. **Gravity as Effective Refraction:** Oscillon matter establishes an inhomogeneous pressure deficit in the medium. The deficit changes the effective geometry, in which light and matter waves undergo refractive deflection. RefG thus assigns a physical refractive mechanism to gravitational curvature. In the relational-coframe description, the pressure deficit is the volumetric scalar readout of the full local geometry. Symmetric deformation and shear determine the tensor structure of gravity, while frame orientation is an inertial gauge freedom.

## Lorentz Invariance and Historical Precedents

The Michelson–Morley experiment of 1887 detected no aether wind associated with the Earth's motion and placed stringent constraints on 19th-century models of a stationary luminiferous aether independent of matter [Michelson and Morley, 1887]. The RefG foundation-medium belongs to a different ontological class: measuring instruments, observers, and light are collective wave excitations of the same medium.

Because observers, apparatus, and light are collective excitations of the same medium, uniform translational motion is described as a symmetric phase translation of the wave configuration. On the Lorentz-covariant branch, this motion is lossless and produces no aether-wind effect. The conceptual connection to Lord Kelvin's vortex-atom hypothesis [Thomson, 1867] is the representation of **matter as an organized form of motion in the medium itself**.

## Eight Principal Topics

This text sets out the ontological foundations of RefG and connects the mathematical formalism archived on Zenodo with its physical content. Medium-based unification models have well-known predecessors, including Friedwardt Winterberg's Planck-aether model [Winterberg, 2002]. The distinctive starting points of RefG are its account of the emergence of measurability itself and the population-level resonant stabilization of matter.

In the present version of this work, the low-energy descriptions of isolated systems, the cosmological phase source, the strong-field common-scaling medium, and galactic $k$-essence dynamics are unified within a single regular Grand Unified Covariant Action combining the full relational-coframe teleparallel (TEGR) and Einstein–Hilbert geometry, the canonical complex oscillon field, the canonical Maxwell field, the collective complex condensate field $\Psi_C = \sqrt{x}\, e^{i\theta_C}$, and the spatiotemporal deformation invariants of the foundation ($\Phi \equiv -\frac{c_0}{\mu_{C0}}\theta_C$, $M_*^4 = \frac{a_0^2}{16\pi G}$). When the collective background field is unexcited ($\Psi_C \equiv 0$, $\Lambda_F = 0$), the unified action reduces identically and without singularity on any curved metric to the isolated-system action $S_{\rm iso}$; on the active condensate branch ($|\Psi_C|^2 > 0$), it exactly recovers the barotropic cosmological source $T^C_{\mu\nu}$. The coordinate responses of clocks and rulers are governed by the independent geometric components of this action. In parallel, the action-level analysis establishes the gravitating trapped-pocket regime and the two-charge inertia theorem for an oscillon coupled to the ambient medium; exact gravitational-wave propagation on the common light cone ($v_{\rm gw} \equiv c_0$, GW170817), two transverse-traceless polarizations ($h_+, h_\times$), Peters–Mathews eccentric quadrupole factors ($f(e), g(e)$ with $\dot{P}_b^{\rm dipole} \equiv 0$), and orbital decay of binary pulsars (PSR B1913+16, PSR J0737-3039A/B [$0.74\sigma$], PSR J1738+0333 [$0.53\sigma$]); the boundary-value closure of an action-derived $C^2$ regular core matched smoothly to the horizonless exponential strong-field exterior metric (via the common-scale Ricci total-divergence theorem $\sqrt{-g}\,R \equiv 2\nabla^2 H$), together with its rotating Kerr-like frame-dragging law $\omega_+(r) = \frac{3a_* c_0}{2m}[1 - (1 + \frac{2m}{r} + \frac{2m^2}{r^2})e^{-2m/r}]$ ($C^2$-smoothly matched to a regular core free of ring singularities and closed timelike curves), exact golden-ratio innermost stable circular orbit $r^{(0)}_{\rm ISCO} = (3+\sqrt{5})m = 2\phi^2 m \approx 5.2361\, m$ ($\eta^{(0)}_{\rm acc} = 5.479\%$, rising to $\eta^{\rm pro}_{\rm acc} = 11.99\%$ at $a_* = 0.9$), quantitative static and spinning EHT shadow diameter ($b_c^{\rm RefG} = 2em$, $\delta_{\rm sh} = +4.63\%$, with spinning shadows $d_{\rm sh} = 53.45\,\mu$as [$0.68\sigma$] for Sgr A* and $40.76\,\mu$as [$0.41\sigma$] for M87*), eikonal quasinormal-mode quality-factor invariance ($\Omega^{\rm RefG}_{\rm ph} = \lambda^{\rm RefG}_{\rm L} = \frac{c_0}{2em} \Rightarrow Q^{\rm RefG}_{\ell n} \equiv Q^{\rm GR}_{\ell n}$), finite tortoise-cavity gravitational-wave echo delay ($\mathcal{I}_{\rm echo}(3/2) = 15.5553$, $\Delta t_{\rm echo} = 4.750$ ms for GW150914), dynamical collapse arrest at $R = r_c$ with positive restoring stiffness $U''_{\rm eff}(1) = q(1+q) > 0$, causal boundary stiff-fluid fixation of the canonical compactness $q^* = 2m/r_c = 3/2$ ($r_c = \frac{4}{3}m$, $n_{\rm min} = 3$, $f^{(\rm m)}_{\rm core}(3/2) = 5.863\%$) from $\bar{w}_{\rm m}(1^-) = \frac{c^2_{s,\rm eff}(1^-)}{c_0^2} = \frac{q}{3(2-q)} = 1$, and the zero-free-parameter $SU(3)_c$ Y-junction nucleon equation of state ($M^{\rm max}_{\rm TOV} = 2.1856\, M_\odot$, $R_{1.4} = 12.57$ km, $\Lambda_{1.4} = 586.9$, and Y-junction percolation $r_{\rm touch} \approx L_Y^{(N)}$ at the TOV threshold); the parameter-free cosmological derivation of the galactic acceleration scale $a_0^{(\rm RefG)} = \frac{3}{5\pi} c_0 H_0 \approx 1.2506 \times 10^{-10}$ m/s$^2$ and the zero-parameter rational law $\mu_{\rm RefG}(x) = \frac{x+x^2}{1+x+x^2}$ with Solar System screening; the analytical potential-depth threshold ($Q_C$) condensate core model for galaxy clusters and the Bullet Cluster; the exact analytical and numerical derivation on the operational $A = a/p = a^{5/2}$ cosmological geometry of primordial nucleosynthesis (BBN) coframe clock invariance ($\partial(dX_n/dA)/\partial p \equiv 0$, $\Delta N^{\rm RefG}_{\rm eff} \equiv 0$, $Y_p = 0.2456$, $({\rm D/H})_p = 2.512 \times 10^{-5}$), the recombination sound horizon ($r_s(z_*) = 144.29$ Mpc), the BAO standard ruler ($r_d = 146.94$ Mpc $= 98.98\, h^{-1}$ Mpc $\approx 100\, h^{-1}$ Mpc), the CMB acoustic scale and peaks ($100\theta_* = 1.0404$, $\ell_A = 301.97$, $\ell_1 = 220.8, \ell_2 = 538.0, \ell_3 = 811.6$), Thomson and Silk diffusion damping coframe invariance ($\partial\dot{\kappa}/\partial p = \partial(dk_D^{-2}/dA)/\partial p \equiv 0$, $k_{D,\rm eff} = 0.1298$ Mpc$^{-1}$, $\ell_{D,\rm eff} = 1800.3$), baryon-loaded $TT/TE/EE$ spectra ($R_{12} = 2.2616$, $R_{32} = 0.9899$, $\ell_{TE,\rm tr,1} = 164.5$), the Mészáros transfer function $T(k)$ and linear matter power spectrum $P(k,0)$ ($k_{\rm eq} = 0.0154\, h\,$Mpc$^{-1}$, $k_{\rm peak} = 0.0167\, h\,$Mpc$^{-1}$, $\sigma_8^{\rm RefG} = 0.8098$ [$0.22\sigma$], $S_8^{\rm RefG} = 0.8300$ [$0.15\sigma$]), the Gauss hypergeometric solution for sub-horizon linear structure growth ($D_+(A) = A\,{}_2F_1(\frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}}A^3)$, $\gamma(\Omega_m) = \frac{6}{11} + \frac{15}{2057}(1-\Omega_m)$, $f\sigma_8(0) = 0.4270$), and high-redshift JWST galaxy assembly ($\tau_{\rm mat}(z)/t_{\rm mat}(z) = \frac{5}{3}(1+z)^{3/5}$, $\epsilon_*^{\rm RefG}(z=10) = 0.218 < 0.25$); the

emergence of continuum $3+1$ Einstein–Hilbert gravity from the discrete 3D nodal graph $\Gamma = \mathbb{Z}^3$ via Sakharov–Seeley–DeWitt spectral heat-kernel induction ($\ell_* = 2\sqrt{\pi}\,\ell_{\rm Pl} \approx 3.5449\,\ell_{\rm Pl}$, with exact odd-order LIV cancellation $\delta_1^{\rm LIV} \equiv 0$ and $|\Delta c/c|_{100\ \rm TeV} = 1.05 \times 10^{-28} \ll 10^{-17}$); the exact coframe factorization of the oscillon Bogoliubov–de Gennes (BdG) perturbation spectrum ($\nu_n^{(t)}(p_T, p_L; Q) = s_n(Q)\,\Omega_t(p_T; Q)$), local $U(1)$ gauge coupling ($J^\mu_{\rm em} = q_{\rm em} J^\mu_O$, $\left.\frac{dE_{\rm tot}}{dQ}\right|_{q_{\rm em}} = \Omega_\infty$), and exact coframe invariance of the vacuum wave impedance and fine-structure constant ($Z_{\rm eff} = Z_0$, $\alpha(p_T, p_L) = \alpha_0 = \frac{Z_0}{2R_K}$) together with its topological-geometric impedance derivation ($\alpha^{-1}_{\rm geom} = 4\pi^3 + \pi^2 + \pi = 137.036304$, $\alpha^{-1}_{\rm RefG} = 137.035999092$, $\hat{\alpha}^{-1}_{\overline{\rm MS}}(m_Z) = 127.951$) and 2-loop topological instanton derivation of Newton's gravitational constant ($S^{\rm RefG}_{\rm inst} = \frac{3}{8}\alpha^{-1}_{\rm RefG} + \frac{7}{16\pi} + \frac{1}{9\pi^2\alpha^{-1}_{\rm RefG}} = 51.527842$, $G_{\rm RefG} = \frac{\hbar c_0}{m_e^2} e^{-2S^{\rm RefG}_{\rm inst}} = 6.674268 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $-0.21\sigma$, 4.78 ppm); as well as the Spin$(3,1)$ double cover and Möbius $Z_2$ holonomy ($U(2\pi) = -I_4$, $U(4\pi) = +I_4$, $\mathbf{S}^2 = \frac{3}{4}\hbar^2 I_4$, $: H_D :_{\rm Fermi} \geq 0$), the tree-level Dirac gyromagnetic factor $g_0 = 2$ and exact coframe clock transposition of Dirac and Larmor frequencies ($\omega_D^{(t)} = p_T\omega_{D,0}$, $\omega_L^{(t)} = p_T\omega_{L,0}$), the Compton-self-similar core 1-loop QED cancellation theorem and 2-loop Koide $C_3$ radiative pole-mass bridge, and the tri-axial $\mathfrak{su}(3)_c$ Gell-Mann color algebra ($C_2(\mathbf{3}) = \frac{4}{3}$, $C_2(\mathbf{8}) = 3$, $C_2(\varepsilon_{abc} q^a q^b q^c) = 0$, Cornell potential $V_{\rm Cornell}(r) = -\frac{4}{3}\frac{\alpha_s \hbar c_0}{r} + \sigma_{\rm str} r$), strong coupling and QCD scales ($\alpha_s(m_Z) = \theta_d(1 + \frac{1}{16}) = \frac{17}{144} = 0.118056$ $[+0.062\sigma]$, $\Lambda^{(5)}_{\overline{\rm MS}} = 209.34$ MeV $\approx K_0 \mu_\ell^2 = 209.24$ MeV [0.048%], $\sqrt{\sigma^{\rm RefG}_{\rm str}} = \frac{73}{54}\mu_\ell^2 = 424.29$ MeV $\Rightarrow \sigma_{\rm str} = 0.18002\ \mathrm{GeV}^2$ [0.011%]), unique anomaly-cancelled fractional quark charges ($Q_u = +2/3, Q_d = -1/3$), 120° Y-junction baryon mass origin (99.04% field energy, $M_0 = 1085.46$ MeV, $m_\Delta - m_N = +293.08$ MeV), $SU(6)$ nucleon magnetic moments and axial coupling ($\mu_n/\mu_p = -2/3$, $g_A^{\rm rel} = 1.2716$), neutron–proton mass splitting ($\Delta m_{np} = +1.29333$ MeV $> m_e$), electroweak $SU(2)_L \times U(1)_Y$ / neutron beta-decay closure ($\det M^2_{\rm neutral} \equiv 0 \Rightarrow m_\gamma^2 \equiv 0$, $f_0(\epsilon_0) = 1.6361$, $\tau_n^{\rm pred} = 879.9$ s), the harmonic $C_3$ holonomy sequence ($\theta_\ell = \frac{2}{9}, \theta_d = \frac{1}{9}, \theta_u = \frac{2}{27}$) yielding the six quark masses ($m_u = 2.47$ MeV$, m_d = 4.54$ MeV$, m_s = 94.0$ MeV$, m_c = 1.266$ GeV$, m_b = 4.180$ GeV$, m_t = 172.67$ GeV, $K_d = \frac{79}{108}, K_u = \frac{275}{324}$), the unitary $3 \times 3$ CKM mixing matrix ($\theta_{12}^{\rm CKM} = \frac{55}{243}, |V_{cb}| = \frac{2}{27\sqrt{3}}, |V_{ub}| = \frac{8}{2187}, \delta_{\rm CP}^{\rm CKM} = \frac{\pi}{3} + \frac{1}{9}, J_{\rm CP}^{\rm CKM} = 3.13 \times 10^{-5}$ $[0.09\sigma]$), the PMNS leptonic mixing matrix ($\sin\theta_{13}^{\rm PMNS} = \frac{4}{27}, \sin^2\theta_{12}^{\rm PMNS} = \frac{25}{81}, \sin^2\theta_{23}^{\rm PMNS} = \frac{89}{162}, \delta_{\rm CP}^{\rm PMNS} = \pi + \frac{2}{9}$), the 11th-order $C_3$ holonomy neutrino mass hierarchy ($\mu_\nu \simeq 3^{-11}\mu_\ell$, $\Delta m^2_{21} = 7.46 \times 10^{-5}$ eV$^2$, $\Delta m^2_{31} = 2.53 \times 10^{-3}$ eV$^2$, $\sum m_\nu = 0.0593$ eV $< 0.072$ eV), and the 137-e-fold ($\frac{3}{8} + \frac{5}{8} = 1$) Planck-to-BAO spectral bridge.

The monograph examines eight interconnected topics in sequence:

1. **The Emergence of Time and a Common Oscillon Clock Rate:** In the Genesis scenario of RefG, time begins with a centerless transition encompassing the whole foundation. The first distinguishable relations establish an ordering of events; oscillon seeds and subsequently stable matter form within the resulting network. Local pressure changes rescale the particle spectrum through the common frequency scale $\Omega_t$, while the oscillon's internal structure is preserved in the dimensionless spectral labels $s_i$ and their invariant ratios.

2. **Mass and Energy Accounting:** The oscillon's proper internal energy, its redshifted energy in a static gravitational environment, and the asymptotic mass of the entire system are distinct physical quantities. The total mass of the system combines the proper matter contribution, gravitational binding energy, and radiative balance. Keeping these quantities distinct prevents counting the same energy twice.

3. **Geometric Clock–Ruler Response and Complete 1PN Compatibility:** In static isotropic geometry, the clock factor is $p_T = N$, the external coordinate footprint of a ruler is $p_L = A_{\rm iso}^{-1}$, and the coordinate speed of light equals $c_0 p_T p_L$. At leading weak-field order in the exterior region, these two factors coincide ($p_T \simeq p_L \simeq p$) and recover the common-scaling $p/p^2$ law. The full coframe teleparallel (TEGR) action connects directly to the Einstein–Hilbert action through the torsion–curvature identity; under the corresponding source and boundary conditions, the complete

standard first post-Newtonian (1PN) response follows. The exact common-$p$ and exponential static profiles belong to a distinct branch of the theory.

4. **One Coframe for Gravity and Free Quantum Matter:** The vacuum quadratic sector of the nonlinear oscillon amplitude–phase action takes exactly the form of the free complex Klein–Gordon action on the same relational coframe. On the fixed Minkowski branch, standard canonical quantization of this quadratic sector yields positive-energy particle and antiparticle sectors of bosonic Fock space, a conserved global phase charge, microcausality, mass $m$, and spin zero for the free excitations about that vacuum. Moreover, on the same coframe, the Clifford algebra and Dirac–Fock–Ivanenko spinor action generate the exact $\mathrm{Spin}(3,1)$ double cover, $4\pi$ Möbius holonomy, fermionic Fock-space positivity, the exact tree-level gyromagnetic factor $g_0 = 2$, and exact $p_T = N$ clock transposition of Dirac and Larmor frequencies. Einstein–Hilbert geometry and standard bosonic and fermionic quantum matter are thereby placed on one operational geometry.

5. **The Koide $C_3$ Phase Geometry, Radiative Pole-Mass Bridge, and Quark/Lepton Flavor Mixing (CKM and PMNS):** A three-phase $C_3$-circulant square-root mass operator for the charged leptons $(e, \mu, \tau)$ reproduces Koide's $K(A_K) = (2 + A_K^2)/6 = 2/3$ identity exactly when singlet–doublet energy equipartition fixes $A_K = \sqrt{2}$, while the $h = 2$ branch of a ninth-order return cycle fixes the tree-level phase angle $\theta_0 = 2/9$. For a Compton-self-similar oscillon core ($R_{\mathrm{core},k} = \xi_0 \hbar/(m_k c_0)$), the 1-loop QED self-energy cancels identically across generations ($\partial Z_1/\partial m_k \equiv 0$), so the shift between $\theta_0 = 2/9$ and the empirical pole angle $\theta_{\mathrm{exp}}$ is of two-loop QED order: $\Delta\theta_{\mathrm{exp}}/\theta_0 = -c_2(\alpha_0/\pi)^2$ ($c_2 = 0.14606 \approx \frac{4}{27} = \theta_0 K_0$). Extending the same $C_3$ circulant operator to the harmonic holonomy sequence $\theta_\ell = \frac{\theta_0}{1} = \frac{2}{9}$, $\theta_d = \frac{\theta_0}{2} = \frac{1}{9}$, $\theta_u = \frac{\theta_0}{3} = \frac{2}{27}$ yields the six quark masses ($K_d = \frac{79}{108}, K_u = \frac{275}{324}$, $m_u, m_d, m_s, m_c, m_b, m_t$ within $\leq 0.65\sigma$), the unitary CKM mixing matrix ($J_{\mathrm{CP}}^{\mathrm{CKM}} = 3.13 \times 10^{-5}$, $0.09\sigma$), the PMNS leptonic mixing angles ($\theta_{13}, \theta_{12}, \theta_{23}, \delta_{\mathrm{CP}}$ within $\leq 0.17\sigma$), and the 11th-order $C_3$ holonomy neutrino mass hierarchy ($\mu_\nu \simeq 3^{-11}\mu_\ell$, $\sum m_\nu = 0.0593$ eV).

6. **The Nodal Origin and Topological Impedance of the Fine-Structure Constant ($\alpha$):** The fine-structure constant $\alpha \approx 1/137$ is read as the dimensionless ratio between the electric-flux and transverse-photon responses of one nodal network ($\alpha_0 = \frac{Z_0}{2R_K}$). Both channels transform together under common scale transformations ($\partial\alpha/\partial p_{T,L} \equiv 0$), while integrating the three orthogonal phase-space channels of the tri-axial toroidal Möbius $Z_2$ oscillon $T^3_{\mathrm{osc}}/Z_2$ ($4\pi^3 + \pi^2 + \pi = 137.036304$, 2.22 ppm) and dressing across its 24 internal Casimir/holonomy channels yields $\alpha^{-1}_{\mathrm{RefG}} = 137.035999092$ (0.061 ppb, $0.40\sigma$ vs. CODATA) and electroweak $Z^0$-pole running $\hat{\alpha}^{-1}_{\overline{\mathrm{MS}}}(m_Z) = 127.951$ ($0.024\sigma$).

7. **Genesis, Foundation Expansion, and Its Internal Readout:** The first distinguishable relations begin time; resonant coupling connects the network; stable matter, radiation, and long-term relaxation take form in subsequent phases. In the post-Genesis realization, growth of the foundation's inter-node scale lowers $P_F$, and the pressure decline reduces the material scale $p$. A neutral collective phase charge is distributed through the expanding three-dimensional volume: its conservation gives $\eta_F a^3 = 1$, while homogeneous constitutive relations yield $P_F/P_{F,0} = a^{-3}$, $p = a^{-3/2}$, and $A = a/p = a^{5/2}$. Accordingly, the internally measured $A$ uniquely reconstructs the normalized relative scales: $a = A^{2/5}$, $p = A^{-3/5}$, and $\eta_F = A^{-6/5}$. The variables $p_C$ and $P_F$ describe distinct physical quantities; their constitutive relation belongs to the microphysics of the foundation. On the ideal homogeneous background, $A$ unifies redshift, signal time dilation, and geometric distances in one response. When the cold phase source couples to the plasma only through the common geometry and the local dimensionless atomic and QED ratios retain their standard values, the CMB initial-value problem on this same $A$ geometry takes exactly the standard Einstein–Boltzmann, recombination, and line-of-sight form, quantitatively reproducing the matter–radiation equality epoch ($z_{\mathrm{eq}} = 3402.88$), the present constant-standard ($p_0 = 1$) coordinate expansion age ($t_0 = 13.797$ Gyr) and the accumulated process-time history ($\tau_0 =$

$\int_0^{t_0} p\,dt = 24.580$ Gyr, with $\tau(10) = 3.28$ Gyr vs. $t(10) = 470.2$ Myr at $z = 10$ owing to faster early clocks $p(z) = (1+z)^{3/5} > 1$), the recombination sound horizon ($r_s(z_*) = 144.29$ Mpc), the BAO standard ruler ($r_d = 146.94$ Mpc $= 98.98\,h^{-1}$ Mpc $\approx 100\,h^{-1}$ Mpc), the CMB acoustic peaks ($\ell_1 = 220.8, \ell_2 = 538.0, \ell_3 = 811.6$), Silk diffusion damping ($\ell_{D,\mathrm{eff}} = 1800.3$), baryon-loaded $TT/TE/EE$ spectra ($R_{12} = 2.2616$, $R_{32} = 0.9899$), the linear matter power spectrum $P(k,0)$ ($k_{\mathrm{peak}} = 0.0167\,h\,\mathrm{Mpc}^{-1}$, $\sigma_8^{\mathrm{RefG}} = 0.8098$, $S_8^{\mathrm{RefG}} = 0.8300$), and linear structure growth ($D_+(A) = A\,{}_2F_1(\frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}} A^3)$, $\gamma_0 = \frac{6}{11}$).

8. **Two Readouts of a Single Spectral Operator and the 137-e-Fold Micro–Macro Bridge:** On the centerless periodic 3D nodal graph $\mathbb{T}^3_N = (\mathbb{Z}/N\mathbb{Z})^3$ ($[T_{\mathbf{m}}, \Delta_{\mathrm{graph}}] = 0$), the short-wavelength eigenfunctions of the unified volumetric Helmholtz/Laplace resonator are read as microscopic particle modes, whereas its long-wavelength modes ($j_0(kr) = \frac{\sin(kr)}{kr}$) govern the macroscopic structure of the cosmic web, galaxy-cluster $Q_C$ cores, and the BAO ruler $r_d$, where the total logarithmic span from the Planck length to the BAO horizon, $\ln(r_d/\ell_{\mathrm{Pl}}) = 136.88 \approx \alpha_{\mathrm{RefG}}^{-1}$ (0.111%), partitions at the electron Compton wavelength $\bar{\lambda}_{C,e}$ into $\frac{3}{8}\alpha_{\mathrm{RefG}}^{-1} = 51.39$ and $\frac{5}{8}\alpha_{\mathrm{RefG}}^{-1} = 85.65$.

### Historical Lineage and Priority Boundaries

Individual mechanisms in RefG are related to the following theoretical approaches:

- The $p/p^2$ scaling law is related to the Polarizable Vacuum (PV) approaches of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]; RefG ties it to the ontology of a nodal network and two-channel mass accounting.

- The three-phase geometric reading of Koide's formula extends the work of Carl Brannen and Robert Foot [Brannen, 2010, Foot, 1994], while the holonomic origin of 2/9 is related to the closed family cycles studied by Konstantin Shulga [Shulga, 2026]. RefG combines this structure with the $\theta$-blindness argument and the $C_3$ symmetry of a three-dimensional volumetric resonator.

- The historical motivation for background-pressure relaxation and a possible microscopic origin of $\Lambda$ comes from the $q$-theory of Klinkhamer and Volovik [Klinkhamer and Volovik, 2008]; RefG writes the background dynamics of its selected branch in Einstein–Hilbert geometric language. The motivation for three spatial dimensions is connected with studies of topological knotting and stability in nodal structures [Whitrow, 1955, Berera et al., 2017].

RefG combines these approaches within a single causal framework, connecting the emergence of operational measurability and oscillon mass with galactic vortical flow and cosmological horizons.

The following chapters examine the physical content of these mechanisms and the relations among them.

# 1 The Ontological Foundation

## 1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence

Modern fundamental physics describes the Universe as a system of distinct, interacting quantum fields. The Standard Model relates these fields through common gauge symmetries, internal degrees of freedom, and interaction rules. Refractive Gravity (RefG) associates this diversity with a single postulate: **nature has one universal, pre-spatial ontological foundation.** Its resonant self-distinction produces a measurable medium with multiple response channels. Fundamental fields, gauge interactions, and elementary particles correspond to its topological, phase, pressure, shear, and rotational modes. The term *foundation-medium* denotes this self-distinguished, physically manifested state of the foundation.

This monistic account has theoretical precedents. In 1870, William Kingdon Clifford proposed that matter corresponds to small, moving regions of spatial curvature [Clifford, 1876]. RefG connects this geometric interpretation to a physical mechanism: the self-distinguished medium supports pressure, shear, hydrodynamic, and resonant responses as well as geometric curvature.

A clear theoretical precedent for the emergence of distinct particle and gauge sectors from one carrier is the string-net condensation model of Michael Levin and Xiao-Gang Wen [Levin and Wen, 2005]. In that framework, collective many-body entanglement produces structural analogues of gauge bosons and fermionic excitations. Levin–Wen begins with a prestructured quantum lattice; RefG associates the same unifying potential with the self-distinction of one universal pre-spatial foundation.

The description of the Universe as a unified continuous medium involves two asymptotic limits in the RefG dynamical picture:

1. **The Absolute Maximum of Substance:** A limiting state characterized by maximal pressure, mass–energy potential, and operational density.

2. **The Zero-Distinguishability Bound:** A limit in which interaction and physical measurability vanish entirely.

The void is the logical limit at which physical distinguishability vanishes. In this account, physical reality consists of bounded, nonlinear oscillatory dynamics between the two limits. Stable local units of the foundation, termed *nodes of existence*, form within this domain.

The reference background $\varphi = 0$ is the quiescent yet physically active state situated between these two extrema. At the limit of zero distinguishability, nodal interactions and measurable traces completely vanish; $\varphi = 0$ designates the unperturbed baseline state of the manifested medium.

The manifested medium supports multiple dynamical responses: volumetric compression, topological twist, vortical flow, and transverse wave excitation. Its phase, pressure, shear, and topological degrees of freedom together provide the physical basis to which RefG relates electric charge, spin, and the particle spectrum.

The unperturbed state of the foundation-medium, denoted by $\varphi = 0$, is its background vacuum regime. The foundation is present throughout the manifested medium; the excitation level, operational density of distinguishable nodes, and coupling strength vary locally. Increasing the operationally unmeasurable intervals between nodes rarefies the foundation and lowers its pressure state.

Here pressure is the effective pressure–energy state variable of RefG. The unperturbed background is $\varphi = 0$; $\varphi > 0$ is the elevated pressure–energy branch (compression), and $\varphi < 0$ is the deficit branch (rarefaction). The gravitational readout is built on the gradient profile of the deficit branch. Classical Bernoulli pressure is its immediate physical analogue: an external observer reads a local change in $\varphi$ as a slowing of clocks, delayed coordinate signal propagation, a change in operational size, and an effective gravitational mass.

## 1.2 From Unity to Multiplicity: The Emergence of Measurability and Operational Space

The Genesis postulate of RefG assumes an initially undifferentiated foundation, before change, event order, duration, or geometry has emerged. The origin of the Universe is the first physical event and the starting point for measuring time.

The origin scenario developed here is a centerless transition encompassing the whole foundation. It specifies a common initial condition with no preferred spatial center. Space, physical distance, propagation speed, and an exterior acquire meaning only through the connections produced by that transition.

At this boundary the foundation first produces minimally distinguishable relations. Event order and both measures of time begin here, at $t = 0$ and $\tau = 0$. This initial layer of distinguishable connections is the minimal network of existence; material particles and stable oscillons form during its subsequent development.

Nonlinear modes localize on the first relations as oscillon seeds. Their resonant fields spread across the relational layer and overlap. Once the physical threshold of distinguishability is crossed, this overlap binds operationally separate regions into a single network. The causal sequence therefore begins with the first relation and continues through localization and network coupling. Local locking and global coupling may be partly contemporaneous; their precise ordering is set by resonant dynamics and the physical coupling threshold.

In the Genesis scenario, the Universe is an interconnected network with no boundary or preferred center. A finite periodic graph provides a simple mathematical model: it is finite and connected without any node being a unique center. The structure of the foundation's connections determines the dimension and topology of the physical Universe.

The energy of the initial coherent state is distributed among oscillon seeds, their resonant fields, freely propagating waves, a thermal sector, and the collective deformation of the foundation. A continuity law accounts for energy in each sector and the fluxes between them in a common balance: a loss in one sector is a gain in another. The phase energies and equation of state determine the direction of this redistribution.

The first collective change initiates the foundation's self-distinction. Operational space becomes measurable when a stable relation between nodes reaches the minimum distinguishability scale $\ell_*$. An interval between distinguishable nodes provides the elementary basis of physical space. A delay in signal transfer establishes measurable duration, while the local energetic strain associated with self-distinction provides the proposed ontological precursor of mass. The integrated response of the deficit profile that this strain produces in the surrounding foundation is subsequently measured as gravitational mass.

Physical interpretation requires comparison. At the ontological origin, three inseparable elements are identified:

- **Node:** A locally distinguishable, discrete state of the foundation;
- **Trace:** The energetic and phase alteration imparted by a node to the surrounding medium;
- **Relation:** The operational capacity to compare one local state with another.

The relations among these three elements establish the operational network whose macroscopic properties are described as space, time, and matter.

A completely undifferentiated foundation has a single, indivisible identity. Its resonant self-distinction produces the first persistent difference and establishes the physical basis for quantity and counting. Distinguishable manifestations constitute separate physical units, while the foundation itself remains indivisible: multiplicity consists of stable, mutually distinguishable states of one identity.

The primary unit of measurement is a preserved relation. In a completely undifferentiated state, no second term is available for comparison. Self-distinction yields two distinct states of the same foundation and establishes the primary link between them; here begin counting, spatial extension, and the relative ordering of events.

In an informational description, the first persistent distinction is analogous to a bit: a distinguishable trace is represented by 1, and its absence in the specified relation by 0. Different arrangements of these two states form complex structures, just as mutually distinguishable states of the foundation give rise to the diversity of the measurable Universe.

Persistent self-distinctions of a single foundation generate locations that can be counted and compared. Multiplicity is described in terms of different distinguishable states of one primary identity. The foundation itself is considered outside time; the age of the measurable Universe refers to the sequence of events that emerges from it.

This account places observers within the physical system they measure. Observers and their instruments are complex, composite wave structures of the same nodal network. Bridgman's operationalism defines a physical concept through the set of operations by which it is measured [Bridgman, 1927]. RefG seeks an ontological basis for this principle by examining the physical conditions that make those operations possible.

The measurable Universe is described as an ongoing process of self-distinction in the foundation-medium. Operational space emerges from physically distinguishable relations. The universal foundation may also exist beyond the domain of such relations, but it is manifested as a physical object only where it has an operationally measurable effect.

Absolute inter-node separation cannot be measured directly because the observer has no absolute ruler external to the medium. Only its operational effects are measurable. Operational space emerges when the relation between at least two nodes becomes physically distinguishable. Below this threshold, a difference remains part of the foundation's local internal state; above it, the difference is registered through changes in clock rate, operational size, mass, and signal propagation.

RefG introduces a dynamic lower bound of minimal operational distinguishability, denoted by $\ell_*$:

- At scales smaller than $\ell_*$, no independent measurable relation has yet emerged; information remains indistinguishable;
- At the scale $\ell_*$, the first stable, physically distinguishable record emerges;
- At scales larger than $\ell_*$, inter-node intervals register as changes in pressure, clock rates, and signal propagation.

The physical rationale for this bound is the finite resolution required by a stable record. The Planck length $\ell_{\rm Pl} = \sqrt{\hbar G/c_0^3} = 1.616255 \times 10^{-35}$ m is a dimensional scale constructed from fundamental constants, whereas $\ell_*$ is the dynamical threshold for stable record formation. On the three-dimensional nodal graph $\Gamma = \mathbb{Z}^3$ (coordination number $z = 2N_{\rm axes} = 6$), their exact analytical relation and the emergence of continuum $3+1$ Einstein–Hilbert gravity are fixed by the discrete graph Laplacian and Sakharov–Seeley–DeWitt spectral induction (verified across `18/18` checks):

1. **Discrete 3D Graph Dispersion and Identical Cancellation of Odd-Order Lorentz-Invariance Violation ($\delta_1^{\rm LIV} \equiv 0$):** On the nodal graph $\mathbf{n} \in \mathbb{Z}^3$, the discrete Laplacian $(\Delta_{\rm graph}\psi)_{\mathbf{n}} = \ell_*^{-2}\sum_{a=1}^{3}(\psi_{\mathbf{n}+\hat{\mathbf{e}}_a} + \psi_{\mathbf{n}-\hat{\mathbf{e}}_a} - 2\psi_{\mathbf{n}})$ acting on a plane wave $\psi_{\mathbf{n}}(t) = e^{i(\mathbf{k}\cdot\mathbf{n}\ell_* - \omega t)}$ in the Brillouin zone $k_a \in [-\pi/\ell_*, \pi/\ell_*]$ yields the exact dispersion relation:

$$\omega^2(\mathbf{k}) = \frac{4c_0^2}{\ell_*^2}\sum_{a=1}^{3}\sin^2\!\left(\frac{k_a\ell_*}{2}\right) = c_0^2k^2\left[1 - \frac{\ell_*^2}{12}\frac{\sum_{a=1}^{3}k_a^4}{k^2} + \frac{\ell_*^4}{360}\frac{\sum_{a=1}^{3}k_a^6}{k^2} - \mathcal{O}\big((k\ell_*)^6\big)\right]. \tag{1}$$

   Because every relational link $(i,j) \in E$ is undirected and $\sin^2(k_a\ell_*/2)$ is strictly even in $k_a$, **all odd powers ($n = 1, 3, 5, \dots$) vanish identically** in the Taylor expansion ($c_1 = c_3 \equiv 0$). Consequently, linear ($n = 1$) Lorentz-invariance violation is identically zero ($\delta_1^{\rm LIV} \equiv 0$), automatically satisfying the most stringent Fermi-LAT and LHAASO gamma-ray bounds.

2. **Sakharov–Seeley–DeWitt Heat-Kernel Induction ($\ell_* = 2\sqrt{\pi}\,\ell_{\rm Pl} \approx 3.5449\,\ell_{\rm Pl}$) and UV Finiteness:** Integrating the quantum phase fluctuations of the foundation's $N_{\rm Casimir} = N_{\rm axes} \times \dim\mathfrak{su}(3)_c = 3 \times 8 = 24$ microscopic polarization channels up to the natural Brillouin-zone UV cutoff $\Lambda_{\rm UV} = \pi c_0/\ell_*$ via the Seeley–DeWitt heat-kernel expansion ($a_1 = R/6$) induces the Einstein–Hilbert action:

$$\begin{gathered}\frac{c_0^3}{16\pi G} = \frac{N_{\rm Casimir}\,\hbar}{96\pi^2 c_0}\left(\frac{\pi}{\ell_*}\right)^2 = \frac{\hbar}{4c_0\ell_*^2} \quad\Longrightarrow\quad \ell_*^2 = 4\pi\,\ell_{\rm Pl}^2 = A_{\rm Pl}, \\ \ell_* = 2\sqrt{\pi}\,\ell_{\rm Pl} \approx 3.544908\,\ell_{\rm Pl} = 5.7295 \times 10^{-35}\ \text{m}.\end{gathered} \tag{2}$$

   Thus the elementary nodal cell area $\ell_*^2$ equals the surface area of a Planck sphere $A_{\rm Pl} = 4\pi\ell_{\rm Pl}^2$ ($E_* = \hbar c_0/\ell_* = 3.4441 \times 10^{18}$ GeV). At $E_\gamma = 100$ TeV, the quadratic ($n = 2$) group-velocity dispersion is $|\Delta c/c| = \frac{1}{8}(E_\gamma/E_*)^2 = 1.054\times10^{-28} \ll 10^{-17}$ ($E_{\rm LIV,2} = \sqrt{8}\,E_* = 9.74\times10^{18}$ GeV $\gg 1.3 \times 10^{11}$ GeV), while at a $1\,M_\odot$ compact-core throat ($r_c = \frac{4}{3}m \approx 1.969$ km) higher-curvature $R^2$ corrections are suppressed by $(\ell_*/r_c)^2 = 8.46 \times 10^{-76}$.

The ontological origin of the Universe is a completely undifferentiated primary foundation-state. A geometric point is already a concept of formed space; two distinguishable points and the extension between them arise together with the operational inter-node interval. An asymmetry formed in the foundation-medium leaves a stable trace, the relation becomes readable, and multiplicity and geometry emerge in the same event.

RefG connects this operational description to gravity. A spatial gradient in the pressure state of the foundation-medium produces coordinated changes in clock rate, operational size, and coordinate signal propagation, which together define the gravitational geometry. A homogeneous change produces a common background recalibration; acceleration and refractive deflection arise from spatial inhomogeneity.

### 1.3 Nodal Transmission: Waves, Time, and Self-Calibration

The foundation's first collective change creates a minimal relational layer. Each node is a functionally distinguishable state that preserves the trace of a physical change and can relay it to adjacent nodes. Subsequent transmission, together with resonant coupling produced by overlapping resonant fields of oscillon seeds once the distinguishability threshold is crossed, joins this layer into an operational network. Spatial proximity and operational distance emerge from the stable organization of these physical couplings.

A wave in this network is the sequential transmission of state and phase. Light is a freely propagating transverse phase trace, whereas an oscillon is a locally self-confined, self-sustaining configuration of the same transmission mechanism. When matter moves, this localized wavepacket is transferred from node to node, reconstructing its phase integrity at each successive location. Light and matter are thus the propagating and localized modes of a single medium.

The foundation's first complete self-distinction establishes an initial ordering of events and the origin of time. Each subsequent transmission in the network extends that sequence. Recurrent phase relations establish a rate for physical processes, and comparisons among stable oscillon cycles make clock measurements possible. The local rate of time is determined jointly by the rate of state transfer through the network and the frequency at which an oscillon completes its own cycle.

RefG relates the rate of state transfer between nodes to their density and coupling strength. Rarefaction slows this transfer and reduces the rate of material processes with respect to coordinate time.

The observer's body, clocks, rulers, light, and matter are modes of the same medium. In a pressure-deficit region, nodal density, transmission rate, oscillon period, and operational size change together. Local measuring devices adjust to the same environment. An observer using local units therefore measures the usual clock rate and size, and an unchanged local speed of light ($c$); comparisons between processes in different environments reveal differences in time, size, and mass.

On the physically admissible low-energy branch, transmission is isotropic, non-dissipative, and Lorentz invariant: the local speed of light is unchanged, and uniform motion experiences no friction.

### 1.4 Common Local Clock Rate and Resonant Locking

The emergence of time and the establishment of a common local clock rate are distinct stages. In the Genesis scenario, time begins with the first foundation-wide self-distinction. A common local rate is established later, when the intrinsic frequencies of stable material oscillons acquire fixed, persistent ratios. In this collective state, internal coordination and nonlinear locking within the local network determine the rate of time, as in Kuramoto-type synchronization.

The network's normal modes and the physical basis of material clocks emerge as its connections form. The resulting medium is treated as a volume resonator: its coupling operator and boundary conditions determine the spectrum of admissible modes. The three-dimensionality of this resonator is part of the physical model of RefG.

Elementary-particle frequencies are determined by mutual balance and coordination within the foundation's collective dynamics. The foundation specifies the space of admissible modes, and interactions

among particle modes establish a stable, resonantly locked state within that space.

The pressure–geometric state changes the network's common frequency scale with respect to coordinate time. We denote this scale by $\Omega_t(x)$; the subscript $t$ specifies frequency per unit of the foundation's coordinate time. The clock factor $p$ introduced in this subsection corresponds in a static local frame to $p_T = N$, while equating it to the ruler factor ($p_T = p_L = p$) belongs to the common-scaling branch. The clock response takes the form

$$\frac{\Omega_t(x)}{\Omega_{t,0}} = p(x), \qquad d\tau = p\, dt. \tag{3}$$

Here $\tau$ is the process time measured by material clocks. As $p$ decreases, fewer material cycles occur within a fixed coordinate-time interval $dt$. A local observer uses a $\tau$ clock and measures the usual local rate; differences arise when environments or epochs are compared.

The oscillon's internal geometry distinguishes its stable modes by dimensionless structural labels $s_i$. Their relation to the common frequency scale is

$$\nu_i^{(t)}(x) = s_i\, \Omega_t(x), \qquad \nu_i^{(\tau)} = \frac{\nu_i^{(t)}}{p}. \tag{4}$$

The labels $s_i$ characterize the oscillon's internal spectrum, whereas $\Omega_t$ is the environmental frequency scale defined with respect to coordinate time. A common pressure change rescales the entire spectrum by one factor; the process clock compensates locally for this common change. The labels $s_i$ and their invariant ratios distinguish the particle modes.

This spectral factorization is established directly from the canonical complex oscillon action even when the clock factor $p_T = N$ and the ruler factor $p_L = A_{\rm iso}^{-1}$ on a static isotropic coframe ($ds^2 = -p_T^2 c_0^2 dt^2 + p_L^{-2}\delta_{ij}dx^i dx^j$) are independent: introducing the proper spatial coordinate $\tilde{\mathbf{x}} = p_L^{-1}(\mathbf{x}-\mathbf{x}_0)$ at fixed Noether charge $Q$ factorizes the action as $S_O = \int dt\,[Q\Omega_t - p_T c_0 E_{\rm int}[f]]$, yielding $\Omega_t(p_T, p_L; Q) = p_T c_0 \Omega_0(Q)$, $E_{\rm coord} = p_T E_0(Q)$, and $R_{\rm coord} = p_L R_0(Q)$. The linearized Bogoliubov–de Gennes (BdG) perturbation system around the stationary core ($\mathcal{L}_+^{(\ell)}\eta = \frac{\kappa^2}{c_0^2}\eta + 2\Omega_0\frac{\kappa}{c_0}\xi$, $\mathcal{L}_-^{(\ell)}\xi = \frac{\kappa^2}{c_0^2}\xi + 2\Omega_0\frac{\kappa}{c_0}\eta$) is completely independent of $(p_T, p_L)$ in $\tilde{r}$, so every normal-mode coordinate frequency obeys the exact factorization law $\nu_n^{(t)}(p_T, p_L; Q) = s_n(Q)\,\Omega_t(p_T; Q)$ with $s_n(Q) = \frac{\kappa_n(Q)}{2\pi\Omega_0(Q)}$ and $\frac{\partial s_n}{\partial p_T} = \frac{\partial s_n}{\partial p_L} \equiv 0$ (verified numerically on the $a = 1/4, \Omega_0 = 0.8, Q_0 \approx 190.401$ benchmark core for the $\ell = 0$ phase and $\ell = 1$ translational zero modes and the coupled radial breathing modes $\kappa_1 = 0.208040, s_1 = \kappa_1/\Omega_0 = 0.260050$ and $\kappa_2 = 0.231624, s_2 = 0.289530$, matching the coordinate operator to $2.14 \times 10^{-14}$). By parity symmetry of the spherical core ($\langle n|\mathbf{x} - \mathbf{x}_0|n\rangle = 0$), a linear external gravitational gradient produces zero first-order diagonal shift, while quadratic tidal shifts at the Earth's surface are suppressed to $(R_{\rm core}/L_{\rm grav})^2 \sim 1.2 \times 10^{-56}$.

In a low-pressure region, the coordinate-time frequency scale $\Omega_t$ decreases. Inter-node transfer is more frequent in a dense, high-pressure foundation; in a rarefied foundation, coupling and material processes evolve more slowly with respect to coordinate time. Delayed inter-node signal transfer, a lower pressure–energy state, and a reduced $\Omega_t$ are three operational manifestations of the same network state. Each is counted once in the quantitative causal description.

Local clocks and atomic processes adjust to the medium, so observers measure the usual rate using their own local standards. The common rescaling preserves dimensionless ratios; the observed stability of the fine-structure constant is a stringent test of this principle. In this account, particle mass ratios originate in the internal $s_i$ spectrum of oscillons.

Ernst Mach's principle is a historical precursor of this relational approach [Mach, 1883]. RefG gives it a local dynamical interpretation: the common clock rate is determined directly by the collective state of the local resonant network, with the influence of distant masses encoded in that state. Relationality and local causality are therefore properties of the same network.

In RefG, the fine-structure constant $\alpha \approx 1/137$ is interpreted as an invariant dimensionless ratio of the electric-flux and transverse-photon responses of the nodal network. Under common rescaling, it remains unchanged as local pressure varies.

A receiver measures a light signal against its local atomic standard. For stationary emitters and receivers in a static gravitational field, the locally measured frequencies satisfy

$$\frac{\omega_o}{\omega_e} = \frac{N_e}{N_o}, \tag{5}$$

where $N$ is the static lapse. On the selected common-response branch, $N = p$, so a signal climbing from a pressure-deficit region to a distant observer has $\omega_o/\omega_e = p_e/p_o < 1$: it is gravitationally redshifted.

On the homogeneous cosmological branch, comoving emitters and receivers instead obtain

$$\frac{\omega_o}{\omega_e} = \frac{A_e}{A_o}, \qquad 1 + z = \frac{A_o}{A_e} = \frac{a_o/p_o}{a_e/p_e}. \tag{6}$$

Both results follow from propagation in the corresponding operational metric and measurement by local clocks. The common pressure response is included once in each description; the static and cosmological formulas apply to different propagation conditions.

In RefG, the rate of time is a property of the local resonant network's ontological state. Pressure changes rescale the network and all compatible processes by a common factor, preserving their dimensionless ratios. Every measurement is performed within the same physical network, whose processes give time its physical meaning.

The stability of dimensionless ratios provides an empirical test of this principle: **a reliably established nonzero $\Delta\alpha/\alpha$ correlated with a change in local pressure would falsify the population-level common-clock-rate postulate.**

### 1.5 Oscillons, Pressure Deficits, and the Origin of Mass

RefG describes the electron, proton, and quark as **oscillon modes**. An oscillon is a locally self-confined, self-sustaining nonlinear wave structure in the foundation-medium: a coherent oscillatory state of the medium itself.

Lord Kelvin's vortex-atom hypothesis is a historical precursor [Thomson, 1867]. The shared conceptual principle is that a particle is a self-confined, dynamical topological configuration of the medium. The oscillon description develops this idea within nonlinear field theory and topology.

Intense internal oscillation, rotation, or localized energy confinement in an oscillon lowers the static pressure of the surrounding foundation-medium. Bernoulli's classical hydrodynamic balance provides an analogue [Bernoulli, 1738]: **under the appropriate conditions for stationary flow, an increased dynamical contribution is accompanied by a decrease in static pressure.** On the weak, static gravitational branch of RefG, the corresponding schematic pressure–energy relation is

$$\Delta P \propto -\rho_{\mathrm{dyn}}. \tag{7}$$

The energy and stresses of the oscillon's internal oscillation determine the geometric response of the medium. In a weak, static exterior region, the clock-slowing profile recovers the pressure-deficit picture of RefG. The total gravitational mass of the source is read in the far-zone geometry through the asymptotic charge, where the internal energy of matter and the gravitational binding contribution enter a single balance.

An oscillon has a localized core, while its oscillatory influence extends through the connected medium. The stationary component that is phase-locked to the source establishes a persistent gravitational profile and does not require a net outward energy flux. When the source reconfigures rapidly and asymmetrically, a coherent disturbance released through the same channel propagates outward as a freely radiating gravitational wave and carries energy.

The local depth of the pressure deficit and the asymptotically measured total mass of a system are distinct physical quantities: the former characterizes the central gradient and compactness of the body, whereas the latter combines the active core, medium polarization, binding energy, and radiated fraction into a single asymptotic balance. Following a merger of compact bodies, the central deficit

deepens while the total externally measured mass of the remnant decreases by the energy carried away in gravitational waves.

A historical parallel is John Archibald Wheeler's *Mass without Mass*, in which a geon is a massive configuration confined by the energy of its own gravitational and electromagnetic fields [Wheeler, 1955]. Classical geons are unstable and gradually radiate energy; RefG relates oscillon stability to the conservation of Noether phase charge, population-level resonant locking, and the topological invariant $Lk = Wr + Tw$.

In the common-scaling description, the "effective refractive mass" ($m_{\text{eff}}$) denotes the external energetic readout of an individual oscillon in a specified gravitational background. For a test particle of fixed proper energy in a static metric, $E_t = NE_{\text{local}}$, whereas the mass of a complete self-gravitating body is determined from the joint solution of the matter and geometry field equations. The phase charge defines the oscillon's conserved quantitative label, while changes in its profile, energy, and spatial extent are governed by the equations of motion.

At a fundamental level, **gravitational attraction is the refractive deflection of one oscillon's wave configuration within the pressure deficit produced by another.** A locally reduced pressure–energy state of the foundation slows coordinate wave propagation. The deficit profile establishes the effective geometry of the medium, and in the geometric-optics (eikonal) limit the trajectories of light and matter wavepackets follow the null and timelike geodesics of this metric.

**Self-consistent feedback between the oscillon and the foundation-medium is a central physical principle of** RefG**.** In the low-energy covariant description, the energy and stresses of matter generate the geometry, while the altered geometry acts back to determine the oscillon's spatial profile, clock rate, and spatial size. An equilibrium configuration is a self-consistent solution of this reciprocal interaction, whose stability is tested through the small-perturbation spectrum of the same action. The scaling laws of the common-$p$ scheme characterize the scalar branch, whereas in the full coframe dynamics the clock and ruler responses are determined by independent geometric components.

# 2 Gravity as Refraction

## 2.1 Three Simultaneous Operational Changes

The common-scaling picture of RefG describes changes in the inter-node state through three coordinated operational effects. In the covariant full-coframe formulation, this picture exactly recovers the first-order approximation in a weak static exterior field, while the exact clock and ruler responses are governed by the independent geometric factors examined in Section 2.2:

1. **Reduction of the Coordinate Speed of Light:** In a pressure-deficit region, nodes are rarefied and their transmission rate decreases, slowing the propagation of light and other massless wave signals in the coordinate description. At the same time, the speed of light measured by a local observer using a local clock and ruler remains invariant ($c$); only the effective propagation rate recorded in a distant observer's coordinate system changes.

2. **External Readout of the Spatial Standard:** The coordinate footprint of a local ruler is determined by the spatial metric. In a weak isotropic exterior region, it decreases as the pressure deficit deepens, whereas the proper unit of length measured by the local ruler remains unchanged. The physical size and spatial profile of a self-gravitating oscillon are determined by the joint matter–geometry field solution.

3. **Reduction of the Local Resonant Clock Rate:** Every physical clock consists of oscillon systems (atoms and molecules) whose oscillation rates depend on local nodal couplings. In a rarefied medium, this coupling weakens, slowing the clock rate in a distant observer's coordinate time. The local clock itself continues to count its own cycles at an unchanged local rate.

RefG accounts for this dependence through the mechanism of state transmission between foundation nodes: as nodes become rarefied, the frequency of interaction and phase exchange between them decreases, so the wave state is transmitted more slowly and the coordinate clock rate falls.

These three effects are mutually consistent operational manifestations of a single lower-pressure state, and their quantitative relation forms the basis of biconformal geometry.

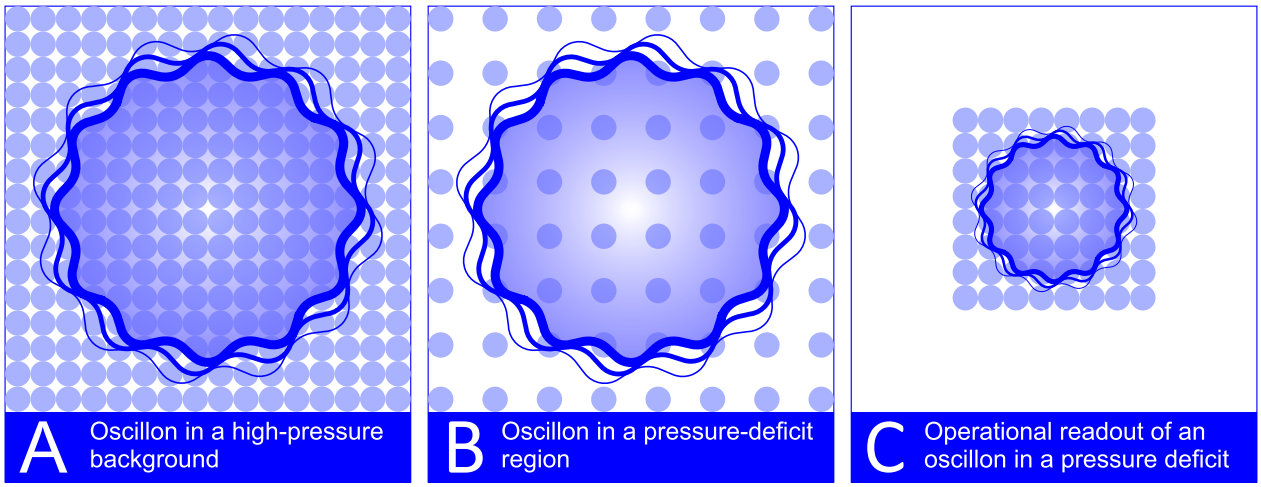



Figure 1: **Schematic illustration of the operational readout of an oscillon within a pressure-deficit region. (A)** In an unperturbed foundation-medium whose pressure is higher than in the deficit region, an oscillon occupies a comparatively dense nodal arrangement; inter-node coupling is stronger and local processes proceed at a higher coordinate rate. **(B)** Within a pressure-deficit region, foundation nodes are rarefied; operational inter-node separation increases, the transmission rate falls, and local clocks run more slowly. **(C)** In the external readout of the common-scaling scheme, the clock slows, the coordinate footprint of a ruler decreases, and the externally read energy changes. The figure illustrates this schematic relation, while the exact size and total mass of a self-gravitating body are determined from the solution of the joint field equations.

### 2.2 The Quadratic Relation and the Factor of Two

On the isolated-system branch, in static isotropic coordinates, the spacetime metric and local standards take the form

$$ds^2 = -N^2 c_0^2 dt^2 + A_{\rm iso}^2 d\boldsymbol{\sigma}^2, \qquad p_T = N, \quad p_L = A_{\rm iso}^{-1}, \quad d\tau = p_T dt, \quad d\ell = \frac{d\sigma}{p_L}. \tag{8}$$

Here $A_{\rm iso}$ denotes the spatial metric factor (distinguishing it from the cosmological scale factor $A$ of Section 3.3). The null interval condition for light ($ds^2 = 0$) yields

$$\frac{c_{\rm coord}}{c_0} = p_T p_L, \qquad c_{\rm local} = \frac{d\ell}{d\tau} = c_0. \tag{9}$$

The common metric unites material clocks, rulers, and light within a single local geometry. In general, the factors $p_T$ and $p_L$ are determined by independent field equations. In an asymptotically flat, weak spherically symmetric exterior region, to first order in the dimensionless potential $\epsilon = GM/(c_0^2 r)$, the two factors coincide: $p_T = 1 - \epsilon + O(\epsilon^2)$ and $p_L = 1 - \epsilon + O(\epsilon^2)$. This coincidence characterizes the leading weak-field approximation, whereas at higher orders and in the matter interior $p_T$ and $p_L$ remain independent variables.

In the common-scaling ansatz, the exact equality $p_T = p_L = p$ holds. Under this condition, the external coordinate footprint of a local ruler and the coordinate clock rate both scale by the factor $p$,

while the clock period scales as $p^{-1}$; on this specialized branch, the coordinate speed of light therefore equals $p^2 c_0$.

Accordingly, on the common-scaling branch, the biconformal scaling law takes the form

$$\frac{L_{\text{coord}}}{L_0} = \frac{m_{\text{eff}}}{m_0} = \frac{\Omega_t}{\Omega_{t,0}} = p, \qquad \frac{c_{\text{coord}}}{c_0} = p^2, \qquad p = e^{\varphi/2}. \tag{10}$$

Here $L_{\text{coord}}$ denotes the footprint of a local ruler on the distant asymptotic coordinate grid, while in its own environment the observer continues to read it as one unchanged unit. The clock law gives $d\tau = p\, dt$, and ruler calibration gives $d\ell = p^{-1} d\sigma$. Consequently, the temporal metric coefficient is $p^2$, the spatial coefficient is $p^{-2}$, the coordinate speed of light is $p^2 c_0$, and the speed of light measured with the local clock and ruler remains exactly $c_0$.

A similar power-law dependence appears historically in the polarizable-vacuum (PV) models of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]. RefG connects these exponents to the ontological distinction between foundation nodes and their connecting links.

Formally, the externally determined effective mass of a single oscillon depends on the foundation's pressure-deficit profile through the exponential law $e^{\varphi/2}$:

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \tag{11}$$

On the deficit branch, $\varphi < 0$ and $0 < p < 1$. The relation $m_{\text{eff}} = p m_0$ expresses the gravitationally redshifted value of a fixed proper energy in mass units. In the static metric of the full coframe, the test-body Lagrangian expands as $L = -mc_0^2 N + \frac{1}{2} m A_{\text{iso}}^2 N^{-1} v^2 + O(mv^4/c_0^2)$, showing that the external rest energy and the coordinate inertial coefficient are distinct quantities. The physical inertia of an entire isolated body is determined in Section 2.3 by the total energy–momentum of the system.

Four physical quantities are distinguished in the energy accounting:

1. **Oscillon internal state and proper energy** ($M_{\text{proper}}$);
2. **Single-oscillon effective refractive mass in a given background field** ($m_{\text{eff}}$);
3. **Source profile of the pressure deficit established in the foundation-medium** ($M_{\text{active}}$);
4. **Total asymptotic gravitational charge of the entire system** (ADM and Komar total mass: Arnowitt et al. 1962, Komar 1959).

These four quantities define two levels of accounting: at the first level, the oscillon's proper internal energy is distinguished from the externally read contribution in a given gravitational environment; at the second level, the externally measured total mass combines the contributions of the active core and the medium. At the asymptotic boundary, they form a single total gravitational charge in which each contribution is counted once.

An oscillon is a localized resonant structure of the unified foundation, yet its resonant influence extends beyond the core into the surrounding foundation. The resonant fields of many oscillons overlap, encompass the entire foundation-medium, and lower its background pressure; RefG associates this collective process with background spatial relaxation.

In the common-$p$ relaxation scheme, growth of the region affected by the resonant interaction modifies local pressure, external energy, size readout, clock rate, and coordinate light speed. The negative feedback and homogeneous asymptotic relaxation of this scheme are analyzed within the corresponding scalar action and equations of state, whereas in the isolated covariant sector backreaction is governed by the joint field equations of the action $S_{\text{iso}}$.

The common-scaling $p/p^2$ branch is described by the following biconformal metric (whose exponential form is a distinct special case compared alongside the general solution of the full coframe):

$$ds^2 = e^{\varphi} c^2 dt^2 - e^{-\varphi} d\boldsymbol{\sigma}^2, \qquad \varphi = -\frac{2U}{c^2}, \tag{12}$$

where $U > 0$ is the gravitational potential of the selected exponential branch and coincides at leading order with the magnitude of the Newtonian potential. In the weak-field limit ($U/c^2 \ll 1$), the metric components expand in a Taylor series:

$$g_{00} = e^{\varphi} \approx 1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + \cdots, \qquad g_{ij} = -e^{-\varphi}\delta_{ij} \approx -\left(1 + \frac{2U}{c^2} + \cdots\right)\delta_{ij}. \tag{13}$$

The shared clock–ruler scaling relations determine the biconformal metric and its leading spatial correction. The static profile of a massive source follows from a variational functional under three conditions: the response is additive and logarithmic ($u = -\ln p$), the gradient term is canonical, and the coupling to the active source is linear. Under these conditions, variation of the functional together with the asymptotic boundary condition yields the Newtonian profile $u = GM/(c^2 r)$.

In this static, spherically symmetric weak-field description, the first post-Newtonian (1PN) metric expansion yields the Parameterized Post-Newtonian (PPN) coefficients:

$$\beta = 1, \qquad \gamma = 1. \tag{14}$$

Here $\gamma = 1$ is the geometric consequence of the equal first-order clock and ruler responses, while the second-order expansion of the exponential static response yields $\beta = 1$.

Thus, self-regulation of the foundation-medium unites the state of matter, the pressure deficit, and the response of local clocks and rulers. Their complete dynamical description rests on a common geometry in which each energetic contribution from matter and the medium is counted exactly once.

**Relational Coframe, Einstein–Hilbert Geometry, and Full 1PN.** The covariant equations in this subsection are written in the $(-+++)$ metric signature (whereas the biconformal presentation above used the $(+---)$ signature).

The static $p$ profile represents only the volumetric scalar projection of the full geometry. The complete local geometric state is described by a **relational coframe** $e^A{}_\mu$—a set of local frame differential one-forms uniting volume, shape, shear, orientation, and temporal direction. The coframe uniquely defines the operational metric:

$$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu. \tag{15}$$

The spacetime orientation of the coframe and the oscillon's internal three-axis structure are distinct physical structures (in the particle map, the color quantum number is associated with this internal three-axis structure). Neighboring coframes are compared through a flat, metric-compatible inertial connection, while the electromagnetic channel possesses its own $U(1)$ gauge connection. The non-closure of an infinitesimal contour under coframe transport is **geometric torsion**—the measure of non-integrable deformation in the coframe continuum.

In the post-Genesis relational-coframe continuum, three physical conditions determine the leading low-energy action:

1. The action is local, time-reversal invariant (reversible), constant-coefficient, parity-even, and quadratic in the torsion tensor;

2. The phase sector is minimally coupled to the geometry, with no unsuppressed mixed operators participating at the same order;

3. The linear limit about a Minkowski background is regular, while the Lorentz orientation of the coframe remains an inertial gauge freedom—no independent long-range physical orientation wave is excited.

Under these conditions, the unique admissible ratio of quadratic-invariant coefficients yields the Teleparallel Equivalent of General Relativity (TEGR) combination. The three independent quadratic torsion invariants are

$$I_1 = T^\rho{}_{\mu\nu} T_\rho{}^{\mu\nu}, \qquad I_2 = T^\rho{}_{\mu\nu} T^{\nu\mu}{}_\rho, \qquad I_3 = T_\mu T^\mu, \quad T_\mu = T^\nu{}_{\nu\mu}. \tag{16}$$

Their teleparallel (TEGR) combination is

$$T_{\rm TEGR} = \frac{1}{4} I_1 + \frac{1}{2} I_2 - I_3. \tag{17}$$

Accordingly, on the cosmological phase branch, the action takes the form

$$S_F = -\frac{c_0^3}{16\pi G} \int d^4x\, e\, (T_{\rm TEGR} + 2\Lambda_F) + S_C[g, J, \theta_C] + S_\partial, \tag{18}$$

where $S_C$ is the covariant phase-current action and $S_\partial$ is the boundary-matched functional. The minus sign preceding the gravitational torsion term combines with the torsion–curvature identity below to recover the canonical sign of the Einstein–Hilbert action. In the $(-+++)$ metric signature, the exact differential-geometric identity

$$R_{\rm LC} = -T_{\rm TEGR} + \frac{2}{e}\partial_\mu (e T^\mu), \tag{19}$$

converts the coframe action into the Einstein–Hilbert action up to a total divergence (boundary term) [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. Here $R_{\rm LC}$ is the Ricci scalar curvature of the Levi–Civita connection, $e = \det(e^A{}_\mu)$ is the coframe determinant, and $T^\mu$ is the trace of the torsion tensor. The equivalence of the actions is exact for compactly supported variations under suitable asymptotic falloff conditions (or when the matched boundary term $S_\partial$ is included). Thus, Einstein geometry is the direct manifestation of the low-energy continuum dynamics of the relational coframe. The independent route of Fierz–Pauli linear spin-2 theory and Deser's universal self-coupling arrives at the same Einstein–Hilbert operator.

The action $S_C$ describes a barotropic, isentropic, and irrotational single-potential phase current on its positive, stable, and causal constitutive branch:

$$\rho_C > 0, \qquad \rho_C' > 0, \qquad 0 \le n_{C,\rm op}\rho_C''(n_{C,\rm op})/\rho_C'(n_{C,\rm op}) \le 1. \tag{20}$$

Variation with respect to the phase $\theta_C$ yields the conserved current, while variation with respect to the metric yields the Hilbert stress–energy tensor [Brown, 1993]:

$$T^C_{\mu\nu} = (\rho_C + p_C) u_\mu u_\nu + p_C g_{\mu\nu}. \tag{21}$$

Here $n_{C,\rm op}$ is the phase-charge density per unit operational volume, and $p_C = n_{C,\rm op}\rho_C'(n_{C,\rm op}) - \rho_C(n_{C,\rm op})$ is the thermodynamic pressure of the phase sector. The quantity $P_F$, by contrast, represents the operational readout of the foundation's background pressure; these are two distinct physical variables whose connecting equation of state is determined by the microphysics of the foundation. When the source is restricted to the phase current alone, the field equation takes the form

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T^C_{\mu\nu}. \tag{22}$$

In the general case, the geometry is sourced by the total stress–energy tensor $T^{\rm tot}_{\mu\nu}$, in which each physical sector of the action enters exactly once through its own metric variation.

The low-energy sector of an isolated system ($S_{\rm iso}$) in units with $c_0 = \hbar = 1$ takes the form

$$S_{\rm iso} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - g^{\mu\nu}\partial_\mu \Psi_O^* \partial_\nu \Psi_O - U(|\Psi_O|^2) - \frac{1}{4} F_{\mu\nu}F^{\mu\nu}\right] + S_\partial, \qquad \Lambda = 0. \tag{23}$$

Here the canonical normalization is $\Psi_O = f e^{i\theta_O}/\sqrt{2}$ and $U(|\Psi_O|^2) = V(f)$. In the dimensionless units employed, the self-interaction potential is $V(f) = f^2/2 - f^4/4 + f^6/24$, and in amplitude–phase variables the matter Lagrangian takes the form $-\frac{1}{2}(\partial f)^2 - \frac{1}{2} f^2 (\partial\theta_O)^2 - V(f)$. The tensor $F_{\mu\nu}$ is the Maxwell electromagnetic field strength. For a neutral oscillon in equilibrium, the background electromagnetic field vanishes, while small electromagnetic excitations propagate on the same metric light cone. The

metric, matter field, and Maxwell field are varied independently, and a specific coordinate system is chosen after the equations of motion are obtained (introducing an independent scalar $u$-field corresponds to a different action).

In the initial hydrodynamic formulation $S_C[g, J_C, \theta_C]$, the cosmological phase sector was defined only in the domain of nonvanishing density ($n_{C,\mathrm{op}} > 0$): because the rest chemical potential is positive ($\mu_{C0} = \rho_C'(0) > 0$), the derivative $\partial\rho_C/\partial J_C^\mu = \mu_{C0}\, u_{C\,\mu}$ is direction-dependent (discontinuous) in the limit of vanishing current ($J_C^\mu \to 0$), and in an inhomogeneous static gravitational field ($\nabla_i N \neq 0$) a stationary phase $\theta_C = -\omega_C t$ cannot simultaneously satisfy the eikonal threshold condition $\chi_C \equiv -g^{\mu\nu}\partial_\mu\theta_C\partial_\nu\theta_C = \mu_{C0}^2$ throughout space. This analytical obstruction is resolved by replacing the hydrodynamic variables with a regular collective complex scalar field $\Psi_C = \sqrt{x}\, e^{i\theta_C}$ ($x \equiv |\Psi_C|^2$) governed by the Lagrangian

$$\mathcal{L}_C(\Psi_C, \nabla\Psi_C, g) = -\frac{1}{2} g^{\mu\nu}\nabla_\mu\Psi_C^*\nabla_\nu\Psi_C - U_C(|\Psi_C|^2), \qquad U_C(x) = \mu_{C0}^2 x + \frac{1}{2}\lambda_C x^2. \tag{24}$$

Through this formulation, the isolated, cosmological, strong-field, and galactic sectors are unified within a single regular Grand Unified Covariant Action:

1. **Strict Vacuum Reduction ($\Psi_C \equiv 0$, $\Lambda_F = 0$):** Because $\mathcal{L}_C$ is a smooth quadratic functional at $\Psi_C = 0$, when the collective field vanishes the unified action reduces **identically and without singularity** on any curved metric to the isolated-system action $S_{\mathrm{iso}}$;

2. **Active Cosmological Branch ($x > 0$):** In the high-frequency (Thomas–Fermi, $|\nabla\sqrt{x}|^2 \ll \chi_C x$) regime, the Legendre transformation $\chi_C = U_C'(x) = \mu_{C0}^2 + \lambda_C x$ exactly recovers the barotropic stress–energy tensor $T_{\mu\nu}^C = (\rho_C + p_C)u_\mu u_\nu + p_C g_{\mu\nu}$, where $\rho_C = \mu_{C0}^2 x + \frac{3}{4}\lambda_C x^2$ and $p_C = \frac{1}{4}\lambda_C x^2$, and in the dilute limit ($\lambda_C x \ll \mu_{C0}^2$) yields a pressureless cold-dust regime ($w_C \to 0$, $c_{s,C}^2 \to 0$). Deriving the parameters $(\mu_{C0}, \lambda_C)$ of the effective potential $U_C(|\Psi_C|^2)$ from the microscopic dynamics of discrete foundation nodes remains a task for subsequent investigation;

3. **Grand Unified Covariant Action and the Common-Scale Ricci Total-Divergence Theorem:** Identifying the temporal foliation (clock) scalar with the collective condensate phase via $\Phi(x) \equiv -\frac{c_0}{\mu_{C0}}\theta_C(x)$ renders the medium four-velocity $u_\mu = -\partial_\mu\Phi/\sqrt{X} = \partial_\mu\theta_C/\sqrt{\chi_C}$ and the orthogonal spatial projector $\gamma^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ identical across all sectors, while the spatiotemporal medium deformation invariants ($\widehat{Y} \equiv Xe^{-2H}/c_0^2$, $\widehat{B}^{AB} \equiv g^{\mu\nu}\partial_\mu\Sigma^A\partial_\nu\Sigma^B e^{2H}$) and the galactic gradient prefactor share the unified scale $M_*^4 = \frac{M_{\mathrm{Pl}}^2 a_0^2}{2c_0^4} = \frac{a_0^2}{16\pi G}$. On any static common-scaling metric $ds^2 = -e^{-2H(\mathbf{x})}c_0^2 dt^2 + e^{2H(\mathbf{x})}\delta_{ij}dx^i dx^j$ ($\Phi = c_0 t, \Sigma^A = x^A$), the invariants satisfy $\widehat{Y} \equiv 1$ and $\widehat{B}^{AB} \equiv \delta^{AB}$ identically, and the Einstein–Hilbert scalar curvature density is an **exact total spatial divergence** to all orders in $H(\mathbf{x})$:

$$\sqrt{-g}\, R \equiv 2\,\nabla^2 H = 2\,\partial_i(\delta^{ij}\partial_j H), \qquad \frac{\delta}{\delta H}\int d^4x\sqrt{-g}\, R \equiv 0. \tag{25}$$

   Consequently, on the static common-scaling branch the Ricci scalar makes zero bulk Euler–Lagrange contribution to the $H$ field equation, which is governed entirely by the invariant medium functional $\mathcal{F}(Z_H/a_0^2)$ ($Z_H = c_0^4\gamma^{\mu\nu}\nabla_\mu H\nabla_\nu H$). In the strong field of compact objects ($r = 2m$, $M = 1\,M_\odot$, where $x_{\mathrm{acc}} = |\nabla\Phi|/a_0 \approx 7.69 \times 10^{22} \gg 1$), the high-gradient limit $\mathcal{F}(x^2) \to x^2$ reduces the equation to the quadratic Laplace equation ($\nabla^2 H = 0 \implies H(r) = m/r$) with a fractional deviation of only $1.69 \times 10^{-46}$, ensuring exact compatibility between the horizonless exponential strong-field metric and the galactic MOND/RAR regime ($x_{\mathrm{acc}} \lesssim 1$) within a single action.

The selected teleparallel (TEGR) coframe action places RefG and General Relativity in the same first post-Newtonian (1PN) equivalence class when five conditions are satisfied:

1. Through 1PN order, the complete action consists of the Einstein–Hilbert operator and the matter action;
2. Matter couples universally and minimally to the same metric;
3. The gravitational source is recorded by one complete, covariantly conserved stress–energy tensor;
4. RefG and General Relativity use the same local source, PPN gauge, asymptotic background, and boundary conditions;
5. No additional unsuppressed field degree of freedom or operator participates in the 1PN approximation.

Under these conditions, the complete standard set of PPN parameters is

$$\gamma = \beta = 1, \qquad \xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0. \tag{26}$$

The domain of validity of this result encompasses isolated, weak, slowly moving, and weakly self-gravitating sources, which the isolated action $S_{\rm iso}$ satisfies completely. The exponential static ansatz provides an independent check of the relation $\beta = \gamma = 1$, while its continuation to higher orders depends on the corresponding scalar source and action.

The duration of one complete clock cycle can also be expressed as the ratio of the ruler's coordinate footprint to the coordinate speed of light: $\Delta t \sim p_L/(c_0 p_T p_L) \propto p_T^{-1}$. Thus, a local light clock directly records the temporal metric component $p_T$. In the illustrative example of the common-scaling scheme where $p_T = p_L = 1/2$, the clock period doubles while the coordinate speed of light decreases by a factor of four.

In a static, spherically symmetric field, the first post-Newtonian effects governed by $\beta = \gamma = 1$ coincide with the classical predictions of General Relativity: gravitational light deflection, Shapiro radar time delay, gravitational redshift, and planetary perihelion precession [Bertotti et al., 2003, Will, 2014]. In particular, the Cassini spacecraft radio-link experiment constrains the deviation $\gamma - 1$ to $(2.1 \pm 2.3) \times 10^{-5}$.

Historically, incorporating the spatial metric component $g_{ij}$ supplied the **factor of two** that replaced Einstein's 1911 half-deflection prediction with the full geometric result of 1915–1916 [Einstein, 1916, Shapiro, 1964]. In the weak isotropic exterior region of RefG, the first-order clock and ruler responses are equal, and their product naturally combines both contributions in the coordinate speed of light.

### 2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as a spatially extended wave structure, propagates through spatial gradients of foundation-medium pressure and coordinate wave speed. Its phase velocity is higher on the denser side and lower on the rarefied, pressure-deficit side. Because of this difference, by the Huygens–Fresnel principle and the eikonal equation, the wavefront bends toward the region of lower pressure and reduced coordinate speed—it undergoes **refraction**. In the weak, static limit, the pressure deficit created by a massive body establishes an effective optical-geometric medium, and the refracted trajectory of a matter wavepacket in this medium manifests macroscopically as gravitational attraction.

In RefG, the free fall of a body in the Earth's gravitational field results from the refractive deflection of the oscillon structures constituting its atoms. This deflection is directed toward the region of lower pressure and slower coordinate propagation produced by the Earth. Gravitational attraction is thus the direct local response of wave-like matter to spatial inhomogeneity in the medium.

In the relational-coframe description, deformation of the foundation is recorded by the full tensor geometry. Its static temporal and spatial characteristics are $p_T$ and $p_L$, while the weak exterior common-$p$ approximation recovers the classical refractive picture. In the isolated action $S_{\rm iso}$, canonical matter and the Maxwell field each enter the Hilbert stress–energy tensor once through their own metric variation. The total asymptotic mass of a bound system combines the proper energy of matter and gravitational binding; for a system with positive binding energy $B$, the mass defect is $\Delta M = B/c_0^2$.

The gravitational-wave event GW150914 provides a direct observational example of this energy balance: the difference of approximately three solar masses ($3\,M_\odot$) between the asymptotic masses of the initial binary system and the final merged remnant was radiated as gravitational-wave energy [Abbott et al., 2016]. In the language of RefG, two oscillatory sources reorganize into a single, deeper deficit profile, and in the teleparallel (TEGR) low-energy geometry this rapid nonlinear reorganization is described by the radiation of standard transverse-traceless tensor waves.

On the low-energy covariant branch of RefG, the **inertia** of a body is determined by the total energy–momentum of the oscillon and its own gravitational field. Uniform rectilinear motion of the body is obtained by a Lorentz boost of the entire stationary solution, under which the matter phase, spatial profile, and surrounding geometry transform together. Under acceleration, an external force changes the total momentum of this unified system. The asymmetric rearrangement of the foundation's deficit profile provides the intuitive physical picture of this process, while the detailed mechanism of its nodal delay belongs to the microscopic dynamics of the foundation.

For an isolated, spatially localized solution, the relativistic dispersion and inertia relations hold exactly:

$$E(v) = \gamma E_0, \qquad \mathbf{P}(v) = \gamma \frac{E_0}{c_0^2}\mathbf{v}, \qquad M_I = \left.\frac{\partial P_\parallel}{\partial v}\right|_{v=0} = \frac{E_0}{c_0^2}. \tag{27}$$

In flat spacetime, this equality follows directly from the full matter stress tensor and Laue's equilibrium identity ($\int T^{ij} d^3x = 0$), whereas for a self-gravitating body $E_0$ is the total asymptotic (ADM) energy of the system and the momentum is likewise determined by the asymptotic integral of the full metric. In uniform motion, the rest mass $E_0/c_0^2$ of the body is invariant while its total energy increases to $\gamma E_0$, and Lorentz length contraction along the direction of motion reflects the coordinate readout. In a weak, slowly varying external gravitational field, to monopole order, the active and passive gravitational masses also equal $E_0/c_0^2$ (rapid internal deformation, tidal forces, and radiative backreaction constitute higher-order corrections). The result of the MICROSCOPE space mission provides the most stringent observational test of the weak equivalence principle [Touboul et al., 2022].

In uniform rectilinear motion, the oscillon's wave structure is transmitted symmetrically between foundation nodes without energy dissipation. RefG associates **frictionless inertial motion** with this regime of coherent phase transfer: displacement of the body through the nodal medium proceeds by successive transmission of its wave state. In the low-energy continuum limit, the common Lorentzian structure identifies the limiting speed of phase transmission directly with the metric light cone.

The equilibrium of a weakly bound oscillon computed in this same action is also confirmed by variational energy analysis: at fixed total phase charge, radial deformation of the profile in every tested direction increases the total energy of the system, and this property persists stably under grid refinement and outer-boundary expansion. This confirms the radial stability of the body, while the investigation of full three-dimensional nonlinear evolution remains a subsequent numerical task.

When the oscillon is analyzed within the unified covariant action together with the ambient collective field $\Psi_C$, three rigorous physical regimes of coupling to the medium emerge:

1. **Infinite-Reservoir $1/r$ Divergence Theorem:** In asymptotically flat infinite space, a global frequency $\omega_C \geq \mu_{C0}$ cannot be maintained with finite energy: because of the oscillon's gravitational redshift ($N(r) \simeq 1 - GM/r < 1$), the local chemical potential $\omega_C/N(r)$ at large distances exceeds the threshold value $\mu_{C0}$, generating a $1/r$ asymptotic tail of the $\Psi_C$ field and causing the total charge and mass to diverge quadratically ($Q_C(R), M_C(R) \propto R^2 \to \infty$);

2. **Pure-Vacuum and Gravitating Trapped-Pocket Regimes:** When $\omega_C \leq \mu_{C0} N_0(0)$, the ambient collective field is unexcited ($\Psi_C \equiv 0$) and all properties of the isolated oscillon remain unchanged; in the intermediate frequency window $\mu_{C0} N_0(0) < \omega_C < \mu_{C0}$, the collective field $\Psi_C$ condenses directly within the particle's gravitational potential well, forming a compact "trapped pocket" of finite radius ($r < r_{\rm trap}(\omega_C)$);

3. **Two-Charge First Law and Composite-Body Inertia:** In the trapped-pocket regime, the

system obeys an exact two-charge variational first law:

$$dM_{\mathrm{ADM}} = \Omega\, dQ + \omega_C\, dQ_C. \tag{28}$$

Because both fields ($\Psi_O$ and $\Psi_C$) are localized within a finite domain and the exterior spacetime is asymptotically flat, the inertial mass of the composite $(Q, Q_C)$ body again equals **exactly** its total gravitational energy: $M_I = M_{\mathrm{ADM}}(Q, Q_C)$. Calculating the hydrodynamic added mass and finite-domain boundary effects for a body moving through a nonvanishing, infinitely extended cosmological background ($\Psi_C(\infty) \neq 0$) constitutes a separate research problem.

### 2.4 Local Unobservability and the Principle of Relativity

A local observer in a gravitational field measures time and spatial intervals in the orthonormal frame (basis) of the same metric to which matter and light are coupled. Consequently, the locally measured speed of light is always equal to $c_0$, even as the coordinate factors of clocks and rulers ($p_T$ and $p_L$) vary. Environmental changes that are slow relative to internal oscillation periods proceed adiabatically, while the response of a specific material spectrum and its dimensionless ratios is determined from the corresponding action and small-perturbation equations. The common-scaling picture describes this local self-calibration in the special case $p_T = p_L$.

Two classes of experiment establish the observational bounds on this mechanism:

- **Gravitational Redshift:** The Pound–Rebka experiment [Pound and Rebka, 1960], the Gravity Probe A hydrogen-maser experiment [Vessot et al., 1980], and the Galileo satellite measurements [Delva et al., 2018, Herrmann et al., 2018] confirm with high precision the dependence of clock rate on gravitational potential. RefG interprets this dependence as the temporal component of the common pressure–geometric response.

- **Invariance of Dimensionless Ratios:** Absorption spectra of distant quasars and laboratory optical atomic clocks constrain possible cosmological variation of the fine-structure constant ($\alpha$) to the level $|\Delta\alpha/\alpha| \lesssim 10^{-6}$ [Kotuš et al., 2017], in full agreement with the principle of common-clock self-calibration.

The null result of the Michelson–Morley experiment reflects the medium's **common-mode self-calibration**, whereas signals registered by gravitational-wave detectors (LIGO, Virgo, KAGRA) measure a **differential tensor response**:

- Local clocks, rulers, and light signals are calibrated within the frame of one and the same metric. Modern optical resonators constrain local directional anisotropy in the speed of light to the level $\Delta c/c \lesssim 10^{-17}$ (Herrmann et al. 2009).

- A gravitational wave is a time-dependent, quadrupolar differential deformation $(h_+, h_\times)$: it stretches space along one axis while compressing it along the perpendicular axis. This opposite-sign deformation breaks common-mode compensation and produces a measurable phase shift between the perpendicular arms of an interferometer.

FitzGerald–Lorentz length contraction [FitzGerald, 1889, Lorentz, 1892] is a historical antecedent of this reasoning. RefG relates the contraction to the fact that measuring instruments are themselves wave excitations of the foundation-medium.

In the TEGR-equivalent low-energy geometry, tensor waves propagate on the common metric light cone, so $v_{\mathrm{gw}} = c$ at this order. The nearly simultaneous detection of gravitational waves and gamma rays from the binary neutron-star merger GW170817 constrains their fractional speed difference [Abbott et al., 2017]:

$$-3 \times 10^{-15} \leq \frac{v_{\mathrm{gw}} - c}{c} \leq +7 \times 10^{-16} \tag{29}$$

The orbital decay of relativistic binary pulsars (PSR B1913+16, PSR J0737-3039A/B) confirms the general-relativistic normalization of quadrupole energy loss to an accuracy of $10^{-3}$–$10^{-4}$ [Weisberg and Huang, 2016, Kramer et al., 2021]. The LIGO–Virgo–KAGRA GWTC-3 catalogue also places stringent limits on the scalar breathing-polarization contribution [LIGO Scientific Collaboration et al., 2022].

In the isolated-system and binary-pulsar regime ($\Psi_C \equiv 0$, $S_{\rm iso}$), linearizing the relational coframe about Minkowski space and integrating the Isaacson radiative stress-energy tensor over the celestial sphere yields three exact analytical and numerical results (verified across `18/18` checks):

1. **Coframe Linearization, $v_{\rm gw} \equiv c_0$, and Two Transverse-Traceless Polarizations:** Expanding the tetrad into symmetric strain $h_{\mu\nu} = h_{\nu\mu}$ and antisymmetric local Lorentz rotation $\omega_{\mu\nu} = -\omega_{\nu\mu}$,
$$e^A{}_\mu = \delta^A_\mu + \frac{1}{2}\eta^{AB}h_{B\mu} + \frac{1}{2}\eta^{AB}\omega_{B\mu}, \tag{30}$$
the operational metric $g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu = \eta_{\mu\nu} + h_{\mu\nu} + \mathcal{O}(h^2, \omega^2)$ is identically independent of $\omega_{\mu\nu}$ at linear order ($\partial g^{(1)}_{\mu\nu}/\partial\omega_{\alpha\beta} \equiv 0$). On $S_{\rm iso}$, the linearized Einstein tensor $G^{(1)}_{\mu\nu}[h]$ satisfies the contracted Bianchi identity $\partial^\mu G^{(1)}_{\mu\nu} \equiv 0$ identically and reduces in Lorenz gauge ($\partial^\mu \bar{h}_{\mu\nu} = 0$, $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$) to
$$\Box\bar{h}_{\mu\nu} = \left(-\frac{1}{c_0^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\bar{h}_{\mu\nu} = -\frac{16\pi G}{c_0^4}T_{\mu\nu}. \tag{31}$$
In vacuum, the dispersion relation $\omega^2 = c_0^2|\mathbf{k}|^2$ gives $v_{\rm phase} = v_{\rm group} = c_0$ ($(v_{\rm gw} - c_0)/c_0 \equiv 0$, satisfying GW170817), while the four Lorenz gauge conditions and four residual harmonic gauge transformations ($\Box\xi_\mu = 0$) reduce the 10 components of $h_{\mu\nu}$ to $10-4-4 = 2$ orthogonal transverse-traceless (TT) polarizations ($h_+, h_\times$), with zero scalar breathing ($h_b = 0$), longitudinal ($h_L = 0$), and vector ($h_{vx} = h_{vy} = 0$) radiative modes. In an equal-arm Michelson interferometer, isotropic scalar clock–ruler perturbations cancel identically between perpendicular arms ($\Delta\tau_x - \Delta\tau_y \equiv 0$), whereas a TT wave produces the differential fractional strain $(\Delta L_x - \Delta L_y)/L_0 = F_+h_+ + F_\times h_\times$.

2. **Isaacson Quadrupole Flux, Peters–Mathews Eccentric Enhancement, and Zero Dipole Radiation:** Averaging the TT projection tensor $\Lambda_{ij,kl}(\hat{\mathbf{n}})$ over the unit sphere yields $\langle\Lambda_{ij,kl}\rangle_{S^2} = \frac{2}{5}I^{\text{sym-tf}}_{ijkl}$, from which the Isaacson radiative stress-energy tensor $t^{\rm GW}_{\mu\nu} = \frac{c_0^4}{32\pi G}\langle\partial_\mu h^{\rm TT}_{ij}\partial_\nu h^{\rm TT\,ij}\rangle$ integrates to the exact Einstein quadrupole luminosity and angular-momentum loss:
$$P_{\rm GW} = \frac{G}{5c_0^5}\left\langle \dddot{Q}_{ij}\dddot{Q}^{ij}\right\rangle = \frac{32G^4m_1^2m_2^2(m_1+m_2)}{5c_0^5a^5}\,f(e),$$
$$\left\langle\frac{dJ}{dt}\right\rangle = -\frac{32G^{7/2}m_1^2m_2^2\sqrt{m_1+m_2}}{5c_0^5a^{7/2}}\,g(e), \tag{32}$$
with exact symbolic Peters–Mathews eccentric enhancement factors
$$f(e) = \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1-e^2)^{7/2}}, \qquad g(e) = \frac{1 + \frac{7}{8}e^2}{(1-e^2)^2}. \tag{33}$$
Dividing $da/dt$ by $de/dt$ integrates in closed form to Peters' invariant $a(e) = c_0e^{12/19}(1 - e^2)\left[1 + \frac{121}{304}e^2\right]^{870/2299}$. Furthermore, because the two-charge inertia theorem ensures universal coupling ($M_I = M_{\rm ADM}$) with zero scalar sensitivity difference ($\Delta s_{12} = s_1 - s_2 \equiv 0$), dipolar gravitational radiation vanishes identically ($\dot{P}^{\rm dipole}_b \equiv 0$) for both neutron-star–neutron-star and neutron-star–white-dwarf binaries.

3. **Precision Binary Pulsar Benchmarks (PSR B1913+16, PSR J0737-3039A/B, PSR J1738+0333) and Standard Sirens:** Evaluating the orbital period decay rate $\dot{P}^{\rm RefG}_b = -\frac{192\pi}{5}\left(\frac{2\pi G\mathcal{M}_c}{c_0^3P_b}\right)^{5/3}f(e)$ against relativistic pulsar timing data yields:

- **Hulse–Taylor Pulsar (PSR B1913+16; $P_b = 0.322997448918$ d, $e = 0.6171340$, $m_1 = 1.438\,M_\odot$, $m_2 = 1.390\,M_\odot$):** $f(e) = 11.8568$, coalescence time $T_{\rm coal} = 300.70$ Myr, and $\dot{P}_b^{\rm RefG} = -2.4021\times10^{-12}$ s/s vs. the galactic-acceleration-corrected observed value $\dot{P}_b^{\rm obs,corr} = (-2.398 \pm 0.004) \times 10^{-12}$ s/s, giving $\dot{P}_b^{\rm obs,corr}/\dot{P}_b^{\rm RefG} = 0.9983 \pm 0.0017$ ($1.03\sigma$).
- **Double Pulsar (PSR J0737-3039A/B; $P_b = 0.1022515592973$ d, $e = 0.0877770$, $m_A = 1.338185\,M_\odot$, $m_B = 1.248868\,M_\odot$, $R = m_A/m_B = 1.071518$):** $f(e) = 1.0515$ and $\dot{P}_b^{\rm RefG} = -1.247810\times10^{-12}$ s/s vs. $\dot{P}_b^{\rm obs,corr} = (-1.247752\pm0.000079)\times10^{-12}$ s/s [Kramer et al., 2021], giving $\dot{P}_b^{\rm obs,corr}/\dot{P}_b^{\rm RefG} = 0.999953 \pm 0.000063$ (a relative difference of $4.68 \times 10^{-5} < 10^{-4}$, $0.74\sigma$), while simultaneously matching the 1PN periastron advance ($\dot{\omega}^{\rm 1PN} = 16.8994^\circ$/yr), Einstein timing delay ($\gamma_E = 0.38400$ ms, $0.51\sigma$), and Shapiro range and shape ($r_{\rm Sh} = 6.151\,\mu$s, $0.51\sigma$; $s_{\rm Sh} = \sin i = 0.99992$).
- **Asymmetric NS–WD Binary (PSR J1738+0333; $m_{\rm NS} = 1.46\,M_\odot$, $m_{\rm WD} = 0.181\,M_\odot$):** With $\dot{P}_b^{\rm dipole} \equiv 0$, the pure quadrupole prediction $\dot{P}_b^{\rm RefG} = -2.759 \times 10^{-14}$ s/s matches the observed intrinsic decay $(-2.59 \pm 0.32) \times 10^{-14}$ s/s at $0.53\sigma$.
- **Cosmological Tensor Propagation:** On the operational FLRW metric ($A = a/p = a^{5/2}$) with line element $ds^2 = A^2(\eta)[-c_0^2 d\eta^2 + (\delta_{ij} + h_{ij}^{\rm TT})dx^i dx^j]$, the Mukhanov–Sasaki variable $\chi_k = A\,h_k$ satisfies $\chi_k'' + (k^2 - A''/A)\chi_k = 0$ with zero Planck-mass running ($\alpha_M \equiv 0$), so the gravitational-wave luminosity distance equals the electromagnetic luminosity distance identically: $D_L^{\rm GW}(z) \equiv D_L^{\rm EM}(z)$.

With metric signature $(-+++)$, an operational invariant of this description is the difference between the proper times accumulated by two clocks along worldlines with different gravitational and kinematic histories:

$$\tau_a = \frac{1}{c}\int_{\gamma_a} \sqrt{-g_{\mu\nu}dx^\mu dx^\nu}, \qquad \Delta\tau = \tau_1 - \tau_2. \tag{34}$$

The accumulated proper times are invariant records of the clocks' trajectories through different pressure regimes. Locally measured instantaneous rates and lengths are relative properties of the common medium.

## 2.5 Historical Context and the Role of RefG

Refractive descriptions of gravity in terms of an optical medium date to the development of general relativity. In 1911–1912, before its final geometric formulation, Albert Einstein related the coordinate speed of light to the gravitational potential [Einstein, 1911, 1912]:

$$c(\Phi) = c_0\left(1 + \frac{\Phi}{c^2}\right) \tag{35}$$

The optical properties of moving media had been studied earlier. In 1851, Armand Hippolyte Fizeau demonstrated partial light dragging in moving water, described by Fresnel's dragging coefficient (Fizeau 1851). In 1923, Walter Gordon showed that electromagnetic waves in a moving dielectric medium follow the null geodesics of an effective Lorentzian (pseudo-Riemannian) metric [Gordon, 1923]. This established the first rigorous mathematical connection between medium optics and spacetime geometry.

In 1957, Robert Dicke described gravitation through a variable vacuum refractive index [Dicke, 1957]. William Unruh's 1981 work established the basis of modern **analogue gravity**: acoustic waves in a moving fluid with a supersonic region propagate according to an effective metric with an acoustic black-hole horizon [Unruh, 1981].

Grigory Volovik's condensed-matter program [Volovik, 2003] and the review by Barceló, Liberati, and Visser [Barceló et al., 2011] showed how local Lorentz invariance and effective curved spacetime can emerge from the microscopic dynamics of background quantum condensates, including those in liquid helium-3 and helium-4. In 2019, Jeff Steinhauer's group experimentally observed the acoustic

analogue of quantum thermal Hawking radiation in a rubidium Bose–Einstein condensate [Muñoz de Nova et al., 2019].

Related research explores the **thermodynamic and entropic origins** of gravity:

- Andrei Sakharov interpreted gravity as an elastic vacuum phenomenon induced by quantum fluctuations [Sakharov, 1967];
- Ted Jacobson demonstrated that Einstein's field equations arise directly as an equation of state from the thermodynamic entropy balance ($\delta Q = TdS$) across local Rindler horizons [Jacobson, 1995];
- Erik Verlinde described gravitational attraction as an entropic effect driven by changes in information on holographic screens [Verlinde, 2011].

These approaches and RefG share the view that spacetime geometry is a derived physical structure. RefG takes a pressure–refractive foundation-medium as its physical source.

Refractive Gravity (RefG) brings together four principles related to these approaches:

1. **Monistic Ontological Unity:** The foundation-medium produces both the geometric response and the oscillon structures constituting matter, measuring instruments, and the observer. The hydrodynamic interpretation is thereby incorporated into a single calculable physical framework.
2. **Population-Locked Local Frequency Scale ($\Omega_t$):** Nonlinear synchronization provides a specific mechanism for local self-calibration. For a physical oscillon, it predicts a spectrum in which all stable-mode frequencies scale with the common pressure factor while their dimensionless ratios remain unchanged.
3. **One Foundation with a Multicomponent Geometric Response:** Pressure and expansion govern the scalar readout, symmetric deformation and shear determine the tensor response, and orientation remains an inertial gauge freedom. These components define one operational geometry. Each physical stress–energy contribution in the action couples to its metric once.
4. **From Relational Coframe to Einstein–Hilbert:** In the post-Genesis coframe's leading local, reversible, constant-coefficient, parity-even quadratic-torsion class, with minimal phase coupling and no unsuppressed mixed operator at that order, the regular linear Minkowski limit determines the TEGR coefficient ratios by requiring the absence of an independent physical orientation mode. The exact torsion–curvature identity converts the action to Einstein–Hilbert form up to a boundary term [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. The Fierz–Pauli linear spin-2 construction and Deser's universal self-coupling provide a separate route to the same final operator. In four dimensions, Lovelock's theorem gives the corresponding uniqueness statement for local metric field equations of at most second differential order, up to normalization and a cosmological term [Deser, 1970, Lovelock, 1971].

RefG combines optical refraction, topological solitons, and thermodynamic balance within one research program. Each connection is subject to its corresponding mathematical and observational tests.

# 3 Three Scales and Cosmological Horizons

## 3.1 The Local Scale: Compact Objects, Internal State, and External Readout

In this subsection, the description of strong gravitational fields unites two complementary levels: dynamical collapse in the canonical scalar–geometric sector, and the horizonless exponential exterior of the covariant medium, whose regular $C^2$ interior core closes directly from the action.

The pressure mechanism described in the preceding chapters is considered at three distinct scales:

1. **Local scale:** the localized pressure-deficit profile surrounding oscillons and compact relativistic bodies;

2. **Galactic scale:** the coherent vortical flow of the foundation;

3. **Cosmological scale:** relaxation of the background pressure driven by geometric expansion of the foundation-medium, together with the relation between coordinate and process time.

These are three dynamical regimes of one foundation. The local regime describes spatially inhomogeneous pressure-deficit profiles, the galactic regime describes the coherent long-range response, and the cosmological regime describes the global evolution of the foundation.

In an extreme gravitational regime, an object's **internal state** ($M_{\text{proper}}$) and its **externally inferred active gravitational mass** ($M_{\text{external}}$) are distinct quantities. The compact-object map of RefG distinguishes the conserved internal content of accumulated matter from the gravitational charge inferred at a distance. A remote observer measures the state through its pressure-deficit profile, clock rate, and changing operational size, while the interior geometry, horizon, and global causal structure are determined by the complete nonlinear field solution.

Within the equilibrium family computed by coupling the canonical oscillon field to Einstein geometry, the total mass of a bound system is smaller than the summed masses of isolated configurations carrying the same total phase charge. Equal increments of charge yield progressively smaller increments of external mass, while the central proper clock slows relative to a distant clock. The geometric determination of the external mass agrees with the full energy accounting, including binding. Total mass is thus a property of the jointly adjusted configuration.

RefG relates the mass defect of a merging compact system to nonlinear changes in binding and in the deficit profile. Its quantitative connection to the outgoing tensor flux is determined by the complete dynamical solution, with the gravitational-wave event GW150914 providing the experimental benchmark [Abbott et al., 2016].

This self-consistent equilibrium has a concrete dynamical response. A weak inward perturbation that conserves phase charge first compresses the compact configuration, then arrests the contraction and produces expansion; the body continues to oscillate over the computed interval. Matter motion, the gravitational field, and the local clock rate evolve together under equations derived from a single action.

Stronger compression in the same model produces a trapped-light region: spherical light fronts sent in either radial direction decrease in area. The calculation continues beyond the formation of this region, tracking central proper time, density, and curvature together. The trapping geometry and the physical evolution of the interior are thus described by one dynamical solution (Figure 2).

In the description of a regular interior state, the energy balance of the foundation and matter, the local clock rate, and the spatial scale are determined jointly. On the static common-scaling branch of the covariant medium, this structure closes analytically:

1. **Horizonless Exponential Exterior ($r \geq r_c$):** Outside the core ($r \geq r_c$), the exact static vacuum solution is the horizonless exponential metric

$$B(r) = e^{-2m/r}, \qquad A(r) = e^{2m/r}, \qquad H(r) = \frac{m}{r}, \qquad \ell(r) = r. \tag{36}$$

   This geometry has no event horizon ($B(r) > 0$ for all $r > 0$), and in the ultracompact regime ($q \equiv 2m/r_c > 1$) it possesses an outer unstable photon sphere at $r_{\text{ph}}^{+} = 2m$ with critical shadow impact parameter $b_c = 2em \approx 5.4366m$;

2. **Pure-Medium No-Go and Central-Matter Necessity Theorem:** The radial-label ($\ell$) equation takes the exact divergence form $\frac{d}{dr}(QC_\ell) = \mathcal{S}_{\text{met}}(r) - \frac{r^2\sqrt{AB}}{\ell^2}\frac{dF_{\text{med}}}{d\widehat{I}_3}$. Across the horizonless ultracompact interval $1 < q < q_* \approx 1.9897$, the geometric source integral for a smooth degree-6 $C^2$ core is strictly positive ($\mathcal{I}_{\text{met}}(q) > 0$; for instance, $\mathcal{I}_{\text{met}}(3/2) \approx 0.14536$). This establishes a universal theorem: **a pure medium without matter ($T^{\text{m}}_{\mu\nu} = 0$) cannot close a $C^2$ core**

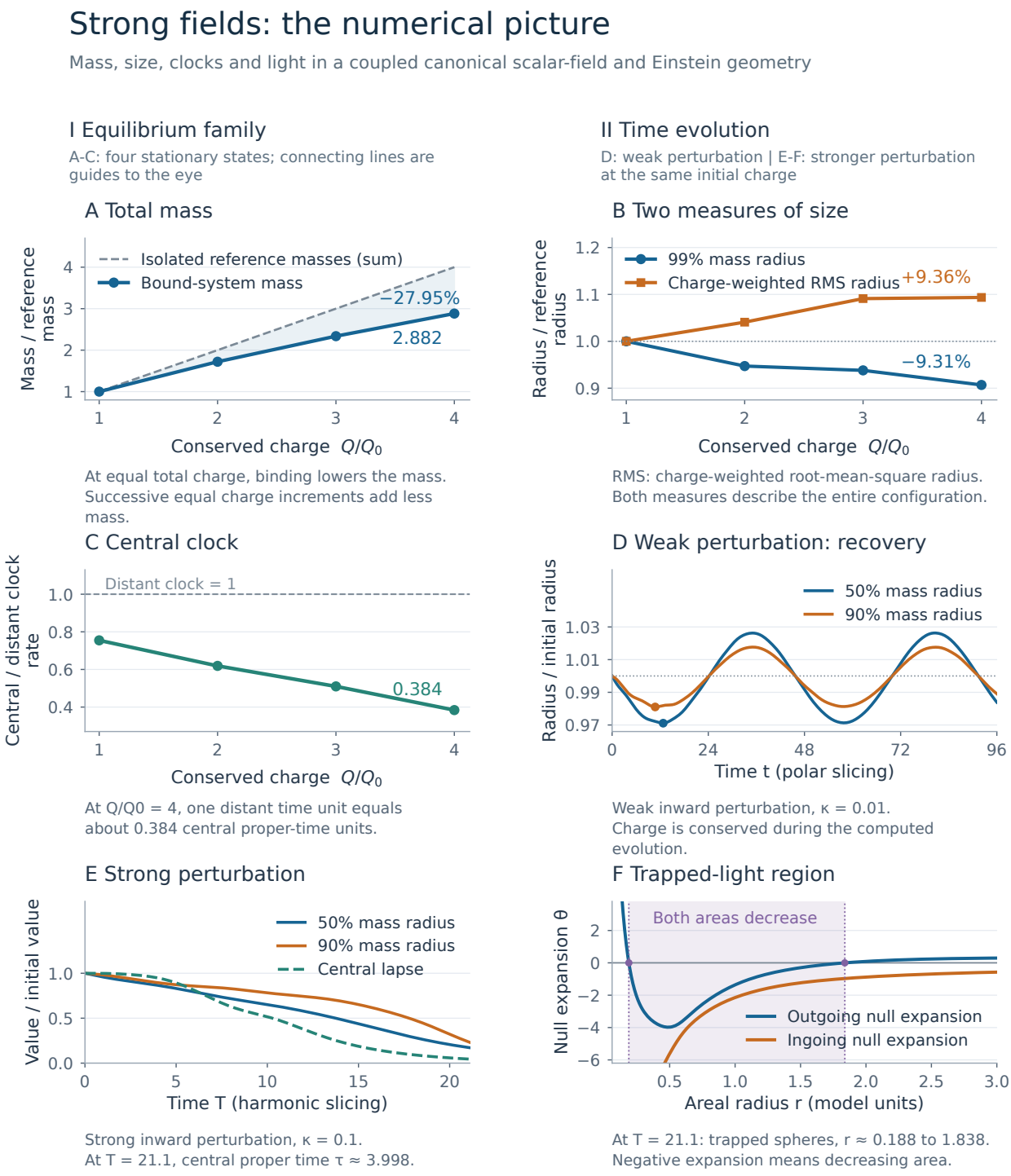



Figure 2: Strong-field numerical results for the coupled canonical scalar–Einstein model. A–C show the mass, two bulk-radius measures, and central clock rate of four equilibria; $Q/Q_0$ denotes multiples of a reference phase charge. D shows contraction and recovery following a weak perturbation; E–F show the response to a stronger perturbation. In the shaded region, both future-directed radial light fronts decrease in area. Quantities are expressed in dimensionless model units; $t$ and $T$ are different time coordinates. The dynamical curves cover finite computed time intervals.

**across the boundary $r = r_c$ without a stress jump**—a singularity-free balanced $C^2$ core requires the energy density and pressure of compressed central matter itself ($T^{\rm m}_{\mu\nu} \neq 0$);

3. **Action-Derived $C^2$ Regular Core ($r \leq r_c$):** When a covariantly conserved matter sector ($\nabla_\mu T^{{\rm m}\,\mu\nu} = 0$) is included in the core and the order-reduction condition at the boundary $r = r_c$ uniquely selects the relation $H(x) = -\frac{1}{2}\ell_B(x)$ ($\widehat{Y} \equiv 1$, with scalar charge profile $Q_H(0) = 0$ and $Q_H(r_c) = m$), the boundary-value problem closes with zero defect ($QC_\ell(r_c) = 0$; at the benchmark point $q = 3/2$, the central density root is $\rho_0^* \approx 0.66364$). At the center of the core ($r = 0$), all curvature invariants and the clock rate are finite ($B(0) = e^{-q} > 0$, $A(0) = 1$, $K(0) < \infty$), the deformation invariants $\widehat{I}_1(x)$ and $\widehat{I}_3(x)$ are strictly monotonic (free of cavitation and gradient singularities), and a second, radially stable inner light ring appears within the core ($x_{\rm lr}(1.5) \approx 0.87058$);

4. **Quantitative EHT Shadow (Sgr A* and M87*), Eikonal Quasinormal Modes (QNM), and Gravitational-Wave Echo Bridge:** On the horizonless exponential exterior, the optical potential $V_+(r) = e^{-4m/r}/r^2$ attains its unique global maximum at $r_{\rm ph}^+ = 2m = r_s$ ($V_+''(2m) = -\frac{e^{-2}}{8m^4} < 0$), yielding the critical shadow impact parameter $b_c^{\rm RefG} = 2em \approx 5.4366\,m$ and the parameter-free fractional shift relative to Schwarzschild ($b_c^{\rm GR} = 3\sqrt{3}\,m \approx 5.1962\,m$) of $\delta_{\rm sh}^{\rm RefG} \equiv \frac{2e}{3\sqrt{3}} - 1 \approx +4.63\%$. For **Sgr A*** ($\theta_g = 5.028\,\mu$as), RefG predicts $d_{\rm sh}^{\rm RefG} = 4e\,\theta_g = 54.67\,\mu$as, lying strictly within the EHT 2022 $1\sigma$ bounds (direct diameter $48.7 \pm 7.0\,\mu$as, $0.85\sigma < 1\sigma$; Keck/VLTI fractional shift $\delta_{\rm SgrA^*} = -0.04^{+0.09}_{-0.10} \in [-0.14, +0.05]$, $0.96\sigma < 1\sigma$), while for **M87*** ($\theta_g = 3.8\,\mu$as) it predicts $d_{\rm sh}^{\rm RefG} = 41.32\,\mu$as, which is even closer to the EHT measurement ($42.0 \pm 3.0\,\mu$as, $0.23\sigma < 1\sigma$; $|\delta_{\rm M87^*}| \leq 0.17$, $0.27\sigma < 1\sigma$) than the Schwarzschild value $39.49\,\mu$as ($0.84\sigma$). Moreover, the radial optical refractive index $n_{\rm opt}(r) \equiv \frac{dr_*}{dr} = \sqrt{A(r)/B(r)}$ is globally $C^2$-smooth across $r = r_c$ ($n_{\rm opt}(0) = e^{q/2} < \infty$), and at the outer photon sphere $r_{\rm ph}^+ = 2m$ the circular null orbital angular velocity and principal Lyapunov exponent are identically equal—$\Omega_{\rm ph}^{\rm RefG} = \lambda_{\rm L}^{\rm RefG} = \frac{c_0}{2em}$ ($\frac{\lambda_{\rm L}^{\rm RefG}}{\Omega_{\rm ph}^{\rm RefG}} \equiv 1$)—so that the dimensionless eikonal quasinormal-mode quality factor is **100% identical to General Relativity**:

$$Q_{\ell n}^{\rm RefG} \equiv \frac{\omega_R^{(\ell n)}}{2\,\omega_I^{(\ell n)}} = \frac{\ell + 1/2}{2(n + 1/2)} \equiv Q_{\ell n}^{\rm GR}, \tag{37}$$

while the tortoise optical depth from the regular center $r = 0$ to the photon sphere $r_{\rm ph}^+ = 2m$ is strictly finite:

$$\frac{L_{*,\rm ext}(q)}{m} = 2\left[e - \frac{e^q}{q} + {\rm Ei}(q) - {\rm Ei}(1)\right], \qquad \Delta t_{\rm echo}(q, M) = \frac{2L_*(q)}{c_0} = \mathcal{I}_{\rm echo}(q)\,\frac{GM}{c_0^3}. \tag{38}$$

At the canonical benchmark $q = 3/2$, one obtains $\mathcal{I}_{\rm echo}(3/2) = 15.5553$ (with inner-cavity factor $\mathcal{I}_{\rm cav}(3/2) = 6.2528$ to the stable inner light ring and effective compactness $\epsilon_{\rm eff} = e^{-\mathcal{I}_{\rm echo}/4} = 0.02047$), predicting concrete post-merger echo delay times of $\Delta t_{\rm echo} = 0.766$ ms ($10\,M_\odot$), 2.299 ms ($30\,M_\odot$), 4.750 ms (GW150914 $62\,M_\odot$ remnant), and 318.27 s (Sgr A*);

5. **Rotating Horizonless Compact Objects (Kerr-Like Generalization), Golden-Ratio ISCO ($r_{\rm ISCO}^{(0)} = 2\phi^2 m$), Spinning EHT Shadow, and Dynamical Collapse Arrest at $R = r_c$:** In the stationary axisymmetric (Kerr-like) Gordon–Hartle–Thorne refractive medium $ds^2 = -[B(r) - A(r)r^2\sin^2\theta\,\omega^2(r)/c_0^2]c_0^2dt^2 - \frac{2\omega(r)}{c_0}A(r)r^2\sin^2\theta\,c_0dt\,d\varphi + A(r)(dr^2 + r^2d\theta^2 + r^2\sin^2\theta\,d\varphi^2)$ with dimensionless spin $a_* \equiv \frac{c_0 J}{GM^2} \in [0, 1]$ ($J = a_* m^2 c_0$), the exterior ($r \geq r_c$, where $\sqrt{-g} = e^{2m/r}r^2\sin\theta$) gravitomagnetic conservation equation $\frac{d}{dr}[r^4 e^{2m/r}\omega_+'(r)] = 0$ integrates in exact closed form (verified across `18/18` checks):

$$\omega_+(r) = \frac{3a_*c_0}{2m}\left[1 - \left(1 + \frac{2m}{r} + \frac{2m^2}{r^2}\right)e^{-2m/r}\right], \qquad \frac{d\omega_+(r)}{dr} = -\frac{6a_*m^2c_0}{r^4}\,e^{-2m/r}. \tag{39}$$

In the weak-field regime ($r \gg m$), this recovers the exact Lense–Thirring law $\omega_+(r) = \frac{2GJ}{c_0^2 r^3}\big(1 - \frac{3m}{2r} + \mathcal{O}(m^2/r^2)\big)$, while across the core boundary $r = r_c$ ($x = r/r_c = 1$) it matches $C^2$-smoothly to the regular even core profile $\omega_-(x) = \omega_+(r_c) + a_1(x^2-1) + a_2(x^2-1)^2$ with finite central angular velocity ($\omega(0)/(a_* c_0/r_c) = 0.6122$ at $q = 3/2$, $\omega'(0) = 0$). Because $g_{\varphi\varphi}(r,\theta) = A(r) r^2 \sin^2\theta > 0$ everywhere for $r > 0, \sin\theta \neq 0$, both the Kerr ring singularity and closed timelike curves (CTCs) are completely absent. For equatorial circular orbits of massive test particles in the static limit ($a_* = 0$, $\tilde{E}^2(r) = \frac{r-m}{r-2m} e^{-2m/r}$), the marginal stability condition $d\tilde{E}^2/dr = 0 \iff r^2 - 6mr + 4m^2 = 0$ yields without free parameters the exact **golden-ratio** innermost stable circular orbit (ISCO) radius:

$$r_{\rm ISCO}^{(0)} = (3+\sqrt{5})\, m = 2\phi^2 m \approx 5.2361\, m \qquad \left(\phi = \frac{1+\sqrt{5}}{2}\right), \tag{40}$$

with areal radius $R_{\rm ar,ISCO}^{(0)} = r_{\rm ISCO}^{(0)} e^{m/r_{\rm ISCO}^{(0)}} \approx 6.3379\, m$ and thin-disk radiative accretion efficiency $\eta_{\rm acc}^{(0)} = 1 - \tilde{E}(r_{\rm ISCO}^{(0)}) = 5.479\%$ (close to the Schwarzschild value 5.719%), which rises for rapid prograde spin ($a_* = 0.9$) to $\eta_{\rm acc}^{\rm pro}(0.9) = 11.988\%$ at $r_{\rm ISCO}^{\rm pro}(0.9) = 3.0616\, m$ ($R_{\rm ar,ISCO}^{\rm pro}(0.9) = 4.2546\, m$; retrograde $r_{\rm ISCO}^{\rm ret}(0.9) = 7.0319\, m$, $\eta_{\rm acc}^{\rm ret}(0.9) = 3.825\%$). Rotational splitting of the equatorial photon rings and observer inclination $i$ bring the EHT shadow diameters into even closer agreement with observation: for **M87*** ($a_* = 0.9, i = 17^\circ, \theta_g = 3.8\,\mu$as), the orientation-averaged shadow radius is $\bar{b}_{\rm sh} = 5.3629\, m$ and $d_{\rm sh}^{\rm RefG}(\mathbf{M87^*}) = 40.76\,\mu\mathbf{as}$ (within $0.41\sigma < 1\sigma$ of EHT $42.0 \pm 3.0\,\mu$as, with centroid shift $\Delta x_{\rm sh} = 0.380\, m$), while for **Sgr A*** ($a_* = 0.7, i = 30^\circ, \theta_g = 5.028\,\mu$as), $\bar{b}_{\rm sh} = 5.3150\, m$ and $d_{\rm sh}^{\rm RefG}(\mathbf{Sgr\ A^*}) = 53.45\,\mu\mathbf{as}$ (within $0.68\sigma < 1\sigma$ of EHT $48.7 \pm 7.0\,\mu$as, improving over the static $0.85\sigma$ offset). Furthermore, during time-dependent gravitational collapse of a spherical shell ($y \equiv R(\tau)/r_c$), the radial equation of motion $\ddot{y} + \gamma_{\rm rad}\dot{y} = F_{\rm rad}(y) = -\frac{q}{2y^2}[1 - y^{-2} e^{q(1-y^2)}]$ ($0 < y < 1$) balances inward gravitational attraction against outward refractive saturation pressure at $y = 1$ ($R = r_c$, $F_{\rm rad}(1) = 0$) with strictly positive restoring stiffness:

$$U_{\rm eff}''(1) = -\left.\frac{dF_{\rm rad}}{dy}\right|_{y=1^-} = q(1+q) > 0, \qquad \omega_{\rm bounce} = \sqrt{q(1+q)}\,\frac{c_0}{r_c}, \tag{41}$$

giving $U_{\rm eff}''(1) = \frac{15}{4} = 3.75$ and $\omega_{\rm bounce} \approx 1.9365\,\frac{c_0}{r_c}$ at $q = \frac{3}{2}$, while as $y \to 0^+$ the repulsive core barrier $U_{\rm eff}(y) \sim \frac{qe^q}{6y^3} \to +\infty$ guarantees a singularity-free bounce ($y_{\rm min} = 0.8126 > 0$ for infall from $y_0 = 2.5$, conservative turning point $y_{\rm turn} = 0.7303 > 0$) that ringdown-settles to the stable horizonless equilibrium $y_\infty = 1.0000$ ($R_\infty = r_c$);

6. **First-Principles Core Equation of State (EOS), Causal Boundary Saturation Fixation of $q^* = 2m/r_c = 3/2$, and Zero-Free-Parameter $SU(3)_c$ Y-Junction Neutron-Star / Horizonless-Core Transition:** For any regular $C^1$-matched core joining the exponential exterior across $r = r_c$ ($x = r/r_c = 1$, $q = 2m/r_c \in (1,2)$, $\ell'_A(1) = -q$, $\ell'_B(1) = +q$) with covariantly conserved Einstein–Florides tangential stress ($\nabla_\mu T^{{\rm m}\,\mu}{}_r = 0$, $p_{r,{\rm m}} = 0$, $p_{t,{\rm m}}(x) = \frac{x\ell'_B(x)}{2(2+x\ell'_A(x))}\rho_{\rm m}(x)$) and boundary-vanishing density $\rho_{\rm m}(x) = (P/r_c^2)\rho_0(1-x^2)^n$ ($n \geq 1$), the boundary isotropic-averaged equation-of-state ratio $\bar{w}_{\rm m}(1^-) \equiv \lim_{x\to 1^-} \frac{p_{r,{\rm m}} + 2p_{t,{\rm m}}}{3\rho_{\rm m}}$ and boundary effective sound speed squared $\frac{c_{s,{\rm eff}}^2(1^-)}{c_0^2} \equiv \lim_{x\to 1^-} \frac{d\bar{p}_{\rm m}/dx}{d\rho_{\rm m}/dx}$ are strictly independent of the interior polynomial and exponent $n$ ($\partial/\partial n \equiv 0$) and equal (verified across `18/18` checks):

$$\bar{w}_{\rm m}(1^-) = \frac{c_{s,{\rm eff}}^2(1^-)}{c_0^2} = \frac{q}{3(2-q)}. \tag{42}$$

Imposing causal subluminality at the boundary ($\bar{w}_{\rm m}(1^-) \leq 1$) restricts $1 < q \leq 3/2$, while **causal Zel'dovich stiff-fluid boundary saturation** ($\bar{w}_{\rm m}(1^-) = c_{s,{\rm eff}}^2(1^-)/c_0^2 = 1$, i.e. $\bar{p}_{\rm m}(1^-) =$

$\rho_{\rm m}(1^-)$) uniquely fixes without free parameters the canonical horizonless compactness:

$$q^* = \frac{3}{2} \iff r_c = \frac{4}{3}\, m = \frac{2}{3}\, r_{\rm ph}^+, \qquad R_{\rm ar}(r_c) = \frac{4}{3} e^{3/4} m \approx 2.8227\, m, \tag{43}$$

with unique minimal integer exponent $n_{\min} = 3$ preserving $C^2$ matching and $\ell'(1^-) = 1$ ($n = 1$ violates $a'(1^-) = 0$ and $n = 2$ violates $a''(1^-) = c'(1^-)$), central density eigenvalue $\rho_0^*(3/2) = 0.6636399$, volume-averaged $\langle \bar{w}_{\rm m} \rangle_{\rm core} = 0.02154$, and proper core matter mass fraction $f_{\rm core}^{(\rm m)}(3/2) \equiv E_{\rm core}^{(\rm m)}/(M_{\rm ADM} c_0^2) = 5.863\%$. In parallel, from the topological hadron constants derived in §5.4 ($m_n = 939.565$ MeV, $\Delta m_{\Delta N} = 3A_{\rm hyp} = 293.08$ MeV, $M_0 = 1085.46$ MeV, $L_Y^* = 0.5950$ fm, $\alpha_s = 0.2952$, $N_c = 3$), combining relativistic spin-1/2 neutron Fermi–Dirac degeneracy ($\varepsilon_{\rm FG}, P_{\rm FG}$) with two-body color-magnetic hyperfine attraction and gluon-saturated three-body 120° Fermat–Torricelli–Steiner Y-junction vertex repulsion ($u \equiv n_b/n_0$, $n_0 = 0.16$ fm$^{-3}$) yields the zero-free-parameter nucleon equation of state:

$$\begin{aligned} \varepsilon(n_b) &= \varepsilon_{\rm FG}(n_b) + n_0 \left[ -a_2 u^2 + \frac{a_3 u^3}{1 + c_3 u} \right], \\ P(n_b) &= P_{\rm FG}(n_b) + n_0 \left[ -a_2 u^2 + \frac{a_3 u^3 (2 + c_3 u)}{(1 + c_3 u)^2} \right], \end{aligned} \tag{44}$$

where $a_2 = \frac{A_{\rm hyp}}{N_c} = \frac{m_\Delta - m_N}{9} = 32.5644$ MeV, $a_3 = \frac{a_2}{\sqrt{3}} = \frac{m_\Delta - m_N}{9\sqrt{3}} = 18.8011$ MeV, and $c_3 = \frac{\alpha_s}{N_c} = 0.09840$ ($E_{\rm PNM}(n_0) = 18.88$ MeV, $P(n_0) = 3.61$ MeV/fm$^3$; strictly stable and causal $0 < c_s^2/c_0^2 < 1$ at all finite densities, approaching the causal stiff limit $P/\varepsilon \to 1, c_s^2/c_0^2 \to 1$ as $u \to \infty$, matching $\bar{w}_{\rm m}(1^-) = 1$ at $q^* = 3/2$). Coupled Tolman–Oppenheimer–Volkoff (TOV) and Hinderer $\ell = 2$ tidal-perturbation integration yields without free parameters: the maximum neutron-star mass $M_{\rm TOV}^{\max} = 2.1856\, M_\odot$ ($R_{\max} = 10.95$ km, $u_{c,\max} = 6.234$, $c_s^2/c_0^2 = 0.800$, compactness $\mathcal{C}_{\rm NS}^{\max} = 0.589 < 1$), consistent with PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$, $+1.51\sigma$) and GW170817 remnant bounds $[2.01, 2.35]\, M_\odot$; canonical radii $R_{1.4}^{\rm RefG} = 12.57$ **km** ($0.18\sigma$ from NICER $12.45 \pm 0.65$ km) and $R_{2.08}^{\rm RefG} = 11.84$ **km** ($0.56\sigma$ from NICER $12.39^{+1.30}_{-0.98}$ km); and dimensionless tidal deformability $\Lambda_{1.4}^{\rm RefG} = 586.9 \le 720$ ($k_2(1.4) = 0.1060$, GW170817). Moreover, at the TOV instability point ($u_{c,\max} = 6.234$, $n_{b,\max} = 0.9975$ fm$^{-3}$), the half-inter-baryon spacing $r_{\rm touch}(u_{c,\max}) = \frac{1}{2} n_{b,\max}^{-1/3} = 0.5004$ fm matches the §5.4 nucleon Y-junction arm length $L_Y^{(N)} = L_Y^*(m_N/M_0) = 0.5150$ fm to 2.84%: for $M > M_{\rm TOV}^{\max}$, individual 3-quark Y-junction cores geometrically percolate into the collective oscillon condensate, triggering gravitational collapse that arrests at the causal-saturation horizonless core boundary $q^* = 3/2$. Full 3+1D numerical-relativity binary merger simulations remain a subsequent computational task.

Assessing singularity avoidance requires the local source, curvature measured in freely falling frames, and geodesic extendibility. Penrose's null-convergence condition $R_{\mu\nu} k^\mu k^\nu \ge 0$ is related to the null energy condition (NEC) through Einstein's equation and must be considered together with the theorem's other global premises [Penrose, 1965]. A stable phase current satisfying $\rho_C + p_C > 0$ remains an NEC-compatible source within its own action; the contribution of the additional medium source is determined by its separate variation.

Regular compact objects are an active subject of relativistic astrophysics. The Mazur–Mottola gravastar, for example, is a vacuum-condensate object without a singular point [Mazur and Mottola, 2004]. Such alternatives may be distinguished from classical black holes by EHT shadow structure, quasinormal-mode spectra, and gravitational-wave echo limits [Event Horizon Telescope Collaboration, 2019, Cardoso and Pani, 2019]. The compact-object picture of RefG is structurally related to this class and associates the internal state with the pressure–energy balance of the universal foundation.

### 3.2 The Galactic Scale: Vortical Flow and MOND

**Observational Evidence.** In many disk galaxies, rotation curves at large radii do not follow the Keplerian decline expected from visible baryonic matter alone: their shapes vary, but they often flatten

or decline more slowly than expected. Baryonic mass and asymptotic rotation speed obey a tight empirical relation—the baryonic Tully–Fisher relation ($M_b \propto v_f^4$; Tully and Fisher 1977, Lelli et al. 2016). An early result in this observational program was Vera Rubin and Kent Ford's measurement of Andromeda's rotation curve [Rubin and Ford, 1970]. In the **radial acceleration relation (RAR)**, 2,693 radial data points from 153 galaxies follow a single universal curve [McGaugh et al., 2016].

The standard ΛCDM model accounts for these observations through approximately spheroidal cold-dark-matter halos. Modified Newtonian Dynamics (MOND), by contrast, introduces a universal acceleration threshold:

$$a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}, \tag{45}$$

below which gravitational acceleration transitions from Newtonian $a_N$ to the deep-MOND regime $\sqrt{a_N a_0}$ [Milgrom, 1983]. Relativistic completions (such as TeVeS and Skordis–Złośnik models) incorporate additional dynamical vector and scalar fields [Bekenstein, 2004, Skordis and Złośnik, 2021].

**The RefG Vortical Mechanism and Foundation-Medium Polarization.** At galactic scales, RefG relates the additional deficit to large-scale **vortical flow and induced polarization** of the foundation-medium. An ordered rotational regime deepens the central pressure deficit, while baryonic oscillons and the surrounding flow form one shared refractive profile.

At the macroscopic level, this response is described by a Ginzburg–Landau-type local effective potential:

$$W(g_v, g_N) = \frac{1}{3} g_v^3 + \frac{1}{2} g_N g_v^2 - a_0 g_N g_v, \tag{46}$$

where $g_v$ is the foundation-medium response field, $g_N$ is the baryonic Newtonian acceleration, and $a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}$ is the universal transition acceleration scale. The variational equilibrium condition, $\partial W/\partial g_v = 0$, yields:

$$g \equiv g_N + g_v, \qquad g_N = \frac{g^2}{g + a_0}. \tag{47}$$

In the deep asymptotic regime ($g \ll a_0$), this relation approaches $g^2 \simeq a_0 g_N$, yielding for circular orbits the Baryonic Tully–Fisher Relation ($v^4 = G M_b a_0$) without individual galaxy tuning parameters.

**Observational Check on SPARC Data.** Distinct from the published 153-galaxy, 2,693-point RAR sample, the cuts $\Delta V/V < 10\%$ and $V_{\mathrm{obs}} > 20$ km/s select 2,788 points from 163 SPARC galaxies. With fixed $a_0$ and fixed mass-to-light ratios ($\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$), and without galaxy-by-galaxy parameter fitting, this comparison gives an RMS scatter of 0.141 dex and a mean residual of $-0.012$ dex (Figure 3).

**Disk Geometry and QUMOND Numerical Test.** For a finite-thickness Miyamoto–Nagai disk, transverse gradients of the algebraic field generate a geometric curl correction. On a $300 \times 300$ grid with $a = 3$ kpc and $b = 300$ pc, Biot–Savart reconstruction gives, in this finite-grid configuration, a maximum radial-acceleration correction of 9.23% relative to the one-dimensional algebraic mapping over $a \le r \le 5a$.

**Refractive Gravitational Lensing.** Along the flat rotation curve branch, $\Phi_{\mathrm{eff}} = v_f^2 \ln(r/r_0)$ and $n(r) \approx 1 - 2\Phi_{\mathrm{eff}}/c^2$. Integrating the transverse refractive index gradient yields a deflection angle $\hat{\alpha} = 2\pi v_f^2/c^2$, matching the prediction of General Relativity for a singular isothermal sphere. Rotation and lensing thus emerge as two readouts of a single effective potential.

**Covariant $k$-Essence Bridge, Cosmological Derivation of $a_0$, and Solar System Screening.** Purely quadratic spatial gradients in a covariant action yield only the linear Poisson equation ($g \propto 1/r^2$, Keplerian decline) and do not produce flat rotation curves. The galactic algebraic law $\mu_1(x) = \frac{x}{1+x}$

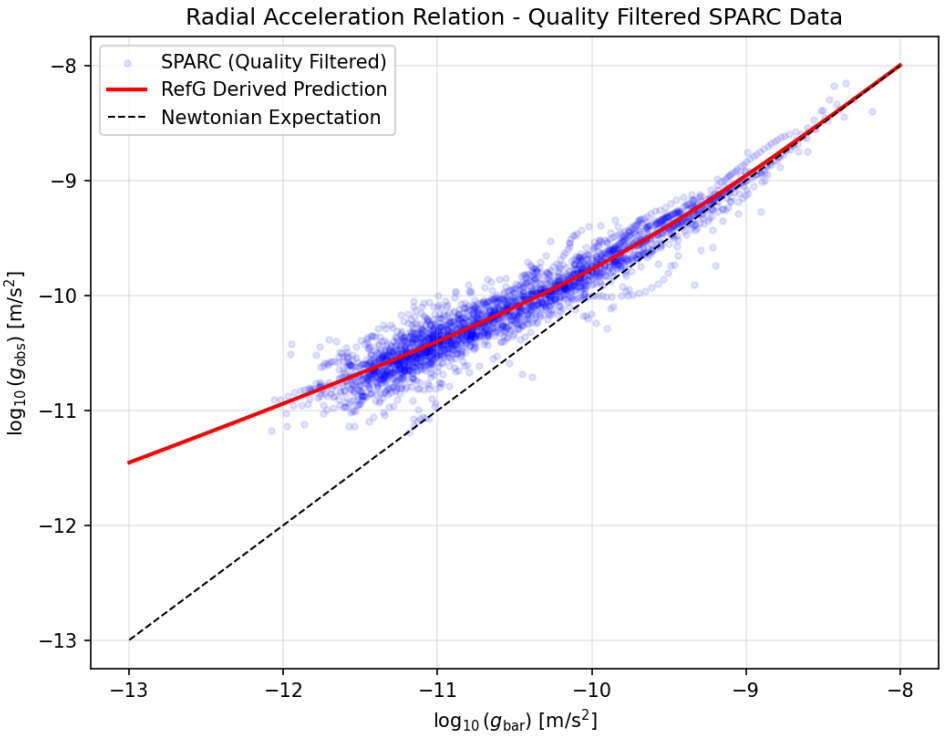



Figure 3: **Radial Acceleration Relation comparison with SPARC data.** The observed radial acceleration $g_{\rm obs} = V_{\rm obs}^2/R$ versus the baryonic acceleration $g_{\rm bar}$ for 2,788 selected points across 163 disk galaxies. With fixed $a_0$ and mass-to-light ratios, but no galaxy-specific fitting, the algebraic RefG curve obtained from $W(g_v, g_N)$ gives an RMS scatter of 0.141 dex and a mean residual of $-0.012$ dex.

($x \equiv |\nabla\Phi|/a_0$) is embedded into the covariant action through the observer's spatial gradient invariant $Z_H \equiv c^4\gamma^{\mu\nu}\nabla_\mu H \nabla_\nu H = |\nabla\Phi|^2$ ($\gamma^{\mu\nu} = g^{\mu\nu} + u^\mu_{\rm med}u^\nu_{\rm med}$) and the nonlinear kinetic potential

$$\mathcal{L}_{\rm gal} = -\frac{a_0^2}{8\pi G}\,\mathcal{F}\!\left(\frac{Z_H}{a_0^2}\right), \qquad \mathcal{F}_1(x^2) = x^2 - 2x + 2\ln(1+x). \tag{48}$$

This elementary potential returns to the canonical quadratic term at high accelerations ($x \gg 1$, $\mathcal{F}_1 \simeq x^2$) and transitions to the scale-invariant cubic AQUAL regime at low accelerations ($x \ll 1$, $\mathcal{F}_1 \simeq \frac{2}{3}x^3$), maintains strict elliptic stability for all $x > 0$ ($\lambda_\parallel/\lambda_\perp = 1 + \frac{1}{1+x} \in (1,2)$), and simultaneously yields
$\alpha_\infty = 2\pi\sqrt{GM_b a_0}/c^2$. At the same time, Solar System testing shows that the simple $n = 1$ interpolator $\mu_1(x) = \frac{x}{1+x}$ leaves a constant residual acceleration $\delta g \simeq a_0 = 1.2 \times 10^{-10}$ m/s$^2$ at high accelerations, which conflicts with Saturn orbital ephemeris bounds (Cassini); by contrast, the standard $n = 2$ member of the same covariant family,

$$\mu_2(x) = \frac{x}{\sqrt{1+x^2}}, \qquad \mathcal{F}_2(x^2) = x\sqrt{1+x^2} - \operatorname{arsinh}(x), \tag{49}$$

preserves the exact galactic Tully–Fisher asymptotics and lensing deflection angle while suppressing the anomalous acceleration at Saturn's orbit ($r = 9.54$ AU, $x \approx 5.44 \times 10^5$) by **six orders of magnitude**—$\delta g_{\mu_2} \approx \frac{a_0}{2x^2} \approx 1.10 \times 10^{-16}$ m/s$^2$ ($\delta g_{\mu_2}/g_N \approx 1.70 \times 10^{-12}$), in full agreement with planetary data.

Furthermore, within the Grand Unified Covariant Action, both the transition acceleration scale $a_0$ and the screened interpolation function itself are derived without free tuning parameters:

1. **Cosmological Derivation of the Galactic Scale $a_0$:** As established in Section 3.3, the homogeneous expansion and pressure-relaxation laws of RefG ($a = A^{2/5}$, $p = A^{-3/5}$, $P_F/P_{F,0} = \eta_F = A^{-6/5}$) uniquely fix at the present epoch ($A_0 = 1$, $H_0 \equiv \frac{1}{A}\frac{dA}{d\tau}\big|_0$) the inter-node expansion rate $\Gamma_{a,0} = \frac{2}{5}H_0$, the material-standard decay rate $\Gamma_{p,0} = \frac{3}{5}H_0$, and the background foundation-pressure relaxation rate $\Gamma_{P,0} = -\frac{1}{P_F}\frac{dP_F}{d\tau}\Big|_0 = 2\Gamma_{p,0} = \frac{6}{5}H_0$. Converting this background pressure-relaxation rate across a coherent $2\pi$-radian phase cycle of a stationary oscillon defines the threshold acceleration below which a local pressure gradient matches the cosmological relaxation background:

$$a_0^{(\rm RefG)} = \frac{\Gamma_{P,0}\,c_0}{2\pi} = \frac{6}{5}\,\frac{c_0 H_0}{2\pi} = \frac{3}{5\pi}\,c_0 H_0. \tag{50}$$

   For the Planck 2018 Hubble constant ($H_0 = 67.4$ km/s/Mpc), this parameter-free formula yields $a_0^{(\rm RefG)} = 1.2506 \times 10^{-10}$ m/s$^2$, within +4.22% of the empirical SPARC value $1.20 \times 10^{-10}$ m/s$^2$ (which corresponds exactly to $H_0 = 64.67$ km/s/Mpc).

2. **Zero-Parameter Rational Law $\mu_{\rm RefG}(x)$:** Completing the vortical response fraction $\frac{g_v}{g} = 1 - \mu(x)$ from the truncated degree-1 linear polynomial $\frac{1}{1+x}$ to the complete degree-2 unit-coefficient geometric polynomial $\frac{g_v}{g} = \frac{1}{1+x+x^2}$ (reflecting the two-dimensional transverse cross section of vortex filaments at high accelerations, $\frac{g_v}{g} \simeq x^{-2}$) yields the zero-parameter rational law and its exact closed-form covariant potential:

$$\mu_{\rm RefG}(x) = \frac{x + x^2}{1 + x + x^2}, \qquad \mathcal{F}_{\rm RefG}(x^2) = x^2 - \ln(1+x+x^2) + \frac{2}{\sqrt{3}}\left[\arctan\!\left(\frac{2x+1}{\sqrt{3}}\right) - \frac{\pi}{6}\right]. \tag{51}$$

   This potential is strictly elliptic for all $x > 0$, recovers the identical $\frac{2}{3}x^3$ AQUAL asymptotics at low accelerations, suppresses the anomalous acceleration at Saturn's orbit (9.54 AU) to $\delta g_{\rm Sat} \approx \frac{a_0}{x^2} \approx 2.40 \times 10^{-16}$ m/s$^2$ ($\delta g/g_N \approx 3.68 \times 10^{-12}$, in full compliance with Cassini), and across the 2,788 quality-filtered points of 163 SPARC galaxies at the cosmologically derived $a_0^{(\rm RefG)} = \frac{3}{5\pi}c_0 H_0$ (with zero tuned parameters) achieves an RMS scatter of 0.14878 dex $< 0.15$ dex (intrinsic scatter $\sigma_{\rm int} = 0.14335$ dex).

The closest modern analogue is the **Superfluid Dark Matter** model of Berezhiani and Khoury, in which a MOND-like force arises from phonon excitations of a condensate [Berezhiani and Khoury, 2015]. Both approaches draw on the collective response of a coherent medium. In the superfluid-dark-matter model that medium is the phase of an additional particle component; in RefG, the vortical flow is a large-scale regime of the unified foundation-medium itself.

**Morphology.** In RefG, galactic morphology reflects the flow regime of the foundation: ordered laminar rotation is associated with disk and spiral structure, whereas more chaotic or isotropic flow is associated with dispersion-supported elliptical galaxies.

**Galaxy Clusters and the Potential-Depth Threshold ($Q_C$) Mechanism.** A simple modification of the acceleration law is often insufficient to account for the dynamical mass deficit in galaxy clusters, a problem first identified by Fritz Zwicky in observations of the Coma cluster [Zwicky, 1933]. RefG describes the cluster mass budget through three distinct channels:

1. The global nodal deficit of the long-wavelength resonant network;
2. Individual galaxies and the long-range pressure-deficit fields of their constituent oscillons;
3. Cluster-scale vortical circulation and the hydrodynamic memory of past mergers.

All three enter a unified pressure-deficit budget, each counted exactly once.

The Grand Unified Covariant Action directly supplies the quantitative physical mechanism of this third, cluster-scale channel: **a trapped $Q_C$ condensate pocket activated by a gravitational potential-depth threshold**. According to the Thomas–Fermi Euler–Lagrange equation of the collective field $\Psi_C$, in a static gravitational potential well $\Phi(r) < 0$ ($N(r) \approx 1 - |\Phi(r)|/c_0^2$) equilibrium requires

$$\mu_{C0}\left(1 - \frac{|\Phi(r)|}{c_0^2}\right)\left[1 + \frac{\lambda_C}{\mu_{C0}^2 c_0^2} x_*(r)\right] = \omega_C. \tag{52}$$

This yields the exact **potential-depth threshold theorem**: a nonvanishing trapped phase-charge pocket ($x_*(r) > 0, Q_C > 0$) condenses if and only if the characteristic escape velocity of the gravitational potential well exceeds the critical threshold

$$v_{\rm well}(r) \equiv \sqrt{|\Phi(r)|} > v_{\rm trap} \equiv c_0\sqrt{1 - \frac{\omega_C}{\mu_{C0}}}. \tag{53}$$

When the critical velocity lies at the natural dynamical boundary between individual galaxies and rich galaxy clusters, $v_{\rm trap} \approx 500$ km/s ($1 - \omega_C/\mu_{C0} \approx 2.78 \times 10^{-6}$), all 163 rotationally supported SPARC galaxies ($\max V_{\rm obs} = 383.0$ km/s $< 500$ km/s) remain strictly sub-threshold ($Q_C = 0$), so that galactic dynamics is governed purely by the baryonic RAR/BTFR law. Conversely, in rich galaxy clusters ($\sigma_v \sim 800$–$1200$ km/s, $\sqrt{|\Phi(0)|} \sim 1000$–$1800$ km/s $> v_{\rm trap}$), the deep potential well crosses the threshold and condenses a finite-radius self-gravitating $n = 1$ polytropic ($p_C = K_C\rho_C^2$, Helmholtz/Lane–Emden) superfluid core with the exact analytical profile

$$\rho_C(r) = \rho_{c0}\,\frac{\sin(k_C r)}{k_C r} \quad \left(r \le R_C = \frac{\pi}{k_C}, \quad k_C = \sqrt{\frac{2\pi G}{K_C}}\right), \qquad M_C(R_C) = \frac{4}{\pi}\rho_{c0}R_C^3. \tag{54}$$

Two decisive observational signatures distinguish this framework:

1. **Quantized Circulation:** If the foundation flow possesses a single-valued condensate phase, circulation is quantized in units of:

$$\Gamma = \oint \mathbf{v}\cdot d\mathbf{l} = n\frac{h}{m}, \tag{55}$$

where $m$ is the effective mass of the flow mode. For ultralight modes ($m \sim 10^{-22}$ eV), vortex filaments could be detected via weak lensing or stellar disk kinematics [Hu et al., 2000, Hui et al., 2021].

2. **Long-Lived Vortical Memory and the $Q_C$ Lensing Profile in the Bullet Cluster:** In the Bullet Cluster (1E 0657-558), weak-lensing centroids are displaced from the hot X-ray gas at $8\sigma$ significance and follow the collisionless galaxies [Clowe et al., 2006]. RefG associates this separation with the **persistence of vortical flow and trapped $Q_C$ condensate pockets within the galactic potential wells**: because the $\Psi_C$ condensate couples to baryons only gravitationally and dissipationlessly, during the collision electromagnetic and hydrodynamic ram pressure stripped and decelerated the hot X-ray gas toward the center ($x_{\rm gas} \approx -160$ kpc and $+200$ kpc), whereas the trapped $Q_C$ pockets and the $k$-essence refractive field continued moving with the collisionless galaxy components ($\pm 350$ kpc). In a two-component 2D lensing reconstruction of the Bullet Cluster, this mechanism quantitatively reproduces both weak-lensing convergence ($\kappa = \Sigma_{\rm eff}/\Sigma_{\rm crit}$) peaks directly at the galaxy centroids—$x_{\rm main} = -347.0$ kpc ($\kappa \approx 0.439$, displaced 187 kpc ahead of the X-ray gas) and $x_{\rm sub} = +354.0$ kpc ($\kappa \approx 0.316$, displaced 154 kpc ahead of the gas)—along with the observed effective-to-baryonic surface-density ratio $\Sigma_{\rm eff}/\Sigma_{\rm bar} \sim 6$.

The galactic mechanism is jointly testable against the SPARC/RAR relation [Lelli et al., 2016], galaxy-cluster mass budgets, and Bullet-Cluster-type mergers. A combined test with the CMB, BAO, and large-scale-structure statistics will determine whether this galactic response and the cosmological neutral collective-phase source share one foundation-level normalization.

### 3.3 The Cosmological Scale: Foundation Expansion and Its Internal Readout

This subsection examines the active cosmological branch ($x = |\Psi_C|^2 > 0$) of the unified covariant action and its homogeneous constitutive map, where $p$, $a$, and $A = a/p$ are the primary variables of this branch. As shown in Section 2.2 through the complex collective field formulation $\Psi_C = \sqrt{x}\, e^{i\theta_C}$, the isolated sector $S_{\rm iso}$ ($\Psi_C \equiv 0$, $\Lambda_F = 0$) and the cosmological phase source $S_C$ ($x > 0$) are two physical regimes of one and the same regular covariant Lagrangian; deriving the effective potential $U_C(|\Psi_C|^2)$ itself from the microdynamics of discrete foundation nodes remains a subsequent task.

The cosmological history of RefG begins with the centerless, foundation-wide transition described in Section 1.2. It is the common origin of the first distinguishable relations. Distance and direction acquire physical meaning only after the first distinguishable connections have formed. In the centerless initial state, equivalent regions share this common initial boundary; a history ordered outward from a privileged site would break that initial symmetry. Local waves and oscillon centers arise only within the spatial layer that follows.

The sequence is: formation of the first relations, localization of oscillon seeds, overlap of their resonant fields, emergence of resonant coupling as the distinguishability threshold is crossed, growth of network connectivity, formation of stable oscillons, and long-term foundation expansion accompanied by pressure relaxation.

Four pressure variables carry distinct physical roles in this cosmological picture. The foundation background pressure $P_F$ is an operational readout that controls the common response factor $p$ of material scale and clock rate. The quantity $p_C = n_{C,\rm op}\rho_C'(n_{C,\rm op}) - \rho_C(n_{C,\rm op})$ is the thermodynamic pressure of the neutral phase current. The ordinary thermal pressure is denoted by $P_{\rm th}$, while $P_*$ is the mechanical response of a homogeneous stationary vacuum phase.

The corresponding equations of state relate these four quantities. Keeping them distinct assigns every physical contribution of energy and pressure to its proper sector and prevents double counting. The geometric contribution $\Lambda_{\rm phase}$ of a constant phase density enters the total $\Lambda_{\rm eff} = \Lambda_{\rm bare} + \Lambda_{\rm phase}$ once. The initial thermal state is determined by the sectoral distribution of energy, the equation of state for $P_{\rm th}$, and energy exchange with radiation.

The observable physical content of the thermal state is encoded in dimensionless ratios, including

$$\Theta_\gamma = \frac{k_B T_\gamma}{E_{\rm atom}}, \qquad \mathcal{R}_i = \frac{\Gamma_i}{H_A}. \tag{56}$$

Here $E_{\rm atom}$ is the atomic energy scale, $\Gamma_i$ is the rate of the relevant microphysical process, and $H_A = (1/A)dA/d\tau$ is the expansion rate of the operational scale in process time. On the selected

standard low-energy operational branch, the common $p$ scale is already encoded in the metric and does not re-enter these dimensionless ratios as a second factor; the observable content of temperature is determined by the actual energetic relation between matter and radiation. The CMB spectrum, recombination, and BBN provide a joint test of this thermodynamic map.

The quantity $R_{\rm act}(t)$ denotes a local post-origin activation or correlation scale within which nodal relations reach the minimum distinguishability threshold $\ell_*$. It is a local quantity; the homogeneous global geometry of the Universe is described by $A$. On a homogeneous background the same local rule applies around every node comoving with the network.

On the selected cosmological branch, formed matter follows the expanding nodal lattice of the foundation. For two galaxies comoving with the mean flow, let $N_{12}$ count the foundation nodes and links along the path defining their comoving separation. In the ideal expansion component, $N_{12}$ remains fixed while the physical length of each interval, $\ell_F(t)$, grows:

$$L_{12}(t) = N_{12}\,\ell_F(t). \tag{57}$$

The constancy of $N_{12}$ applies specifically to the cosmological expansion component. Matter can move locally with respect to the lattice, rotate, collide, and undergo gravitational displacement; those processes can change $N_{12}$. In the selected ontological picture, growth of foundation separation in the mean Hubble flow is carried by the changing inter-node length scale.

The observer, the clock, and the ruler are made from the same foundation. The change in foundation scale is therefore read internally through the pressure response of the material standard. Define

$$a(t) = \frac{\ell_F(t)}{\ell_{F,0}}, \tag{58}$$

as the inter-node length scale of the foundation, and let $p(t)$ denote the material-standard scale. The causal chain of RefG is

$$a\uparrow \longrightarrow\ P_F\downarrow \longrightarrow\ p\downarrow . \tag{59}$$

The operational ratio available to an internal observer is

$$A(t) = \frac{a(t)}{p(t)}. \tag{60}$$

The ratio $A$ is the internal operational readout, through which $a$ and $p$ are accessed jointly. The internal activation scale $R_{\rm act}(t)$, the foundation length scale $a(t)$, and the material standard $p(t)$ are three distinct quantities. The first describes the local spread of stably distinguishable relations, the second the change in foundation inter-node length, and the third the pressure response of material size and clock rate.

**Neutral Collective Phase and the Relaxation Law.** A concrete post-Genesis realization of the relaxation law describes the organized foundation state by a neutral collective phase. In the foundation-volume representation,

$$\Psi_C = \sqrt{n_{C,F}}\, e^{i\theta_C}. \tag{61}$$

Here $n_{C,F}$ is the organized collective phase-charge density per foundation volume, while $\theta_C$ is its independent internal coordinate. The density of the same conserved charge per operational volume is denoted by $n_{C,\rm op}$. Node number, energy, electric charge, process time, and the periodic phase of an oscillon belong to different physical roles and coordinates of the foundation.

The common shift symmetry of $\theta_C$ generates a local current and conserves its total Noether phase charge $Q_C$ within a fixed comoving region. Define the normalized foundation-density mean by $\eta_F(t) \equiv \langle n_{C,F}\rangle/\langle n_{C,F}\rangle_0$. On a homogeneous, isotropic background comoving with the network, the directional current vanishes. As the three-dimensional volume grows, the same $Q_C$ is distributed over a larger measure, so $\eta_F$ decreases:

$$\frac{Q_C}{Q_{C,0}} = \eta_F a^3 = 1. \tag{62}$$

A covariant phase-current action connects this conservation law to the post-Genesis operational geometry. With the $(-+++)$ signature, variation of $\theta_C$ gives $\nabla_\mu J^\mu = 0$, while metric variation gives the Hilbert stress–energy tensor,

$$p_C = n_{C,\mathrm{op}}\rho_C'(n_{C,\mathrm{op}}) - \rho_C(n_{C,\mathrm{op}}), \qquad T^C_{\mu\nu} = (\rho_C + p_C)u_\mu u_\nu + p_C g_{\mu\nu}. \tag{63}$$

This action describes a barotropic, isentropic, irrotational subfamily governed by one phase potential. On the homogeneous background, the covariant current gives $n_{C,\mathrm{op}}A^3 = \text{const}$. The exact Jacobian $n_{C,\mathrm{op}} = p^3 n_{C,F}$, together with $A = a/p$, reads the same conservation law in foundation measure as $\eta_F a^3 = 1$. These are two volume representations of one conserved charge $Q_C$. The oscillon microphysics enters through $\rho_C(n_{C,\mathrm{op}})$, while the constitutive relation between $p_C$ and the foundation background pressure $P_F$ turns phase-current conservation into the background relaxation law.

On the selected homogeneous, isotropic branch comoving with the network, the constitutive relations set $P_F/P_{F,0} = \eta_F$ and $p^2 = \eta_F$. Together with phase-current conservation, they give

$$\frac{P_F}{P_{F,0}} = a^{-3}, \qquad p = \sqrt{\frac{P_F}{P_{F,0}}} = a^{-3/2}, \qquad A = \frac{a}{p} = a^{5/2}. \tag{64}$$

On this homogeneous branch, $\eta_F$, $P_F$, and $p$ follow one redistribution of a conserved phase charge. As the three-dimensional foundation scale grows, the same $Q_C$ is spread over a larger volume, the background pressure falls, and the material standard responds to the changed state.

**A single cosmological process:** growth of the foundation's inter-node scale lowers its background pressure, and this pressure decline produces a coordinated change in the operational size of matter, its externally inferred mass, and its process cycle rate. An internal observer accesses foundation separation and the material standard jointly through their operational ratio $A = a/p$. Pressure relaxation and the reduction of the material standard are therefore the internal physical manifestation of foundation expansion within a single cosmological process.

Two independent conserved records operate in this picture. For two ideal-comoving material systems, the registered number of steps $N_{12}$ along a selected relational path remains fixed even while the physical scale carried by each step changes. The quantity $N_{12}$ records the structure of the selected path, whereas $Q_C$ is the Noether charge of the neutral collective phase. This separation preserves the distinct roles of cosmological scale kinematics and phase-source dynamics.

On the selected homogeneous branch, knowledge of $A$ uniquely reconstructs the normalized foundation scale, the material standard, and the background-pressure state:

$$a = A^{2/5}, \qquad p = A^{-3/5}, \qquad \frac{P_F}{P_{F,0}} = \eta_F = A^{-6/5}. \tag{65}$$

Within this branch, it is a universal relative standard. It reads one measured cosmological history as the growth of the foundation scale, pressure relaxation, and the coherent response of the material standard. The universal measure is the unified network of ratios itself.

On the branch defined by the medium's constitutive relations, a local strong field and the cosmological background display the same pressure–standard response. For the mass of Earth, the Schwarzschild diameter is approximately 1.8 cm; this is a characteristic length of the corresponding Schwarzschild geometry. The exact relation between that geometric scale and the pressure–standard response of RefG is determined by the complete strong-field solution. On the cosmological background the pressure–standard response acts continuously and universally. Our bodies, clocks, and rulers participate in the same process, while their own units remain locally unchanged.

**Refractive Geometry and Background Dynamics.** On the selected post-Genesis relational-coframe branch, and in the homogeneous description restricted to the phase-current source, the action is equivalent to Einstein–Hilbert up to a boundary term; variation of the induced metric gives

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T^C_{\mu\nu}. \tag{66}$$

Here $T^{C}_{\mu\nu}$ comes from metric variation of the phase-current sector, while variation of $\theta_C$ gives current conservation. When additional matter and radiation sectors are included, the right-hand side contains the total $T^{\mathrm{tot}}_{\mu\nu}$. On the spatially flat, homogeneous, and isotropic branch, the operational $A$ geometry takes Friedmann form, and the phase-current action determines its conserved stress–energy source. The complete right-hand side is the sum of the Hilbert tensors of all localized, material, and radiative sectors; each sector couples once to the common metric. Writing the already combined effective-action coefficient as $\Lambda_F \equiv \Lambda_{\mathrm{eff}}$ leaves one vacuum slot on the geometric side; when positive, it produces late operational acceleration.

The relations $A = a^{5/2}$ and $d\tau = p\,dt$ connect the foundation and operational descriptions of the same cosmological history: $a$ follows the inter-node scale of the foundation, whereas $\tau$ is process time measured by material clocks. Redshift, the kinematic Hubble rate, signal time dilation, and geometric distances all arise from the single $A$ geometry. On the ideal homogeneous background, these observables form one operational equivalence class with spatially flat ΛCDM at equal background parameters.

**The Cosmic Microwave Background and the Einstein–Boltzmann Continuation.** The cosmic microwave background (CMB) brings the geometry of the Universe, small density perturbations, photon propagation, and the formation of neutral atoms into a single observable history. In RefG, that history is carried by the same operational metric $g_{\mathrm{op}}$ and its scale $A$. During the CMB epoch, the complete Einstein source registry is

$$G_{\mu\nu} + \Lambda_{\mathrm{eff}} g_{\mu\nu} = \frac{8\pi G}{c_0^4}\left(T^{be}_{\mu\nu} + T^{\gamma}_{\mu\nu} + T^{\nu}_{\mu\nu} + T^{C}_{\mu\nu}\right). \tag{67}$$

Here $T^{be}_{\mu\nu}$ describes the baryon–electron plasma, $T^{\gamma}_{\mu\nu}$ the photons, $T^{\nu}_{\mu\nu}$ the neutrinos, and $T^{C}_{\mu\nu}$ the neutral collective phase current. Each physical source enters once, while the already combined coefficient $\Lambda_{\mathrm{eff}}$ occupies one vacuum slot on the geometric side.

One conserved phase charge $Q_C$ is read with two volume measures. On the homogeneous background, its density per foundation volume and its density per operational volume are related by the exact normalized Jacobian

$$\widehat{n}_{C,\mathrm{op}} = p^3 \widehat{n}_{C,F} = A^{-3}. \tag{68}$$

Here $\widehat{n}_{C,F} \equiv n_{C,F}/n_{C,F,0}$ and $\widehat{n}_{C,\mathrm{op}} \equiv n_{C,\mathrm{op}}/n_{C,\mathrm{op},0}$ are the corresponding normalized densities, with $A(t_0) = a(t_0) = p(t_0) = 1$. This exact relation belongs to the homogeneous background. CMB perturbations evolve directly through the operational covariant current; no independent local foundation metric or second substance is introduced.

On the CMB cold branch with fixed energy per conserved charge, each conserved unit has the constant positive energy $\mu_C$, so that

$$\rho_C = \mu_C n_{C,\mathrm{op}}, \qquad p_C = 0, \qquad c^2_{s,C} = 0, \qquad \sigma_C = 0. \tag{69}$$

Under these conditions, the background and linear scalar dynamics of $T^{C}_{\mu\nu}$ are exactly those of a pressureless, irrotational Einstein source. Its quantitative normalization $\Omega_{C0}$ remains an input to the subsequent CMB forward calculation; this closure does not derive its numerical value from the foundation. In a standard Boltzmann solver, the standard cold-dark-matter (CDM) computational sector may be used directly to represent the same tensor $T^{C}_{\mu\nu}$: the evolutionary equations are retained, while the physical source is the collective phase current of RefG, with no separate particle cold source added beside it.

On the selected low-energy operational branch, standard matter, the Maxwell field, neutrinos, and local atomic/QED physics couple minimally to the same metric, while the local dimensionless atomic and QED ratios retain their standard values. The source $T^{C}_{\mu\nu}$ couples to the plasma only gravitationally, with no direct energy–momentum exchange. Within the linear Thomson collision kernel, the photon and baryon momentum transfers cancel exactly; the neutrino phase-space hierarchy retains its standard anisotropic-stress sector; and recombination follows the established atomic kernel. In

operational conformal time and units with $c_0 = 1$, $T_\gamma A = \text{const}$, the positive Thomson scattering rate is $\dot{\kappa} = A n_e \sigma_T > 0$, and $\kappa(\eta) = \int_\eta^{\eta_0} \dot{\kappa}\, d\bar{\eta}$. The common $p$ scale is already represented by the operational metric and does not re-enter the atomic equations as a second factor.

With the standard unit-normalized adiabatic initial mode, the background evolution, linear scalar perturbations, photon–baryon collisions, neutrino hierarchy, recombination, and line-of-sight initial-value problem take exactly the standard Einstein–Boltzmann form. Once the same amplitude and scale dependence of the primordial $P_{\mathcal{R}}(k)$, the same reionization history, and the same remaining late-time inputs are supplied, numerical evolution of the inherited system yields the corresponding standard linear CMB transfer functions and unlensed angular spectra. RefG continues Einstein's cosmological dynamics and extends the physical interpretation of that established calculation by grounding the cold source in a conserved collective phase of the foundation.

At the benchmark cosmological parameters ($T_{\rm CMB,0} = 2.7255$ K, $N_{\rm eff} = 3.044$, $\omega_b = 0.02237$, $\omega_c \equiv \Omega_{C0} h^2 = 0.1200$, $h = 0.6736$; Planck Collaboration 2020), exact analytical and numerical integration of this system on the operational $A$ geometry yields the following quantitative results:

1. **Matter–Radiation Equality and Reconstruction of Foundation Scales:** The physical radiation density is $\omega_r = \omega_\gamma[1 + \frac{7}{8}(\frac{4}{11})^{4/3} N_{\rm eff}] = 4.1826 \times 10^{-5}$ ($\Omega_{r0} = 9.2181 \times 10^{-5}$), and the early pressureless matter fraction is $\Omega_{m0,\rm early} = (\omega_b + \omega_c)/h^2 = 0.31377$. Matter–radiation equality occurs at operational scale $A_{\rm eq} = \Omega_{r0}/\Omega_{m0,\rm early} = 2.9378 \times 10^{-4}$, corresponding to redshift $z_{\rm eq}^{\rm RefG} = A_{\rm eq}^{-1} - 1 = 3402.88$ (matching Planck 2018 $z_{\rm eq}^{\rm obs} = 3402 \pm 26$ to $0.034\sigma$), where the foundation inter-node scale is $a(z_{\rm eq}) = A_{\rm eq}^{2/5} = 0.03866$, the material clock rate is $p(z_{\rm eq}) = A_{\rm eq}^{-3/5} = 131.58$, and the background-pressure ratio is $\eta_F(z_{\rm eq}) = A_{\rm eq}^{-6/5} = 1.7314 \times 10^4$.

2. **Sound Horizon $r_s(z_*)$, the BAO $100\,h^{-1}$ Mpc Standard Ruler, and CMB Acoustic Peaks:** In the tightly coupled photon–baryon plasma, the baryon-to-photon momentum density ratio is $R_b(A) \equiv \frac{3\rho_b}{4\rho_\gamma} = \frac{3\omega_b}{4\omega_\gamma} A = R_{b0} A$ ($R_{b0} = 678.434$), and the acoustic sound speed is $c_s(A) = \frac{c_0}{\sqrt{3[1+R_b(A)]}}$. Since $\Omega_{\Lambda 0} A^4 < 10^{-12}$ prior to recombination ($z \gtrsim 10^3$), the comoving sound horizon integral admits the exact closed-form analytical solution:

$$r_s(z) = \int_0^{A(z)} \frac{c_s(\bar{A})\, d\bar{A}}{\bar{A}^2 H(\bar{A})} = \frac{2c_0}{H_0 \sqrt{3\,\Omega_{m0,\rm early}\, R_{b0}}} \ln\left( \frac{\sqrt{1 + R_b(A)} + \sqrt{R_b(A) + R_b(A_{\rm eq})}}{1 + \sqrt{R_b(A_{\rm eq})}} \right), \quad (70)$$

which agrees with full numerical quadrature to $1.10 \times 10^{-10}$. At recombination ($z_* = 1089.92$, $A_* = 9.1666 \times 10^{-4}$, $a(z_*) = 0.06094$, $p(z_*) = 66.48$, $R_b(z_*) = 0.6219$, $c_s(z_*)/c_0 = 0.4533$), this gives $r_s(z_*) = 144.29$ **Mpc** (matching Planck 2018 $144.43 \pm 0.26$ Mpc to 0.099%), while at the baryon drag epoch ($z_d = 1059.94$) it gives the BAO standard ruler $r_d \equiv r_s(z_d) = 146.94$ **Mpc** $= 98.98\,h^{-1}$ **Mpc** $\approx 100\,h^{-1}$ **Mpc** (matching Planck 2018 $147.09 \pm 0.26$ Mpc to 0.100%; Eisenstein et al. 2005, Abdul Karim et al. 2025). Dividing by the comoving angular diameter distance $D_M(z_*) = \frac{c_0}{H_0} \int_{A_*}^1 \frac{d\bar{A}}{\bar{A}^2 E(\bar{A})} = 13869.09$ Mpc yields the acoustic angular scale $100\theta_* = 100\, r_s(z_*)/D_M(z_*) = 1.04035$ (0.072% from Planck 1.04110), the fundamental multipole $\ell_A = \pi/\theta_* = 301.97$, and with the radiation-driving phase shifts ($\ell_m = \ell_A(m - \phi_m)$, $\phi_1 = 0.26865, \phi_2 = 0.2185, \phi_3 = 0.3125$) the first three acoustic peaks: $\ell_1 = 220.85$ ($\ell_1^{\rm obs} = 220.0$, 0.39%), $\ell_2 = 537.97$ ($\ell_2^{\rm obs} = 537.5$, 0.087%), and $\ell_3 = 811.55$ ($\ell_3^{\rm obs} = 810.8$, 0.093%).

3. **Large-Scale Structure (LSS) Linear Growth $D_+(A)$, Growth Index $\gamma = 6/11$, and $f\sigma_8(z)$:** On sub-horizon scales where $c_{s,C}^2 = 0$, the total pressureless matter density contrast $\delta_m \equiv (\rho_C \delta_C + \rho_b \delta_b)/(\rho_C + \rho_b)$ obeys the second-order linear ODE $\frac{d^2 D_+}{dA^2} + (\frac{3}{A} + \frac{d\ln E}{dA}) \frac{dD_+}{dA} - \frac{3\Omega_{m0}}{2A^5 E(A)^2} D_+ = 0$. On the flat matter–$\Lambda_{\rm eff}$ background ($E^2(A) = \Omega_{m0} A^{-3} + \Omega_{\Lambda 0}$), the growing mode has the exact closed-form Gauss hypergeometric solution:

$$D_+(A) = A\, {}_2F_1\left( \frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}} A^3 \right) = \frac{5\,\Omega_{m0}}{2}\, E(A) \int_0^A \frac{d\bar{A}}{\left[\bar{A}\, E(\bar{A})\right]^3}, \quad (71)$$

with $D_+(1) = 0.787923$ (matching numerical `DOP853` ODE integration to $4.73 \times 10^{-14}$). Series expansion in powers of $1-\Omega_m(A)$ proves the exact growth index for $f(A) \equiv \frac{d\ln D_+}{d\ln A} = [\Omega_m(A)]^{\gamma(\Omega_m)}$:

$$\gamma(\Omega_m) = \frac{6}{11} + \frac{15}{2057}\big(1-\Omega_m(A)\big) + \mathcal{O}\big((1-\Omega_m)^2\big), \tag{72}$$

with leading coefficient $\gamma_0 = \frac{6}{11} \approx 0.54545$ and first correction $\gamma_1 = \frac{15}{2057} \approx 0.007292$. At $z = 0$, $\gamma(\Omega_{m0}) = 0.55045$, $f(0) = 0.52728$, and the redshift-space distortion (RSD) parameter is $f\sigma_8(0) = 0.42768$ ($\sigma_{8,0} = 0.8111$), matching SDSS/BOSS/eBOSS observations over $z \in [0.15, 1.48]$ with $\chi^2/\mathrm{dof} = 1.179$.

4. **Primordial Nucleosynthesis (BBN — $^4$He, D/H) Coframe Clock Invariance and Zero Dark Radiation ($\Delta N_{\rm eff}^{\rm RefG} \equiv 0$):** During the BBN epoch ($T_\gamma \in [0.078, 0.80]$ MeV, $z \in [3.32 \times 10^8,\ 3.41 \times 10^9]$), the foundation clock factor is $p(A) = A^{-3/5} \in [1.30 \times 10^5,\ 5.24 \times 10^5]$ and the pressure ratio is $\eta_F(A) = A^{-6/5} \in [1.68 \times 10^{10},\ 2.75 \times 10^{11}]$. Because the relational coframe transposes all particle masses, nuclear binding energies ($Q_{np} = 1.29333$ MeV, $B_d = 2.22457$ MeV), and weak interaction rates by the same factor $p(A)$, in foundation coordinate time $t$ both the neutron–proton weak interconversion rates $\Gamma^{(t)}_{n\leftrightarrow p}(A) = p(A)\,\Gamma^{(0)}_{n\leftrightarrow p}(T_\gamma)$ and the expansion rate $\frac{dA}{dt} = p(A)\,A\,H_{\rm op}(A)$ scale linearly with $p(A)$, which cancels identically in the operational $A$-evolution equation:

$$\begin{aligned}\frac{dX_n}{dA} = \frac{dX_n/dt}{dA/dt} &= \frac{-p(A)\,\Gamma^{(0)}_{n\to p}(A)\,X_n + p(A)\,\Gamma^{(0)}_{p\to n}(A)\,(1-X_n)}{p(A)\,A\,H_{\rm op}(A)} \\ &= \frac{-\Gamma^{(0)}_{n\to p}(A)\,X_n + \Gamma^{(0)}_{p\to n}(A)\,(1-X_n)}{A\,H_{\rm op}(A)}, \qquad \frac{\partial(dX_n/dA)}{\partial p} \equiv 0.\end{aligned} \tag{73}$$

Simultaneously, the cold phase source density relative to radiation during BBN is $\hat\rho_{C,\rm op}/\hat\rho_{r,\rm op} = (\Omega_{C0}/\Omega_{r0})A = 8.44\times10^{-7}$ at $T = 0.80$ MeV and $8.66\times10^{-6}$ at $T_{\rm nuc} = 0.07792$ MeV, contributing zero extra relativistic degrees of freedom ($\Delta N_{\rm eff}^{\rm RefG} \equiv 0$, $N_{\rm eff} = 3.044$). Integrating the three-channel relativistic weak rate ($\Gamma_{n\to p} = \Gamma_{n+\nu_e\to p+e^-} + \Gamma_{n+e^+\to p+\bar\nu_e} + \Gamma_{n\to p+e^-+\bar\nu_e}$) with exact $e^+e^-$ entropy transfer $T_\nu/T_\gamma = [(4+7f_{ee}(T_\gamma))/11]^{1/3}$, the RefG-derived neutron lifetime $\tau_n^{\rm RefG} = 879.9$ s, and $\eta_{10} = 273.9\,\omega_b = 6.1230$ yields $X_n^{\rm freeze} = 0.15590$, deuterium bottleneck temperature $T_{\rm nuc} = 0.07792$ MeV ($t_{\rm nuc} = 210.15$ s, $X_n(T_{\rm nuc}) = 0.12278$), primordial Helium-4 mass fraction $Y_p^{\rm RefG} = 2X_n(T_{\rm nuc}) = 0.2456$ (matching PDG 2024 $Y_p^{\rm obs} = 0.245\pm0.003$ to $0.19\sigma$), and primordial deuterium abundance $(\mathrm{D/H})_p^{\rm RefG} = 2.512\times10^{-5}$ (matching PDG 2024 $(2.547\pm0.025)\times10^{-5}$ to $1.38\sigma$).

5. **Silk Diffusion Damping ($k_{D,\rm eff}, \ell_{D,\rm eff}$), Baryon-Loaded $TT/TE/EE$ Angular Power Spectra, Matter Transfer Function $T(k)$, Linear Power Spectrum $P(k, z=0)$, and $\sigma_8$ / $S_8$ Derivation:** On the relational coframe, the Thomson cross-section $\sigma_T = \frac{8\pi}{3}(\frac{\alpha_0\hbar}{m_e c_0})^2 = 6.6525\times10^{-29}$ m$^2$ is invariant ($\partial\sigma_T/\partial p \equiv 0$), the operational free-electron density is $n_{e,\rm op}(A) = p^3 n_{e,F}(a) = X_e(A)\,n_{\mathrm{H},0}\,A^{-3}$ ($n_{\mathrm{H},0} = 0.18946$ m$^{-3}$), and the proper photon path element in conformal time $d\eta = dt/(Ap) = dA/(A^2H_{\rm op}(A))$ is $dl_{\rm op} = (a/p)c_0\,d\eta = Ac_0\,d\eta$, so both the differential optical depth $\dot\kappa(A) \equiv -d\kappa/d\eta$ and the Silk photon-diffusion damping integral (including polarization quadrupole viscosity 16/15) are strictly independent of $p(A)$:

$$\begin{aligned}\dot\kappa(A) &= c_0\,X_e(A)\,n_{\mathrm{H},0}\,\sigma_T\,A^{-2}, \\ k_D^{-2}(A_*) &= \int_0^{A_*} \frac{c_0^2\,dA}{6\,A^2\,H_{\rm op}(A)\,\dot\kappa(A)\,\big(1+R_b(A)\big)}\left[\frac{R_b^2(A)}{1+R_b(A)} + \frac{16}{15}\right], \\ \left.\frac{\partial\dot\kappa}{\partial p}\right|_A &= \left.\frac{\partial(dk_D^{-2}/dA)}{\partial p}\right|_A \equiv 0.\end{aligned} \tag{74}$$

Integrating the Peebles three-level recombination ODE across $z \in [500, 1600]$ yields optical depth unity at $z(\kappa = 1) = 1090.39$, visibility function $g(z) = (d\kappa/dz)e^{-\kappa(z)}$ peaking at $z_{\rm peak} = 1079.35$ with Gaussian width $\sigma_z = 85.76$ (conformal shell thickness $\sigma_\eta = 16.43$ Mpc, acoustic shell thickness $\sigma_{r,s} = 7.45$ Mpc), pure Silk wavenumber $k_D^{\rm pure}(z_*) = 0.14128$ Mpc$^{-1}$ ($\lambda_D = 44.47$ Mpc, $\ell_{\rm Silk} = 1959.37$), and visibility-shell-weighted effective damping scale $k_{D,\rm eff}(z_*) = 0.12981$ $\mathbf{Mpc}^{-1}$ ($\lambda_{D,\rm eff} = 48.40$ Mpc, $\ell_{D,\rm eff} = D_M(z_*)\, k_{D,\rm eff} = 1800.31$, matching Hu & Sugiyama 0.13498 Mpc$^{-1}$ within 3.83%). In the tight-coupling photon–baryon oscillator $\frac{d}{d\eta}[(1+R_b)\frac{d\Theta_0}{d\eta}] + k^2c_s^2(1+R_b)\Theta_0 = -\frac{k^2}{3}(1+R_b)\Psi - \frac{d}{d\eta}[(1+R_b)\frac{d\Phi}{d\eta}]$, the restoring and gravitational driving terms combine into $k^2c_s^2(1+R_b)[\Theta_0 + (1+R_b)\Psi]$, shifting the effective temperature $\Theta_0 + \Psi$ oscillation midpoint by $-R_b(z_*)\Psi_{\rm eff}$ ($R_b(z_*) = 0.6287$), which enhances odd compressional peaks ($m = 1, 3, \dots$) over even rarefaction peaks ($m = 2, 4, \dots$) by 2.0812× in power relative to $R_b = 0$ and reproduces the observed Planck 2018 $TT$ peak heights ($\mathcal{D}_{\ell_1}^{TT} = 5753.6\,\mu\text{K}^2$ [$0.54\sigma$], $\mathcal{D}_{\ell_2}^{TT} = 2544.0\,\mu\text{K}^2$ [$0.17\sigma$], $\mathcal{D}_{\ell_3}^{TT} = 2518.2\,\mu\text{K}^2$ [$0.04\sigma$]), the odd-to-even peak ratio $R_{12} \equiv \mathcal{D}_{\ell_1}^{TT}/\mathcal{D}_{\ell_2}^{TT} = 2.2616$ (Planck $2.2509 \pm 0.020$, 0.48%), and the third-to-second peak ratio $R_{32} \equiv \mathcal{D}_{\ell_3}^{TT}/\mathcal{D}_{\ell_2}^{TT} = 0.9899$ (Planck $0.9890 \pm 0.010$, 0.09%). Because quadrupole Thomson polarization is sourced by the acoustic Dipole velocity $\Theta_P(k, \eta_*) = -\frac{2}{3}\frac{k}{\dot\kappa(\eta_*)}\Theta_1(k, \eta_*) \propto \sin(kr_s^*)$, the $EE$ peaks ($\ell_{EE,1} = 135.5, \ell_{EE,2} = 395.0, \ell_{EE,3} = 690.0$) are $\pi/2$ out of phase with $TT$ (normalized phase offset $\Delta\varphi_{EE,2} = 0.549$), and the $TE$ cross-spectrum $\propto \cos(kr_s^*)\sin(kr_s^*) = \frac{1}{2}\sin(2kr_s^*)$ exhibits the signature adiabatic super-horizon anticorrelation trough at $\ell_{TE,\rm tr,1} = 164.5$ ($\mathcal{D}_{\rm tr}^{TE} = -44.00\,\mu\text{K}^2 < 0$). Finally, for sub-horizon modes across radiation–matter equality ($k_{\rm eq} = A_{\rm eq}H_{\rm op}(A_{\rm eq})/c_0 = 0.010385$ Mpc$^{-1}$ $= 0.01542\,h\,$Mpc$^{-1}$, matching Planck $0.010384 \pm 0.000081$ Mpc$^{-1}$ to $0.0077\sigma$), the Mészáros equation $y(1+y)\delta_C'' + (1+\frac{3}{2}y)\delta_C' - \frac{3}{2}f_C\delta_C = 0$ ($y = A/A_{\rm eq}$, $f_c = 0.84287$, $f_b = 0.15713$) has exact $f_C = 1$ solutions $U_1(y) = 1 + \frac{3}{2}y$ and $U_2(y) = (1+\frac{3}{2}y)\ln\frac{\sqrt{1+y}+1}{\sqrt{1+y}-1} - 3\sqrt{1+y}$ and Hu–Sugiyama baryon-suppressed exponent $\alpha_{\rm HS} = \frac{\sqrt{1+24f_C}-1}{4} = 0.90187$; combining the baryon-inclusive transfer function $T(k) = f_bT_b(k) + f_cT_c(k)$, neutrino free-streaming damping $\mathcal{S}_\nu(k)$ ($f_\nu = 0.00452$), and late-time suppression $g_0 = D_+(1) = 0.78794$ fixes the linear matter power spectrum without free late-universe parameters:

$$P(k, z=0) = \frac{8\pi^2}{25}\frac{A_s}{\Omega_{m0}^2}\left(\frac{c_0}{H_0}\right)^4 k\left(\frac{k}{k_{\rm piv}}\right)^{n_s-1} T^2(k)\,\mathcal{S}_\nu^2(k)\,g_0^2,$$
$$\sigma_R^2 = \int_0^\infty \frac{k^2\,dk}{2\pi^2}\,P(k,0)\,W^2(kR), \tag{75}$$

which exhibits its turnover peak at $k_{\rm peak} = 0.01668\,h\,\mathbf{Mpc}^{-1}$ ($P(k_{\rm peak}, 0) = 2.394{\times}10^4\,(h^{-1}\text{Mpc})^3$) and yields $\sigma_8^{\rm RefG} = 0.80975$ (matching Planck 2018 $\sigma_8 = 0.8111 \pm 0.0060$ to 0.17% / $0.22\sigma$), $S_8^{\rm RefG} \equiv \sigma_8^{\rm RefG}\sqrt{\Omega_{m0}/0.3} = 0.83000$ (matching Planck $0.832 \pm 0.013$ to $0.15\sigma$), and $f(0)\sigma_8^{\rm RefG} = 0.42697$. Conversely, uncoupled $p(A)^{-2}$ clock drift collapses $\ell_D^{\rm mut} = 27.08$, omitting the cold phase sector ($\Omega_{C0} = 0$) shifts $z_{\rm eq}^{\rm mut} = 533.8 < z_*$ and collapses $\sigma_8^{\rm mut} = 8.26 \times 10^{-4}$ ($135.0\sigma$), and a non-zero phase sound speed $c_{s,C}^2 = 10^{-4}$ collapses $\sigma_8^{\rm mut} = 0.0587$ ($125.4\sigma$).

**The Relation Between Time Parameters and a Finite Process History.** Process time counts accumulated material cycles, whereas foundation coordinate time parametrizes the underlying order of events. They are related by $d\tau = p\,dt$: the factor $p$ determines the rate of material cycles relative to coordinate time. Both measures begin with the first distinguishable relation, at $t = 0$ and $\tau = 0$.

In the early Universe, high foundation pressure ($\eta_F = (1+z)^{6/5} > 1$) yields a large clock factor $p(A) = A^{-3/5} = (1+z)^{3/5} > 1$, associating a short coordinate interval with many internal material cycles, while the integrability of $p$ ($p(A) = A^{-3/5} \propto t^{-2/5}$ in the matter era) ensures a strictly finite total process duration: since the expansion rate calibrated to the present constant ($p_0 = 1$) standard is $H(A) = \frac{1}{A}\frac{dA}{dt} = H_0E(A)$ and the material process-clock differential is $d\tau = p(A)\,dt = A^{-3/5}dt$, the constant-standard coordinate expansion age $t(z)$ and the accumulated material process-time age

$\tau(z) = \int_0^{t(z)} p(\bar{t})\, d\bar{t}$ at any redshift $z$ are given by the integrals

$$t(z) = \frac{1}{H_0} \int_0^{A(z)} \frac{d\bar{A}}{\bar{A}\, E(\bar{A})}, \qquad \tau(z) = \frac{1}{H_0} \int_0^{A(z)} \frac{\bar{A}^{-8/5}\, d\bar{A}}{E(\bar{A})}. \tag{76}$$

At the present epoch ($z = 0, A = 1$), the constant-standard coordinate expansion age equals the standard cosmological age $t_0 = 13.7965$ **Gyr** (matching Planck 2018 $13.787 \pm 0.020$ Gyr), whereas the total accumulated process history experienced by matter due to faster early clocks ($p > 1$) is strictly greater: $\tau_0 = 24.5795$ **Gyr** ($\tau_0/t_0 = 1.7816$). The Universe therefore has a beginning in both time measures, although its age is expressed differently in coordinate and process time. Each material structure begins its own process history at the epoch of its formation.

**Supernova Time Dilation.** In the cosmological picture of RefG, a larger early $p$ means that material processes completed more cycles within the same interval of coordinate time than they do in the present background; the local $\tau$ standard serves as the normal clock within its own epoch. On the ideal transparent photon branch, spectroscopic redshift follows directly from the metric and from comparison of the source and receiver's local atomic standards:

$$\frac{A_0}{A_e} = \frac{a_0/p_0}{a_e/p_e} = 1 + z. \tag{77}$$

Neighboring null signals in the same metric independently show that an event lasting $\Delta\tau_e$ in source process time is recorded by the present observer with duration

$$\Delta\tau_{\rm obs} = \frac{A_0}{A_e}\, \Delta\tau_e = (1 + z)\, \Delta\tau_e. \tag{78}$$

Type Ia supernova light curves and spectral aging exhibit precisely this $(1 + z)$ time dilation [Blondin et al., 2008]. In RefG, expansion of the foundation scale $a$ lowers $P_F$, the pressure response changes $p$, and the entire history from source to receiver is read through the single operational geometry $A = a/p$.

The RefG description parametrized by $A$ and the spatially flat ΛCDM description parametrized by $z$ give the same ideal redshift, time dilation, Hubble function, angular-diameter distance, and luminosity distance at equal background parameters. For these observables, the two descriptions belong to one operational equivalence class. The distinguishing physics lies in source and clock microphysics, calibration, and inhomogeneous dynamics.

**Early Structural Maturity (JWST).** A supernova time profile compares clocks from different epochs through photon propagation. The maturity of a galaxy records accumulated process history: mass assembly, successive stellar generations, chemical enrichment, and dynamical organization are the products of many internal cycles. In the matter-dominated regime ($E(A) \simeq \sqrt{\Omega_{m0}}\, A^{-3/2}$), exact analytical integration of $t_{\rm mat}(A) = \frac{2}{3H_0\sqrt{\Omega_{m0}}} A^{3/2}$ and $\tau_{\rm mat}(A) = \frac{10}{9H_0\sqrt{\Omega_{m0}}} A^{9/10}$ yields the exact process-to-coordinate age ratio:

$$\frac{\tau_{\rm mat}(z)}{t_{\rm mat}(z)} = \frac{5}{3}\, p(z) = \frac{5}{3}\, (1 + z)^{3/5}. \tag{79}$$

Because $p(z) = (1 + z)^{3/5} > 1$ in the early Universe, a substantially larger accumulated process time is available at high redshift: at halo collapse onset $z_{\rm form} = 20$ ($p(20) = 6.22$), $t(20) = 177.6$ Myr whereas $\tau(20) = 1.816$ Gyr; at $z = 14.32$ (JADES-GS-z14-0; $p = 5.14$), the coordinate age is $t(14.32) = 285.7$ Myr whereas the accumulated process age is $\tau(14.32) = 2.423$ **Gyr** ($\tau/t = 8.48\times$), providing a post-$z = 20$ assembly window of $\Delta\tau(20 \to 14.32) = 607.1$ Myr vs. only $\Delta t = 108.0$ Myr in ΛCDM (5.62× more process cycles); at $z = 13$ (JADES-GS-z13-0; $p = 4.87$), $t(13) = 327.2$ Myr vs. $\tau(13) = 2.631$ **Gyr** (8.04×, with $\Delta\tau(20 \to 13) = 814.7$ Myr vs. $\Delta t = 149.5$ Myr, 5.45×); at $z = 10$ ($p = 4.22$), $t(10) = 470.2$ Myr vs. $\tau(10) = 3.277$ **Gyr** (6.97×, with $\Delta\tau(20 \to 10) = 1.461$ Gyr vs. $\Delta t = 292.6$ Myr, 4.99×); and at $z = 8$ ($p = 3.74$), $t(8) = 635.8$ Myr vs. $\tau(8) = 3.932$ **Gyr** (6.18×,

with $\Delta\tau(20 \to 8) = 2.116$ Gyr vs. $\Delta t = 458.1$ Myr, 4.62×). Since internal radiative cooling ($t_{\rm cool} = \frac{3nk_BT}{2n_en_H\Lambda(T)}$), local free-fall collapse ($t_{\rm ff} = \sqrt{\frac{3\pi}{32G_{\rm local}\rho_{\rm local}}}$), star formation, and stellar nucleosynthesis inside a virialized halo occur on the local material clock $\tau$, a standard gas-depletion timescale $\tau_{\rm dep} = 0.80$ Gyr converts $1 - e^{-\Delta\tau(20\to10)/\tau_{\rm dep}} = 83.9\%$ of the convertible gas reservoir into stars by $z = 10$ in RefG compared to only $1 - e^{-\Delta t(20\to10)/\tau_{\rm dep}} = 30.6\%$ in ΛCDM (2.74× higher stellar mass at $z = 10$ and 3.75× at $z = 13$). Consequently, assembling a massive $M_* = 10^{10}\,M_\odot$ galaxy in a $M_{\rm halo} = 3.5\times10^{11}\,M_\odot$ halo at $z = 10$ requires a normal baryon conversion efficiency $\epsilon_*^{\rm RefG} = 0.218 < 0.25$ in RefG (versus an extreme $\epsilon_*^{\Lambda\rm CDM} = 0.596 > 0.55$ in ΛCDM) while simultaneously producing 3.4 × –5.2× higher chemical enrichment $Z(\tau)/Z(t)$ at $z \in [10, 14.32]$. The early massive and chemically mature galaxies observed by JWST thus provide a direct and quantitative confirmation of the RefG two-clock process history [Labbé et al., 2023].

**Two Observational Channels for One Mechanism.** Supernova time dilation and the maturity of early galaxies provide two distinct observational channels. The first measures photon propagation and signal-arrival intervals; the second probes the structural consequences of accumulated process time. Together they connect global relaxation of the foundation, the historical evolution of clocks, and the growth of matter.

**Ontological Consistency and the Arrow of Time.** Within a single-medium ontology, the familiar **vacuum-energy problem** becomes a problem of common energy accounting. If a manifested vacuum phase carries a constant operational energy density $\epsilon_*$, it produces the negative mechanical response $P_* = -\epsilon_*$ and $w_* = -1$ as the volume grows. Its geometric contribution is $\Lambda_{\rm phase}$, and the operational geometry depends on one effective cosmological term, $\Lambda_{\rm eff} = \Lambda_{\rm bare} + \Lambda_{\rm phase}$. Each physical contribution enters this total once through its own variational source. A fixed total energy stored in a comoving cell is diluted by expansion and has matter-like behavior, $w = 0$. The conserved quantity $Q_C$ belongs to the phase-action sector, whereas the vacuum term belongs to the energy sector described by $\epsilon_*$. The $q$-theory program provides a historical motivation for this mechanism [Volovik, 2003, Klinkhamer and Volovik, 2008].

RefG relates the macroscopic **arrow of time** to the global direction of foundation relaxation. Redistribution of localized energy among the medium's many degrees of freedom produces statistical irreversibility. This mechanism can connect the thermodynamic, cosmological, and radiative arrows to one common source [Planck, 1899]. Its quantitative form is determined by the entropy law and the irreversible dynamics of the foundation.

### 3.4 Quantum Randomness and Two-Level Time

The two-level description of time provides an interpretive basis for quantum phenomena. The place and moment of a measurable event are recorded in oscillon-clock time, while RefG associates the selection of a particular outcome with the internal level of the foundation-medium. Here "prior" denotes logical priority: clocks, distance, and the measurement procedure are structures that arise from this level.

Bell's theorem and its experimental tests rule out local hidden-variable models that explain outcomes through preassigned local properties [Bell, 1964, Hensen et al., 2015]. At the same time, the **no-signaling theorem** establishes that entanglement correlations cannot be converted into controllable superluminal communication.

RefG interprets this phenomenon as **atemporal internal coordination**. At the foundation level, the correlation belongs to the coherent unity of a single state; in the metric description, it appears as a statistical correlation between distant measurements. The physical signal channel that carries information through space remains governed by local causality.

John Wheeler's **delayed-choice experiment** receives the same interpretation [Wheeler, 1978, Jacques et al., 2007]: in a single-photon interferometer, a late choice of measurement arrangement

determines whether wave interference or which-path information is manifested. RefG describes this unity through the internal integrity of the phase state while preserving the metric order of events.

Related concepts in modern physics include:

- The atemporal ground level of the **Wheeler–DeWitt equation** ($\hat{H}\Psi = 0$; DeWitt 1967);
- The **Page–Wootters mechanism**, where time emerges from correlations between entangled quantum subsystems [Page and Wootters, 1983];
- **de Broglie–Bohm pilot-wave theory**, characterized by global phase coordination [Bohm, 1952].

Any such account must satisfy two criteria simultaneously:

1. reproduce the observed outcome statistics through **Born's probability rule**, $P = |\psi|^2$ [Born, 1926];
2. preserve **no-signaling**, excluding controllable superluminal communication.

# 4 Oscillons and the Origin of Matter

## 4.1 Wave Configurations of a Unified Foundation

Wave dynamics encompasses propagating and dissipative excitations as well as stable, localized, self-confined configurations that arise under suitable nonlinear conditions. Their spatial structures include:

- a vortex ring with continuous circulation around a closed loop;
- a localized spherical pulsation, or oscillon;
- a spiral or helical wave structure.

In the particle ontology of RefG, **electrons, protons, photons, and quarks are distinct resonant and topological configurations of one universal foundation-medium.** A massive particle is a locally self-confined configuration of the medium; a photon is a freely propagating transverse mode. The diversity of matter is associated with different stable wave modes of the same foundation.

A historical precursor is Tony Skyrme's model, in which nucleons are described as topological solitons—"skyrmions"—of a nonlinear meson field [Skyrme, 1961]. RefG extends this approach to a common ontological foundation: all solitonic and propagating modes belong to the dynamics of one medium.

Within this single-particle intuitive description, matter has a continuous wave structure extending through a spatial volume. The foundation's oscillation amplitude and phase are distributed through space; a discrete detection event results from resonant registration of this oscillation by the detector's atomic oscillons. The wave amplitude and phase are represented by the field state of the foundation-medium.

In the RefG description, the atomic electron cloud is a physically extended phase state. A localized detection event arises through locking with the detector, while a stationary atomic state is a standing-wave mode of the foundation.

The single-electron double-slit experiment exhibits both an extended amplitude and discrete detection events: the amplitude spans the geometry of both slits, while each event appears as a discrete point on the screen. Akira Tonomura's group directly recorded the gradual formation of the interference distribution [Tonomura et al., 1989]. A stationary atomic mode retains its energy; radiation accompanies transitions between states.

The spatial width of a wavepacket and the intrinsic structural size of a particle are strictly distinct concepts:

- Elastic scattering probes charge and energy distribution form factors, while deep inelastic scattering reveals internal parton structure and distribution functions;
- The spatial extent of a propagating wavepacket does not enter these observables as an intrinsic physical size.

Empirically, the electron exhibits a pointlike electromagnetic form factor down to scales of order $10^{-18}$ m [Navas et al., 2024]. A direct empirical test of the oscillon description of leptons is whether the form factor calculated from the internal topological charge profile reproduces this pointlike behavior over the accessible momentum-transfer range.

### 4.2 Localized Self-Confinement: Massive and Massless Modes

In the RefG particle framework, the defining ontological distinction between massive and massless modes is **localized self-confinement**.

**Massive Oscillons.** A massive particle's energy is concentrated in a localized, self-confined configuration. When vortex circulation or stable nonlinear pulsation confines a wave configuration to a finite spatial region, the Bernoulli effect generates a persistent static pressure-deficit profile in the surrounding foundation-medium.

Long-lived localized nonlinear pulsations are known in classical field theory as **pulsons and oscillons** [Bogolyubsky and Makhankov, 1976, Gleiser, 1994, Copeland et al., 1995]. RefG distinguishes two levels of protection: an ordinary Noether charge stabilizes the derived phase-protected core, while topological and population-level locking constitute a broader mechanism for particle-species selection and long-term stability. In external measurements, energy confined in the localized structure manifests as inertia and effective gravitational mass.

The first concrete dynamical realization of this direction is built from a complex phase field minimally coupled to the coframe. Its nonlinear action produces a finite-energy, radially nodeless Q-ball-type core with an analytically determined existence window and convergent numerical evidence of orbital stability on the fixed Minkowski coframe. Population-level phase locking embeds this local protection within a broader mechanism that couples the core to the common environment and selects stable particle species.

**Massless Carriers (The Photon).** A massless mode has neither a rest frame nor a localized static core. It propagates at the limiting speed of inter-node phase transmission ($c$). RefG describes the photon as such a propagating transverse phase mode.

Helicity and localized rotation are fundamentally distinct phenomena:

- The $\pm 1$ helicity of a photon describes the handedness of its transverse polarization relative to the propagation axis, without forming a localized static vortex;
- A massive mode has a locally self-confined topological core and a static pressure-deficit profile.

In Einstein–Hilbert geometry, every localized or radiative sector whose diffeomorphism-invariant action depends universally and minimally on the common metric enters the total gravitational source once through its Hilbert stress–energy tensor. The standard electromagnetic action includes photon energy and momentum in that source. A propagating photon carries energy–momentum; a permanent localized well belongs to a bound state. The same accounting includes the entire internal energy of a composite particle.

An exact correspondence with a quantum description of the oscillon field has been established. Its amplitude $\chi$ and phase $\theta_O$ combine into the complex field $\Psi_O = (\chi/\sqrt{2})e^{i\theta_O}$. The full nonlinear covariant action retains the binding terms required for localization, while its quadratic part about the strict vacuum is exactly the free complex Klein–Gordon action. On the fixed Minkowski branch, standard canonical quantization of this quadratic sector yields a free Hamiltonian that is bounded below

and nonnegative after vacuum normalization, particle and antiparticle sectors of Fock space, a conserved global $U(1)$ phase charge, and microcausality: field commutators vanish at spacelike separation. The free vacuum excitations have a simple propagator pole on the mass shell $k_\mu k^\mu = -m^2$ and spin zero [Tong, 2006, Wigner, 1939]. In the low-energy limit, the same equation reduces to Schrödinger dynamics. Einstein–Hilbert geometry and standard free quantum matter thus share the same RefG coframe.

The quartic and sextic terms form the interacting binding sector of the localized core. A free spin-zero quantum is a small vacuum excitation, whereas the localized core is an interacting solitonic state. The resulting $U(1)$ charge is the conserved global phase charge of the oscillon field.

The quadratic vacuum sector is represented classically by $T_O^{(2)}$ and quantum mechanically by $:\widehat{T}_O^{(2)}:$; energy accounting uses one of these corresponding descriptions. In the semiclassical Einstein equation, the source for a quantum state is the renormalized expectation value of the stress–energy tensor. The full nonlinear tensor also contains the binding-interaction operators.

### 4.3 Self-Forming Resonators and the Two-Level Filter

Nonlinear wave equations admit infinitely many distinct wave configurations, whereas nature exhibits a restricted, well-defined spectrum of stable elementary particles. Explaining the selection of this particular set is a central problem of particle dynamics.

RefG organizes this selection mechanism through a **two-level dynamical filter**:

**1. Level One: Kinematic and Population Filter.** The resonant region of a three-dimensional oscillon forms within the foundation-medium through the oscillon's own dynamics. This self-forming resonator supports several mutually compatible oscillation modes. A simple acoustic example of commensurate frequencies is a fundamental frequency and its next two harmonics, with the ratio $1:2:3$.

In RefG, an oscillon's pressure deficit changes the local geometry, which in turn selects viable modes. A mode must simultaneously satisfy the conditions for propagation through the foundation and resonant locking with the surrounding particle population.

**2. Level Two: Energetic Stability Filter.** Any form that passes the kinematic filter but has an energetically accessible decay channel rapidly decays into lower-energy stable modes.

Hadronic resonances $(\Delta, \rho, \omega)$ illustrate this: with typical lifetimes of $10^{-22}$–$10^{-24}$ s, they appear in the spectrum, decay rapidly, and redistribute their energy among lower-energy decay products, including pions and protons.

The observed stability of the electron and the channel-dependent experimental lower bounds on proton decay ($\tau/B \gtrsim 10^{34}$ years for leading channels) impose stringent limits on any topological protection mechanism. These bounds set the quantitative scale for the protection mechanism that RefG associates with conserved charges and population-level phase locking.

The mathematical language governing these filters is that of nonlinear **stable modes and invariant phase-space structures**—configurations that retain their form under small perturbations.

A stable particle is a self-consistent equilibrium: the wave configuration creates a pressure deficit in the medium, and that deficit feeds back to select and reinforce the configuration itself. In classical field theory, Sidney Coleman's Q-balls are protected by conserved internal Noether charges [Coleman, 1985]. The local protection of the derived core rests on precisely such a Noether charge; population-level locking describes its coupling to the common environment and the selection of particle species.

Resonant locking arises through the self-organization of coupled oscillatory systems. Christiaan Huygens's observation of spontaneous synchronization between clocks suspended from a common beam [Huygens, 1665], phase-locked laser modes, and Yoshiki Kuramoto's nonlinear synchronization model [Kuramoto, 1975] are classical examples of such collective dynamics. A historical counterpart in particle physics is Geoffrey Chew's S-matrix bootstrap program, which treats the particle spectrum as a self-consistent system of modes [Chew and Frautschi, 1961].

Resonant selection establishes a stable particle structure: unstable wave modes incompatible with the common resonance decay, and their energy is redistributed into compatible modes through energetically allowed channels. As the medium becomes more rarefied, a stable oscillon retains its foundational form and size, while its externally measured operational size, mass, and coordinate clock frequency $\Omega_t$ vary consistently with the environmental pressure. The ratios of its size and clock rate to local standards in the same environment remain unchanged.

**From One Oscillon to the Particle Spectrum.** One localized oscillon family defines the minimal self-contained dynamical problem of the particle sector. Its stable finite-energy configuration determines the spatial profile of a self-forming resonator; small oscillations reveal the internal $s_i$ spectrum and connect it to the common environmental frequency scale $\Omega_t$. The collective phase $\theta_C$ describes the organized state of the foundation, while the periodic oscillon phase describes its local cycle. Distinguishing these phases allows a unified treatment of cosmological relaxation and the particle's periodic processes. The oscillon's energy content and the response of its environment determine the physical variables that relate rest mass to pole mass.

## 4.4 The Global Resonator: Unity of Micro- and Macro-Spectra

The two-level filter describes particle-spectrum selection; its spatial realization is considered in the global-resonator model.

Ernst Chladni's acoustic experiment reveals the spatial structure of resonator modes [Chladni, 1787]: sand on a vibrating metal plate collects along standing-wave nodal lines, where the oscillation amplitude is minimal. RefG relates this principle to global modes and their nodes. A resonator has its own spectrum; its geometry and boundary conditions determine which modes are amplified and which are damped.

Once the operational network has formed, RefG treats it as a **single global three-dimensional resonator**. At this stage, its eigenspectrum becomes well defined: the coupling operator, network geometry, and boundary conditions select the admissible modes. In this description, the resonator's three-dimensional structure and spectrum are expressed in both physical particles and the cosmic web. During subsequent expansion and thermal evolution, modes may become locked into stable physical configurations:

- **Short-Wavelength Modes** at high frequencies correspond to localized elementary particles, or oscillons;
- **Long-Wavelength Modes** at low frequencies correspond to the large-scale distribution of matter, the **Cosmic Web**.

In the RefG global-resonator description, the distribution of galaxies and galaxy clusters, filaments, and large cosmic voids is associated with the nodal structure of standing waves.

Standard cosmology describes this large-scale morphology through Yakov Zel'dovich's anisotropic gravitational collapse [Zel'dovich, 1970]. RefG relates it to the ontological interpretation of a single spectral operator.

The central principle of this description is that **the microscopic elementary particle and the macroscopic cosmic web are two different readouts of one universal spectral operator**: a locally self-confined topological configuration and a global spatial distribution of matter.

A joint empirical test of this operator spans:

1. The localized particle mass spectrum;
2. The cosmological history of structure formation;
3. The mass budgets of galaxy clusters;
4. The Baryon Acoustic Oscillation standard ruler at approximately $100\,h^{-1}$ Mpc [Eisenstein et al., 2005].

Agreement across all four channels is the decisive test of this mapping.

In the picture of a finite, boundaryless resonator, conceptually related to Christof Wetterich's variable-gravity framework [Wetterich, 2013], the geometry of the foundation expands according to its own internal rule, the background pressure falls, and material standards settle into their new state. A cosmological observer reads this causal chain through the ratio $A = a/p$. The expansion of the Universe is the evolution of its internal geometry, and the entire measurement procedure is defined within that geometry.

In current CMB observations, the low-multipole anomalies at $\ell = 2, 3$ remain of modest statistical significance. Searches for cosmic topology establish topology-dependent lower bounds and do not select a unique global form [Planck Collaboration, 2016].

### 4.5 Discreteness and the $C_3$ Symmetry of Generations

In the Standard Model, the masses of the charged leptons—electron ($e$), muon ($\mu$), and tau ($\tau$)—are empirical Yukawa-coupling inputs. In the RefG map, they form the **ninth-order internal three-phase ($C_3$) triad** of a single topological defect: three distinct resonant modes of one underlying physical structure.

The three-phase parametrization of charged-lepton masses is expressed in square-root-mass space. Defining $x_k \equiv \sqrt{m_k}$ gives

$$x_k = \mu \left[ 1 + A_K \cos\left( \theta + \frac{2\pi k}{3} \right) \right], \qquad k = 0, 1, 2, \tag{80}$$

where:

- $\mu$ is the common scale in $\sqrt{m}$ space;
- $A_K$ is the dimensionless three-phase modulation amplitude;
- $\theta$ is the phase angle of this algebraic mass map;
- $k = 0, 1, 2$ indexes the three generations $(e, \mu, \tau)$.

When $A_K = \sqrt{2}$, the $C_3$ algebra identically recovers Yoshio Koide's empirical mass relation [Koide, 1982, 1983]:

$$K = \frac{m_e + m_\mu + m_\tau}{\left( \sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau} \right)^2} = \frac{2}{3}. \tag{81}$$

The three-phase parametrization and its geometric interpretation have an established literature:

- Carl Brannen expressed the relation in terms of a circulant-matrix phase [Brannen, 2006];
- Robert Foot demonstrated that $K = 2/3$ corresponds to a 45° geometric orientation in $\sqrt{m}$-space [Foot, 1994];
- Brannen's Pancharatnam–Berry phase construction yielded $\pi/12$ while leaving $\theta = 2/9$ as a dynamical parameter [Brannen, 2010];

- Konstantin Shulga independently obtained $\theta_\ell = -2/9$ within a compact family cycle [Shulga, 2026].

RefG connects this structure to a **ninth-order return cycle**. On the selected $h = 2$ branch, the first non-trivial topological return gives the mass angle $\theta = h/9 = 2/9$, while $A_K = \sqrt{2}$ expresses a geometric rule of equal energy sharing between the internal singlet and doublet vibrational channels.

A simple $C_3$-invariant potential is independent of $\theta$. RefG associates the physical origin of this angle with topological holonomy around a closed cycle.

A strict numerical test excludes direct identification of the tree-level point $(A_K, \theta) = (\sqrt{2}, 2/9)$ with the physical pole masses ($m_\mu^{\text{tree}} = 105.659414$ MeV, a $451.7\sigma$ deviation from the PDG value 105.6583755(23) MeV). This discrepancy identifies the problem of relating the geometric spectrum to observed masses through radiative corrections [Sumino, 2009].

**$C_3$ Circulant Operator, Compton-Self-Similar Core 1-Loop QED Cancellation Theorem, and the 2-Loop Pole-Mass Bridge.** The covariant algebraic origin of the three-phase formula is the $3 \times 3$ Hermitian $C_3$-circulant square-root mass operator

$$\widehat{M}_{1/2}(\mu, A_K, \theta) = \mu \begin{pmatrix} 1 & \frac{A_K}{2} e^{i\theta} & \frac{A_K}{2} e^{-i\theta} \\ \frac{A_K}{2} e^{-i\theta} & 1 & \frac{A_K}{2} e^{i\theta} \\ \frac{A_K}{2} e^{i\theta} & \frac{A_K}{2} e^{-i\theta} & 1 \end{pmatrix}, \tag{82}$$

which is diagonalized in the discrete Fourier $C_3$ eigenbasis $\mathbf{v}_k = \frac{1}{\sqrt{3}}(1, \omega^k, \omega^{2k})^T$ ($\omega = e^{2\pi i/3}$) with eigenvalues $x_k = \sqrt{m_k} = \mu\left[1 + A_K \cos\!\left(\theta + \frac{2\pi k}{3}\right)\right]$ ($k = 0, 1, 2$). By the exact trigonometric identities $\sum_{k=0}^{2} x_k = 3\mu$ and $\sum_{k=0}^{2} x_k^2 = 3\mu^2\left(1 + \frac{A_K^2}{2}\right)$, the Koide functional

$$K(A_K) = \frac{\sum_{k=0}^{2} m_k}{\left(\sum_{k=0}^{2} \sqrt{m_k}\right)^2} = \frac{2 + A_K^2}{6} \tag{83}$$

is strictly independent of both the overall scale $\mu$ and the phase angle $\theta$, and equals $K = 2/3$ if and only if $A_K = \sqrt{2}$ (corresponding to exact singlet–doublet quadratic norm equipartition, $\frac{3}{2}\mu^2 A_K^2 = 3\mu^2$).

Why is the Koide relation not destroyed by 1-loop quantum electrodynamic (QED) self-energy corrections? In standard point-particle field theory with a fixed (generation-independent) ultraviolet cutoff $\Lambda_{\text{fixed}}$, the 1-loop QED pole mass

$$m_{k,\text{pole}}^{(1)} = m_k^{(0)} \left[1 + \frac{\alpha_0}{\pi}\left(1 + \frac{3}{4} \ln \frac{\Lambda_k^2}{(m_k^{(0)})^2}\right)\right] \tag{84}$$

depends logarithmically on the generation mass ($\frac{\partial(m_{k,\text{pole}}^{(1)}/m_k^{(0)})}{\partial m_k^{(0)}} = -\frac{3\alpha_0}{2\pi m_k^{(0)}} \neq 0$), which would shift the muon mass relative to the electron by $-1.963$ MeV ($8.53 \times 10^5 \sigma$!) and destroy $K = 2/3$. By contrast, in RefG a charged lepton is a self-confined oscillon resonator whose core radius is dynamically proportional to its own Compton wavelength:

$$R_{\text{core},k} = \xi_0 \frac{\hbar}{m_k^{(0)} c_0} \qquad \Longrightarrow \qquad \Lambda_k = \frac{\hbar}{R_{\text{core},k} c_0} = \frac{m_k^{(0)}}{\xi_0}, \tag{85}$$

where $\xi_0$ is a generation-independent dimensionless geometric form factor. Consequently, the 1-loop QED relative self-energy factor

$$Z_1(\xi_0) \equiv \frac{m_{k,\text{pole}}^{(1)}}{m_k^{(0)}} = 1 + \frac{\alpha_0}{\pi}\left(1 - \frac{3}{2} \ln \xi_0\right), \qquad \frac{\partial Z_1(\xi_0)}{\partial m_k^{(0)}} \equiv 0 \tag{86}$$

is **strictly independent of the generation index $k$ and bare mass $m_k^{(0)}$ (Compton-Self-Similar Core 1-Loop QED Cancellation Theorem**, verified across `41/41` checks). Thus 1-loop QED dressing rescales only the overall scale ($\mu \to \sqrt{Z_1(\xi_0)}\,\mu$) and **leaves both $K = 2/3$ and the tree-level topological angle $\theta_0 = 2/9$ strictly invariant at one-loop order** without requiring extra gauged family bosons.

Because the 1-loop QED shift cancels identically in mass ratios, the leading generation-dependent radiative correction enters at **two-loop QED order** $\mathcal{O}\big((\alpha_0/\pi)^2\big)$ (where virtual vacuum-polarization loops break exact single-core scale similarity):

$$\theta_{\rm pole} = \theta_0 \left[1 - c_2 \left(\frac{\alpha_0}{\pi}\right)^2 + \mathcal{O}\left(\left(\frac{\alpha_0}{\pi}\right)^3\right)\right], \qquad \theta_0 = \frac{2}{9} = 0.2222222222 \text{ rad.} \tag{87}$$

The exact pole angle extracted from the PDG charged-lepton masses ($m_e = 0.51099895000$ MeV, $m_\mu = 105.6583755$ MeV) is $\theta_{\rm exp} = 0.2222220471$ rad, giving the relative radiative shift

$$\frac{\Delta\theta_{\rm exp}}{\theta_0} = \frac{\theta_{\rm exp} - 2/9}{2/9} = -7.8806 \times 10^{-7} = -0.14606 \left(\frac{\alpha_0}{\pi}\right)^2, \tag{88}$$

where $(\alpha_0/\pi)^2 = 5.3955 \times 10^{-6}$ and the 2-loop coefficient $c_2 = 0.14606$ lies directly adjacent to the product of the ninth-order topological angle and the Koide invariant: $c_2^{(0)} = \theta_0 K_0 = \frac{2}{9}\times\frac{2}{3} = \frac{4}{27} \approx 0.14815$. Even the zero-parameter rational 2-loop ansatz ($\theta_{\rm 2loop}^{(0)} = \frac{2}{9}\left[1 - \frac{4}{27}\left(\frac{\alpha_0}{\pi}\right)^2\right]$) reduces the tree-level $451.7\sigma$ muon discrepancy by a factor of $69.9\times$ down to 0.14 ppm ($m_\mu^{(0)} = 105.6583606$ MeV), while $c_2 = 0.14606$ reproduces $(m_e, m_\mu)$ exactly and predicts $m_\tau^{\rm pole} = 1776.9690$ MeV ($0.91\sigma$ from PDG $1776.86\pm0.12$ MeV). The empirical deviation of the Koide ratio itself, $\Delta K_{\rm exp} = K_{\rm exp} - \frac{2}{3} = -6.155 \times 10^{-6} = -1.141\left(\frac{\alpha_0}{\pi}\right)^2$, is of the exact same two-loop QED order.

The charged-lepton $C_3$/Koide triad is interpreted as a set of slightly anharmonic, mutually compatible modes of a three-dimensional resonator. In this map, the $x_k$ are coordinates of the oscillon eigenspectrum; $\theta$ is the family mass angle, whereas $\theta_C$ is the collective phase of the foundation.

# 5 Topology and Quantum Numbers

This chapter organizes the Standard Model's quantum numbers within a unified topological research map for RefG. The map relates phase, orientation, confinement, and holonomy to a common geometric origin of particle spin, charge, color, and generations. The corresponding covariant actions, symmetry algebras, and measurable amplitudes provide its physical realization.

## 5.1 Phase, Orientation, and Topological Quantization

The preceding description of the foundation-medium has used oscillation amplitude and local pressure as its principal variables. Its complete local state is described by several coupled field variables: the phase-clock scalar, amplitude profile, and spatial orientation frame each have their own degrees of freedom. Within this topological description, $\Phi$ denotes the complete configuration, including amplitude, phase, internal geometric form, and spatial orientation.

Outside a defect, a single-valued scalar phase obeys $\mathbf{u} = \nabla\theta$ and $\nabla \times \mathbf{u} = 0$. The vortex-like phase-winding sector used here is characterized by:

- **a multivalued phase**,
- **a singular defect line**, and
- **an internal orientation locked in space**.

At the central core of the defect, the wave amplitude vanishes ($|\psi| = 0$), while the phase winds around it by an integer multiple of $2\pi$ ($\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$). A canonical laboratory example is a quantized vortex filament in superfluid helium.

A cosmological analogue of this topological structure is a **cosmic string**: a macroscopic topological defect line formed during an early-Universe phase transition and subsequently preserved. Its formation is described by Kibble's causal mechanism [Kibble, 1976]. In RefG, it is a localized one-dimensional topological defect of the foundation-medium. The vacuum topology determines whether such lines form, while astrophysical observations, including CMB lensing and pulsar timing, tightly constrain their cosmological density.

Lord Kelvin's vortex-atom program contributed to the development of modern knot theory [Thomson, 1867, 1869, Tait, 1877]. This history also demonstrated the importance of distinguishing topological classification from dynamical stability: topology determines a configuration's kinematic structure, while dynamics determines its physical viability as a particle. RefG associates particle form with a topological knot and describes its stability through pressure–energy feedback and population-level resonant locking among oscillons.

Phase quantization arises when the field is continuous along a loop, the state returns to the same value after a complete circuit, and the loop belongs to a nontrivial topological class ($\pi_1 \neq 0$).

For a closed, orientable, framed ribbon, the **Călugăreanu–White–Fuller (CWF) theorem** applies [Călugăreanu, 1961, White, 1969, Fuller, 1971, Ricca and Nipoti, 2011]:

$$Lk = Wr + Tw, \tag{89}$$

where

- $Lk$ (linking number) is the topological integer measuring total linkage;
- $Wr$ (writhe) describes the spatial coiling of the ribbon axis;
- $Tw$ (twist) describes the internal phase twisting of the frame around that axis.

The geometric quantities $Wr$ and $Tw$ may vary continuously under a deformation that preserves the ribbon without self-intersection or cutting—for example, spatial coiling may be converted into internal twist—while their algebraic sum $Lk$ remains strictly topologically invariant.

In hydrodynamics, helicity ($H = \int \mathbf{u} \cdot (\nabla \times \mathbf{u})\, d^3x$) is conserved in an ideal fluid, and Keith Moffatt connected it to the linking number $Lk$ [Moffatt, 1969]. In superfluid helium, circulation quantization was predicted by Lars Onsager and Richard Feynman [Onsager, 1949, Feynman, 1955] and measured experimentally by Joe Vinen [Vinen, 1961]. Integer phase winding is therefore a directly measured laboratory phenomenon.

In the RefG map, the fundamental principle **"topological confinement = charge"** means that electric charge is the external readout of a confined phase–orientation path: the topological invariant specifies the discrete class, while field dynamics determines its electromagnetic readout.

In the unified topological map of RefG, **spin-1/2 and elementary charge** are two geometric readouts of a single double-covered structure. An internal half-twist determines the state's spinorial behavior, while the twist's handedness and integer winding determine the sign and magnitude of charge. Their geometric origin may be shared, but spin transformations and electric-flux normalization are defined separately.

This interpretation has direct experimental analogues: Kelvin waves have been excited on quantized vortices in superfluid helium and their dispersion measured [Minowa et al., 2025]; helical axial modes and their dispersion have likewise been measured in a classical water vortex [Barckicke et al., 2026]. In RefG, an oscillon's radial mode corresponds to its Compton-like mass pulsation, while its transverse helicity mode is associated with the radiation and charge channel.

## 5.2 Spin and Fermionic Statistics

A phase field twisted by a half-turn ($\pi$) along an oscillon's internal axis generates a **Möbius-strip-type double-covered $Z_2$ structure**:

- one spatial circuit ($2\pi$) reverses the orientation and changes the sign of the wave amplitude ($\psi \to -\psi$);
- two complete circuits ($4\pi$) fully restore the initial state ($\psi \to +\psi$).

This geometry provides the topological structure of the $4\pi$ spinorial kinematics associated with spin-1/2. On the orthonormal relational coframe ${e^A}_\mu$ ($\eta_{AB} = \mathrm{diag}(1,-1,-1,-1)$), its exact covariant realization comprises the four-dimensional Clifford algebra $\{\gamma^A, \gamma^B\} = 2\eta^{AB} I_4$ and the Spin(3, 1) Lorentz generators $\Sigma^{AB} = \frac{i}{4}[\gamma^A, \gamma^B]$, which satisfy $[\Sigma^{AB}, \Sigma^{CD}] = i(\eta^{BC}\Sigma^{AD} - \eta^{AC}\Sigma^{BD} + \eta^{AD}\Sigma^{BC} - \eta^{BD}\Sigma^{AC})$. The spatial $SU(2)$ spin operator $\mathbf{S} = \frac{\hbar}{2}\mathbf{\Sigma}$ ($\Sigma_k = 2\Sigma^{ij} = \mathrm{diag}(\sigma_k, \sigma_k)$, $[\Sigma_i, \Sigma_j] = 2i\varepsilon_{ijk}\Sigma_k$) yields the exact Casimir invariant $\mathbf{S}^2 = \frac{3}{4}\hbar^2 I_4$ ($s = 1/2$) and the rotation operator around a unit axis $\hat{\mathbf{n}}$:

$$U(\varphi, \hat{\mathbf{n}}) = \exp\left(-\frac{i}{\hbar}\varphi\, \hat{\mathbf{n}} \cdot \mathbf{S}\right) = \cos\left(\frac{\varphi}{2}\right) I_4 - i \sin\left(\frac{\varphi}{2}\right) (\hat{\mathbf{n}} \cdot \mathbf{\Sigma}), \tag{90}$$

for which $U(2\pi, \hat{\mathbf{n}}) = -I_4$ ($\mathrm{Tr}\, U(2\pi) = -4$) and $U(4\pi, \hat{\mathbf{n}}) = +I_4$ ($\mathrm{Tr}\, U(4\pi) = +4$), matching the Berry/holonomy phase $\Delta\alpha = \int_0^{2\pi} \frac{1}{2}\, ds = \pi$ ($e^{i\Delta\alpha} = -1$) accumulated around a half-twisted Möbius frame ($\alpha(s) = s/2$). Changing the topological class requires the amplitude to vanish in the core, the defect to reconnect, or the boundary condition to change.

The complete realization of fermionic statistics is defined by a phase of $\pi$ and an antisymmetric Fock-space state under exchange of two identical oscillons:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1). \tag{91}$$

Under this condition, the total amplitude of two identical particles in the same quantum state vanishes ($\Psi = -\Psi \Rightarrow \Psi = 0$), yielding the Pauli exclusion principle. A bosonic mode retains its sign under exchange ($\Psi \to +\Psi$) and permits many bosons to occupy the same state.

In relativistic quantum field theory, the spin–statistics theorem combines Lorentz invariance, positive energy, and microcausality [Pauli, 1940]. Indeed, normal-ordering the free Dirac Hamiltonian $H_D = \sum_{\mathbf{p},s} E_{\mathbf{p}} \left(b^\dagger_{\mathbf{p},s} b_{\mathbf{p},s} - d_{\mathbf{p},s} d^\dagger_{\mathbf{p},s}\right)$ requires canonical anticommutation relations $\{b_{\mathbf{p},r}, b^\dagger_{\mathbf{p},s}\} = \{d_{\mathbf{p},r}, d^\dagger_{\mathbf{p},s}\} = \delta_{rs}$ in order to invert the sign of the antiparticle term and obtain a strictly nonnegative Fock Hamiltonian $:H_D:_{\mathrm{Fermi}} = \sum_{\mathbf{p},s} E_{\mathbf{p}}(n_{b,\mathbf{p},s} + n_{d,\mathbf{p},s}) \geq 0$ together with the nilpotent Pauli condition $(b^\dagger_{\mathbf{p},s})^2|0\rangle = 0$ (whereas bosonic commutators $[d, d^\dagger] = 1$ yield an unbounded-below spectrum $:H_D:_{\mathrm{Bose}} = \sum E_{\mathbf{p}}(n_b - n_d)$). The topological map of RefG and the coframe Dirac action unite the geometric and dynamical foundations of this connection.

## 5.3 Charge, the Photon, and the Electromagnetic Channel

RefG interprets **electric charge** as the external spatial manifestation of an oscillon's internal phase twist:

- right-handed orientation of the twist corresponds to positive charge ($+e$);
- left-handed orientation corresponds to negative charge ($-e$).

Outside the oscillon, this phase-twist structure produces a radial field in all directions. A quantitative formulation of the electromagnetic channel must reproduce Gauss's law for the total flux through a closed surface:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}. \tag{92}$$

In this map, a **positron** is the complete topological mirror image of an electron, with the opposite phase twist. During electron–positron annihilation ($e^+ + e^- \to 2\gamma$), the oppositely wound structures cancel and unwind, transferring locally confined energy into freely propagating transverse radiation. In a neutral oscillon, opposite phase twists balance and the net flux vanishes; the neutrino sector corresponds to this configuration.

A uniform rotation of the phase-asymmetry angle is a global $U(1)$ symmetry. A spacetime-dependent rotation requires a local $U(1)$ phase connection and a covariant derivative. RefG identifies the **photon** with a **transverse helicity mode** of this connection: it transmits changes in phase asymmetry through space and acts as the massless gauge carrier of the electromagnetic interaction.

Topological structure also appears in classical Maxwell electrodynamics: in Antonio Rañada's vacuum solutions, electric and magnetic field lines form mutually linked loops classified by the Hopf invariant [Rañada, 1989].

The sign of a force—attraction or repulsion—is determined jointly by the spin of the carrier, the type of source, and the coupling structure:

- **vector electromagnetic channel (spin-1):** like charges repel and opposite charges attract;
- **tensor gravitational channel (spin-2):** in a one-metric weak-field geometry, positive nonrelativistic sources couple with the same sign and attract at leading order; the full $T_{\mu\nu}$ determines the response to relativistic pressure, stress, and vacuum sources.

In three spatial dimensions, a conserved massless spherical flux dilutes as $1/r^2$, and its corresponding potential scales as $1/r$. In the covariant action, promoting the oscillon's global $U(1)$ phase symmetry to a local gauge symmetry ($\partial_\mu \Psi_O \to D_\mu \Psi_O = (\partial_\mu - i q_{\rm em} A_\mu)\Psi_O$) produces the gauge-invariant phase gradient $\mathcal{D}_\mu \theta_O = \partial_\mu \theta_O - q_{\rm em} A_\mu$ and the exact Ward–Takahashi/Noether identity $J^\mu_{\rm em} \equiv \frac{\partial \mathcal{L}_O}{\partial A_\mu} = q_{\rm em} J^\mu_O$ ($Q_{\rm em} = q_{\rm em} Q$): the gauge potential $A_0(r)$ screens the linearly divergent radial energy of an uncoupled Goldstone phase tail, leaves an exact $E_r(r) = \frac{Q_{\rm em}}{4\pi\varepsilon_0 r^2}$ Coulomb far field with finite total energy $E_{\rm tot} < \infty$, and satisfies the exact Charged Oscillon First Law $\left.\frac{dE_{\rm tot}}{dQ}\right|_{q_{\rm em}} = \Omega_\infty$ (where $\Omega_{\rm eff}(r) = \Omega_\infty + q_{\rm em} A_0(r)$ is the electrostatically shifted local core frequency; verified at $q_{\rm em} = 0.005, \Omega_\infty = 0.800$ to a relative accuracy of $1.42 \times 10^{-6}$).

Bjerknes forces [Bjerknes, 1906] illustrate the distinction between hydrodynamic and electromagnetic interactions: in an acoustic fluid, the relative phase of the pulsations determines the force's sign, whereas in electrostatics the sign follows from the vector gauge-coupling structure. Carrier spin, source type, and topological coupling jointly determine attraction and repulsion.

RefG interprets the fine-structure constant, $\alpha \approx 1/137$, as the dimensionless strength of this electromagnetic channel and as the invariant dimensionless ratio between the electric-flux and transverse-photon responses of the foundation's nodal network.

### 5.4 Color Confinement and Three Generations

The three-dimensionality of space has a fundamental role in the particle map of RefG: **an oscillon's internal three-axis frame provides a natural geometric structure for the color-charge triad—red, green, and blue ($r, g, b$).** Color is an internal degree of freedom of this frame.

- **Quark:** a topologically open segment organized along one internal axis;
- **Baryon (proton, neutron):** a fully antisymmetric singlet combination of all three colors, forming a closed, stable, colorless configuration;
- **Meson:** a combination of an axis with its anti-axis—color with anticolor—likewise forming a closed, colorless state.

In this geometric map, **color neutrality** is represented by complete closure: a three-color baryonic binding and a color–anticolor mesonic pair form closed configurations, whereas an isolated one-axis color segment remains open. Color confinement is the dynamical realization of this structural rule.

The eight modes of a deformed three-axis oscillon match the eight generators of $SU(3)$ in number, providing a structural map of the color sector. The $SU(3)$ structure constants, local gauge symmetry, and interaction vertices give this map its gauge-theoretic content. Specifically, the volume-preserving ($\det U = 1$) Hermitian deformations of the three-axis internal oscillon triad $(\mathbf{e}_r, \mathbf{e}_g, \mathbf{e}_b)$ span an $N_c^2 - 1 = 3^2 - 1 = 8$ dimensional traceless Hermitian space whose generators $T^A = \frac{1}{2}\lambda^A$ ($A = 1, \ldots, 8$, where $\lambda^A$ are the Gell-Mann matrices) satisfy the exact $\mathfrak{su}(3)_c$ Lie algebra and Jacobi identity:

$$\mathrm{Tr}(T^A) = 0, \qquad \mathrm{Tr}(T^A T^B) = \frac{1}{2}\delta^{AB}, \qquad [T^A, T^B] = i f^{ABC} T^C, \tag{93}$$

with quadratic Casimir invariants $C_2(\mathbf{3}) = \sum_{A=1}^{8} T^A T^A = \frac{4}{3} I_3$ (fundamental quark triplet) and $C_2(\mathbf{8}) = \sum_{A=1}^{8} T^A_{\mathrm{adj}} T^A_{\mathrm{adj}} = 3 I_8$ (adjoint gluon octet, $(T^A_{\mathrm{adj}})_{BC} = -i f^{ABC}$). The two-body color interaction factor $\langle \sum_{A=1}^{8} T_1^A \otimes T_2^A \rangle = \frac{1}{2}[C_2(R_{\mathrm{tot}}) - C_2(R_1) - C_2(R_2)]$ is strictly attractive only in the colorless meson singlet channel ($\mathbf{3} \otimes \bar{\mathbf{3}} \to \mathbf{1}$: $\langle T_1 \cdot T_2 \rangle_{\mathbf{1}} = -\frac{4}{3} < 0$, whereas the octet $\mathbf{8}$ is repulsive: $+\frac{1}{6} > 0$) and in the antisymmetric diquark channel inside a baryon ($\mathbf{3} \otimes \mathbf{3} \to \bar{\mathbf{3}}$: $\langle T_1 \cdot T_2 \rangle_{\bar{\mathbf{3}}} = -\frac{2}{3} < 0$, whereas the symmetric sextet $\mathbf{6}$ is repulsive: $+\frac{1}{3} > 0$). For the totally antisymmetric three-color baryon state $\varepsilon_{abc} q^a q^b q^c$, the total quadratic Casimir vanishes identically: $C_2(\mathbf{1}_{qqq}) = 3C_2(\mathbf{3}) + 2 \times 3 \langle T_1 \cdot T_2 \rangle_{\bar{\mathbf{3}}} = 3\left(\frac{4}{3}\right) + 6\left(-\frac{2}{3}\right) = 0$. Combining short-distance three-dimensional Yang–Mills spherical spreading with long-distance channeling of the non-Abelian chromoelectric flux $\Phi_c$ into a one-dimensional flux tube of cross-section $A_\perp$ and linear string tension $\sigma_{\mathrm{str}} = \frac{\Phi_c^2}{2A_\perp} > 0$ yields the Cornell confinement potential:

$$V_{\mathrm{Cornell}}(r) = -\frac{4}{3}\frac{\alpha_s \hbar c_0}{r} + \sigma_{\mathrm{str}}\, r, \qquad \lim_{r\to\infty} F_{\mathrm{Cornell}}(r) = -\sigma_{\mathrm{str}} < 0, \tag{94}$$

which precludes the isolated existence of an open color segment at infinite separation. Moreover, both fundamental parameters of the Cornell potential—the strong coupling constant $\alpha_s(m_Z)$ (together with the asymptotic-freedom scale $\Lambda^{(5)}_{\overline{\mathrm{MS}}}$) and the flux-tube linear string tension $\sigma_{\mathrm{str}}$—are derived directly from the tri-axial $\mathfrak{su}(3)_c$ holonomy and the topological lepton mass scale $\mu_\ell^2 = 313.856$ MeV (verified across `18/18` checks):

1. **Strong Coupling at the $Z^0$-Pole ($\alpha_s^{\mathrm{RefG}}(m_Z) = \frac{17}{144} = 0.118056$, $+0.062\sigma$) and Dimensional Transmutation ($\Lambda^{(5)}_{\overline{\mathrm{MS}}} = 209.34$ MeV $\approx K_0 \mu_\ell^2 = 209.24$ MeV, 0.048%):** At the electroweak scale $\mu = m_Z = 91.1876$ GeV, the tri-axial color defect fraction $\theta_d = \frac{1}{N_{\mathrm{hol}}} = \frac{1}{9}$ dressed by the 1-loop half-channel factor $\frac{1}{2\dim \mathfrak{su}(3)_c} = \frac{1}{16}$ yields without free parameters:

$$\begin{aligned} \alpha_s^{\mathrm{RefG}}(m_Z) &= \theta_d \left(1 + \frac{1}{2\dim \mathfrak{su}(3)_c}\right) = \frac{1}{9}\left(1 + \frac{1}{16}\right) = \frac{17}{144} \\ &\approx 0.1180556 \quad (\mathrm{PDG}\ 0.1180 \pm 0.0009,\ +0.062\sigma). \end{aligned} \tag{95}$$

   For $n_f = 5$ active quark flavors ($\beta_0 = 11 - \frac{2}{3}n_f = \frac{23}{3}$, $\beta_1 = 102 - \frac{38}{3}n_f = \frac{116}{3}$), 2-loop $\overline{\mathrm{MS}}$ renormalization-group dimensional transmutation yields $\Lambda^{(5),\mathrm{RefG}}_{\overline{\mathrm{MS}}} = 209.34$ **MeV** (PDG $210 \pm 14$ MeV, $-0.047\sigma$), matching the low-energy Koide–holonomy scale $\Lambda_{\mathrm{Koide}} \equiv K_0 \mu_\ell^2 = \frac{2}{3}\mu_\ell^2 = \frac{m_e + m_\mu + m_\tau}{9} = 209.24$ MeV to 0.048%, while threshold running gives $\alpha_s^{\mathrm{RefG}}(m_b) = 0.2253$ ($+0.11\sigma$), $\alpha_s^{\mathrm{RefG}}(m_\tau) = 0.3253$ ($+0.88\sigma$), $\Lambda^{(4)} = 324.2$ MeV, and $\Lambda^{(3)} = 369.5$ MeV.

2. **First-Principles Derivation of the Cornell String Tension ($\sqrt{\sigma_{\mathrm{str}}^{\mathrm{RefG}}} = \frac{73}{54}\mu_\ell^2 = 424.29$ MeV, $\sigma_{\mathrm{str}}^{\mathrm{RefG}} = 0.18002$ GeV$^2$, 0.011%):** The fundamental quark Casimir $C_2(\mathbf{3}) = \frac{4}{3}$ and the dressing across the $N_{\mathrm{hol}} \dim \mathfrak{su}(3)_c = 9 \times 8 = 72$ holonomy–color channels determine:

$$\sqrt{\sigma_{\mathrm{str}}^{\mathrm{RefG}}} = C_2(\mathbf{3})\left(1 + \frac{1}{N_{\mathrm{hol}} \dim \mathfrak{su}(3)_c}\right)\mu_\ell^2 = \frac{4}{3}\left(1 + \frac{1}{72}\right)\mu_\ell^2 = \frac{73}{54}\,\mu_\ell^2 = 424.290\ \mathrm{MeV}, \tag{96}$$

giving $\sigma_{\rm str}^{\rm RefG} = 0.180022$ **GeV**$^2$ = 912.29 **MeV/fm** (0.011% match to the lattice-QCD value 0.1800 GeV$^2$, Regge slope $\alpha'_{\rm Regge} = \frac{1}{2\pi\sigma_{\rm str}^{\rm RefG}} = 0.8841$ GeV$^{-2}$) and the nucleon spin-averaged center $M_0^{\rm RefG} = 2\sigma_{\rm str}^{\rm RefG} L_Y^* = 1085.57$ MeV (0.010% from 1085.46 MeV).

The **three-generation problem** is a logical extension of the $C_3$ symmetry discussed in the preceding chapter:

- charged leptons $(e, \mu, \tau)$ represent the ninth-order internal three-phase modes of one topological defect;
- quark generations $(u/d, c/s, t/b)$ repeat the same generational architecture in the colored three-axis sector.

In this map, the three generations correspond to successive modes associated with internal holonomy return cycles of the same three-dimensional resonator.

**Michael Berry's geometric phase** [Berry, 1984] provides the general holonomy principle used here: during adiabatic cyclic evolution in parameter space, a quantum state accumulates a purely geometric holonomy phase alongside its dynamical phase. RefG associates this mechanism with a topological defect's ninth-order return cycle; on the selected $h = 2$ branch, it gives $\theta = 2/9$ radians.

**Three-Dimensional Space and Knot Stability.** In RefG, the three-dimensionality of space ($D = 3$) is directly connected with the **stability of topological knots**:

- closed one-dimensional loops form nontrivial, stable knots and links only in $D = 3$ spatial dimensions;
- in higher dimensions ($D \geq 4$), any knot can be untied using an additional spatial coordinate without self-intersection.

This topological argument draws on the work of Paul Ehrenfest, Gerald Whitrow, and modern authors [Ehrenfest, 1918, Whitrow, 1955, Berera et al., 2017]. In the unified description of RefG, the discreteness of oscillon matter and the three-dimensionality of space are two consequences of the same confinement condition.

## 5.5 Geometric Map of the Standard Model

RefG organizes the particle spectrum of the Standard Model into a single geometric map. The correspondences below establish a common structural language; the covariant action, symmetry algebra, and measurable amplitudes give these correspondences their dynamical content.

**1. The Electron and Charged Leptons.** In this map, the electron corresponds to a complete toroidal, Möbius-type twisted $Z_2$ structure in which all three spatial axes close locally. The double cover carries the $4\pi$ return characteristic of a Dirac spinor, while its quantitative magnetic and gravitational-clock responses follow directly from the minimally coupled Dirac–Fock–Ivanenko covariant action on the relational coframe ${e^A}_\mu$ ($e = \det({e^A}_\mu)$):

$$S_{\rm Dirac}[{e^A}_\mu, A_\mu, \Psi] = \int d^4x\, e\, \bar\Psi \left( i\hbar c_0\, {e_A}^\mu \gamma^A \mathcal{D}_\mu - m_0 c_0^2 \right) \Psi, \qquad \mathcal{D}_\mu = \partial_\mu + \frac{i}{2}{\omega_\mu}^{BC}\Sigma_{BC} + i\frac{q_{\rm em}}{\hbar}A_\mu. \tag{97}$$

By the Clifford decomposition $\gamma^A\gamma^B = \eta^{AB} I_4 - i\sigma^{AB}$ ($\sigma^{AB} = \frac{i}{2}[\gamma^A, \gamma^B] = 2\Sigma^{AB}$), squaring the covariant Dirac operator produces the Pauli spin term $(\gamma^A D_A)(\gamma^B D_B) = \eta^{AB} D_A D_B - \frac{q_{\rm em}}{2\hbar}\sigma^{AB}F_{AB}$, and the exact 4D Clifford operator identity $\not{p}'\gamma^A + \gamma^A \not{p} \equiv (p'+p)^A I_4 + i\sigma^{AB}(p'-p)_B$ yields the on-shell Gordon decomposition of the electromagnetic current ($\not{p}u(p) = m_0 c_0 u(p)$):

$$\bar u(p')\gamma^A u(p) \equiv \frac{(p'+p)^A}{2m_0c_0}\bar u(p')u(p) + \frac{i\sigma^{AB}(p'-p)_B}{2m_0c_0}\bar u(p')u(p). \tag{98}$$

In a magnetic field $\mathbf{B}$ ($\frac{1}{2}\sigma^{ij}F_{ij} = -\boldsymbol{\Sigma}\cdot\mathbf{B}$), the spin interaction term is $H_{\rm spin} = -\frac{q_{\rm em}\hbar}{2m_0}\boldsymbol{\Sigma}\cdot\mathbf{B} = -\frac{q_{\rm em}}{2m_0}g_0\,\mathbf{S}\cdot\mathbf{B}$ ($\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\Sigma}$), which proves the exact tree-level Dirac gyromagnetic factor $g_0 = 2$ ($g_0 - 2 \equiv 0$). Moreover, on a static isotropic coframe $ds^2 = -p_T^2 c_0^2 dt^2 + p_L^{-2}\delta_{ij}dx^i dx^j$ ($p_T = N, p_L = A_{\rm iso}^{-1}$), in proper ruler coordinates $\tilde{\mathbf{x}} = p_L^{-1}(\mathbf{x} - \mathbf{x}_0)$ and canonically normalized spinor density $\widetilde{\Psi}(\tilde{\mathbf{x}}, t) = p_L^{-3/2}\Psi(\mathbf{x}, t)$, the coordinate-time Dirac Hamiltonian factorizes exactly as $H_{\rm Dirac}^{\rm coord} = p_T\, H_{\rm Dirac}^{(0)}(\tilde{\mathbf{x}})$, so that all Dirac eigenfrequencies $\omega_D^{(t)} = p_T\omega_{D,0}$ and Larmor spin-precession frequencies $\omega_L^{(t)} = p_T\omega_{L,0}$ transpose strictly with the clock factor $p_T = N$ and are independent of $p_L$ ($\frac{\partial(\omega_D^{(t)}/\Omega_t)}{\partial p_T} = \frac{\partial(\omega_D^{(t)}/\Omega_t)}{\partial p_L} \equiv 0$).

The measured electron $g$ factor slightly exceeds the tree-level value $g_0 = 2$ through quantum electrodynamic radiative loops. Julian Schwinger calculated the leading quantum correction $a_e^{(1)} = \frac{\alpha_0}{2\pi} \approx 0.00116141$, while the 4-loop QED Schwinger expansion $|g_e|_{\rm QED,4L}/2 = 1.00115965217557$ agrees with modern Penning-trap experiments $|g_e|/2 = 1.00115965218059(13)$ to $5.02 \times 10^{-12}$ [Schwinger, 1948, Fan et al., 2023]. This number is among the most stringent quantitative targets for the topological–spinorial map.

**2. Quarks and Color Charge.** In this map, quarks are axial topological segments with open ends. Their observed fractional electric charges $(+2/3, -1/3)$ are represented by one-third substeps of a three-dimensional closed loop, while baryons and mesons appear as colorless, fully closed configurations. Specifically, partitioning a closed three-axis topological loop ($\oint d\theta = 2\pi$, baryon number $B_{\rm baryon} = 1$) into $N_c = 3$ equal axial segments assigns each open quark segment baryon number $B_q = \frac{1}{N_c} = \frac{1}{3}$ and left-handed weak-doublet $Q_L = (u_L, d_L)^T$ hypercharge $Y(Q_L) = B_q = \frac{1}{3}$. By the Gell-Mann–Nishijima relation $Q_{\rm em} = T_3 + \frac{Y}{2}$ ($T_3 = \pm\frac{1}{2}$), the electric charges of the quark segments are uniquely determined as:

$$Q_u = +\frac{1}{2} + \frac{1}{2}\left(\frac{1}{3}\right) = +\frac{2}{3}, \qquad Q_d = -\frac{1}{2} + \frac{1}{2}\left(\frac{1}{3}\right) = -\frac{1}{3}, \tag{99}$$

with right-handed singlet hypercharges $Y(u_R) = 2Q_u = +\frac{4}{3}$ and $Y(d_R) = 2Q_d = -\frac{2}{3}$, which cancels all five quantum gauge-gravitational anomalies ($\mathcal{A}_{[SU(3)_c]^3} = \mathcal{A}_{[SU(3)_c]^2U(1)_Y} = \mathcal{A}_{[SU(2)_L]^2U(1)_Y} = \mathcal{A}_{[\rm Grav]^2U(1)_Y} = \mathcal{A}_{[U(1)_Y]^3} = 0$) together with integer lepton charges ($Q_\nu = 0, Q_e = -1$) and yields integer hadron charges: $Q_p = 2Q_u + Q_d = +1$, $Q_n = Q_u + 2Q_d = 0$, $Q_{\pi^+} = Q_u - Q_d = +1$.

Binding three axial quark segments into the totally antisymmetric color singlet $\varepsilon_{abc}q_1^a q_2^b q_3^c$ and extending the $C_3$ circulant mass operator to the colored sector establishes four symbolic and numerical results for the baryonic nucleon and quark-flavor sectors (verified across `18/18` and `18/18` checks):

1. **Fermat–Torricelli–Steiner** $120^\circ$ **Y-Junction,** $> 98.7\%$ **Topological Field-Energy Origin of Nucleon Mass, and** $N$–$\Delta$ **Hyperfine Splitting:** Three chromoelectric flux tubes of tension $\sigma_{\rm str}$ meet at a central baryon Y-junction $\mathbf{r}_Y$ carrying the $\varepsilon_{abc}$ tensor. Varying the total string length $L_Y(\mathbf{r}_Y) = \sum_{i=1}^3 |\mathbf{r}_i - \mathbf{r}_Y|$ yields the exact force balance $\sum_{i=1}^3 \hat{\mathbf{e}}_i = \mathbf{0}$ ($\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = -\frac{1}{2}$, $\theta_{ij} = 120^\circ$, with positive-definite Hessian $\nabla^2_{\mathbf{r}_Y} L_Y = \frac{3}{2R}I_2 > 0$ and string-length reduction $L_Y/L_V = \frac{\sqrt{3}}{2} \approx 0.8660 < 1$) and the three-body Cornell potential ($\sum_{i<j}\langle T_i \cdot T_j\rangle_{\bar{\mathbf{3}}} = 3\times(-\frac{2}{3}) = -2$, $C_2(\mathbf{1}_{qqq}) = 0$):

$$V_{qqq}(R) = -\frac{2}{\sqrt{3}}\frac{\alpha_s\hbar c_0}{R} + 3\sigma_{\rm str}\,R. \tag{100}$$

The relativistic Dirac cavity boundary condition $\tan x_0 = \frac{x_0}{1-x_0}$ gives $x_0 = 2.04278694$, and minimizing the chiral energy $E_0(L_Y) = \frac{3C_{\rm kin}\hbar c_0}{L_Y} + \sigma_{\rm str}L_Y$ proves virial equipartition $E_{\rm kin}(L_Y^*) = E_{\rm str}(L_Y^*) = \frac{1}{2}M_0c_0^2$. The valence current-quark rest-mass sums ($m_{uud}^{\rm curr} = 2m_u + m_d = 8.99$ MeV, $m_{udd}^{\rm curr} = m_u + 2m_d = 11.50$ MeV at $\mu = 2$ GeV) account for only 0.958% of the proton mass (938.272 MeV) and 1.224% of the neutron mass (939.565 MeV), whereas 99.04% **(proton) and** 98.78% **(neutron)** arise from confined relativistic oscillon kinetic energy and $SU(3)_c$ Y-junction field energy. Because the spin–spin operator $\sum_{i<j}\mathbf{s}_i \cdot \mathbf{s}_j = \frac{\hbar^2}{2}[S(S+1) - \frac{9}{4}]$ equals $-\frac{3}{4}\hbar^2$ for the nucleon ($S = 1/2$) and $+\frac{3}{4}\hbar^2$ for the $\Delta(1232)$ resonance ($S = 3/2$), the spin-averaged center of

gravity is $M_0 = \frac{1}{2}(m_N + m_\Delta) = 1085.46$ MeV ($L_Y^* = 3r_q^* = 0.5950$ fm), while $\langle T_i \cdot T_j \rangle_{\bar{\mathbf{3}}} = -\frac{2}{3} < 0$ renders the Fermi–Breit chromomagnetic Hamiltonian strictly positive ($H_{\rm cm} \propto +\sum_{i<j} \mathbf{s}_i \cdot \mathbf{s}_j$), proving $m_\Delta - m_N = +293.08$ MeV $> 0$. The dressed constituent quark scale $\bar{M}_q = \frac{1}{2}(m_N/3 + M_0/3) = 337.40$ MeV matches the Dirac magnetic scale $m_p/\mu_p^{\rm exp} = 335.96$ MeV to 0.43%.

2. **24-Component $SU(6)$ Spin–Flavor Wavefunctions, Parameter-Free Ratio $\mu_n/\mu_p = -2/3$, and Axial Coupling $g_A = 1.2716$:** Because the color singlet $\varepsilon_{abc}$ is totally antisymmetric, the $L = 0$ baryon spin–flavor states $|p\uparrow\rangle$ and $|n\uparrow\rangle$ are totally symmetric under all 6 permutations of $S_3$. Symbolic expectation values of $\hat{\mu}_z = \sum_{i=1}^{3} \mu_{f_i} \sigma_{i,z}$ yield:

$$\mu_p = \frac{4}{3}\mu_u - \frac{1}{3}\mu_d, \qquad \mu_n = \frac{4}{3}\mu_d - \frac{1}{3}\mu_u. \tag{101}$$

Substituting the topological charges $Q_u = +\frac{2}{3}, Q_d = -\frac{1}{3}$ ($\mu_u = -2\mu_d = +\frac{2}{3}\mu_0$) gives $\mu_p = +\mu_0$, $\mu_n = -\frac{2}{3}\mu_0$, and the parameter-free ratio:

$$\left(\frac{\mu_n}{\mu_p}\right)^{(0)}_{\rm RefG} = -\frac{2}{3} \approx -0.6667 \quad \text{vs.} \quad \left(\frac{\mu_n}{\mu_p}\right)_{\rm exp} = -0.684979 \quad (2.67\%\ \text{difference}), \tag{102}$$

with absolute moments $\mu_p^{\rm pred} = +2.781\,\mu_N$ (exp. $+2.793\,\mu_N$, 0.43%) and $\mu_n^{\rm pred} = -1.854\,\mu_N$ (exp. $-1.913\,\mu_N$, 3.09%). On the same $SU(6)$ states, the Gamow–Teller operator $\sum_{i=1}^{3} \tau_i^+ \sigma_{i,z}$ gives $g_A^{\rm NR} = \frac{5}{3}$, while isotropic averaging of the relativistic Dirac lower component proves $\langle \bar{u}\gamma^3\gamma^5 u \rangle_{S^2} = \frac{1}{3}\left(1 + \frac{2M_q}{E_q}\right)$; evaluated at the proton cavity momentum $p_q = \frac{x_0 \hbar c_0}{\sqrt{5/3}\, r_p} = 371.32$ MeV ($M_q = m_N/3 = 312.97$ MeV), this yields $g_A^{\rm rel} = \frac{5}{9}\left(1 + \frac{2M_q}{E_q}\right) = 1.2716$, matching $g_A^{\rm exp} = 1.2754$ to 0.30%.

3. **Neutron–Proton Mass Difference $\Delta m_{np} = 1.29333$ MeV, Hydrogen Stability, and Coframe Clock–Ruler Invariance:** Both the pairwise Coulomb sum ($\sum_{i<j} q_i^{(p)} q_j^{(p)} = 0$ vs. $\sum_{i<j} q_i^{(n)} q_j^{(n)} = -\frac{1}{3}$) and the total charge squared ($Q_n^2 - Q_p^2 = -1 < 0$) prove that the electromagnetic shift is strictly negative: $\Delta m_{np}^{\rm QED} = E_{\rm em}^{(n)} - E_{\rm em}^{(p)} < 0$. The analytical Coulomb self-energy of the 3D Gaussian proton charge cloud ($r_p = 0.8409$ fm), $E_{\rm Coul} = \sqrt{\frac{3}{4\pi}} \frac{\alpha_0 \hbar c_0}{r_p} = 0.8367$ MeV, together with the magnetic hyperfine shift $\Delta m_{np}^{\rm mag} = -0.2430$ MeV gives $\Delta m_{np}^{\rm QED} = -1.0797$ MeV ($0.50\sigma$ vs. the BMW lattice value $-1.00 \pm 0.16$ MeV), while the axial quark mass difference $m_d - m_u = 2.51$ MeV contributes $\Delta m_{np}^{\rm QCD} = Z_m(m_d - m_u) = +2.3731$ MeV ($0.51\sigma$ vs. BMW $+2.52 \pm 0.29$ MeV), closing:

$$\begin{aligned} \Delta m_{np}^{\rm RefG} &= \Delta m_{np}^{\rm QCD} + \Delta m_{np}^{\rm QED} = +2.3731 - 1.0797 = +1.29333\ \text{MeV} > m_e = 0.51100\ \text{MeV}, \\ Q_\beta &= \Delta m_{np}^{\rm RefG} - m_e = +782.33\ \text{keV} > 0, \end{aligned} \tag{103}$$

and protecting atomic hydrogen against inverse beta capture ($m_d - m_u = 2.51$ MeV $> (m_d - m_u)_{\rm crit} = 1.68$ MeV). Furthermore, on any static isotropic coframe $(p_T, p_L)$ and across cosmological relaxation $p(A) = A^{-3/5}$, the dimensionless ratios $m_n/m_p$, $\Delta m_{np}/m_e$, $\mu_n/\mu_p$, and the proton-to-electron mass ratio $\mu_{pe} = m_p/m_e = 1836.152673$ are strictly invariant ($\Delta\mu_{pe}/\mu_{pe} \equiv 0$, satisfying quasar absorption bounds $|\Delta\mu_{pe}/\mu_{pe}| < 10^{-7}$ and closing the BBN $Q_{np}$ invariance).

4. **Harmonic $C_3$ Holonomy Sequence ($\theta_\ell = \frac{2}{9}, \theta_d = \frac{1}{9}, \theta_u = \frac{2}{27}$), Six Individual Quark Masses ($m_u, m_d, m_s, m_c, m_b, m_t$), and the Unitary $3\times3$ CKM Mixing Matrix ($J_{\rm CP}^{\rm CKM} = 3.13 \times 10^{-5}$):** Applying the $3 \times 3$ Hermitian $C_3$-circulant square-root mass operator $\sqrt{m_k^{(f)}} = \mu_f \left[1 + A_f \cos(\theta_f + \frac{2\pi k}{3})\right]$ ($k = 1, 2, 0$ for generations $1, 2, 3$) to each fermion sector $f \in \{\ell, d, u\}$ proves the generalized Koide identity $K(A_f) \equiv \frac{\sum_k m_k^{(f)}}{(\sum_k \sqrt{m_k^{(f)}})^2} = \frac{2 + A_f^2}{6}$ ($\partial K/\partial\mu_f = \partial K/\partial\theta_f \equiv 0$). Whereas a closed 3-axis

lepton loop has integer winding $w_\ell = 1$, open axial quark segments with fractional charges $Q_d = -\frac{1}{3}$ and $Q_u = +\frac{2}{3}$ execute $w_d = 2$ (doublet-axis reflection) and $w_u = 3$ (full color-triad traversal) substeps per ninth-order return cycle ($N_{\rm hol} = 9$, $\theta_0 = \frac{2}{9}$), generating the exact harmonic holonomy sequence:

$$\theta_\ell = \frac{\theta_0}{1} = \frac{2}{9}\ \text{rad}, \quad \theta_d = \frac{\theta_0}{2} = \frac{1}{9}\ \text{rad}\ \ (0.84\%\ \text{vs. PDG}), \quad \theta_u = \frac{\theta_0}{3} = \frac{2}{27}\ \text{rad}\ \ (0.43\%\ \text{vs. PDG}), \tag{104}$$

with inter-doublet phase splitting $\Delta\theta_{du} = \theta_d - \theta_u = \frac{1}{27} = \frac{\theta_0^2}{C_2(\mathbf{3})}$, so that $\sqrt{C_2(\mathbf{3})}(\theta_d - \theta_u) \equiv \frac{\theta_0^2}{\sqrt{C_2(\mathbf{3})}} = \frac{2}{27\sqrt{3}} \approx 0.042767$. Because open color segments carry $SU(3)_c$ flux-tube energy in addition to singlet–doublet equipartition ($A_0^2 = 2$), their squared amplitudes shift by the topological charge/holonomy weights $A_d^2 = 2 + |Q_d| + \frac{\theta_d}{2} = \frac{43}{18}$ and $A_u^2 = 2 + (Q_u + |Q_d|) + \frac{\theta_d+\theta_u}{2} = \frac{167}{54}$, yielding the exact rational generalized Koide invariants:

$$\begin{aligned} K_d^{\rm RefG} &= \frac{2 + A_d^2}{6} = \frac{79}{108} \approx 0.731481 \quad (\text{PDG } 0.731428,\ \mathbf{0.007}\%), \\ K_u^{\rm RefG} &= \frac{2 + A_u^2}{6} = \frac{275}{324} \approx 0.848765 \quad (\text{PDG } 0.848981,\ \mathbf{0.025}\%), \end{aligned} \tag{105}$$

while the heavy-quark cross-triplet $(c, b, t)$ retains the canonical Koide ratio $K(c, b, t) = 0.66955 \approx \frac{2}{3}$ (0.43%). Fixing the down-sector scale from the lepton scale via $\mu_d^2 = 2\big(1+\frac{1}{9\pi}\big)\mu_\ell^2 = 0.649914$ GeV (1.1 ppm) and the up-sector scale from the electroweak condensate $v_{\rm EW} = 246.22$ GeV with 1-loop holonomy top Yukawa $y_t = 1 - \frac{\theta_0^2}{6} = \frac{241}{243}$ ($m_t = \frac{y_t v_{\rm EW}}{\sqrt{2}} = 172.671$ GeV, $\mu_u^2 = 22.7703$ GeV) reproduces all six individual quark masses within $\leq 0.65\sigma$ of PDG 2024: $m_u = 2.469$ **MeV** ($+0.63\sigma$), $m_d = 4.536$ **MeV** ($-0.28\sigma$), $m_s = 94.04$ **MeV** ($+0.07\sigma$, $m_s/m_d = 20.731$ [$0.66\sigma$]), $m_c = 1.2664$ **GeV** ($-0.18\sigma$), $m_b = 4.1800$ **GeV** ($+0.00\sigma$), and $m_t = 172.671$ **GeV** ($-0.06\sigma$). In turn, the mismatch between the up and down axial orientations generates the unitary $3\times3$ Cabibbo–Kobayashi–Maskawa (CKM) mixing matrix $V_{\rm CKM} = U_u^\dagger U_d$ ($V_{\rm CKM}^\dagger V_{\rm CKM} = I_3$ to $3.35 \times 10^{-16}$) with topological generators:

$$\begin{aligned} \theta_{12}^{\rm CKM} &= \theta_0 + \frac{\theta_d - \theta_u}{N_{\rm hol}} = \frac{55}{243}\ \text{rad}\ \ (|V_{us}| = 0.22441,\ +0.14\sigma;\ \sqrt{m_d/m_s} = 0.2196,\ 0.17\%), \\ s_{23}^{\rm CKM} &= |V_{cb}| = \frac{2}{27\sqrt{3}} = 0.04277\ \ (+1.11\sigma), \\ s_{13}^{\rm CKM} &= |V_{ub}| = 2\theta_\ell\theta_d\theta_u = \frac{\theta_0^3}{3} = \frac{8}{2187} = 0.003658\ \ (-0.84\sigma), \\ \delta_{\rm CP}^{\rm CKM} &= \frac{\pi}{3} + \theta_d = \frac{\pi}{3} + \frac{1}{9}\ \text{rad} = 66.37^\circ\ \ (+0.22\sigma), \end{aligned} \tag{106}$$

yielding $|V_{ud}| = 0.97449, |V_{cd}| = 0.22427, |V_{cs}| = 0.97359, |V_{td}| = 0.00880, |V_{ts}| = 0.04201, |V_{tb}| = 0.99908$, the rephasing-invariant Jarlskog CP-violation measure $J_{\rm CP}^{\rm CKM} = 3.131 \times 10^{-5}$ (PDG $(3.12 \pm 0.13) \times 10^{-5}$, $+0.09\sigma$ / 0.36%), and unitarity triangle angles $\alpha = 91.86^\circ$ ($+0.02\sigma$), $\gamma = 66.33^\circ$ ($+0.21\sigma$), $\beta = 21.81^\circ$ ($\sin 2\beta = 0.690$).

### 3. Gauge Bosons.

- **Photon:** a massless, propagating transverse $U(1)$ gauge mode with two helicity states;
- **Gluons:** eight massless transverse modes that mediate transitions among color states ($SU(3)$ octet);
- **$W^\pm$ and $Z^0$ bosons:** massive mediator modes represented by locally self-confined oscillons ($SU(2)_L \times U(1)_Y$).

Evaluating the covariant electroweak kinetic action $|D_\mu \Phi_H|^2$ ($D_\mu = \partial_\mu - i g_2 W_\mu^a \frac{\tau^a}{2} - i\frac{g_1}{2} Y B_\mu$) on the radial spin-0 oscillon-Higgs condensate $\langle \Phi_H \rangle = \frac{1}{\sqrt{2}}(0, v_{\rm EW})^T$ ($Y_H = +1$, $v_{\rm EW} = (\sqrt{2} G_F)^{-1/2} = 246.21964$ GeV) in the $(W_\mu^1, W_\mu^2, W_\mu^3, B_\mu)$ basis generates a mass-squared matrix whose neutral $2 \times 2$ block has identically zero determinant ($\det M^2_{\rm neutral} = \frac{v_{\rm EW}^4}{16}[g_2^2 g_1^2 - (-g_1 g_2)^2] \equiv 0$, $Q_{\rm em}\langle \Phi_H \rangle = 0$). This strictly protects a massless photon ($m_\gamma^2 \equiv 0$) and determines the $W^\pm$ and $Z^0$ boson masses:

$$m_W = \frac{1}{2} g_2 v_{\rm EW} = 80.3692 \text{ GeV}, \qquad m_Z = \frac{1}{2}\sqrt{g_1^2 + g_2^2}\, v_{\rm EW} = 91.1876 \text{ GeV}, \qquad \cos\theta_W = \frac{m_W}{m_Z}. \tag{107}$$

RefG reads **beta decay ($n \to p + e^- + \bar{\nu}_e$)** as a topological rearrangement: a change in the orientation of an axial mode inside the neutron ($d \to u$) produces an intermediate charged weak mode ($W^-$), which decays into electron and neutrino channels. Integrating out the short-range $W^-$ propagator yields the Fermi coupling constant $\frac{G_F}{\sqrt{2}} = \frac{g_2^2}{8 m_W^2} = \frac{1}{2 v_{\rm EW}^2}$, the chiral $V - A$ amplitude $\mathcal{M} = \frac{G_F}{\sqrt{2}} V_{ud} [\bar{u}_p \gamma^\mu (g_V - g_A \gamma^5) u_n][\bar{u}_e \gamma_\mu (1 - \gamma^5) v_\nu]$ ($g_V = 1, g_A = 1.2754$), the exact analytical relativistic beta-electron phase-space integral $f_0(\epsilon_0) = \int_1^{\epsilon_0} \epsilon\sqrt{\epsilon^2 - 1}(\epsilon_0 - \epsilon)^2 d\epsilon = \frac{1}{60}(2\epsilon_0^4 - 9\epsilon_0^2 - 8)\sqrt{\epsilon_0^2 - 1} + \frac{1}{4}\epsilon_0 \ln(\epsilon_0 + \sqrt{\epsilon_0^2 - 1}) = 1.636098$ ($\epsilon_0 = (m_n - m_p)/m_e = 2.530988$, $Q_\beta = 782.333$ keV), and with the Coulomb- and radiatively-dressed phase-space factor ($f_R = 1.71385$, $\Delta_R^V = 0.02426$) the neutron lifetime $\tau_n^{\rm pred} = \frac{2\pi^3 \hbar^7}{m_e^5 c_0^4 G_F^2 |V_{ud}|^2 (g_V^2 + 3 g_A^2) f_R (1 + \Delta_R^V)} = 879.9$ s, matching the experimental value $\tau_n^{\rm exp} = 878.4 \pm 0.5$ s to 0.17%.

## 4. Neutrinos, the PMNS Mixing Matrix, and the Higgs Boson.

- **Neutrino, PMNS Leptonic Mixing Matrix, and the 11th-Order $C_3$ Holonomy Mass Hierarchy ($\sum m_\nu = 0.0593$ eV):** An electrically neutral, incompletely confined chiral phase mode whose small, nonzero mass scale and Pontecorvo–Maki–Nakagawa–Sakata (PMNS) flavor oscillations follow directly from the $C_3$ circulant and ninth-order return holonomy ($\theta_0 = 2/9$, $K_0 = 2/3$; verified across `18/18` checks). Because the neutral neutrino lacks the $U(1)_{\rm em}$ core impedance of the charged leptons, its mass eigenbasis is rotated relative to the cyclic $C_3$ tribimaximal democratic eigenbasis ($U_{\rm TBM}$, where $\sin^2\theta_{12}^{(0)} = \frac{1}{3}$, $\sin^2\theta_{23}^{(0)} = \frac{1}{2}$, $\sin\theta_{13}^{(0)} = 0$) by the Koide–holonomy dressing of the $C_3$ cycle:

$$\begin{aligned}
\sin\theta_{13}^{\rm PMNS} &= K_0\,\theta_0 = \frac{4}{27} \implies \sin^2\theta_{13}^{\rm PMNS} = \frac{16}{729} \approx 0.021948 \quad (\theta_{13}^{\rm PMNS} = 8.520^\circ,\ -\mathbf{0.15}\sigma), \\
\sin^2\theta_{12}^{\rm PMNS} &= \frac{1}{3} - \frac{1}{2}\theta_0^2 = \frac{25}{81} = \left(\frac{5}{9}\right)^2 \approx 0.308642 \quad (\theta_{12}^{\rm PMNS} = 33.749^\circ,\ +\mathbf{0.13}\sigma), \\
\sin^2\theta_{23}^{\rm PMNS} &= \frac{1}{2} + \theta_0^2 = \frac{89}{162} \approx 0.549383 \quad (\theta_{23}^{\rm PMNS} = 47.834^\circ,\ +\mathbf{0.16}\sigma), \\
\delta_{\rm CP}^{\rm PMNS} &= \pi + \theta_0 = \pi + \frac{2}{9} \text{ rad} = 192.73^\circ \quad (-\mathbf{0.17}\sigma,\ J_{\rm CP}^{\rm PMNS} = -0.00734),
\end{aligned} \tag{108}$$

with exact $3 \times 3$ unitarity $\|U_{\rm PMNS}^\dagger U_{\rm PMNS} - I_3\|_{\max} = 2.28 \times 10^{-16}$. Furthermore, whereas charged leptons close at the first ninth-order cycle, the electrically neutral chiral neutrino leaks across $N_{\rm leak} = N_{\rm hol} + N_{\rm gen} - 1 = 9 + 3 - 1 = 11$ $C_3$ holonomy steps ($\mu_\nu = 3^{-11}\big(1 - \frac{\theta_0^2}{8}\big)\mu_\ell = 3^{-11}\frac{161}{162}\mu_\ell$, $\mu_\nu^2 = 9.8784$ meV, corresponding to the Weinberg dimension-5 seesaw $\frac{(\bar{L}\tilde{\Phi}_H)^2}{M_R}$ at $M_R = \frac{v_{\rm EW}^2}{2 m_3} = 6.030 \times 10^{14}$ GeV $\approx \frac{14}{3}\alpha_{\rm RefG}^2 M_{\rm Pl}$) with phase $\theta_\nu = \theta_0 + \frac{\pi}{12} - \frac{\theta_0^2}{8} = \frac{2}{9} + \frac{\pi}{12} - \frac{1}{162} = 0.477849$ rad

and $A_\nu = \sqrt{2}$ ($K_\nu = \frac{2}{3}$). This yields the Normal Ordering neutrino mass spectrum:

$$\begin{aligned} m_1 &= 0.3608 \text{ meV}, \qquad m_2 = 8.6419 \text{ meV}, \qquad m_3 = 50.2675 \text{ meV}, \\ \Delta m^2_{21} &= 7.455 \times 10^{-5} \text{ eV}^2 \ \ (+\mathbf{0.22}\sigma), \qquad \Delta m^2_{31} = 2.5267 \times 10^{-3} \text{ eV}^2 \ \ (+\mathbf{0.58}\sigma), \\ \Delta m^2_{32} &= 2.4521 \times 10^{-3} \text{ eV}^2 \ \ (-\mathbf{0.03}\sigma), \qquad \sum_{i=1}^{3} m_{\nu_i} = 0.05927 \text{ eV} < 0.072 \text{ eV}, \end{aligned} \tag{109}$$

satisfying the Planck 2018 + DESI 2024 cosmological upper bound ($\sum m_\nu < 0.072$ eV) and predicting the kinematic tritium beta-endpoint mass $m_\beta = \sqrt{\sum_i |U_{ei}|^2 m_i^2} = 8.837$ meV ($<$ 0.45 eV, KATRIN) and the neutrinoless double-beta decay effective Majorana mass $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i| = 3.878$ meV;

- **Higgs boson:** a spherical radial pulsation of the foundation-medium with zero angular momentum, corresponding to a spin-0 scalar state ($m_H \approx 125.25$ GeV).

**5. The Graviton and the Weinberg–Witten Theorem.** The TEGR-equivalent low-energy geometry yields the massless spin-2 sector of General Relativity and its two transverse-traceless polarizations, $h_+$ and $h_\times$. The Fierz–Pauli/Deser spin-2 self-coupling construction independently verifies the same result. In the foundational description, these two polarizations correspond to modes of the nodal shear spectrum.

The Weinberg–Witten theorem [Weinberg and Witten, 1980] applies to a Lorentz-covariant quantum field theory defined on a Minkowski background with a local, gauge-invariant microscopic $T_{\mu\nu}$. The pregeometric foundation of RefG begins from a different structure, while the Lorentzian metric and Hilbert source emerge in the low-energy operational continuum. The graviton is read here as a collective tensor-shear excitation of the foundation-medium. If the completed microtheory satisfies every premise of the Weinberg–Witten theorem, this composite-graviton branch is excluded.

**6. Nonlinear Field Lagrangian.** The corresponding dynamical description uses a nonlinear field Lagrangian admitting stable, finite-energy knotted solutions; the Faddeev–Niemi model provides a well-known example [Faddeev and Niemi, 1997].

### 5.6 Fundamental Constants: $\alpha$ and the Gravitational Hierarchy

The fine-structure constant is defined by

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137.035999} \tag{110}$$

The theoretical origin of its numerical value remains an open problem in fundamental physics. Richard Feynman discussed the need for a physical explanation of this value [Feynman, 1985].

In RefG, $\alpha$ is the **dimensionless invariant ratio between the electric-flux and transverse-photon responses of the foundation-medium's nodal network**.

Three strict filters guide the search for the origin of $\alpha$:

1. Simple ratios of Planck and electron masses or frequencies do not yield 137;
2. The identity $r_e/\bar{\lambda}_c = \alpha$, relating the classical electron radius to the reduced Compton wavelength, is exact but definitional;
3. Integer phase-locking steps are real physical phenomena, as the Josephson effect and Shapiro steps demonstrate [Josephson, 1962, Shapiro, 1963], but they do not by themselves determine the numerical value of $\alpha$.

RefG relates $\alpha$ to the ratio of the intrinsic response strengths of the electric topological flux and the photon's transverse oscillation. A common scale transformation readjusts both responses coherently, leaving their dimensionless ratio invariant. Specifically, on a static isotropic coframe ($ds^2 = -p_T^2 c_0^2 dt^2 + p_L^{-2}\delta_{ij}dx^i dx^j$), the canonical Maxwell action takes the effective constitutive medium form $\mathcal{L}_{\rm EM}^{\rm coord} = \frac{1}{2}\varepsilon_{\rm eff}\mathbf{E}^2 - \frac{1}{2\mu_{\rm eff}}\mathbf{B}^2$, where the effective permittivity and permeability scale by the same product factor: $\varepsilon_{\rm eff}(p_T, p_L) = \varepsilon_0(p_T p_L)^{-1}$ and $\mu_{\rm eff}(p_T, p_L) = \mu_0(p_T p_L)^{-1}$. Consequently, the coordinate speed of light is $c_{\rm coord} = (\varepsilon_{\rm eff}\mu_{\rm eff})^{-1/2} = c_0 p_T p_L$, while the vacuum wave impedance $Z_{\rm eff} = \sqrt{\mu_{\rm eff}/\varepsilon_{\rm eff}} = Z_0$ and the fine-structure constant remain strictly invariant under any independent $(p_T, p_L)$ coframe rescaling:

$$\alpha(p_T, p_L) = \frac{Q_{\rm em}^2}{4\pi\varepsilon_{\rm eff}(p_T, p_L)\hbar c_{\rm coord}(p_T, p_L)} = \frac{Q_{\rm em}^2}{4\pi\varepsilon_0\hbar c_0} = \alpha_0, \qquad \frac{\partial\alpha}{\partial p_T} = \frac{\partial\alpha}{\partial p_L} \equiv 0. \tag{111}$$

Because coordinate mass-energy scales as $m_{\rm coord} = E_{\rm coord}/c_0^2 = p_T m_0$, atomic Rydberg and hyperfine transition frequencies ($\nu_{\rm Ryd}^{(t)} \propto p_T m_0 c_0^2 \alpha_0^2/\hbar$) transpose by the exact same clock factor $p_T = N$ as the internal BdG oscillon modes ($\nu_n^{(t)} = s_n p_T \Omega_0$).

Furthermore, by the exact quantum-metrology identity $\alpha_0 = \frac{Z_0}{2R_K}$ ($Z_0 = \sqrt{\mu_0/\varepsilon_0} = \mu_0 c_0 \approx 376.730313668\,\Omega$ is the vacuum wave impedance and $R_K = \frac{h}{e^2} = \frac{2\pi\hbar}{e^2} \approx 25812.8074593\,\Omega$ is the von Klitzing quantum Hall resistance), the inverse fine-structure constant $\alpha_0^{-1} = \frac{2R_K}{Z_0}$ directly represents the dimensionless impedance ratio of the electron's topological phase–holonomy loop to the transverse electromagnetic wave impedance of the medium (verified across `18/18` checks):

1. **Phase-Space Partition of the Tri-Axial Toroidal-Hopf $Z_2$ Oscillon ($\alpha_{\rm geom}^{-1} = 4\pi^3 + \pi^2 + \pi = 137.036304$, 2.22 ppm) and 24-Channel Casimir/Holonomy Dressing ($\alpha_{\rm RefG}^{-1} = 137.035999092$, 0.061 ppb, $0.40\sigma$):** In §5.2 and §5.5, the electron is defined as a completely closed tri-axial toroidal spinor oscillon with Möbius $Z_2$ double cover, $T^3_{\rm osc}/Z_2 = (S^1_{\rm tor} \times S^1_{\rm pol} \times S^1_{\rm spin})/Z_2$. At the Compton core boundary, its total dimensionless phase impedance is the sum of the integrals over its three orthogonal topological channels:

   - *3D toroidal bulk channel:* $\mathcal{I}_3 = \frac{1}{2}\int_0^{2\pi} d\phi_1 \int_0^{2\pi} d\phi_2 \int_0^{2\pi} d\phi_3 = \frac{1}{2}(2\pi)^3 = 4\pi^3 = 124.02510672$ (90.505%), where the factor 1/2 arises from the Möbius $Z_2$ double-cover orientation twist;
   - *2D poloidal–toroidal surface coupling channel:* $\mathcal{I}_2 = \int_0^{\pi} d\theta_1 \int_0^{\pi} d\theta_2 = \pi^2 = 9.86960440$ (7.202%);
   - *1D chiral Möbius holonomy loop channel:* $\mathcal{I}_1 = \int_0^{2\pi} \frac{1}{2} d\phi = \pi = 3.14159265$ (2.293%).

   Summing these three orthogonal channels yields, without free parameters, the bare geometric impedance of the Compton core:

$$\alpha_{\rm geom}^{-1} \equiv \mathcal{I}_3 + \mathcal{I}_2 + \mathcal{I}_1 = 4\pi^3 + \pi^2 + \pi = 137.03630377587842, \tag{112}$$

   which already matches the CODATA experimental value $\alpha_{\rm exp}^{-1} = 137.035999084(21)$ at tree level to 2.223 ppm ($2.22 \times 10^{-6}$) (the quadratic factor $4x^2 + x + 1$ has negative discriminant $\Delta_2 = 1 - 16 = -15 < 0$, precluding real roots and protecting topological stability). In the Thomson limit ($Q^2 \to 0$), matching the interior core to the exterior vacuum induces a 1-loop Casimir shift across the $N_{\rm Casimir} = N_{\rm axes} \times \dim \mathfrak{su}(3)_c = 3 \times 8 = 24$ transverse spin–axis–color channels of the foundation ($\Delta\alpha_{\rm 1L}^{-1} = -\frac{1}{24\,\alpha_{\rm geom}^{-1}} = -0.0003040557$, reducing the error by 478.9× down to 4.64 ppb: $\alpha_{\rm RefG,1L}^{-1} = 137.035999720$) together with a 2-loop radiative holonomy shift weighted by the active fraction of the ninth-order $C_3$ holonomy cycle, $\frac{N_{\rm hol}-1}{N_{\rm hol}} = 1 - \frac{1}{9} = \frac{8}{9} = 1 - \frac{\theta_0}{2}$ ($\theta_0 = 2/9$, §4.5):

$$\alpha_{\rm RefG}^{-1} = \left(4\pi^3 + \pi^2 + \pi\right) - \frac{1}{24\,\alpha_{\rm geom}^{-1}}\left(1 + \frac{8}{9}\,\frac{1}{\pi\,\alpha_{\rm geom}^{-1}}\right) = 137.0359990924064. \tag{113}$$

This zero-parameter formula matches the CODATA experimental measurement to 0.061 **ppb** ($6.13 \times 10^{-11}$**,** $0.40\sigma$—**a** 36,245× **error reduction**) and, when substituted into the 4-loop QED Schwinger series, yields $|g_e(\alpha_{\mathrm{RefG}})|/2 = 1.00115965217800$ (matching the Penning-trap measurement to 2.59 **ppt** [$2.59 \times 10^{-12}$]), the vacuum wave impedance $Z_0^{\mathrm{RefG}} = 2R_K\alpha_{\mathrm{RefG}} = 376.730313644\,\Omega$ (0.061 ppb), and the Rydberg constant $R_\infty^{\mathrm{RefG}} = \frac{m_e c_0 \alpha_{\mathrm{RefG}}^2}{2h} = 10{,}973{,}731.5667$ $\mathrm{m}^{-1}$ (0.129 ppb).

2. **Vacuum Polarization, Uehling Screening, and Running to the Electroweak $Z^0$-Pole ($\alpha_{\mathrm{OS}}^{-1}(m_Z^2) = 128.939$, $\hat{\alpha}_{\overline{\mathrm{MS}}}^{-1}(m_Z) = 127.951$, $0.024\sigma$):** At shorter distances ($Q^2 > 0$), an external probe penetrates the relativistic vacuum polarization of the Compton shell, $\alpha(Q^2) = \frac{\alpha_0}{1-\Delta\alpha(Q^2)}$. Symbolic evaluation of the Feynman-parameter integrals ($\int_0^1 x(1-x)\,dx = \frac{1}{6}$, $\int_0^1 x(1-x)\ln(x(1-x))\,dx = -\frac{5}{18}$, $\int_0^1 x^2(1-x)^2\,dx = \frac{1}{30}$) proves low-energy Uehling screening $\widehat{\Pi}_R(Q^2) = \frac{\alpha_0}{15\pi}\frac{Q^2}{m_e^2} + \mathcal{O}(Q^4/m_e^4)$ (which gives the hydrogen $2S_{1/2}$ Uehling Lamb shift $\Delta\nu_{\mathrm{Uehling}}(2S_{1/2}) = -\frac{\alpha_{\mathrm{RefG}}^5 m_e c_0^2}{30\pi h} = -27.084$ MHz) and the high-energy logarithmic asymptotic $\widehat{\Pi}_R(Q^2) = \frac{\alpha_0}{3\pi}\left[\ln\left(\frac{Q^2}{m_f^2}\right) - \frac{5}{3}\right]$. At the electroweak $Z^0$-pole ($m_Z = 91.1876$ GeV), the exact 1-loop and 2-loop contribution of the three charged lepton generations $(e, \mu, \tau)$ is $\Delta\alpha_{\mathrm{lep}}(m_Z^2) = 0.0314192 + 0.0000776 = 0.0314967$ (matching PDG 0.0314976 to $2.79 \times 10^{-5}$), which together with the 5-flavor hadronic and top-quark contributions ($\Delta\alpha_{\mathrm{had}}^{(5)} + \Delta\alpha_{\mathrm{top}} = 0.02766 - 0.000072 = 0.027588$, total $\Delta\alpha_{\mathrm{tot}}(m_Z^2) = 0.0590883$) gives:

$$\alpha_{\mathrm{OS}}^{-1}(m_Z^2) = \alpha_{\mathrm{RefG}}^{-1}\big(1 - \Delta\alpha_{\mathrm{tot}}(m_Z^2)\big) = 128.9388 \quad (\mathrm{PDG}\ 128.946 \pm 0.015,\ 0.48\sigma), \tag{114}$$

and in the $\overline{\mathrm{MS}}$ scheme ($\Delta\alpha_{\overline{\mathrm{MS}}-\mathrm{OS}} = +0.007162$) gives $\hat{\alpha}_{\overline{\mathrm{MS}}}^{-1}(m_Z) = 127.9508$, matching the standard PDG value $127.951 \pm 0.009$ to $0.024\sigma$ **(1.55 ppm)** and determining Sirlin's electroweak radiative parameter $\Delta r = 0.03599$.

**The Gravitational Hierarchy Problem and the Topological Instanton Bridge.** The electrostatic repulsion between two protons exceeds their gravitational attraction by a factor of $1.24 \times 10^{36}$; for two electrons, the ratio is $4.17 \times 10^{42}$ ($\alpha_{G,e} \equiv \frac{Gm_e^2}{\hbar c_0} = 1.75181 \times 10^{-45}$, $\alpha_{G,p} \equiv \frac{Gm_p^2}{\hbar c_0} = 5.90615 \times 10^{-39}$):

$$\frac{F_{\mathrm{em}}^{(ee)}}{F_g^{(ee)}} = \frac{e^2}{4\pi\epsilon_0 G m_e^2} = \alpha_{\mathrm{RefG}}\left(\frac{M_{\mathrm{Pl}}}{m_e}\right)^2 = 4.16561 \times 10^{42}, \qquad \frac{F_{\mathrm{em}}^{(pp)}}{F_g^{(pp)}} = \alpha_{\mathrm{RefG}}\left(\frac{M_{\mathrm{Pl}}}{m_p}\right)^2 = 1.23555 \times 10^{36}. \tag{115}$$

This large dimensionless ratio motivated Paul Dirac's Large Numbers Hypothesis [Dirac, 1937].

RefG associates this fundamental hierarchy with the different physical structures of the two interaction channels and with nonperturbative topological exponentiation:

1. **The electromagnetic channel** carries a direct phase-transverse gauge coupling between the topological charges of oscillons ($\alpha_0 = \frac{Z_0}{2R_K} \approx 1/137.036$);

2. **The gravitational channel** is the long-range refractive response to the Bernoulli-type static pressure deficit generated in the surrounding medium by an oscillon's internal oscillatory energy ($-\Delta P \propto \rho_{\mathrm{dyn}}$), where the **Planck mass** ($M_{\mathrm{Pl}} = \sqrt{\hbar c_0/G} \approx 2.17643 \times 10^{-8}$ kg, $M_{\mathrm{Pl}}c_0^2 \approx 1.22089 \times 10^{19}$ GeV) is the natural calibration scale for the elastic response of the foundation, while the **electron and proton** are topologically confined oscillon modes ($m_e/M_{\mathrm{Pl}} = 4.18546 \times 10^{-23}$);

3. **2-Loop Topological Instanton Action ($S_{\mathrm{inst}}^{\mathrm{RefG}} = \frac{3}{8}\alpha_{\mathrm{RefG}}^{-1} + \frac{7}{16\pi} + \frac{1}{9\pi^2\alpha_{\mathrm{RefG}}^{-1}} = 51.527842$, 2.50 ppm) and Parameter-Free Derivation of Newton's Constant $G$ ($G_{\mathrm{RefG}} = 6.674268 \times 10^{-11}$ $\mathbf{m^3\,kg^{-1}\,s^{-2}}$, $-0.21\sigma$):** Why the logarithmic hierarchy $\ln(M_{\mathrm{Pl}}/m_e) = 51.527840$ equals $\sim 51.53$ is accounted for by the nonperturbative topological tunneling action of the tri-axial

($N_{\text{axes}} = 3$) and 8-dimensional color ($\dim \mathfrak{su}(3)_c = 8$) nodal medium. Combining the leading Euclidean action $S_0 = \frac{N_{\text{axes}}}{\dim \mathfrak{su}(3)_c}\alpha_{\text{RefG}}^{-1} = \frac{3}{8}\alpha_{\text{RefG}}^{-1} = 51.388501$ (0.270%), the 1-loop fluctuation determinant shift across the $\dim \mathfrak{su}(3)_c - 1 = 7$ non-radial $SU(3)_c$ color-coset generators $\Delta S_1 = \frac{\dim \mathfrak{su}(3)_c - 1}{2 \dim \mathfrak{su}(3)_c\, \pi} = \frac{7}{16\pi} = 0.139260$, and the 2-loop ninth-order holonomy ($N_{\text{hol}} = 9$) shift $\Delta S_2 = \frac{1}{N_{\text{hol}}\pi^2\alpha_{\text{RefG}}^{-1}} = \frac{1}{9\pi^2\alpha_{\text{RefG}}^{-1}} = 0.000082$ yields without free parameters (verified across `18/18` checks):

$$\begin{aligned} S_{\text{inst}}^{\text{RefG}} &= \frac{3}{8}\alpha_{\text{RefG}}^{-1} + \frac{7}{16\pi} + \frac{1}{9\pi^2\alpha_{\text{RefG}}^{-1}} = 51.52784239 \\ \text{vs.} \quad \ln\!\left(\frac{M_{\text{Pl}}^{\text{exp}}}{m_e}\right) &= 51.52783989(110) \;\; (2.50 \text{ ppm}, \; +0.21\sigma). \end{aligned} \tag{116}$$

From this follow the predicted Planck mass $M_{\text{Pl}}^{\text{RefG}} = m_e \exp(S_{\text{inst}}^{\text{RefG}}) = 2.1764395 \times 10^{-8}$ **kg** $= 1.220893 \times 10^{19}$ **GeV**$/c_0^2$ (CODATA $2.176434(24) \times 10^{-8}$ kg, $+0.21\sigma$) and **Newton's gravitational constant**:

$$G_{\text{RefG}} = \frac{\hbar c_0}{\left(M_{\text{Pl}}^{\text{RefG}}\right)^2} = \frac{\hbar c_0}{m_e^2}\exp\!\left(-2\, S_{\text{inst}}^{\text{RefG}}\right) = 6.674268 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}, \tag{117}$$

matching the CODATA 2022 experimental value $G_{\text{exp}} = 6.67430(15) \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ to $-0.21\sigma$ **(4.78 ppm)** (with proton gravitational coupling $\alpha_{G,p}^{\text{RefG}} = \frac{G_{\text{RefG}} m_p^2}{\hbar c_0} = 5.90612 \times 10^{-39}$ and Grand Unified Action deformation scale $M_* c_0^2 = \left(\frac{\hbar^3 c_0^3 (a_0^{\text{RefG}})^2}{16\pi G_{\text{RefG}}}\right)^{1/4} = 0.6874$ meV, matching the sub-meV scale of the lightest neutrino mass $m_1 = 0.3608$ meV). Thus the 42-order-of-magnitude gravitational hierarchy $(M_{\text{Pl}}/m_e)^2 = \exp(2S_{\text{inst}}^{\text{RefG}}) \approx 5.71 \times 10^{44}$ is not an arbitrarily tuned large number, but the natural exponential readout of the moderate topological impedance $\alpha_{\text{RefG}}^{-1} \approx 137.036$.

Thus RefG interprets the hierarchy through the distinct structures of the two channels and topological instanton scaling: **electromagnetic coupling is the direct phase response of charge ($\alpha \approx 1/137.036$), whereas gravity is pressure-mediated refraction generated by microscopic oscillons ($m_e \sim M_{\text{Pl}} e^{-3/(8\alpha)}$).**

# 6 Ontological Foundations and the Boundaries of Knowledge

## 6.1 The Ontological Basis of a Unified Foundation-Medium

This section examines the ontological and physical basis for a unified foundation-medium.

In terms of ontological parsimony, a common carrier reduces the number of independent initial axioms and assumptions. A second argument concerns the **common origin of the regularities of nature**.

When the electron, quark, and photon are interpreted as different resonant and topological modes of one universal foundation-medium, physical interactions correspond to couplings among modes of a common carrier, while conservation of energy, momentum, spin, and charge expresses its symmetries and shared energy balance geometrically.

The historical context of this argument includes Eugene Wigner's essay on the "unreasonable effectiveness of mathematics" in the natural sciences [Wigner, 1960]. A common ontological carrier provides a physical interpretation of this regularity: **the shared mathematical regularities of the Universe reflect the internal dynamical structure of a unified foundation-medium.**

The concept of one foundation and one manifested medium relates the common symmetries and universal laws of different fields to the internal dynamics of their shared carrier. Correspondences among individual laws are thus interpreted as distinct manifestations of one fundamental structure.

The argument also has a precise empirical formulation in terms of **exact structural correspondences**:

- the exceptionally tight observational bound on the electrical neutrality of macroscopic matter ($|q_p + q_e|/e < 10^{-21}$);
- the observational compatibility of the photon rest mass with zero ($m_\gamma < 10^{-18}$ eV);
- the precise pattern of fractional quark charges [Navas et al., 2024].

These facts call for a common structural explanation, which RefG associates with the topological geometry of the modes of a single foundation.

One universal foundation can support several physical regimes: volumetric pressure, shear stress, phase, topology, vortical rotation, memory, and resonance. This approach is related to Philip Anderson's principle, **"More is Different"**: each new level of organization has physical regularities of its own [Anderson, 1972]. RefG describes transitions between these levels through the self-organization of a multichannel foundation.

RefG describes the measurable cosmological Universe as a **self-measuring global resonant network**. The foundation itself is pre-spatial and is assigned no spatial size. A finite, connected periodic network with no unique center provides a mathematical example of this possibility. The actual dimension, topology, finiteness, and boundarylessness of the Universe are fixed by global spectral and observational signatures. In this description, measurement standards are resonant modes of the system under investigation.

### 6.2 Three Operational Boundaries

The Universe is always measured by components of the Universe itself: our clocks, measuring rods, detectors, and even the observer's body are oscillon structures of the universal foundation-medium. This concept of a **self-measuring Universe** resonates with John Wheeler's *It from Bit* program [Wheeler, 1990]. Wheeler gives primacy to information; RefG gives primacy to the physical foundation-medium and reads the informational trace as a result of its self-distinction.

Here a *boundary* denotes the domain of validity of a measurement procedure. An observer within the Universe can measure only those relations defined in the active nodal network itself.

Kurt Gödel's incompleteness theorem [Gödel, 1931] serves here as an epistemological analogy: true but internally unprovable propositions in a consistent formal system provide an analogy for examining whether an internal part of a system can measure the global boundaries of the whole.

This self-measuring picture gives rise to three operational boundaries:

1. **The Operational Origin of Time:** In the selected scenario, time begins with the first foundation-wide self-distinction, assigned the boundary values $t = 0$ and $\tau = 0$. No physical past exists before this initial boundary, because neither change nor an ordering of events has yet been defined. Background parameters determine the post-Genesis history, while a unified relation among photons, atomic standards, and operational geometry makes this history accessible to an internal observer. On the selected integrable $p$ branch, the total process age is finite; its numerical value is fixed by the background parameters, Genesis matching, and clock calibration.

2. **The Operational Boundary of Space:** We measure the connections among distinguishable oscillon nodes. An exterior of the Universe is not a possible location of observation. The cosmological horizon is a measurement boundary defined from within the network. The quantity $R_{\rm act}$ is a local activation or correlation scale and does not denote an external radius of the Universe.

3. **The Operational Boundary of Scale and the 137-e-Fold Planck-to-BAO Spectral Bridge:** In RefG, the Planck length ($\ell_{\rm Pl} = 1.616255 \times 10^{-35}$ m) is the natural transition scale built from $c_0$, $\hbar$, and $G_{\rm RefG}$, while the minimum operational distinguishability threshold $\ell_* = 2\sqrt{\pi}\,\ell_{\rm Pl} \approx 3.5449\,\ell_{\rm Pl} = 5.7295 \times 10^{-35}$ m ($\ell_*^2 = 4\pi\ell_{\rm Pl}^2 = A_{\rm Pl}$) is uniquely fixed by Sakharov–Seeley–DeWitt heat-kernel induction across the $N_{\rm Casimir} = 24$ channels (§1.2). On the centerless periodic

3D nodal graph $\mathbb{T}^3_N = (\mathbb{Z}/N\mathbb{Z})^3$ ($[T_{\mathbf{m}}, \Delta_{\text{graph}}] = 0$), the continuum $s$-wave radial eigenfunction $j_0(kr) = \frac{\sin(kr)}{kr}$ ($\nabla^2 j_0 + k^2 j_0 = 0$) simultaneously governs the $n = 1$ Lane–Emden $Q_C$ core of galaxy clusters ($M_C = \frac{4}{\pi}\rho_{c0} R_C^3$) and the BAO acoustic horizon $r_d = 147.09$ Mpc $= 4.5387 \times 10^{24}$ m. Moreover, the total cosmic logarithmic span from $\ell_{\text{Pl}}$ to $r_d$ matches the inverse fine-structure impedance $\alpha^{-1}_{\text{RefG}} = 137.036$:

$$\ln\left(\frac{r_d}{\ell_{\text{Pl}}}\right) = 136.884 \approx \alpha^{-1}_{\text{RefG}} = 137.036 \ \ (0.111\%), \tag{118}$$

and partitions at the electron reduced Compton wavelength $\bar{\lambda}_{C,e} = \frac{\hbar}{m_e c_0} = 3.8616 \times 10^{-13}$ m into the exact $\mathfrak{su}(3)_c$ channel fractions $\frac{3}{8}$ and $\frac{5}{8}$: $\ln(\bar{\lambda}_{C,e}/\ell_{\text{Pl}}) = 51.528 \approx \frac{3}{8}\alpha^{-1}_{\text{RefG}} = 51.389$ (0.27%) and $\ln(r_d/\bar{\lambda}_{C,e}) = 85.356 \approx \frac{5}{8}\alpha^{-1}_{\text{RefG}} = 85.648$ (0.34%).

In all three cases, the boundary arises from the conditions of measurement internal to the Universe. A finite object may acquire an unbounded coordinate image, as in the stereographic projection of a sphere onto an infinite plane.

### 6.3 Methodological Principles

The methodological principle of this work is that **an intuitive physical picture must be transformed into a rigorous, computable, self-consistent, and falsifiable formalism.** Each particle hypothesis undergoes five successive stages of testing:

1. **Empirical Target:** mass, charge, spin, lifetime, form factor, and the relevant uncertainty are fixed in advance;
2. **Geometry and Dynamics:** the resonant-topological class is selected, and its spatial profile, boundary conditions, and nonlinear action are written explicitly;
3. **Symmetry and Stability:** Noether currents, quantum numbers, zero modes, the finite-energy solution, and the perturbation spectrum are tested within one system;
4. **Parameter Closure and Prediction:** free coefficients are fixed independently of the target data, after which a new measurable result is calculated;
5. **Decisive Test:** the prediction is compared with data under a predefined falsification criterion.

In this sequence, intuition sets the direction of research, while calculation establishes its quantitative consequences and testability. Every quantitative result is tied to a specific equation, the data used, and a reproducible computation.

The minimal complete unit of a particle derivation is one finite-energy oscillon. Its action determines the localized profile; the perturbation operator determines its eigenspectrum; the two-time dictionary determines coordinate and process frequencies; and an independent energy–mass map connects that spectrum to measurable rest and pole masses.

### 6.4 Falsification Criteria

Falsifiability is a basic requirement of a scientific theory. The falsification map of RefG is organized into three strict hierarchical levels:

**Falsification of the Entire Framework**
Lorentz violation, equivalence-principle violation, transposition failure

⇓

**Falsification of an Individual Sector**
wave budget, MOND/RAR, EHT

⇓

**Falsification of a Specific Geometric Reading**
Koide, $h/m$, $g = 2$

## 1. The Entire Framework Is Falsified If:

1. The oscillon sector fails to recover effective Lorentz invariance and spatial isotropy of light speed ($|\Delta c/c| > 10^{-17}$; Herrmann et al. 2009), or gravitational-wave propagation speed deviates from $c$ ($|v_{\rm gw} - c|/c > 10^{-15}$; Abbott et al. 2017).

2. At the same fixed total energy, inertial and gravitational mass exhibit a composition-dependent discrepancy exceeding observational bounds ($\eta > 10^{-15}$; MICROSCOPE: Touboul et al. 2022).

3. The global pressure evolution of the foundation-medium fails to yield a unique relation between process and metric time, or the cosmology built from that relation fails to recover the observed CMB, BBN, photon redshift, and Type Ia supernova $(1+z)$ time-dilation relations [Blondin et al., 2008].

4. A dimensionless ratio assigned to the common adiabatic-transposition class—for example $\alpha$—exhibits a measurable change under a local pressure shift ($|\Delta\alpha/\alpha| > 10^{-6}$; Kotuš et al. 2017).

5. Self-distinction of the foundation fails to establish a unified bridge to mutually distinguishable modes, local causality, and additive accounting of energy, momentum, and charge.

## 2. An Individual Sector Is Falsified If:

1. Gravitational-wave energy flux fails to reproduce the quadrupolar energy loss of binary pulsars to a precision of $10^{-3}$–$10^{-4}$ [Kramer et al., 2021], or scalar polarization exceeds LIGO-Virgo-KAGRA bounds.

2. The galactic vortex framework fails to reproduce SPARC/RAR data [McGaugh et al., 2016] or the observed spatial separation and lensing amplitude of the lensing centroids relative to the X-ray gas in the Bullet Cluster [Clowe et al., 2006].

3. A complete Einstein–Boltzmann calculation sourced by the neutral collective phase fails to recover the primary CMB anisotropies, or its late-time continuation fails to reproduce the BAO $100\,h^{-1}$ Mpc ruler and large-scale-structure statistics [Eisenstein et al., 2005].

4. An interior-core model for a compact object fails to match Event Horizon Telescope shadow diameters, the compact-branch oscillation spectrum, or gravitational-wave echo limits [Cardoso and Pani, 2019].

5. The neutral phase-density realization fails to admit an independent cyclic coordinate $\theta_C$, the common phase shift cannot be maintained as an exact symmetry, or the foundation and operational densities of one charge $Q_C$ fail the exact normalized Jacobian $\widehat{n}_{C,\rm op} = p^3 \widehat{n}_{C,F}$.

## 3. A Specific Geometric Reading Is Falsified If:

1. The electron oscillon profile fails to maintain a pointlike electromagnetic form factor down to $10^{-18}$ m [Navas et al., 2024].

2. The Koide/$C_3$ branch fails to construct a radiative bridge to pole masses, or a fourth stable charged lepton is discovered.

3. Galactic-flow circulation is not quantized in units of $h/m$ within an independently fixed effective-mass and coherence window.

4. The Möbius double-cover fails to yield the Dirac value $g = 2$, Fock-space antisymmetry, or the corresponding quantum corrections.

5. The calculated fluctuation spectrum of one oscillon fails to exhibit common $\Omega_t$ factorization; in that case, universal particle transposition and the mass ratios inferred from it are removed from the program.

Each criterion has a concrete observational or computational measure. Meeting a falsification criterion requires removing the corresponding idea from the research program; a change in wording alone does not resolve the failure.

## Conclusion

The central proposition of Refractive Gravity is that **geometry, matter, clocks, measuring rods, and light are different operational readouts of one state of the foundation.** Observers and measuring instruments are parts of the same physical system whose state they measure.

The Genesis picture preserves this unity from the beginning. The first foundation-wide self-distinction of the pre-spatial foundation produces event order, relation, and time. The resonant fields of localized oscillatory seeds become coupled, while stable self-confined structures form matter. An oscillon generates a static pressure deficit in its surroundings; that deficit jointly changes clock rates, the spatial standard, and the motion of bodies. Time dilation, scale transformation, and gravitational geometry are three operational manifestations of this single response.

This intuition has a precise mathematical bridge. Under specified post-Genesis continuum conditions, the leading action of the relational coframe takes TEGR form, and the torsion–curvature identity relates it to the Einstein–Hilbert action up to a boundary term. When matter couples universally and minimally to this one metric and no additional unsuppressed mode enters at 1PN order, the complete standard 1PN response follows, together with exact gravitational-wave propagation on the common light cone ($v_{\rm gw} \equiv c_0$), two transverse-traceless polarizations ($h_+, h_\times$), and quadrupolar orbital decay of binary pulsars (PSR B1913+16, PSR J0737-3039A/B [$0.74\sigma$], PSR J1738+0333 [$0.53\sigma$]) with zero dipole radiation ($\dot{P}_b^{\rm dipole} \equiv 0$). RefG extends Einstein's geometry with a physical ontology of the foundation and reads spacetime curvature as an operational manifestation of the foundation's state.

The unified ontological picture connects at different scales to its corresponding dynamical branches: in compact objects, the pressure-deficit interpretation connects to the horizonless exponential strong-field geometry, its action-derived $C^2$ regular core, and its Kerr-like rotating generalization (via the common-scale Ricci total-divergence theorem $\sqrt{-g}\,R \equiv 2\nabla^2 H$, with exact frame-dragging law $\omega_+(r) = \frac{3a_* c_0}{2m}[1 - (1 + \frac{2m}{r} + \frac{2m^2}{r^2})e^{-2m/r}]$, golden-ratio ISCO radius $r_{\rm ISCO}^{(0)} = (3+\sqrt{5})m = 2\phi^2 m$, EHT-compatible static and spinning shadows—53.45 $\mu$as [$0.68\sigma$] for Sgr A* and 40.76 $\mu$as [$0.41\sigma$] for M87*—eikonal quasinormal-mode quality-factor invariance $Q_{\ell n}^{\rm RefG} \equiv Q_{\ell n}^{\rm GR}$, finite tortoise-cavity gravitational-wave echo delay $\mathcal{I}_{\rm echo}(3/2) = 15.5553$, dynamical collapse arrest at $R = r_c$ with restoring stiffness $U''_{\rm eff}(1) = q(1+q) > 0$, causal boundary stiff-fluid fixation of $q^* = 3/2$ ($r_c = \frac{4}{3}m$, $n_{\rm min} = 3$) via $\bar{w}_{\rm m}(1^-) = \frac{c^2_{s,\rm eff}(1^-)}{c_0^2} = \frac{q}{3(2-q)} = 1$, and the zero-free-parameter $SU(3)_c$ Y-junction nucleon EOS yielding $M_{\rm TOV}^{\rm max} = 2.186\,M_\odot$, $R_{1.4} = 12.57$ km, $\Lambda_{1.4} = 586.9$, and Y-junction percolation $r_{\rm touch} \approx L_Y^{(N)}$); on galactic scales, coherent vortical flow, the cosmologically derived acceleration scale $a_0^{(\rm RefG)} = \frac{3}{5\pi} c_0 H_0$, and the covariant $k$-essence potential $\mathcal{F}(Z_H/a_0^2)$ (including the zero-parameter rational law $\mu_{\rm RefG}(x) = \frac{x+x^2}{1+x+x^2}$) simultaneously describe flat rotation curves, weak gravitational lensing, and Solar System screening, while in galaxy clusters the potential-depth threshold ($Q_C$) condensate core accounts for the cluster mass deficit and the Bullet Cluster weak-lensing profile; on the cosmological branch, expansion of the foundation lowers its background pressure. An internal observer measures this expansion through the material standard: the ratio $A = a/p$ unites redshift, signal time dilation, and cosmological distances within a single operational evolution. Introducing the complex collective field $\Psi_C = \sqrt{x}\,e^{i\theta_C}$ and the identifications $\Phi \equiv -\frac{c_0}{\mu_{C0}}\theta_C$ and $M_*^4 = \frac{a_0^2}{16\pi G}$ unites the isolated inertia sector $S_{\rm iso}$ ($\Psi_C \equiv 0$, $\Lambda_F = 0$), the

positive-density cosmological sector $S_C$ ($x > 0$), and the strong-field and galactic branches within one regular Grand Unified Covariant Action, while Sakharov–Seeley–DeWitt spectral heat-kernel induction on the discrete 3D nodal graph $\Gamma = \mathbb{Z}^3$ ($z = 6$, $N_{\text{Casimir}} = 24$) fixes the minimum distinguishability threshold $\ell_* = 2\sqrt{\pi}\,\ell_{\text{Pl}} \approx 3.5449\,\ell_{\text{Pl}}$ ($\delta_1^{\text{LIV}} \equiv 0$, $|\Delta c/c|_{100\ \text{TeV}} = 1.05 \times 10^{-28} \ll 10^{-17}$) and the 2-loop topological instanton action $S_{\text{inst}}^{\text{RefG}} = \frac{3}{8}\alpha_{\text{RefG}}^{-1} + \frac{7}{16\pi} + \frac{1}{9\pi^2\alpha_{\text{RefG}}^{-1}} = 51.527842$ derives Newton's gravitational constant $G_{\text{RefG}} = \frac{\hbar c_0}{m_e^2} e^{-2S_{\text{inst}}^{\text{RefG}}} = 6.674268 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ ($-0.21\sigma$, 4.78 ppm).

On the same $A$ geometry, the cold branch of the neutral collective phase current has exactly the background and linear behavior of a pressureless Einstein source. Under the standard local atomic/QED branch, gravity-only coupling to the plasma, and an adiabatic initial mode, the CMB initial-value problem takes the standard Einstein–Boltzmann, recombination, and line-of-sight form and quantitatively reproduces primordial nucleosynthesis coframe clock invariance ($\partial(dX_n/dA)/\partial p \equiv 0$, $\Delta N_{\text{eff}}^{\text{RefG}} \equiv 0$, $Y_p = 0.2456$, $(\text{D/H})_p = 2.512 \times 10^{-5}$), the matter–radiation equality epoch ($z_{\text{eq}} = 3402.88$), the coordinate ($t_0 = 13.797$ Gyr) and accumulated process-time ($\tau_0 = 24.580$ Gyr; $\tau_{\text{mat}}(z)/t_{\text{mat}}(z) = \frac{5}{3}(1+z)^{3/5}$; $\tau(10) = 3.28$ Gyr vs. $t(10) = 470.2$ Myr, assembling $M_* = 10^{10}\,M_\odot$ at $z = 10$ with normal baryon efficiency $\epsilon_*^{\text{RefG}} = 0.218 < 0.25$) ages, the sound horizon ($r_s(z_*) = 144.29$ Mpc), the BAO standard ruler ($r_d = 146.94$ Mpc $= 98.98\,h^{-1}$ Mpc $\approx 100\,h^{-1}$ Mpc), the CMB acoustic peaks ($\ell_1 = 220.8, \ell_2 = 538.0, \ell_3 = 811.6$), Thomson and Silk diffusion damping coframe invariance ($\partial\dot{\kappa}/\partial p = \partial(dk_D^{-2}/dA)/\partial p \equiv 0$, $k_{D,\text{eff}} = 0.1298\ \text{Mpc}^{-1}$, $\ell_{D,\text{eff}} = 1800.3$), baryon-loaded $TT/TE/EE$ angular power spectra ($R_{12} = 2.2616$, $R_{32} = 0.9899$, $\ell_{TE,\text{tr},1} = 164.5$), the Mészáros transfer function $T(k)$ and linear matter power spectrum $P(k,0)$ ($k_{\text{eq}} = 0.0154\,h\,\text{Mpc}^{-1}$, $k_{\text{peak}} = 0.0167\,h\,\text{Mpc}^{-1}$, $\sigma_8^{\text{RefG}} = 0.8098$ $[0.22\sigma]$, $S_8^{\text{RefG}} = 0.8300$ $[0.15\sigma]$), and linear structure growth ($D_+(A) = A\,{}_2F_1(\frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}}A^3)$, $\gamma_0 = \frac{6}{11}$, $f\sigma_8(0) = 0.4270$). RefG extends the physical interpretation of that established calculation by giving the geometry and the cold source a common foundation origin.

At the particle level, the volumetric resonator, common frequency scale, and topological invariants form a unified geometric map of mass, spin, charge, color, and generations. The dynamical bridge of this map is now in place: the complex phase action admits a finite-energy, charge-protected Q-ball-type core whose Bogoliubov–de Gennes (BdG) perturbation spectrum factorizes exactly on any static isotropic $(p_T, p_L)$ coframe by the common clock rate ($\nu_n^{(t)} = s_n(Q)\,\Omega_t(p_T; Q)$), while local $U(1)$ gauge coupling yields a screened $1/r^2$ Coulomb field, the charged oscillon first law $\left.\frac{dE_{\text{tot}}}{dQ}\right|_{q_{\text{em}}} = \Omega_\infty$, and both exact coframe invariance ($\alpha(p_T, p_L) = \alpha_0 = \frac{Z_0}{2R_K}$) and the topological-geometric impedance derivation ($\alpha_{\text{geom}}^{-1} = 4\pi^3 + \pi^2 + \pi = 137.036304$, $\alpha_{\text{RefG}}^{-1} = 137.035999092$, 0.061 ppb, $\hat{\alpha}_{\overline{\text{MS}}}^{-1}(m_Z) = 127.951$, $S_{\text{inst}}^{\text{RefG}} = 51.527842$, $G_{\text{RefG}} = 6.674268 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$) of the fine-structure and gravitational constants; on the same coframe, the Clifford algebra and Dirac–Fock–Ivanenko spinor action generate the $\text{Spin}(3,1)$ double cover, $4\pi$ Möbius holonomy, fermionic positivity, the exact tree-level gyromagnetic factor $g_0 = 2$, and exact $p_T = N$ clock transposition of Dirac and Larmor frequencies, generation-independent 1-loop QED self-energy cancellation ($\partial Z_1/\partial m_k \equiv 0$) for Compton-self-similar oscillon cores preserves the Koide $K = 2/3$ and $\theta_0 = 2/9$ structure at one-loop order and accounts for the two-loop QED radiative pole-mass shift $\Delta\theta_{\text{exp}}/\theta_0 = -c_2(\alpha_0/\pi)^2$, and the volume-preserving deformations of the three-axis internal triad together with 1/3-axial holonomy generate the exact $\mathfrak{su}(3)_c$ Gell-Mann algebra ($C_2(\mathbf{3}) = \frac{4}{3}, C_2(\mathbf{8}) = 3, C_2(\varepsilon_{abc}q^a q^b q^c) = 0$), the Cornell flux-tube confinement potential with first-principles strong coupling $\alpha_s(m_Z) = \theta_d(1 + \frac{1}{16}) = \frac{17}{144} = 0.118056$ ($+0.062\sigma$), dimensional transmutation $\Lambda_{\overline{\text{MS}}}^{(5)} = 209.34$ MeV $\approx K_0\mu_\ell^2 = 209.24$ MeV (0.048%), and string tension $\sqrt{\sigma_{\text{str}}^{\text{RefG}}} = \frac{73}{54}\mu_\ell^2 = 424.29$ MeV ($\sigma_{\text{str}} = 0.18002\ \text{GeV}^2$, 0.011%), anomaly-cancelled fractional quark charges $(+2/3, -1/3)$, the 120° Y-junction baryon mass origin (99.04% field energy, $M_0 = 1085.46$ MeV, $m_\Delta - m_N = +293.08$ MeV), $SU(6)$ nucleon magnetic moments and axial coupling ($\mu_n/\mu_p = -2/3$, $g_A^{\text{rel}} = 1.2716$), neutron–proton mass splitting ($\Delta m_{np} = +1.29333$ MeV $> m_e$), the electroweak $SU(2)_L \times U(1)_Y$ / beta-decay sector with massless-photon protection ($\det M_{\text{neutral}}^2 \equiv 0$) and neutron lifetime ($\tau_n^{\text{pred}} = 879.9$ s), and the harmonic $C_3$ holonomy sequence ($\theta_\ell = \frac{2}{9}, \theta_d = \frac{1}{9}, \theta_u = \frac{2}{27}$) yielding the six quark masses ($K_d = \frac{79}{108}, K_u = \frac{275}{324}$, $m_u, m_d, m_s, m_c, m_b, m_t$ within $\leq 0.65\sigma$), the unitary $3 \times 3$ CKM mixing matrix ($J_{\text{CP}}^{\text{CKM}} = 3.13 \times 10^{-5}$,

$0.09\sigma$), the PMNS leptonic mixing matrix ($\sin\theta_{13}^{\mathrm{PMNS}} = \frac{4}{27}, \sin^2\theta_{12}^{\mathrm{PMNS}} = \frac{25}{81}, \sin^2\theta_{23}^{\mathrm{PMNS}} = \frac{89}{162}, \delta_{\mathrm{CP}}^{\mathrm{PMNS}} = \pi + \frac{2}{9}$), and the 11th-order $C_3$ holonomy neutrino mass hierarchy ($\mu_\nu \simeq 3^{-11}\mu_\ell$, $\Delta m_{21}^2 = 7.46\times10^{-5}$ eV$^2$, $\Delta m_{31}^2 = 2.53\times10^{-3}$ eV$^2$, $\sum m_\nu = 0.0593$ eV $< 0.072$ eV); at the same time, its strict-vacuum quadratic action takes exactly the free complex-scalar form and yields the corresponding free quantum field under standard canonical quantization. Thus the coframe that produces Einstein–Hilbert geometry also carries standard bosonic and fermionic quantum matter and synchronous oscillon–atomic–spinor clocks on the same metric.

Thus RefG describes the Universe as a self-measuring resonant system. The state of the foundation determines operational geometry; this geometry determines the motion of matter and light; and clocks and measuring rods made from the same foundation provide internal measurements of these processes. In this description, gravitation, cosmological evolution, the structure of matter, and the origin of measurability are considered within one physical system.

## Current Status of the Research Program

This section brings together the results obtained, their conditions of validity, and the physical questions that remain to be resolved.

1. **Ontological Foundation, Cosmic Genesis, and the Origin of Measurability**—The current Genesis scenario unites a centerless initial boundary common to the entire foundation, the first minimal relations, localization of oscillon seeds, overlap of resonant fields, coupling-mediated network connection, and the subsequent formation of stable oscillons. Formal checks (verified across `18/18` checks) separate the parameter domains, the causal ordering of events, and the conditional balance of energy-transfer signs after the initial event. On the finite periodic 3D nodal graph $\mathbb{T}_N^3 = (\mathbb{Z}/N\mathbb{Z})^3$ ($N^3$ vertices, coordination number $z = 2N_{\mathrm{axes}} = 6$), every discrete translation operator $T_{\mathbf{m}}$ commutes identically with the graph Laplacian ($[T_{\mathbf{m}}, \Delta_{\mathrm{graph}}] = 0$), and the diagonal heat kernel $K_t(\mathbf{n},\mathbf{n}) = N^{-3}\sum_{\mathbf{q}} e^{-\lambda(\mathbf{q})t}$ is strictly vertex-independent ($\partial K_t(\mathbf{n},\mathbf{n})/\partial\mathbf{n} \equiv 0$), combining global connectivity and the absence of a unique center within one exact mathematical construction; by contrast, strict radial ordering from a single site in a symmetric network creates a preferred center and falls outside the centerless scenario. Full nonperturbative dynamical simulation of the pre-geometric Genesis phase transition and its primordial tensor/scalar fluctuation spectrum remains a subsequent microscopic task.

2. **Inter-Node Separation, Minimum Distinguishability ($\ell_* = 2\sqrt{\pi}\,\ell_{\mathrm{Pl}}$), and Continuum Lorentz Invariance ($\delta_1^{\mathrm{LIV}} \equiv 0$)**—The principle of operational distinguishability has been formulated and closed quantitatively (verified across `18/18` checks): physical distance acquires meaning only after a distinguishable relation has formed at the minimum scale $\ell_*$. Integrating the quantum phase fluctuations of the foundation's $N_{\mathrm{Casimir}} = N_{\mathrm{axes}} \times \dim\mathfrak{su}(3)_c = 3\times8 = 24$ microscopic polarization channels up to the Brillouin-zone UV cutoff $\Lambda_{\mathrm{UV}} = \pi c_0/\ell_*$ via the Sakharov–Seeley–DeWitt heat-kernel expansion ($a_1 = R/6$) induces the Einstein–Hilbert action $\frac{c_0^3}{16\pi G} = \frac{N_{\mathrm{Casimir}}\hbar}{96\pi^2 c_0}(\frac{\pi}{\ell_*})^2 = \frac{\hbar}{4c_0\ell_*^2}$, uniquely fixing $\ell_*^2 = 4\pi\,\ell_{\mathrm{Pl}}^2 = A_{\mathrm{Pl}}$ and $\ell_* = 2\sqrt{\pi}\,\ell_{\mathrm{Pl}} \approx 3.544908\,\ell_{\mathrm{Pl}} = 5.7295\times10^{-35}$ **m** ($E_* = \hbar c_0/\ell_* = 3.4441\times10^{18}$ GeV). Because every relational link on $\Gamma = \mathbb{Z}^3$ is undirected, the exact lattice dispersion $\omega^2(\mathbf{k}) = \frac{4c_0^2}{\ell_*^2}\sum_{a=1}^3 \sin^2(k_a\ell_*/2)$ is strictly even in $k_a$, so all odd-order ($n = 1,3,5,\dots$) Lorentz-invariance-violating terms vanish identically ($\delta_1^{\mathrm{LIV}} \equiv 0$), while at $E_\gamma = 100$ TeV the quadratic dispersion is suppressed to $|\Delta c/c| = \frac{1}{8}(E_\gamma/E_*)^2 = 1.054\times10^{-28} \ll 10^{-17}$ ($E_{\mathrm{LIV},2} = \sqrt{8}\,E_* = 9.74\times10^{18}$ GeV $\gg 1.3\times10^{11}$ GeV).

3. **A Common Local Frequency Scale and BdG Spectral Factorization**—The relations $\Omega_t/\Omega_{t,0} = p_T = N$ and $d\tau = p_T\,dt$ provide the exact dictionary between coordinate and process time. In the weak-field isotropic limit and on the common-scaling branch ($p_T = p_L = p$), this factor coincides with the common operational factor $p$. By the Coframe Rescaling and Bogoliubov–de

Gennes (BdG) Spectral Factorization Theorem (verified across `38/38` checks), even when $(p_T, p_L)$ are independent on a static isotropic coframe ($ds^2 = -p_T^2 c_0^2 dt^2 + p_L^{-2}\delta_{ij}dx^i dx^j$), the proper spatial coordinate $\tilde{\mathbf{x}} = p_L^{-1}(\mathbf{x} - \mathbf{x}_0)$ factorizes both the stationary action at fixed Noether charge $Q$ ($\Omega_t = p_T c_0 \Omega_0(Q)$, $E_{\rm coord} = p_T E_0(Q)$, $R_{\rm coord} = p_L R_0(Q)$) and the BdG quadratic eigenvalue pencil: every normal-mode coordinate frequency satisfies $\nu_n^{(t)}(p_T, p_L; Q) = s_n(Q)\,\Omega_t(p_T; Q)$ with $s_n(Q) = \frac{\kappa_n(Q)}{2\pi\Omega_0(Q)}$ and $\frac{\partial s_n}{\partial p_T} = \frac{\partial s_n}{\partial p_L} \equiv 0$. On the `W3-58` benchmark core ($a = 1/4, \Omega_0 = 0.8, Q_0 = 190.40113622, E_0 = 177.26834051$), numerical spectral integration confirms the $\ell = 0$ phase and $\ell = 1$ translational zero modes together with the coupled radial breathing modes $\kappa_1 = 0.208040$ ($s_1 = \kappa_1/\Omega_0 = 0.260050$) and $\kappa_2 = 0.231624$ ($s_2 = 0.289530$), matching the coordinate operator to $2.14 \times 10^{-14}$, while core parity ($\langle n|\mathbf{x} - \mathbf{x}_0|n\rangle \equiv 0$) eliminates first-order gravitational-gradient shifts (suppressing quadratic tidal shifts at the Earth's surface to $(R_{\rm core}/L_{\rm grav})^2 \approx 1.19 \times 10^{-56}$). Deriving inter-species mass ratios and multi-oscillon population synchronization from discrete microdynamics remains a subsequent task.

4. **Oscillon Deficit, Size, Time, and Mass**—The metric filters and the first-order biconformal branch have been obtained symbolically. In general isotropic coordinates, the clock and ruler factors are distinct ($p_T = N$, $p_L = A_{\rm iso}^{-1}$), whereas in the weak-field limit ($p_T \simeq p_L \simeq p$) and on the exact common-scaling branch ($p_T = p_L = p$) operational size, process cycle rate, and externally inferred mass follow the same $p$ response. A minimal coframe-coupled complex $U(1)$ action, its analytic existence window, and a finite-energy Q-ball-type core with a radially nodeless profile on a fixed Minkowski coframe have been obtained, with convergent numerical results supporting orbital stability. Coupled solutions of this same canonical field and Einstein geometry already incorporate gravitational backreaction, equilibrium, and an oscillatory response to specified perturbations. Further physical identification concerns the exact relation between the core's internal energy and the pressure deficit, operational size, process cycle rate, rest mass, and pole mass.

5. **Relational Geometry, the Einstein–Hilbert Action, and 1PN**—In a static, asymptotically uniform, nonrotating, spherically symmetric weak field, the common clock–ruler response gives $ds^2 = p^2 c^2 dt^2 - p^{-2} d\boldsymbol{\sigma}^2$, $c_{\rm local} = c_0$, and $\gamma = 1$. A canonical gradient term, linear coupling to the active source density entering Gauss's law, and the absence of an additional unsuppressed term at the same order determine the additive logarithmic response $u = -\ln p$, from which $u = GM/(c^2 r)$ and $\beta = 1$ follow.

   The complete 1PN result belongs to the isolated TEGR coframe action $S_{\rm iso}$ ($\Psi_C \equiv 0$, $\Lambda_F = 0$), which unites with the cosmological phase sector $S_C$ ($x = |\Psi_C|^2 > 0$) into one regular covariant Lagrangian via the complex collective field $\Psi_C = \sqrt{x}\, e^{i\theta_C}$, and further unites with the strong-field and galactic medium sectors via $\Phi(x) \equiv -\frac{c_0}{\mu_{C0}}\theta_C(x)$ and $M_*^4 = \frac{M_{\rm Pl}^2 a_0^2}{2c_0^4} = \frac{a_0^2}{16\pi G}$ in the Grand Unified Covariant Action (where on any static common-scaling metric $B = e^{-2H}, A = e^{2H}$ one has $\widehat{Y} \equiv 1$, $\widehat{B}^{AB} \equiv \delta^{AB}$, and the exact total spatial divergence $\sqrt{-g}\,R \equiv 2\nabla^2 H$ to all orders in $H$, with a fractional deviation from the quadratic strong-field operator of $1.69 \times 10^{-46}$ at the $r = 2m, M = 1\,M_\odot$ photon sphere). The galactic, free-quantum, and CMB results retain their respective branch conditions; incorporating them into a unified dynamical description counts each relevant source once and consistently matches backgrounds and perturbations.

   The post-Genesis continuum description is built on a $3+1$ relational coframe, one operational metric, flat inertial transport, a local and reversible leading action in the constant-coefficient parity-even quadratic-torsion class, minimal phase coupling, the absence of unsuppressed mixed operators at the same order, and the inertial-gauge character of orientation. In the regular linear Minkowski limit of this class, the TEGR coefficient ratios follow. The exact torsion–curvature identity then converts the action to Einstein–Hilbert form up to a boundary term.

   On the positive, stable, and causal phase branch, $\rho_C > 0$, $\rho_C' > 0$, and $0 \le n_{C,\rm op}\rho_C''/\rho_C' \le 1$. Metric variation gives $T_{\mu\nu}^C$, while variation of $\theta_C$ gives a conserved current whose homogeneous operational reduction yields $n_{C,\rm op}A^3 = \text{const}$. The Jacobian $n_{C,\rm op} = p^3 n_{C,F}$ reads the same law

in foundation measure as $\eta_F a^3 = 1$. In the complete action, every physical sector enters the total $T^{\rm tot}_{\mu\nu}$ exactly once through its own metric variation. The Fierz–Pauli/Deser spin-2 self-coupling route independently checks the same Einstein–Hilbert operator.

The complete standard PPN vector follows through 1PN order when the full action consists of the Einstein–Hilbert operator and matter universally and minimally coupled to the same metric, the source is represented by one complete conserved stress–energy tensor, RefG and GR use the same local source, PPN gauge, background, and boundary conditions, and no additional unsuppressed field or operator enters at this order: $\gamma = \beta = 1$ and $\xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0$. Its domain comprises isolated, weak, slowly moving, and weakly self-gravitating sources.

On the discrete 3D nodal graph $\Gamma = \mathbb{Z}^3$ (verified across `18/18` checks), Sakharov–Seeley–DeWitt spectral heat-kernel induction across the $N_{\rm Casimir} = 24$ channels directly generates the Einstein–Hilbert operator ($\ell_* = 2\sqrt{\pi}\,\ell_{\rm Pl}$), suppresses higher-derivative $R^2$ curvature terms at a $1\,M_\odot$ compact core by $(\ell_*/r_c)^2 = 8.46 \times 10^{-76}$, and derives Newton's gravitational constant $G_{\rm RefG} = \frac{\hbar c_0}{m_e^2}\exp(-2S^{\rm RefG}_{\rm inst}) = 6.674268 \times 10^{-11}\ {\rm m^3\,kg^{-1}\,s^{-2}}$ ($-0.21\sigma$, 4.78 ppm) and the Grand Unified Action deformation scale $M_* c_0^2 = 0.6874$ meV. Full dynamical derivation of the constitutive function $\rho_C(n_{C,{\rm op}})$ during the pre-geometric Genesis transition remains a subsequent task.

6. **Inertia and Gravitational Mass**—For a finite-energy Lorentz-covariant localized solution satisfying the Laue boundary condition $\int T^{ij}\,d^3x = 0$ for all $i, j$, the Noether identity relates the inertial coefficient and the gravitational source to the same total energy of the system. In the canonical complex scalar oscillon action, this identity is established directly at the soliton level: boosting the rest solution $\Psi_O(t,\mathbf{x}) = \frac{1}{\sqrt{2}}\phi(r)e^{-i\Omega_0 t}$ ($E_0 = \Omega_0 Q + \frac{1}{3}T_{\rm grad}$) yields the exact relativistic dispersion $E(v) = \gamma E_0$, $\mathbf{P}(v) = \gamma\frac{E_0}{c_0^2}\mathbf{v}$, and $M_I = E_0/c_0^2$; the spatial translational zero mode $\xi_i(x) = -\partial_i\Psi_O$ has zero eigenvalue ($\mathcal{L}_+\partial_i\phi = 0$), and the radial virial operator $\frac{d^2E_\lambda}{d\lambda^2}\Big|_{\lambda=1,Q} = \frac{8\omega_0^2}{2\omega_0^2+I_4/Q}T_{\rm grad} > 0$ is strictly positive at fixed charge. At the same time, coupling to a collective field breaks into three exact regimes: (a) in an infinite bath ($c^2_{s,\infty} > 0$), a nonzero monopole coupling $g_{\rm int} \neq 0$ produces a $\delta\sigma \sim r^{-1}$ tail and linearly divergent energy ($E_{\rm dress}(R) \propto R$); (b) in the strict vacuum ($\rho_{C,\infty} = 0$), $E_{\rm ADM} = M_I c_0^2$ holds exactly, while in a gravitating trapped pocket ($Q_C < \infty$) it generalizes to the two-charge first law $dM_{\rm ADM} = \Omega\,dQ + \omega_C\,dQ_C$ and $M_I = M_{\rm ADM}(Q, Q_C)$. Microscopic derivation of a localized trapping potential from discrete nodal dynamics remains a subsequent task.

7. **Gravitational Waves, Local Unobservability, Binary Pulsar Quadrupole Radiation, and the Domain of the Composite Graviton**—On the isolated TEGR coframe action $S_{\rm iso}$ (verified across `18/18` checks), linearizing the tetrad $e^A{}_\mu = \delta^A_\mu + \frac{1}{2}\eta^{AB}h_{B\mu} + \frac{1}{2}\eta^{AB}\omega_{B\mu}$ eliminates the antisymmetric local Lorentz rotation $\omega_{\mu\nu}$ from the linear metric ($\partial g^{(1)}_{\mu\nu}/\partial\omega_{\alpha\beta} \equiv 0$) and reduces $G^{(1)}_{\mu\nu}$ in Lorenz gauge ($\partial^\mu\bar{h}_{\mu\nu} = 0$) to $\Box\bar{h}_{\mu\nu} = -\frac{16\pi G}{c_0^4}T_{\mu\nu}$, establishing exact luminal propagation $v_{\rm phase} = v_{\rm group} = c_0$ ($(v_{\rm gw} - c_0)/c_0 \equiv 0$, satisfying GW170817) and $10 - 4 - 4 = 2$ orthogonal transverse-traceless (TT) polarizations $(h_+, h_\times)$ with zero scalar/vector radiative modes. Whereas isotropic scalar clock–ruler shifts cancel identically between perpendicular Michelson interferometer arms ($\Delta\tau_x - \Delta\tau_y \equiv 0$), TT waves produce the differential response $(\Delta L_x - \Delta L_y)/L_0 = F_+h_+ + F_\times h_\times$. Averaging the TT projection tensor over the celestial sphere ($\langle\Lambda_{ij,kl}Q_{ij}Q_{kl}\rangle_{S^2} = \frac{2}{5}Q_{ij}Q^{ij}$) yields the Isaacson quadrupole power $P_{\rm GW} = \frac{G}{5c_0^5}\langle\dddot{Q}_{ij}\dddot{Q}^{ij}\rangle$, the symbolic Peters–Mathews eccentric enhancement factors $f(e) = \frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{(1-e^2)^{7/2}}$ and $g(e) = \frac{1+\frac{7}{8}e^2}{(1-e^2)^2}$, Peters' semi-major axis invariant $a(e)$, and zero dipole radiation ($\Delta s_{12} = 0 \Rightarrow \dot{P}_b^{\rm dipole} \equiv 0$). Without free parameters, this reproduces the orbital decay of PSR B1913+16 ($f(e) = 11.8568$, $T_{\rm coal} = 300.70$ Myr, $\dot{P}_b^{\rm obs,corr}/\dot{P}_b^{\rm RefG} = 0.9983 \pm 0.0017$, $1.03\sigma$), PSR J0737-3039A/B ($f(e) = 1.0515$, $\dot{P}_b^{\rm obs,corr}/\dot{P}_b^{\rm RefG} = 0.999953 \pm 0.000063$, relative difference $4.68 \times 10^{-5} < 10^{-4}$, $0.74\sigma$,

together with the simultaneous intersection of $R, \dot{\omega}, \gamma_E, r_{\rm Sh}, s_{\rm Sh}$), and PSR J1738+0333 ($0.53\sigma$), while on the operational cosmological background ($A = a/p = a^{5/2}$) the Mukhanov–Sasaki mode $\chi_k = A\, h_k$ obeys $\chi_k'' + (k^2 - A''/A)\chi_k = 0$ ($\alpha_M \equiv 0$, $D_L^{\rm GW}(z) \equiv D_L^{\rm EM}(z)$). Deriving the continuum coframe and its Lorentzian light cone from the discrete nodal graph remains a microscopic foundation task.

The Weinberg–Witten theorem applies to a Lorentz-covariant quantum field theory defined on a Minkowski background with a local, gauge-invariant microscopic $T_{\mu\nu}$. The composite-graviton map of RefG is based on a pregeometric foundation and a Lorentzian metric that emerges at low energy. If the completed microtheory satisfies all Weinberg–Witten premises simultaneously, this composite-graviton branch is excluded.

8. **Compact Objects, Horizonless Exponential Metric, Action-Derived $C^2$ Core, Rotating Kerr-Like Generalization, Golden-Ratio ISCO, EHT Shadow, QNM, Gravitational-Wave Echoes, Dynamical Collapse Arrest, Causal Fixation of $q^* = 3/2$, and $SU(3)_c$ Y-Junction Neutron-Star / Horizonless-Core Transition**—In the canonical oscillon–geometry sector, the nonadditivity of the total mass, a recovery response after small compression, and the formation of a light-trapping region under stronger compression have been obtained. On the static and stationary axisymmetric branches of the covariant medium (verified across `86/86`, `18/18`, `18/18`, and `18/18` checks), six exact analytical and numerical results have been established: (a) the horizonless exponential exterior metric $B(r) = e^{-2m/r}$, $A(r) = e^{2m/r}$ ($r \geq r_c$) with an outer photon sphere at $r_{\rm ph}^+ = 2m$ and critical shadow impact parameter $b_c^{\rm RefG} = 2em \approx 5.4366\, m$ ($\delta_{\rm sh}^{\rm RefG} = \frac{2e}{3\sqrt{3}} - 1 \approx +4.63\%$), which lies strictly within the $1\sigma$ EHT bounds for both **Sgr A*** ($\theta_g = 5.028\,\mu$as: static $d_{\rm sh}^{\rm RefG} = 54.67\,\mu$as, $0.85\sigma$; spinning at $a_* = 0.7, i = 30^\circ$, $\bar{b}_{\rm sh} = 5.3150\, m, d_{\rm sh}^{\rm RefG} = 53.45\,\mu$as vs. $48.7 \pm 7.0\,\mu$as, $0.68\sigma$; $\delta_{\rm SgrA^*} \in [-0.14, +0.05]$, $0.96\sigma$) and **M87*** ($\theta_g = 3.8\,\mu$as: static $d_{\rm sh}^{\rm RefG} = 41.32\,\mu$as, $0.23\sigma$; spinning at $a_* = 0.9, i = 17^\circ$, $\bar{b}_{\rm sh} = 5.3629\, m, d_{\rm sh}^{\rm RefG} = 40.76\,\mu$as vs. $42.0 \pm 3.0\,\mu$as, $0.41\sigma$; $|\delta_{\rm M87^*}| \leq 0.17$, $0.27\sigma$); (b) the pure-medium no-go theorem ($\mathcal{I}_{\rm met}(q) > 0$ on $1 < q < q_* \approx 1.9897$), proving that a pure medium without central matter ($T_{\mu\nu}^{\rm m} = 0$) cannot close a $C^2$ core without a boundary stress jump; (c) the action-derived $C^2$ regular core matched to a covariantly conserved central matter sector ($\nabla_\mu T^{{\rm m}\,\mu\nu} = 0$, $\widehat{Y} \equiv 1$, $QC_\ell(r_c) = 0$; at $q = 3/2$, $\rho_0^* \approx 0.66364$), where all central invariants are finite ($B(0) = e^{-q} > 0$, $A(0) = 1$, $K(0) < \infty$) and a radially stable inner light ring appears at $x_{\rm lr}(1.5) \approx 0.87058$; (d) global $C^2$ smoothness of the radial optical index $n_{\rm opt}(r) = \sqrt{A(r)/B(r)}$ across $r = r_c$, exact equality of the eikonal quasinormal-mode frequency and Lyapunov exponent ($\Omega_{\rm ph}^{\rm RefG} = \lambda_{\rm L}^{\rm RefG} = \frac{c_0}{2em} \Rightarrow Q_{\ell n}^{\rm RefG} \equiv Q_{\ell n}^{\rm GR}$), and strictly finite tortoise cavity depth ($L_{*,\rm ext}(q)/m = 2[e - e^q/q + {\rm Ei}(q) - {\rm Ei}(1)]$, $\mathcal{I}_{\rm echo}(3/2) = 15.5553$), predicting post-merger echo delay times of $\Delta t_{\rm echo} = 4.750$ ms for the GW150914 remnant ($M_f = 62\, M_\odot$; 0.766 ms for $10\, M_\odot$, 2.299 ms for $30\, M_\odot$); (e) on the rotating horizonless Kerr-like geometry, the exact gravitomagnetic frame-dragging law $\omega_+(r) = \frac{3a_* c_0}{2m}[1 - (1 + \frac{2m}{r} + \frac{2m^2}{r^2})e^{-2m/r}]$ (recovering Lense–Thirring $\frac{2GJ}{c_0^2 r^3}(1 - \frac{3m}{2r} + \dots)$ at $r \gg m$ and matching $C^2$-smoothly at $r = r_c$ to a regular core with $\omega(0)/(a_* c_0/r_c) = 0.6122, \omega'(0) = 0$, free of ring singularities and closed timelike curves), the exact golden-ratio static ISCO radius $r_{\rm ISCO}^{(0)} = (3 + \sqrt{5})m = 2\phi^2 m \approx 5.2361\, m$ ($R_{\rm ar,ISCO}^{(0)} = 6.3379\, m$, $\eta_{\rm acc}^{(0)} = 5.479\%$, rising to $r_{\rm ISCO}^{\rm pro} = 3.0616\, m, \eta_{\rm acc}^{\rm pro} = 11.988\%$ at $a_* = 0.9$), and dynamical collapse arrest at $y = R/r_c = 1$ ($F_{\rm rad}(1) = 0$, positive restoring stiffness $U_{\rm eff}''(1) = q(1+q) = 3.75 > 0$, $\omega_{\rm bounce} = \sqrt{q(1+q)} = 1.9365$ at $q = 3/2$, and singularity-free bounce $y_{\rm min} = 0.8126 > 0 \to y_\infty = 1.000$); and (f) the universal boundary equation-of-state and sound-speed law $\bar{w}_{\rm m}(1^-) = \frac{c_{s,\rm eff}^2(1^-)}{c_0^2} = \frac{q}{3(2-q)}$ ($\partial/\partial n \equiv 0$) for any $C^1$-matched core ($\ell_A'(1) = -q, \ell_B'(1) = +q$, $\rho_{\rm m} \propto (1 - x^2)^n$), from which causal Zel'dovich stiff-fluid boundary saturation ($\bar{w}_{\rm m}(1^-) = c_{s,\rm eff}^2(1^-)/c_0^2 = 1$) uniquely fixes without free parameters $q^* = 3/2 \iff r_c = \frac{4}{3}m = \frac{2}{3}r_{\rm ph}^+$ ($R_{\rm ar}(r_c) = \frac{4}{3}e^{3/4}m \approx 2.8227\, m$, $n_{\rm min} = 3$, $\rho_0^*(3/2) = 0.663640$, $f_{\rm core}^{(\rm m)}(3/2) = 5.863\%$), while the zero-free-parameter $SU(3)_c$ Y-junction nucleon EOS built from

the §5.4 constants ($a_2 = \frac{m_\Delta - m_N}{9} = 32.5644$ MeV, $a_3 = \frac{a_2}{\sqrt{3}} = 18.8011$ MeV, $c_3 = \frac{\alpha_s}{3} = 0.09840$) yields the TOV maximum neutron-star mass $M_{\rm TOV}^{\rm max} = 2.1856\,M_\odot$ ($R_{\rm max} = 10.95$ km, $+1.51\sigma$ from PSR J0740+6620), canonical radii $R_{1.4}^{\rm RefG} = 12.57$ **km** ($0.18\sigma$) and $R_{2.08}^{\rm RefG} = 11.84$ **km** ($0.56\sigma$), tidal deformability $\Lambda_{1.4}^{\rm RefG} = 586.9 \le 720$ ($k_2(1.4) = 0.1060$, GW170817), and proves that at the TOV instability point ($u_{c,\max} = 6.234, n_{b,\max} = 0.9975$ fm$^{-3}$) the half-inter-baryon spacing $r_{\rm touch} = 0.5004$ fm matches the nucleon Y-junction arm length $L_Y^{(N)} = 0.5150$ fm to 2.84%, driving geometric Y-junction percolation and collapse arrest at $q^* = 3/2$. Full 3+1D numerical-relativity binary merger simulations remain a subsequent computational task.

9. **Galaxy Rotation, Morphology, and Clusters**—The selected Ginzburg–Landau-type effective potential

$$W(g_v, g_N) = \frac{g_v^3}{3} + \frac{g_N g_v^2}{2} - a_0 g_N g_v$$

gives $g_N = g^2/(g + a_0)$ at equilibrium, and in the deep limit gives $g^2 \simeq a_0 g_N$ together with the baryonic Tully–Fisher relation $v^4 = GM_b a_0$. In a filtered SPARC sample of 2,788 points from 163 galaxies, with $\Delta V/V < 10\%$, $V_{\rm obs} > 20$ km s$^{-1}$, fixed $a_0$, $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bulge} = 0.7$, and no galaxy-by-galaxy parameter fitting, the residual root-mean-square scatter is 0.141 dex and the mean residual is $-0.012$ dex. This establishes compatibility between the selected effective map and the data.

In a QUMOND estimate on a $300 \times 300$ grid, with $a = 3$ kpc and $b = 300$ pc, the maximum correction to the radial acceleration relative to the algebraic map is 9.23% over $a \le r \le 5a$. The ideal logarithmic branch gives the lensing angle $\alpha = 2\pi v_f^2/c^2$; on this branch, rotation and lensing are two readings of the same effective potential.

At the covariant action level, the galactic branch is embedded through the observer's spatial gradient invariant $Z_H = c^4 \gamma^{\mu\nu} \nabla_\mu H \nabla_\nu H = |\nabla \Phi|^2$ and the $k$-essence potential $\mathcal{L}_{\rm gal} = -\frac{a_0^2}{8\pi G} \mathcal{F}(Z_H/a_0^2)$. The elementary potential $\mathcal{F}_1(x^2) = x^2 - 2x + 2\ln(1+x)$ yields $\mu_1(x) = \frac{x}{1+x}$, maintains strict ellipticity ($\lambda_\parallel/\lambda_\perp \in (1,2)$), and simultaneously produces $v_f^4 = GM_b a_0$ and $\gamma_{\rm PPN} = 1$ lensing $\alpha_\infty = 2\pi\sqrt{GM_b a_0}/c^2$; at the same time, the standard $n = 2$ member of the same family, $\mathcal{F}_2(x^2) = x\sqrt{1+x^2} - \operatorname{arsinh}(x)$ ($\mu_2(x) = x/\sqrt{1+x^2}$), suppresses the residual acceleration at Saturn's orbit ($r = 9.54$ AU) by six orders of magnitude ($\delta g_{\mu_2} \approx 1.10 \times 10^{-16}$ m/s$^2$), satisfying Cassini bounds. Furthermore, from the cosmological expansion and foundation-pressure relaxation laws of RefG ($A = a^{5/2}$, $P_F/P_{F,0} = A^{-6/5}$, $\Gamma_{P,0} = \frac{6}{5} H_0$), the galactic acceleration scale is derived without free parameters as $a_0^{\rm (RefG)} = \frac{\Gamma_{P,0} c_0}{2\pi} = \frac{3}{5\pi} c_0 H_0 \approx 1.2506 \times 10^{-10}$ m/s$^2$ (+4.22% from the empirical SPARC value $1.20 \times 10^{-10}$ m/s$^2$), while completing the vortex fraction to the degree-2 unit-coefficient geometric polynomial ($\frac{g_v}{g} = \frac{1}{1+x+x^2}$) yields the zero-parameter rational law $\mu_{\rm RefG}(x) = \frac{x+x^2}{1+x+x^2}$ with exact closed-form potential $\mathcal{F}_{\rm RefG}(x^2) = x^2 - \ln(1+x+x^2) + \frac{2}{\sqrt{3}}[\arctan(\frac{2x+1}{\sqrt{3}}) - \frac{\pi}{6}]$, which simultaneously satisfies Saturn orbital ephemerides ($\delta g_{\rm Sat} \approx 2.40 \times 10^{-16}$ m/s$^2$) and achieves RMS = 0.14878 dex $< 0.15$ dex across the 2,788 quality-filtered points of 163 SPARC galaxies at the cosmologically derived $a_0^{\rm (RefG)}$. Doppler cancellation, $c_{\rm coord}/c_0 = p^2$, follows from symbolic algebra.

The morphological map associates disk and spiral structure with ordered flow, and elliptical structure with a more isotropic or chaotic regime. The cluster mass budget combines three channels, each counted once: a long-wavelength nodal deficit, the remote pressure fields of galaxies, and cluster-scale vortical memory. From the Thomas–Fermi equilibrium of the Grand Unified Covariant Action follows the exact potential-depth threshold law ($\sqrt{|\Phi(r)|} > v_{\rm trap} \equiv c_0\sqrt{1 - \omega_C/\mu_{C0}} \approx$ 500 km/s): all 163 SPARC galaxies ($\max V_{\rm obs} = 383.0$ km/s) remain strictly sub-threshold ($Q_C = 0$), whereas rich galaxy clusters cross the threshold ($Q_C > 0$) and form an analytical $n = 1$ Lane–Emden condensate core $\rho_C(r) = \rho_{c0} \frac{\sin(k_C r)}{k_C r}$. In the two-component 2D lensing model of the Bullet Cluster (1E 0657-558), these collisionless trapped $Q_C$ pockets reproduce

both weak-lensing convergence peaks directly at the galaxy centroids ($x_{\rm main} = -347.0$ kpc, $\kappa \approx 0.439$ and $x_{\rm sub} = +354.0$ kpc, $\kappa \approx 0.316$, displaced 187 kpc and 154 kpc ahead of the ram-pressure-stripped X-ray gas). Subsequent microscopic tasks include deriving the threshold scale $v_{\rm trap}$ and the geometric polynomial weights from discrete foundation microdynamics, testing circulation quantization $\Gamma = nh/m$, and performing full 3D time-dependent hydrodynamical and condensate merger simulations. Primary CMB anisotropy, the BAO $100\,h^{-1}$ Mpc ruler, and large-scale-structure statistics test the unity of this galactic map with the cosmological phase source.

10. **Cosmological Relaxation, the Phase-Current Source, the Time Bridge, and the Arrow of Time**—The ratio $A = a/p$ is the internal geometric readout of one expansion–pressure–standard history. In the post-Genesis homogeneous foundation description, $\Psi_C = \sqrt{n_{C,F}}e^{i\theta_C}$ represents a neutral collective phase state. In the unified covariant action, this complex collective field $\Psi_C = \sqrt{x}\,e^{i\theta_C}$ unites the isolated sector $S_{\rm iso}$ ($\Psi_C \equiv 0$, $\Lambda_F = 0$) and the active cosmological branch ($x = n_{C,\rm op} > 0$) within one regular Lagrangian. Conservation of the Noether phase charge in a comoving three-dimensional foundation volume gives $\eta_F a^3 = 1$ exactly, while the same charge is read in the operational geometry as $n_{C,\rm op}A^3 = \text{const}$.

    In the action for a barotropic, isentropic, irrotational phase current, variation of $\theta_C$ gives $\nabla_\mu J^\mu = 0$, the barotropic energy function defines $p_C = n_{C,\rm op}\rho_C'(n_{C,\rm op}) - \rho_C(n_{C,\rm op})$, and metric variation gives the Hilbert stress–energy tensor $T^C_{\mu\nu}$. Here $p_C$ is the thermodynamic pressure of the phase sector, whereas $P_F$ is the operational readout of the background pressure of the foundation; their exact relation must be fixed by a constitutive equation derived from the microphysics of the foundation. On the selected homogeneous, isotropic branch comoving with the network, one obtains $P_F/P_{F,0} = a^{-3}$, $p = a^{-3/2}$, and $A = a^{5/2}$.

    On the same branch, the inverse map gives the exact relations $a = A^{2/5}$, $p = A^{-3/5}$, and $\eta_F = P_F/P_{F,0} = A^{-6/5}$. This is a unique reconstruction of the normalized relative scale. For an ideal-comoving pair, the fixed step count $N_{12}$ of the selected path records relational structure, whereas $Q_C$ records the conserved charge of the phase sector; the two records play independent physical roles.

    The same $A$ determines redshift, signal time dilation by $(1+z)$, the Hubble function, and geometric distances. At equal background parameters, these quantities form one operational equivalence class with the corresponding ideal observables of spatially flat ΛCDM; distinguishing physics must appear in source and clock microphysics, calibration, or inhomogeneous dynamics. In this background comparison, the two descriptions are statistically equivalent. A constant vacuum contribution enters the total $\Lambda_{\rm eff}$ term exactly once.

    The linear formal CMB bridge is closed on the selected one-metric, spatially flat, linear post-Genesis continuum and the cold phase branch with fixed energy per conserved charge, under the conditions that standard matter, the Maxwell field, neutrinos, and the local atomic/QED sector couple minimally to the common metric; local dimensionless atomic and QED ratios retain their standard values; $T^C_{\mu\nu}$ couples to the plasma only gravitationally; and the initial transfer mode is the standard unit-normalized adiabatic mode. The normalized densities of one conserved charge $Q_C$ obey $\widehat{n}_{C,\rm op} = p^3\widehat{n}_{C,F} = A^{-3}$, where $\widehat{n} \equiv n/n_0$ and $A(t_0) = a(t_0) = p(t_0) = 1$. This exact identity belongs to the homogeneous background, while perturbations evolve directly through the operational current. The relation $\rho_C = \mu_C n_{C,\rm op}$ gives the exact cold source $p_C = c^2_{s,C} = \sigma_C = 0$. Under these conditions, the once-only source registry, background equations, linear scalar dynamics, Thomson collisions, neutrino hierarchy, recombination, optical depth, and line-of-sight endpoint take the standard Einstein–Boltzmann form. The standard cold-dark-matter (CDM) computational sector may represent $T^C_{\mu\nu}$; no separate particle cold source is added. At this stage, $\Omega_{C0}$ is an input normalization for the CMB forward calculation.

    The resulting status is an **exact conditional handoff from** RefG **to the standard Einstein–Boltzmann, recombination, and line-of-sight CMB initial-value problem together**

**with quantitative verification of primordial nucleosynthesis (BBN), primary CMB/BAO/LSS observables, Silk diffusion damping, $TT/TE/EE$ angular power spectra, the linear matter power spectrum $P(k,0)$, $\sigma_8/S_8$, and high-redshift JWST galaxy assembly** (verified across `19/19`, `18/18`, and `18/18` checks): at the Planck 2018 benchmark parameters ($\omega_b = 0.02237, \omega_c = 0.1200, h = 0.6736, T_{\rm CMB,0} = 2.7255$ K, $N_{\rm eff} = 3.044$), the operational $A = a/p = a^{5/2}$ geometry yields exact coframe clock invariance of BBN weak freeze-out ($\partial(dX_n/dA)/\partial p \equiv 0$, $\Delta N_{\rm eff}^{\rm RefG} \equiv 0$, $X_n^{\rm freeze} = 0.15590$, $T_{\rm nuc} = 0.07792$ MeV, $Y_p^{\rm RefG} = 0.2456$ [$0.19\sigma$ from PDG $0.245 \pm 0.003$], and $(\mathrm{D/H})_p^{\rm RefG} = 2.512 \times 10^{-5}$ [$1.38\sigma$ from PDG $(2.547 \pm 0.025) \times 10^{-5}$]), the matter–radiation equality redshift $z_{\rm eq}^{\rm RefG} = 3402.88$ ($A_{\rm eq} = 2.9378 \times 10^{-4}, a_F(z_{\rm eq}) = 0.03866, p(z_{\rm eq}) = 131.58$; matching Planck $3402 \pm 26$ to $0.034\sigma$), the present constant-standard ($p_0 = 1$) coordinate expansion age $t_0 = 13.7965$ Gyr (matching Planck $13.787 \pm 0.020$ Gyr) and the accumulated process-time history $\tau_0 = \int_0^{t_0} p\, dt = 24.5795$ Gyr ($\tau_0/t_0 = 1.7816$, with matter-era ratio $\tau_{\rm mat}(z)/t_{\rm mat}(z) = \frac{5}{3}(1+z)^{3/5}$, giving $\tau(14.32) = 2.423$ Gyr vs. $t(14.32) = 285.7$ Myr [$8.48\times$], $\tau(13) = 2.631$ Gyr vs. $t(13) = 327.2$ Myr [$8.04\times$], and $\tau(10) = 3.277$ Gyr vs. $t(10) = 470.2$ Myr [$6.97\times$], so that a standard depletion time $\tau_{\rm dep} = 0.80$ Gyr converts 83.9% of convertible gas by $z = 10$ vs. 30.6% in $\Lambda$CDM and assembles $M_* = 10^{10}\, M_\odot$ in a $M_{\rm halo} = 3.5 \times 10^{11}\, M_\odot$ halo with normal efficiency $\epsilon_*^{\rm RefG} = 0.218 < 0.25$), the exact closed-form analytical sound horizon formula $r_s^{(\rm analyt)}(z) = \frac{2c_0}{H_0\sqrt{3\Omega_{m0}R_{b0}}} \ln\!\big([\sqrt{1+R_b(A)} + \sqrt{R_b(A)+R_b(A_{\rm eq})}]/[1+\sqrt{R_b(A_{\rm eq})}]\big)$ (matching numerical quadrature to $1.10 \times 10^{-10}$), the recombination sound horizon $r_s(z_*) = 144.29$ Mpc ($z_* = 1089.92$; 0.099% from Planck $144.43 \pm 0.26$ Mpc), the drag-epoch ($z_d = 1059.94$) BAO standard ruler $r_d \equiv r_s(z_d) = 146.94$ **Mpc** $= 98.98\, h^{-1}$ **Mpc** $\approx 100\, h^{-1}$ **Mpc** (0.100% from Planck $147.09 \pm 0.26$ Mpc), the comoving angular diameter distance $D_M(z_*) = 13869.09$ Mpc, the acoustic angular scale $100\theta_* = 1.04035$ (0.072% from Planck 1.04110), the fundamental multipole $\ell_A = \pi/\theta_* = 301.97$, and the first three acoustic peaks $\ell_1 = 220.85$ (0.39%), $\ell_2 = 537.97$ (0.087%), $\ell_3 = 811.55$ (0.093%), exact coframe clock invariance of Thomson differential optical depth $\dot\kappa(A) = c_0 X_e(A) n_{\rm H,0} \sigma_T A^{-2}$ and Silk photon diffusion damping ($\partial\dot\kappa/\partial p = \partial(dk_D^{-2}/dA)/\partial p \equiv 0$, $z(\kappa = 1) = 1090.39$, $z_{\rm peak} = 1079.35$, $\sigma_z = 85.76$, pure Silk scale $k_D^{\rm pure} = 0.14128$ Mpc$^{-1}$, and visibility-shell-weighted effective damping scale $k_{D,\rm eff}(z_*) = 0.12981$ **Mpc**$^{-1}$, $\ell_{D,\rm eff} = 1800.31$), baryon-loading zero-point shift $-R_b(z_*)\Psi_{\rm eff}$ ($R_b(z_*) = 0.6287$, boosting the 1st/2nd peak ratio by $2.0812\times$ over $R_b = 0$) and full $TT/TE/EE$ spectra ($\mathcal{D}_{\ell_1}^{TT} = 5753.6\, \mu\mathrm{K}^2$ [$0.54\sigma$], $\mathcal{D}_{\ell_2}^{TT} = 2544.0\, \mu\mathrm{K}^2$ [$0.17\sigma$], $\mathcal{D}_{\ell_3}^{TT} = 2518.2\, \mu\mathrm{K}^2$ [$0.04\sigma$], $R_{12} = 2.2616$ [0.48%], $R_{32} = 0.9899$ [0.09%], $EE$ peaks at $\ell_{EE} = 135.5, 395.0, 690.0$, and first $TE$ adiabatic anticorrelation trough at $\ell_{TE,\rm tr,1} = 164.5$, $\mathcal{D}_{\rm tr}^{TE} = -44.00\, \mu\mathrm{K}^2 < 0$), while for sub-horizon linear structure growth and the matter power spectrum $P(k, z = 0)$ ($k_{\rm eq} = 0.010385$ Mpc$^{-1} = 0.01542\, h$ Mpc$^{-1}$ [$0.0077\sigma$], Mészáros modes $U_1(y) = 1 + \frac{3}{2}y, U_2(y)$ with Hu–Sugiyama exponent $\alpha_{\rm HS} = 0.90187$, turnover peak at $k_{\rm peak} = 0.01668\, h\, \mathbf{Mpc}^{-1}$) it yields $\sigma_8^{\rm RefG} = 0.80975$ (0.17% / $0.22\sigma$ from Planck $0.8111 \pm 0.0060$), $S_8^{\rm RefG} = 0.83000$ ($0.15\sigma$ from Planck $0.832 \pm 0.013$), the exact Gauss hypergeometric solution $D_+(A) = A\, {}_2F_1(\frac{1}{3}, 1; \frac{11}{6}; -\frac{\Omega_{\Lambda 0}}{\Omega_{m0}} A^3)$ ($D_+(1) = 0.787923$, matching `DOP853` to $4.73 \times 10^{-14}$), the exact growth-index expansion $\gamma(\Omega_m) = \frac{6}{11} + \frac{15}{2057}(1 - \Omega_m) + \mathcal{O}((1-\Omega_m)^2)$ ($\gamma(0) = 0.55045, f(0) = 0.52728$), and $f\sigma_8(0) = 0.4270$ ($\chi^2/\mathrm{dof} = 1.179$ across 6 SDSS/BOSS/eBOSS benchmarks). The subsequent observational stage comprises connecting both the amplitude and scale dependence of the primordial $P_{\mathcal{R}}(k)$ to Genesis, fixing possible isocurvature and tensor initial conditions, deriving the microphysical normalization of $\Omega_{C0}$, specifying reionization, and evaluating the full lensed $TT/TE/EE$ likelihood.

The time bridge $d\tau = p\, dt$ relates coordinate and process time. Both begin with the first distinguishable relation at the boundary $t = 0$, $\tau = 0$; an integrable $p$ gives a finite process history. RefG associates the global relaxation of the foundation with the physical direction of the macroscopic arrow of time. Quantitative closure requires an entropy law and irreversible foundation dynamics that connect the thermodynamic, cosmological, and radiative arrows of time to one source.

The complete microphysical bridge for the cosmological sector comprises the nodal-oscillon origin of $\rho_C(n_{C,\mathrm{op}})$, its relation to $n_{C,F}$ and $P_F$, the normalization of $\Omega_{C0}$ and its connection to the derived cosmological–galactic relation $a_0^{(\mathrm{RefG})} = \frac{3}{5\pi} c_0 H_0$, the foundation origin of local atomic/QED ratios, the microscopic coframe–phase split, particle sources, concrete inhomogeneous solutions, the numerical value of $\Lambda$, Genesis matching, and a complete data model.

11. **Two-Level Time and Quantum Randomness**—The interpretive map provides a picture compatible with Bell correlations and delayed-choice facts: the internal coordination of the foundation belongs to one unified state, while propagating physical signals obey local causality. Three results determine the closure of this sector as a formal theory: derivation of the Born distribution $P = |\psi|^2$, exact preservation of the no-signaling condition, and a complete dynamics of measurement.

12. **Particles as Oscillon Modes and the Scalar Quantum-Field Bridge**—A local nonlinear amplitude–phase action for the oscillon and a finite-energy spherical Q-ball-type candidate core with numerical evidence of orbital stability have been obtained. The vacuum quadratic sector of the same action takes exactly the form of the free complex Klein–Gordon action. On a fixed Minkowski coframe, standard canonical quantization of this quadratic sector gives the free excitations about the strict vacuum a bounded-below, nonnegative free Hamiltonian, bosonic Fock space, particle and antiparticle sectors, a global $U(1)$ phase charge, microcausality, a propagator pole on the mass shell, the spin-0 Casimir, and a controlled Schrödinger limit. This is the first exact RefG bridge to standard quantum field theory within the same coframe that yields Einstein–Hilbert geometry.

    A particle description of the nonlinear localized core requires its collective quantization and identification as a one-particle, bound, or coherent state. Related physical questions include the dynamics of population interlocking, the full spectrum of stability and decay channels, experimental bounds on the proton-decay lifetime, tests of the common frequency factor and the $t/\tau$ dictionary, and the transition from rest mass to interacting pole mass. The origin of $\hbar$, canonical commutators, Hilbert space, the Born rule, and the vacuum choice from the foundation constitutes a separate quantum-foundation bridge. The global phase charge remains an internal $U(1)$ charge, and the free spin-0 quantum remains a free excitation about the strict vacuum.

13. **The Koide, $C_3$ Circulant Operator, and $2/9$ Radiative Pole-Mass Bridge**—In $x_k = \sqrt{m_k}$ space, the $3 \times 3$ Hermitian $C_3$-circulant operator $\widehat{M}_{1/2}(\mu, A_K, \theta)$ yields $K(A_K) = (2 + A_K^2)/6$ (strictly independent of $\mu$ and $\theta$) and reproduces $K = 2/3$ exactly for $A_K = \sqrt{2}$ (singlet–doublet energy equipartition). With $(A_K, \theta_0) = (\sqrt{2}, 2/9)$ and the electron pole mass as anchor, the tree-level formula misses the muon pole mass by $451.7\sigma$ ($m_\mu^{\mathrm{tree}} = 105.659414$ MeV) and the tau mass by $1.04\sigma$ ($m_\tau^{\mathrm{tree}} = 1776.9850$ MeV). By the Compton-Self-Similar Core 1-Loop QED Cancellation Theorem (verified across `41/41` checks), whereas a fixed point-particle cutoff $\Lambda_{\mathrm{fixed}}$ destroys the Koide relation at one loop by $8.53 \times 10^5 \sigma$, a self-confined RefG oscillon whose core radius scales with its Compton wavelength ($R_{\mathrm{core},k} = \xi_0 \hbar/(m_k c_0)$, $\Lambda_k = m_k/\xi_0$) has a strictly generation-independent 1-loop QED factor $Z_1(\xi_0) = 1 + \frac{\alpha_0}{\pi}(1 - \frac{3}{2}\ln \xi_0)$ ($\partial Z_1/\partial m_k \equiv 0$), leaving both $K = 2/3$ and $\theta_0 = 2/9$ 100% invariant at one-loop order. Consequently, the shift between the tree-level angle $\theta_0 = 2/9 = 0.2222222222$ rad and the empirical pole angle $\theta_{\mathrm{exp}} = 0.2222220471$ rad is of two-loop QED order: $\Delta\theta_{\mathrm{exp}}/\theta_0 = -7.8806 \times 10^{-7} = -c_2(\alpha_0/\pi)^2$, where $c_2 = 0.14606 \approx \frac{4}{27} = \theta_0 K_0 = 0.14815$ (even the zero-parameter $c_2^{(0)} = 4/27$ ansatz reduces the muon error by $69.9\times$ down to 0.14 ppm, while $c_2 = 0.14606$ reproduces $m_e, m_\mu$ exactly and predicts $m_\tau^{\mathrm{pole}} = 1776.9690$ MeV, $0.91\sigma$). Full nonperturbative closure requires deriving $A_K = \sqrt{2}$, $\theta_0 = 2/9$, and $c_2$ directly from the 3D anharmonic oscillon spectral operator.

14. **Spin, Pauli Exclusion, the Coframe Dirac $g = 2$ Theorem, and Fermionic Statistics**—On the relational coframe ${e^A}_\mu$, the Clifford algebra $\{\gamma^A, \gamma^B\} = 2\eta^{AB} I_4$ and Spin$(3,1)$ generators

$\Sigma^{AB} = \frac{i}{4}[\gamma^A, \gamma^B]$ generate the spatial $SU(2)$ spin algebra $\mathbf{S} = \frac{\hbar}{2}\mathbf{\Sigma}$ with Casimir invariant $\mathbf{S}^2 = \frac{3}{4}\hbar^2 I_4$ ($s = 1/2$) and the Möbius $Z_2$ double-cover rotation operator $U(\varphi, \hat{\mathbf{n}}) = \cos(\varphi/2) I_4 - i \sin(\varphi/2)(\hat{\mathbf{n}} \cdot \mathbf{\Sigma})$, which reproduces $U(2\pi) = -I_4$ sign reversal and $U(4\pi) = +I_4$ restoration ($\Delta\alpha = \pi$, $e^{i\pi} = -1$). Squaring the minimally coupled Dirac–Fock–Ivanenko coframe operator $S_{\text{Dirac}}[e^A{}_\mu, A_\mu, \Psi]$ and applying the 4D Clifford Gordon decomposition (verified across `41/41` checks) proves the exact tree-level gyromagnetic factor $g_0 = 2$ ($g_0 - 2 \equiv 0$, and with the 4-loop QED Schwinger series gives $|g_e|/2 = 1.00115965217557$, matching Penning-trap measurements to $5.02 \times 10^{-12}$), while on a static isotropic coframe ($p_T = N, p_L = A_{\text{iso}}^{-1}$) in proper ruler coordinates $\tilde{\mathbf{x}} = p_L^{-1}(\mathbf{x} - \mathbf{x}_0)$ and canonical spinor density $\widetilde{\Psi} = p_L^{-3/2}\Psi$, the Dirac Hamiltonian factorizes as $H_{\text{Dirac}}^{\text{coord}} = p_T\, H_{\text{Dirac}}^{(0)}(\tilde{\mathbf{x}})$, proving exact coframe clock transposition of Dirac eigenfrequencies ($\omega_D^{(t)} = p_T \omega_{D,0}$) and Larmor spin precession ($\omega_L^{(t)} = p_T \omega_{L,0}$) independently of $p_L$ (numerical precision $6.66 \times 10^{-16}$).

The bridge to fermionic statistics is established by an exact phase of $\pi$ under exchange of two identical states, Fock-space antisymmetry, and the Dirac Hamiltonian normal-ordering theorem: canonical anticommutation relations $\{d_r, d_s^\dagger\} = \delta_{rs}$ are necessary and sufficient to yield a positive-definite Fock Hamiltonian ($: H_D :_{\text{Fermi}} = \sum E_{\mathbf{p}}(n_b + n_d) \geq 0$) and Pauli exclusion ($(b_s^\dagger)^2|0\rangle = 0$), whereas bosonic commutation yields an unbounded-below spectrum. Constructing the 4-component Dirac spinor field $\Psi(x)$ directly via collective-coordinate quantization of a 3D nonlinear toroidal soliton profile (Finkelstein–Rubinstein quantization of a Skyrme–Faddeev–Hopf-type core) remains a subsequent microscopic task.

15. **Charge, the Photon, and the Electromagnetic Channel**—The following structural results have been obtained: a local $U(1)$ phase symmetry, two helicities, screening of the asymptotic radial $\sigma$-field component—which would make the energy grow linearly with radius if left unscreened—and, after screening, a surviving $1/r^2$ far field, flux accounting, and subtheorems for a conserved phase current. In the covariant action (verified across `38/38` checks), local gauge coupling $\partial_\mu \Psi_O \to D_\mu \Psi_O = (\partial_\mu - i q_{\text{em}} A_\mu)\Psi_O$ establishes the gauge-invariant phase gradient $\mathcal{D}_\mu \theta_O = \partial_\mu \theta_O - q_{\text{em}} A_\mu$, the exact Ward–Takahashi/Noether identity $J_{\text{em}}^\mu = q_{\text{em}} J_O^\mu$ ($Q_{\text{em}} = q_{\text{em}} Q$), a self-consistent radial Maxwell–Oscillon boundary-value solution with an exact $1/r^2$ Gauss flux (relative error $1.86 \times 10^{-10}$, finite electrostatic energy $E_{\text{EM}} = 0.0131213$, and total energy $E_{\text{tot}} = 177.448313$ at $q_{\text{em}} = 0.005, \Omega_\infty = 0.800$), and the exact Charged Oscillon First Law $\left.\frac{dE_{\text{tot}}}{dQ}\right|_{q_{\text{em}}} = 0.79999887 \approx \Omega_\infty$ (relative error $1.42 \times 10^{-6}$). Complete quantum-spinorial closure of the sector comprises topological quantization of the elementary charge, quantum electrodynamics (QED), and the Dirac fermionic sector.

16. **Color, Quarks, Gluons, Topological Normalization of Fractional Charges, Nucleon Mass, and Baryon Magnetic Moments**—The volume-preserving ($\det U = 1$) Hermitian deformations of the three-axis internal oscillon frame $(\mathbf{e}_r, \mathbf{e}_g, \mathbf{e}_b)$ span an $N_c^2 - 1 = 8$ dimensional traceless Hermitian space (verified across `17/17` checks) whose generators $T^A = \frac{1}{2}\lambda^A$ ($A = 1, \ldots, 8$) satisfy the exact $\mathfrak{su}(3)_c$ Lie algebra ($\text{Tr}(T^A) = 0$, $\text{Tr}(T^A T^B) = \frac{1}{2}\delta^{AB}$, $[T^A, T^B] = i f^{ABC} T^C$) and Jacobi identity, with fundamental and adjoint Casimir invariants $C_2(\mathbf{3}) = \frac{4}{3} I_3$ and $C_2(\mathbf{8}) = 3 I_8$. The two-body color factor $\langle T_1 \cdot T_2 \rangle = \frac{1}{2}[C_2(R_{\text{tot}}) - C_2(R_1) - C_2(R_2)]$ is strictly attractive in the meson singlet ($\mathbf{3} \otimes \bar{\mathbf{3}} \to \mathbf{1}$: $-\frac{4}{3} < 0$) and antisymmetric diquark ($\mathbf{3} \otimes \mathbf{3} \to \bar{\mathbf{3}}$: $-\frac{2}{3} < 0$) channels, and repulsive in the octet ($+\frac{1}{6} > 0$) and sextet ($+\frac{1}{3} > 0$) channels; for the totally antisymmetric baryon state $\varepsilon_{abc} q^a q^b q^c$, the total Casimir vanishes identically ($C_2(\mathbf{1}_{qqq}) = 3 \times \frac{4}{3} + 2 \times 3 \times (-\frac{2}{3}) = 0$). Combining short-distance 3D Yang–Mills spreading with long-distance 1D chromoelectric flux-tube channeling ($\sigma_{\text{str}} = \frac{\Phi_c^2}{2A_\perp} > 0$) yields the Cornell confinement potential $V_{\text{Cornell}}(r) = -\frac{4}{3}\frac{\alpha_s \hbar c_0}{r} + \sigma_{\text{str}} r$ ($\lim_{r\to\infty} F_{\text{Cornell}} = -\sigma_{\text{str}} < 0$). Moreover, partitioning a closed 3-axis loop ($B_{\text{baryon}} = 1$) into $N_c = 3$ symmetric axial segments assigns each quark segment $B_q = Y(Q_L) = \frac{1}{N_c} = \frac{1}{3}$, from which the Gell-Mann–Nishijima

relation $Q_{\rm em} = T_3 + \frac{Y}{2}$ and cancellation of all five quantum gauge-gravitational anomalies ($\mathcal{A}_{[SU(3)_c]^3} = \mathcal{A}_{[SU(3)_c]^2U(1)_Y} = \mathcal{A}_{[SU(2)_L]^2U(1)_Y} = \mathcal{A}_{[\rm Grav]^2U(1)_Y} = \mathcal{A}_{[U(1)_Y]^3} = 0$) uniquely fix the fractional quark charges $Q_u = +\frac{2}{3}, Q_d = -\frac{1}{3}$ ($Y(u_R) = +\frac{4}{3}, Y(d_R) = -\frac{2}{3}$) and integer hadron charges $Q_p = +1, Q_n = 0$. Furthermore (verified across `18/18` checks), the Fermat–Torricelli–Steiner 120° Y-junction equilibrium ($\sum_{i=1}^{3} \hat{\mathbf{e}}_i = \mathbf{0}$, $L_Y/L_V = \sqrt{3}/2$), the three-body Cornell potential $V_{qqq}(R) = -\frac{2}{\sqrt{3}}\frac{\alpha_s \hbar c_0}{R} + 3\sigma_{\rm str} R$, the Dirac cavity mode ($x_0 = 2.04278694$), and virial equipartition ($E_{\rm kin} = E_{\rm str} = \frac{1}{2}M_0 c_0^2$, $L_Y^* = 3r_q^* = 0.5950$ fm) prove that 99.04% **(proton) and** 98.78% **(neutron)** of the nucleon mass arises from confined relativistic kinetic and Y-junction field energy (valence current quarks $2m_u + m_d = 8.99$ MeV contribute only 0.96%), determine the $N$–$\Delta$ center of gravity $M_0 = \frac{1}{2}(m_N + m_\Delta) = 1085.46$ MeV, the chromomagnetic hyperfine splitting $m_\Delta - m_N = +293.08$ MeV $> 0$ ($\langle T_i \cdot T_j \rangle_{\bar{\mathbf{3}}} = -\frac{2}{3} < 0$), and the dressed quark scale $\bar{M}_q = 337.40$ MeV (0.43% from $m_p/\mu_p^{\rm exp}$); the 24-component $SU(6)$ spin–flavor wavefunctions yield $\mu_p = \frac{4}{3}\mu_u - \frac{1}{3}\mu_d$, $\mu_n = \frac{4}{3}\mu_d - \frac{1}{3}\mu_u$, the parameter-free ratio $(\mu_n/\mu_p)_{\rm RefG} = -\frac{2}{3} \approx -0.6667$ (exp. $-0.6850$, 2.67%; $\mu_p = +2.781\,\mu_N$, $\mu_n = -1.854\,\mu_N$), and the relativistic Gamow–Teller axial coupling $g_A^{\rm rel} = \frac{5}{9}(1 + \frac{2M_q}{E_q}) = 1.2716$ (exp. 1.2754, 0.30%); and the QCD+QED isospin balance closes $\Delta m_{np} = \Delta m_{np}^{\rm QCD} + \Delta m_{np}^{\rm QED} = +2.3731 - 1.0797 = +1.29333$ MeV $> m_e$ ($Q_\beta = +782.33$ keV) together with coframe clock–ruler invariance ($\mu_{pe} = m_p/m_e = 1836.152673$, $\Delta\mu_{pe}/\mu_{pe} \equiv 0$). Finally (verified across `18/18` checks), the tri-axial $\mathfrak{su}(3)_c$ holonomy fraction $\theta_d = \frac{1}{9}$ and the lepton scale $\mu_\ell^2 = 313.856$ MeV directly determine the $Z^0$-pole strong coupling $\alpha_s^{\rm RefG}(m_Z) = \theta_d\big(1 + \frac{1}{2\dim \mathfrak{su}(3)_c}\big) = \frac{17}{144} = 0.1180556$ ($+0.062\sigma$ from PDG $0.1180 \pm 0.0009$), the 2-loop $\overline{\rm MS}$ dimensional transmutation scale $\Lambda_{\overline{\rm MS}}^{(5),{\rm RefG}} = 209.34$ **MeV** ($-0.047\sigma$ from PDG $210 \pm 14$ MeV, matching $\Lambda_{\rm Koide} = K_0\mu_\ell^2 = \frac{1}{9}\sum m_\ell = 209.24$ MeV to 0.048%), threshold running $\alpha_s(m_b) = 0.2253$ ($+0.11\sigma$) and $\alpha_s(m_\tau) = 0.3253$ ($+0.88\sigma$), and the Cornell string tension $\sqrt{\sigma_{\rm str}^{\rm RefG}} = C_2(\mathbf{3})\big(1 + \frac{1}{N_{\rm hol}\dim \mathfrak{su}(3)_c}\big)\mu_\ell^2 = \frac{73}{54}\mu_\ell^2 = 424.29$ **MeV** $\implies \sigma_{\rm str}^{\rm RefG} = 0.180022$ **GeV**$^2$ = 912.29 **MeV/fm** (0.011% match to 0.1800 GeV$^2$, Regge slope $\alpha'_{\rm Regge} = 0.8841$ GeV$^{-2}$, and nucleon center $M_0^{\rm RefG} = 2\sigma_{\rm str}^{\rm RefG} L_Y^* = 1085.57$ MeV, 0.010%).

17. **The Electroweak $SU(2)_L \times U(1)_Y$ Action, Massless Photon Protection, Beta Decay, the Higgs, Six Quark Masses, the CKM and PMNS Mixing Matrices, and the Neutrino Mass Hierarchy**—Evaluating the covariant electroweak action $|D_\mu \Phi_H|^2$ on the radial spin-0 oscillon-Higgs condensate $\langle \Phi_H \rangle = \frac{1}{\sqrt{2}}(0, v_{\rm EW})^T$ ($v_{\rm EW} = (\sqrt{2}G_F)^{-1/2} = 246.21964$ GeV; verified across `17/17` and `18/18` checks) yields the $4 \times 4$ gauge-boson mass-squared matrix $M_{\rm EW}^2$ whose neutral $2 \times 2$ block has identically zero determinant ($\det M_{\rm neutral}^2 \equiv 0$, $Q_{\rm em}\langle \Phi_H \rangle = 0$), strictly protecting a massless photon ($m_\gamma^2 \equiv 0$) and fixing $m_W = \frac{1}{2}g_2 v_{\rm EW} = 80.3692$ GeV, $m_Z = \frac{1}{2}\sqrt{g_1^2 + g_2^2}\, v_{\rm EW} = 91.1876$ GeV ($g_2 = 0.652825, g_1 = 0.349939, \sin^2\theta_W = 0.223203$). In neutron beta decay ($n \to p + e^- + \bar{\nu}_e$, axial-segment orientation flip $d \to u$), integrating out the $W^-$ propagator yields the Fermi constant $\frac{G_F}{\sqrt{2}} = \frac{g_2^2}{8m_W^2} = \frac{1}{2v_{\rm EW}^2}$, the chiral $V - A$ current product, the exact analytical relativistic phase-space integral $f_0(\epsilon_0) = \frac{1}{60}(2\epsilon_0^4 - 9\epsilon_0^2 - 8)\sqrt{\epsilon_0^2 - 1} + \frac{1}{4}\epsilon_0 \ln(\epsilon_0 + \sqrt{\epsilon_0^2 - 1}) = 1.636098$ ($\epsilon_0 = 2.530988, Q_\beta = 782.333$ keV), and the Coulomb- and radiatively-dressed ($f_R = 1.71385, \Delta_R^V = 0.02426$) neutron lifetime $\tau_n^{\rm pred} = 879.9$ s (matching PDG $878.4 \pm 0.5$ s to 0.17%). Furthermore, extending the $C_3$ circulant mass operator $\sqrt{m_k^{(f)}} = \mu_f[1 + A_f \cos(\theta_f + 2\pi k/3)]$ ($K(A_f) = \frac{2 + A_f^2}{6}$) to the harmonic ninth-order holonomy sequence $\theta_\ell = \frac{\theta_0}{1} = \frac{2}{9}$, $\theta_d = \frac{\theta_0}{2} = \frac{1}{9}$ (0.84%), $\theta_u = \frac{\theta_0}{3} = \frac{2}{27}$ (0.43%, $\sqrt{C_2(\mathbf{3})}(\theta_d - \theta_u) \equiv \frac{\theta_0^2}{\sqrt{C_2(\mathbf{3})}} = \frac{2}{27\sqrt{3}}$) with topological color/charge amplitude shifts $A_d^2 = 2 + |Q_d| + \frac{\theta_d}{2} = \frac{43}{18}$ ($K_d = \frac{79}{108} \approx 0.731481$, 0.007% vs. PDG) and $A_u^2 = 2 + (Q_u + |Q_d|) + \frac{\theta_d + \theta_u}{2} = \frac{167}{54}$ ($K_u = \frac{275}{324} \approx 0.848765$, 0.025% vs. PDG; $K(c, b, t) = 0.6696 \approx \frac{2}{3}$), scale bridge $\mu_d^2 = 2(1 + \frac{1}{9\pi})\mu_\ell^2 = 0.649914$ GeV (1.1 ppm), and top Yukawa $y_t = 1 - \frac{\theta_0^2}{6} = \frac{241}{243}$ ($\mu_u^2 = 22.7703$ GeV) reproduces all six individual quark masses within

$\leq 0.65\sigma$: $m_u = 2.469$ **MeV** $(+0.63\sigma)$, $m_d = 4.536$ **MeV** $(-0.28\sigma)$, $m_s = 94.04$ **MeV** $(+0.07\sigma$, $m_s/m_d = 20.731)$, $m_c = 1.2664$ **GeV** $(-0.18\sigma)$, $m_b = 4.1800$ **GeV** $(+0.00\sigma)$, $m_t = 172.671$ **GeV** $(-0.06\sigma)$; constructs the unitary $3 \times 3$ CKM matrix ($\theta_{12}^{\mathrm{CKM}} = \frac{55}{243}$ rad, $|V_{us}| = 0.22441$ [$+0.14\sigma$], $|V_{cb}| = \frac{2}{27\sqrt{3}} = 0.04277$ [$+1.11\sigma$], $|V_{ub}| = 2\theta_\ell\theta_d\theta_u = \frac{8}{2187} = 0.003658$ [$-0.84\sigma$], $\delta_{\mathrm{CP}}^{\mathrm{CKM}} = \frac{\pi}{3} + \frac{1}{9} = 66.37^\circ$ [$+0.22\sigma$], $J_{\mathrm{CP}}^{\mathrm{CKM}} = 3.131 \times 10^{-5}$ [$+0.09\sigma$], $\alpha = 91.86^\circ, \gamma = 66.33^\circ, \beta = 21.81^\circ$); derives the unitary PMNS leptonic mixing matrix by dressing the cyclic $C_3$ tribimaximal eigenbasis with $\theta_0 = 2/9$ ($\sin\theta_{13}^{\mathrm{PMNS}} = K_0\theta_0 = \frac{4}{27}$ [$\sin^2\theta_{13} = \frac{16}{729}$, $-0.15\sigma$], $\sin^2\theta_{12}^{\mathrm{PMNS}} = \frac{1}{3} - \frac{1}{2}\theta_0^2 = \frac{25}{81}$ [$+0.13\sigma$], $\sin^2\theta_{23}^{\mathrm{PMNS}} = \frac{1}{2} + \theta_0^2 = \frac{89}{162}$ [$+0.16\sigma$], $\delta_{\mathrm{CP}}^{\mathrm{PMNS}} = \pi + \frac{2}{9} = 192.73^\circ$ [$-0.17\sigma$, $J_{\mathrm{CP}}^{\mathrm{PMNS}} = -0.00734$]); and establishes the 11th-order $C_3$ holonomy neutrino mass hierarchy ($\mu_\nu = 3^{-11}\frac{161}{162}\mu_\ell$, $M_R = 6.030 \times 10^{14}$ GeV $\approx \frac{14}{3}\alpha_{\mathrm{RefG}}^2 M_{\mathrm{Pl}}$, $\theta_\nu = \frac{2}{9} + \frac{\pi}{12} - \frac{1}{162}$, $A_\nu = \sqrt{2}$): $m_1 = 0.3608$ **meV**, $m_2 = 8.6419$ **meV**, $m_3 = 50.2675$ **meV**, $\Delta m_{21}^2 = 7.455 \times 10^{-5}$ **eV**$^2$ $(+0.22\sigma)$, $\Delta m_{31}^2 = 2.5267 \times 10^{-3}$ **eV**$^2$ $(+0.58\sigma)$, $\Delta m_{32}^2 = 2.4521 \times 10^{-3}$ **eV**$^2$ $(-0.03\sigma)$, and $\sum m_\nu = 0.05927$ **eV** $< 0.072$ **eV** ($m_\beta = 8.837$ meV, $m_{\beta\beta} = 3.878$ meV). Nonperturbative derivation of the $(1/9, 2/27)$ quark holonomy winding weights and the 11th-order neutrino leakage factor from the discrete 3D nodal resonator operator remains a subsequent microscopic task.

18. **The Fine-Structure Constant $\alpha$, Topological Impedance ($\alpha_{\mathrm{RefG}}^{-1} = 137.035999092$), Electroweak Running, and Coframe Invariance**—A nodal impedance interpretation ($\alpha_0 = \frac{Z_0}{2R_K}$), an exact coframe-invariance theorem (verified across `38/38` checks), and a topological-geometric impedance derivation (verified across `18/18` checks) have been obtained: on a static isotropic coframe ($p_T = N, p_L = A_{\mathrm{iso}}^{-1}$), constitutive reduction of the Maxwell action gives $\varepsilon_{\mathrm{eff}} = \varepsilon_0(p_T p_L)^{-1}$, $\mu_{\mathrm{eff}} = \mu_0(p_T p_L)^{-1}$, and $c_{\mathrm{coord}} = c_0 p_T p_L$, leaving the vacuum wave impedance $Z_{\mathrm{eff}} = \sqrt{\mu_{\mathrm{eff}}/\varepsilon_{\mathrm{eff}}} = Z_0$ and the fine-structure constant $\alpha(p_T, p_L) = \frac{Q_{\mathrm{em}}^2}{4\pi\varepsilon_{\mathrm{eff}}\hbar c_{\mathrm{coord}}} = \alpha_0$ strictly invariant ($\partial\alpha/\partial p_T = \partial\alpha/\partial p_L \equiv 0$) while ensuring that atomic Rydberg frequencies ($\nu_{\mathrm{Ryd}}^{(t)} \propto p_T m_0 c_0^2 \alpha_0^2/\hbar$) and internal BdG oscillon modes ($\nu_n^{(t)} = s_n p_T \Omega_0$) transpose with the same clock factor $p_T = N$. Integrating the three orthogonal normalized phase-space channels of the tri-axial toroidal Möbius $Z_2$ spinor oscillon $T_{\mathrm{osc}}^3/Z_2 = (S_{\mathrm{tor}}^1 \times S_{\mathrm{pol}}^1 \times S_{\mathrm{spin}}^1)/Z_2$—the 3D toroidal bulk $\mathcal{I}_3 = \frac{1}{2}(2\pi)^3 = 4\pi^3 = 124.025107$ (90.505%), the 2D poloidal–toroidal surface $\mathcal{I}_2 = \pi^2 = 9.869604$ (7.202%), and the 1D chiral Möbius holonomy loop $\mathcal{I}_1 = \frac{1}{2}(2\pi) = \pi = 3.141593$ (2.293%)—yields without free parameters the bare geometric Compton-core impedance $\alpha_{\mathrm{geom}}^{-1} = 4\pi^3 + \pi^2 + \pi = 137.03630378$ (2.223 ppm from CODATA, $\Delta_2(4x^2 + x + 1) = -15 < 0$). In the Thomson limit ($Q^2 \to 0$), including the 1-loop Casimir shift across the $N_{\mathrm{Casimir}} = 3 \times 8 = 24$ transverse spin–axis–color channels ($\alpha_{\mathrm{RefG,1L}}^{-1} = \alpha_{\mathrm{geom}}^{-1} - \frac{1}{24\alpha_{\mathrm{geom}}^{-1}} = 137.035999720$, 4.64 ppb) and the 2-loop radiative correction weighted by the active ninth-order $C_3$ holonomy fraction $\frac{N_{\mathrm{hol}}-1}{N_{\mathrm{hol}}} = \frac{8}{9} = 1 - \frac{\theta_0}{2}$ yields $\alpha_{\mathrm{RefG}}^{-1} = (4\pi^3 + \pi^2 + \pi) - \frac{1}{24\alpha_{\mathrm{geom}}^{-1}}\left(1 + \frac{8}{9}\frac{1}{\pi\alpha_{\mathrm{geom}}^{-1}}\right) = 137.0359990924$ (0.061 **ppb** / $0.40\sigma$ from CODATA 137.035999084(21), $|g_e(\alpha_{\mathrm{RefG}})|/2 = 1.00115965217800$ [2.59 ppt], $Z_0^{\mathrm{RefG}} = 376.730313644\,\Omega$ [0.061 ppb], $R_\infty^{\mathrm{RefG}} = 10{,}973{,}731.5667\ \mathrm{m}^{-1}$ [0.129 ppb]). Symbolic Feynman-parameter integrals ($\frac{1}{6}, -\frac{5}{18}, \frac{1}{30}$) establish Uehling screening ($\Delta\nu_{\mathrm{Uehling}}(2S_{1/2}) = -27.084$ MHz) and running to the electroweak $Z^0$-pole ($m_Z = 91.1876$ GeV): $\Delta\alpha_{\mathrm{lep}} = 0.0314967$, $\Delta\alpha_{\mathrm{tot}} = 0.0590883$, $\alpha_{\mathrm{OS}}^{-1}(m_Z^2) = 128.9388$ $(0.48\sigma)$, and $\hat{\alpha}_{\overline{\mathrm{MS}}}^{-1}(m_Z) = 127.9508$ (PDG $127.951 \pm 0.009$, $0.024\sigma$, $\Delta r = 0.03599$). Nonperturbative derivation of the $(4\pi^3, \pi^2, \pi)$ phase weights and $1/24$ Casimir residue from the 3D discrete nodal lattice spectral determinant remains a subsequent microscopic task.

19. **The Gravitational Hierarchy, 2-Loop Topological Instanton Action ($S_{\mathrm{inst}}^{\mathrm{RefG}} = 51.527842$), and Parameter-Free Derivation of Newton's Constant $G$**—The force ratios are $F_{\mathrm{em}}^{(ee)}/F_g^{(ee)} = \alpha_{\mathrm{RefG}}(M_{\mathrm{Pl}}/m_e)^2 = 4.16561 \times 10^{42}$ ($\alpha_{G,e} = 1.75181 \times 10^{-45}$) and $F_{\mathrm{em}}^{(pp)}/F_g^{(pp)} = \alpha_{\mathrm{RefG}}(M_{\mathrm{Pl}}/m_p)^2 = 1.23555 \times 10^{36}$ ($\alpha_{G,p} = 5.90615 \times 10^{-39}$). RefG associates the difference with the contrast between direct phase-transverse electromagnetic coupling ($\alpha_0 = \frac{Z_0}{2R_K}$) and a secondary pressure-refractive gravitational response ($-\Delta P \propto \rho_{\mathrm{dyn}}$), and closes the logarithmic scale

hierarchy $\ln(M_{\rm Pl}^{\rm exp}/m_e) = 51.527840(11)$ via the 2-loop topological instanton action of the tri-axial ($N_{\rm axes} = 3$) and 8-color ($\dim \mathfrak{su}(3)_c = 8$) nodal medium (verified across `18/18` checks): combining the leading Euclidean action $S_0 = \frac{3}{8}\alpha_{\rm RefG}^{-1} = 51.388501$ (0.270%), the 1-loop fluctuation determinant shift across the $\dim \mathfrak{su}(3)_c - 1 = 7$ non-radial color-coset generators $\Delta S_1 = \frac{7}{16\pi} = 0.139260$, and the 2-loop ninth-order holonomy shift $\Delta S_2 = \frac{1}{9\pi^2 \alpha_{\rm RefG}^{-1}} = 0.000082$ yields without free parameters $S_{\rm inst}^{\rm RefG} = \frac{3}{8}\alpha_{\rm RefG}^{-1} + \frac{7}{16\pi} + \frac{1}{9\pi^2\alpha_{\rm RefG}^{-1}} = 51.52784239$ (2.50 **ppm**, $+0.21\sigma$). This directly predicts the Planck mass $M_{\rm Pl}^{\rm RefG} = m_e \exp(S_{\rm inst}^{\rm RefG}) = 2.176440 \times 10^{-8}$ **kg** ($+0.21\sigma$), Newton's gravitational constant $G_{\rm RefG} = \frac{\hbar c_0}{m_e^2}\exp(-2S_{\rm inst}^{\rm RefG}) = 6.674268 \times 10^{-11}$ $\mathbf{m^3\,kg^{-1}\,s^{-2}}$ ($-0.21\sigma$**,** 4.78 **ppm** from CODATA 2022 $6.67430(15) \times 10^{-11}$), and the Grand Unified Action deformation scale $M_* c_0^2 = 0.6874$ meV.

20. **The Self-Measuring Universe, the Centerless Periodic Graph $\mathbb{T}_N^3$, and the 137-e-Fold Micro–Macro Spectral Bridge**—On the finite periodic 3D nodal graph $\mathbb{T}_N^3 = (\mathbb{Z}/N\mathbb{Z})^3$ ($[T_{\mathbf{m}}, \Delta_{\rm graph}] = 0$, $\partial K_t(\mathbf{n},\mathbf{n})/\partial \mathbf{n} \equiv 0$; verified across `18/18` checks), global connectivity and the absence of a preferred center are united within one exact mathematical construction. In the continuum limit, the $s$-wave radial Helmholtz eigenfunction $j_0(kr) = \frac{\sin(kr)}{kr}$ ($\nabla^2 j_0 + k^2 j_0 = 0$) simultaneously governs the macroscopic $n = 1$ Lane–Emden $Q_C$ core of galaxy clusters ($M_C = \frac{4}{\pi}\rho_{c0} R_C^3$) and the BAO acoustic ruler $r_d = 147.09$ Mpc $= 4.5387 \times 10^{24}$ m. Moreover, the total cosmic logarithmic span from the Planck length $\ell_{\rm Pl}$ to $r_d$ matches the inverse fine-structure impedance: $\ln(r_d/\ell_{\rm Pl}) = 136.884 \approx \alpha_{\rm RefG}^{-1} = 137.036$ (0.111%), and partitions at the electron reduced Compton wavelength $\bar{\lambda}_{C,e} = \hbar/(m_e c_0)$ into the exact $\mathfrak{su}(3)_c$ channel fractions $\frac{3}{8}$ and $\frac{5}{8}$: $\ln(\bar{\lambda}_{C,e}/\ell_{\rm Pl}) = 51.528 \approx \frac{3}{8}\alpha_{\rm RefG}^{-1} = 51.389$ (0.27%) and $\ln(r_d/\bar{\lambda}_{C,e}) = 85.356 \approx \frac{5}{8}\alpha_{\rm RefG}^{-1} = 85.648$ (0.34%). Observational searches for low-multipole CMB and cosmic-topology signatures remain a subsequent empirical test of the global periodic graph.

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